

**The data file AMMONIAEX
(based on AMUJEL format):
Additional Atomic and Molecular Data for EIRENE for
AMMONIA**

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Introduction

Record:

- April, 17: First version
- July 3, 17: Adding cross sections
- July 31, 17: Database update. Thanks to Ms S. TAMODARIN and Mr S. KETHESWARAN for their work.

to be done:

- To be done

Additional atomic data fits, read by EIRENE Specific reaction for AMMONIAC
Format as HYDHEL DATEN [2] or METHANE DATEN [1] or AMJUEL DATEN
SPECIFIC BIBLIOGRAPHY IS AVAILBLE

End of preface:

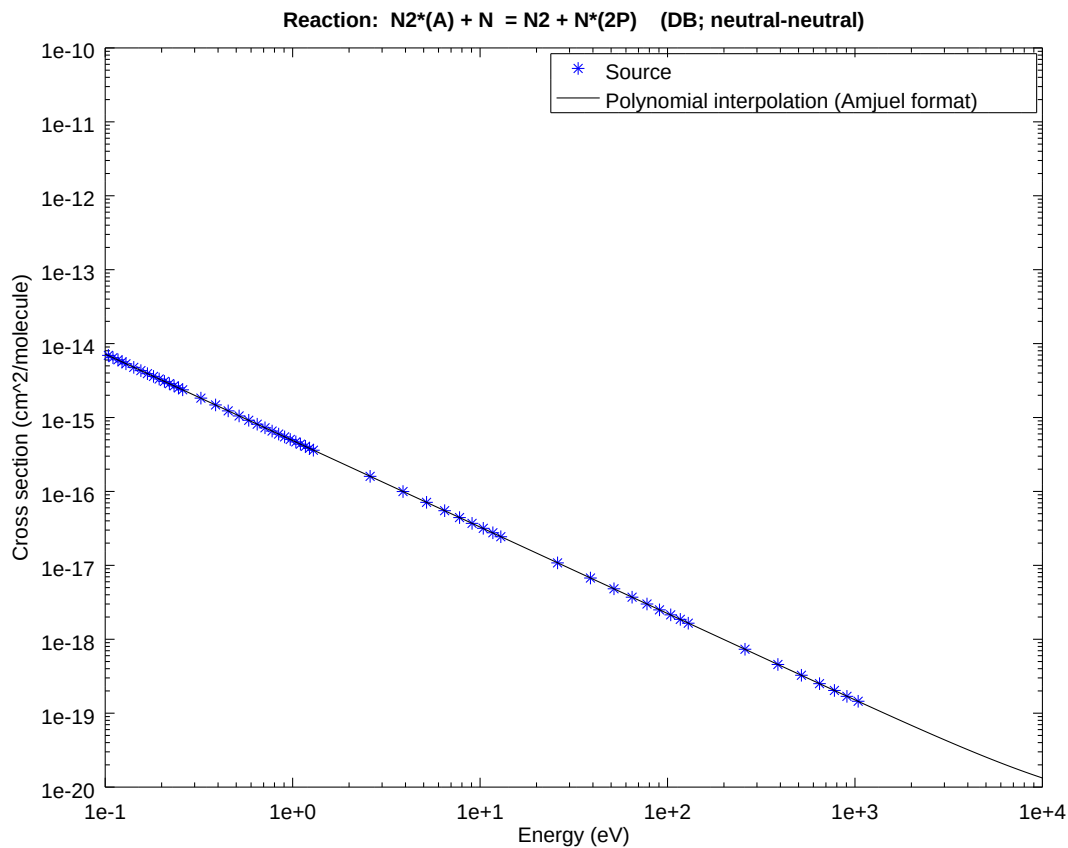
This next string is searched by EIRENE in subroutine SLREAC to initialize search for a particular set of fit coefficients. From here on, a character string ‘**H.n**’ , n an integer, must only appear in the section title, but not in the text. Likewise: identifiers p0, a0, b0, ...,h0, k0 are used in SLREAC and must not appear in the text elsewhere, from here on.

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.....  
.  
.      ##BEGIN DATA HERE##  
.  
.....
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1 H.1 : Fits for cross section

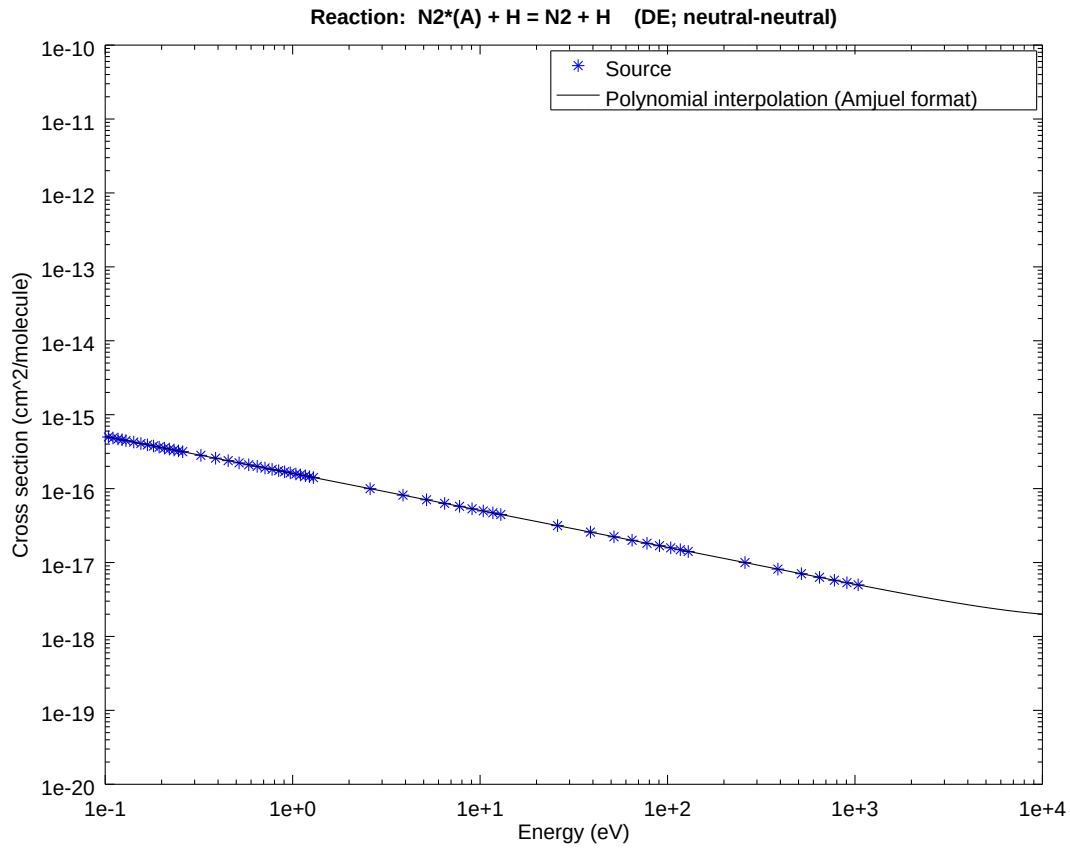
1.1 Reaction DB N2*(A) + N = N2 + N*(2P)

a0	-3.525604931656E+01	a1	-1.169467047039E+00	a2	-3.016619218433E-04
a3	-1.131053596850E-04	a4	5.075664503871E-05	a5	5.186112097513E-06
a6	-2.519799999214E-06	a7	4.025963385058E-08	a8	1.886635246743E-08



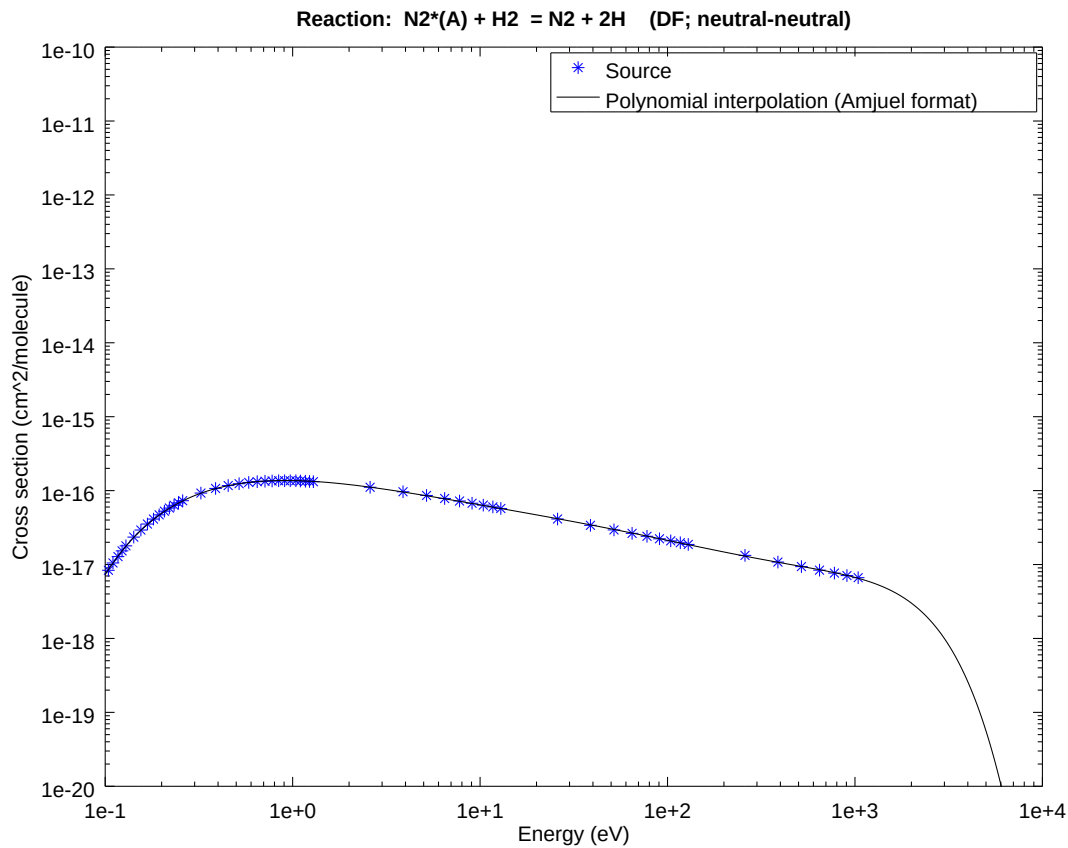
1.2 Reaction DE $\text{N}_2^*(\text{A}) + \text{H} = \text{N}_2 + \text{H}$

a0	-3.636724579729E+01	a1	-4.996921680518E-01	a2	-3.004238048443E-04
a3	-7.027601268270E-05	a4	4.639562577142E-05	a5	4.000298458661E-06
a6	-2.200508309113E-06	a7	1.127902891757E-08	a8	1.890104031525E-08



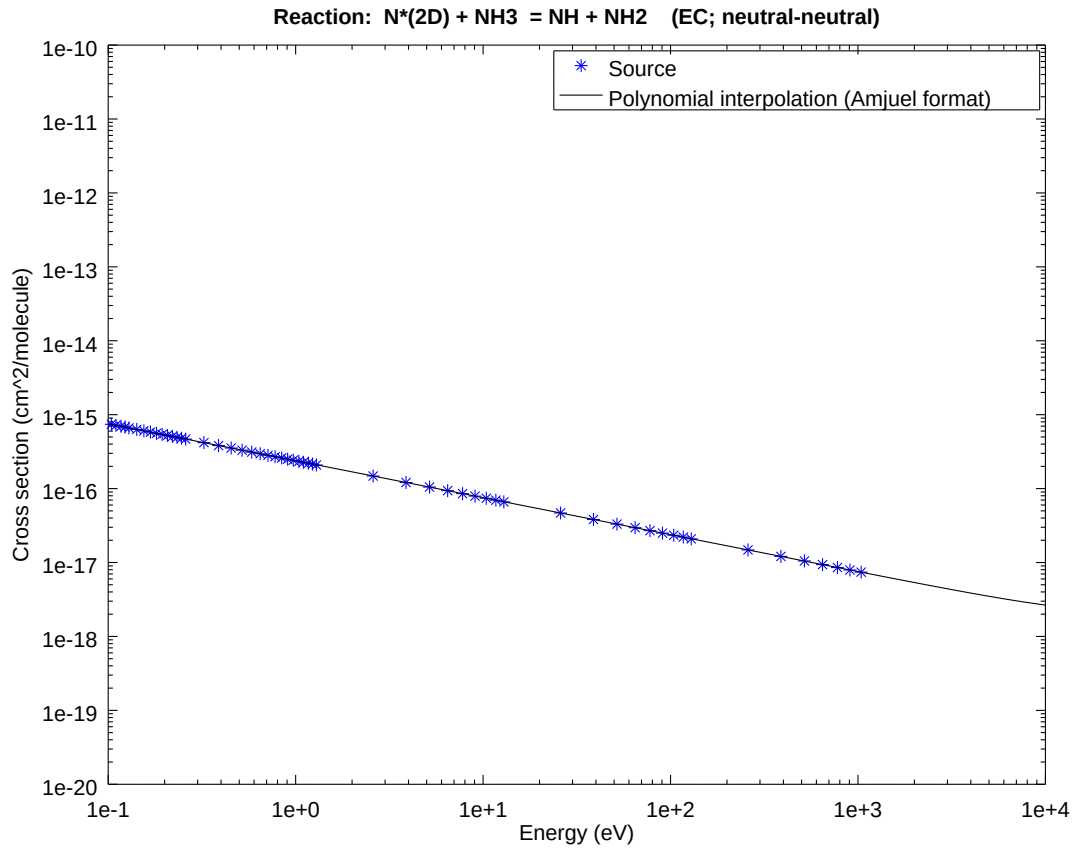
1.3 Reaction DF $\text{N}_2^*(\text{A}) + \text{H}_2 = \text{N}_2 + 2\text{H}$

a0	-3.652800397725E+01	a1	-4.578607537346E-02	a2	-2.399741909925E-01
a3	7.519012714366E-02	a4	-1.537136580981E-02	a5	3.557144220075E-03
a6	-9.047956443652E-04	a7	1.282478836314E-04	a8	-6.737601321279E-06



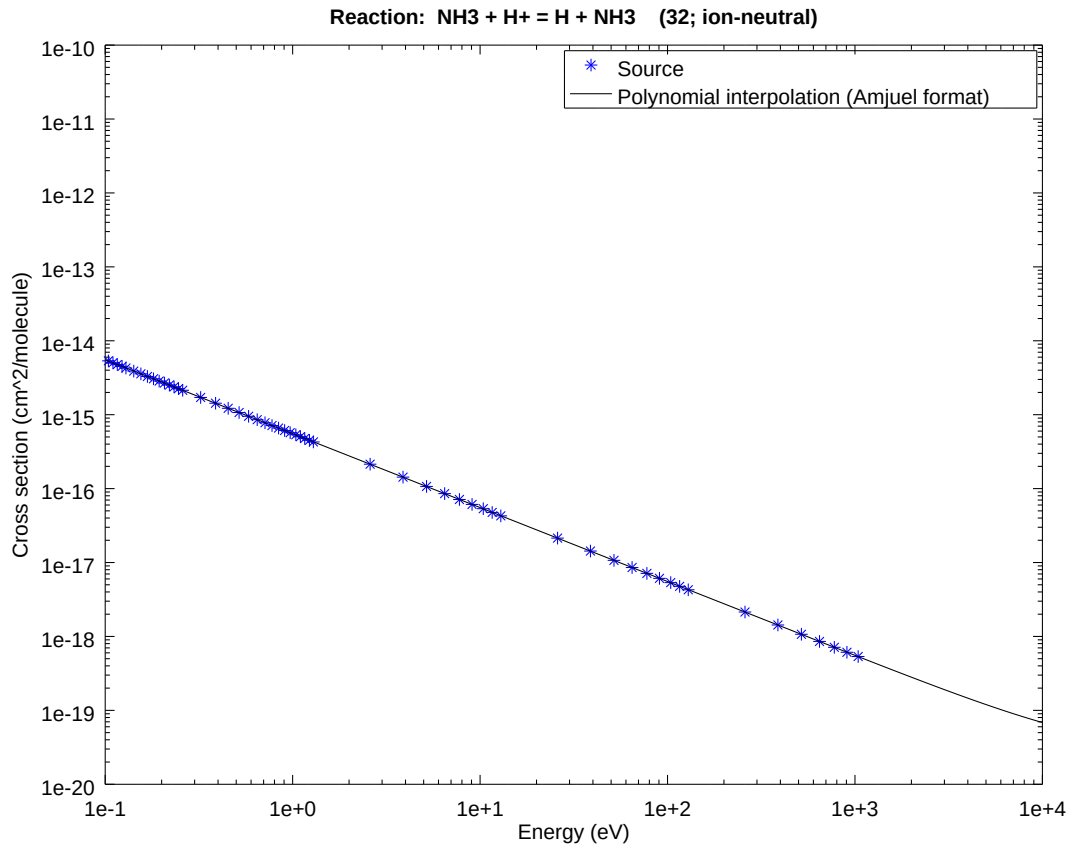
1.4 Reaction EC N*(2D) + NH3 = NH + NH2

a0	-3.597312484632E+01	a1	-4.992860132291E-01	a2	2.221724976694E-04
a3	-2.185431927439E-04	a4	-1.875235453436E-05	a5	1.801683905709E-05
a6	-5.631533912406E-07	a7	-4.004005884577E-07	a8	3.456994046962E-08



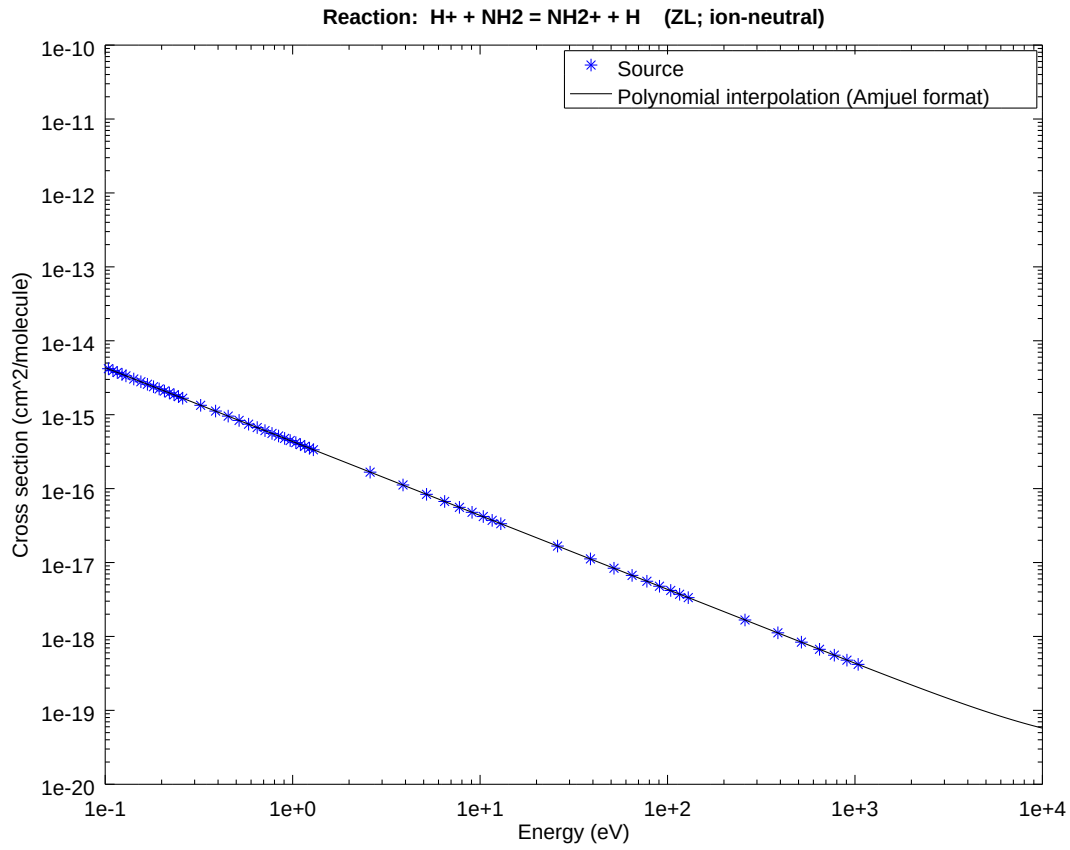
1.5 Reaction 32 $\text{NH}_3 + \text{H}^+ = \text{H} + \text{NH}_3$

a0	-3.512950843118E+01	a1	-9.995197673296E-01	a2	-1.089341894829E-04
a3	-1.324296798829E-04	a4	1.991180950364E-05	a5	1.021163226029E-05
a6	-1.537210572679E-06	a7	-1.883169541811E-07	a8	2.892314580928E-08



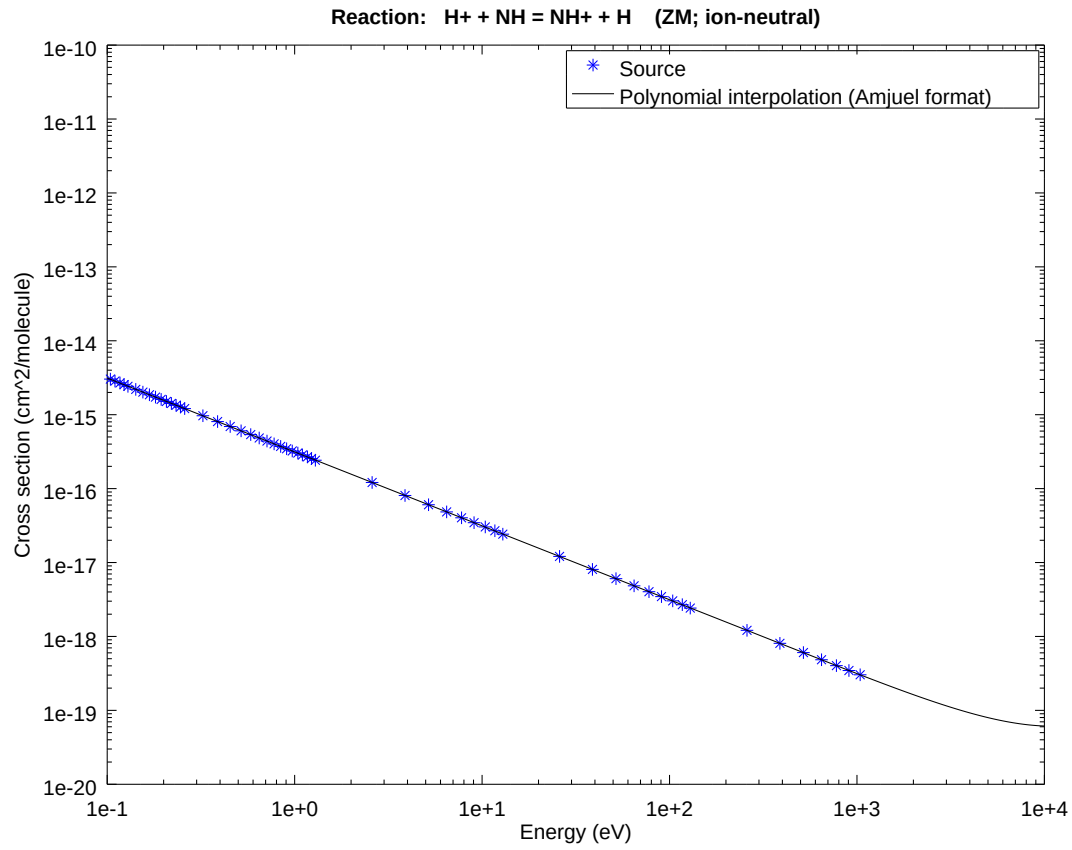
1.6 Reaction ZL $\text{H}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}$

a0	-3.537490331955E+01	a1	-9.993461448873E-01	a2	-1.949720016317E-04
a3	-1.729087654282E-04	a4	3.569519275369E-05	a5	1.165878099393E-05
a6	-2.244838297527E-06	a7	-1.767355711604E-07	a8	3.403726702521E-08



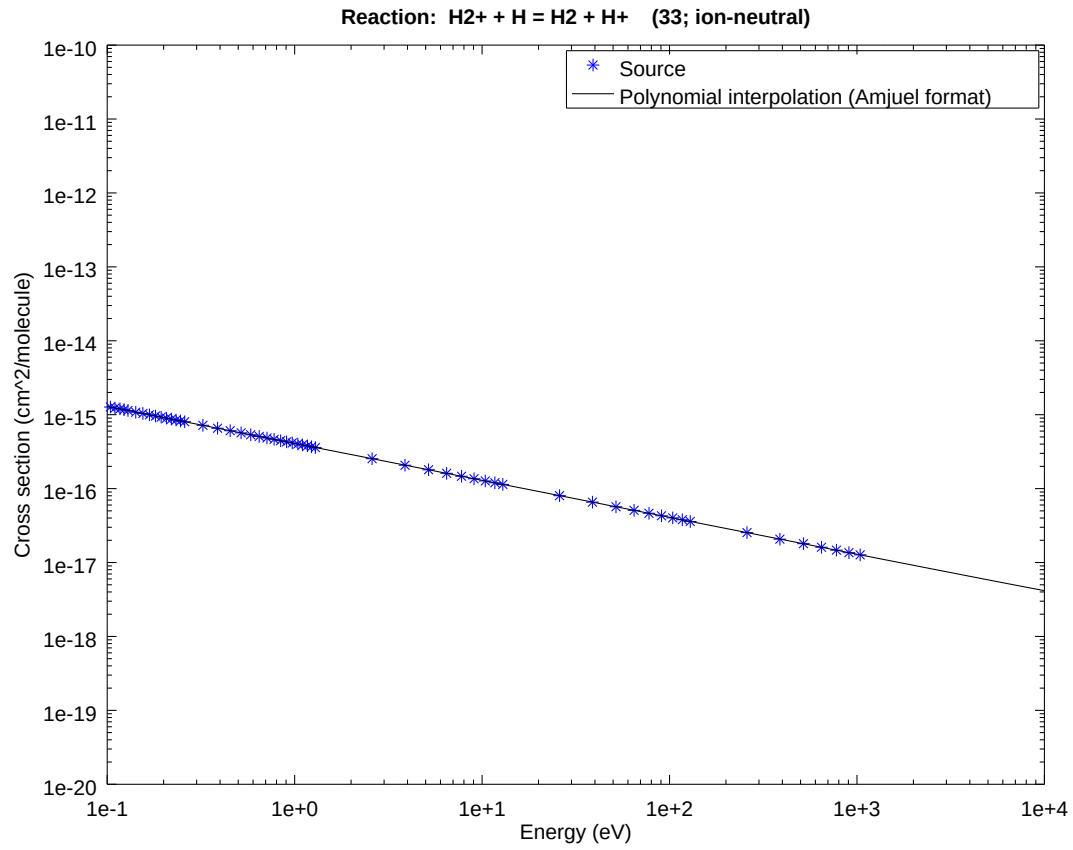
1.7 Reaction ZM $\text{H}^+ + \text{NH} = \text{NH}^+ + \text{H}$

a0	-3.569871167432E+01	a1	-9.983552723791E-01	a2	-2.774822166591E-04
a3	-4.556629385908E-04	a4	5.969049346956E-05	a5	3.538344846415E-05
a6	-4.659439894078E-06	a7	-7.556905829568E-07	a8	1.034211511159E-07



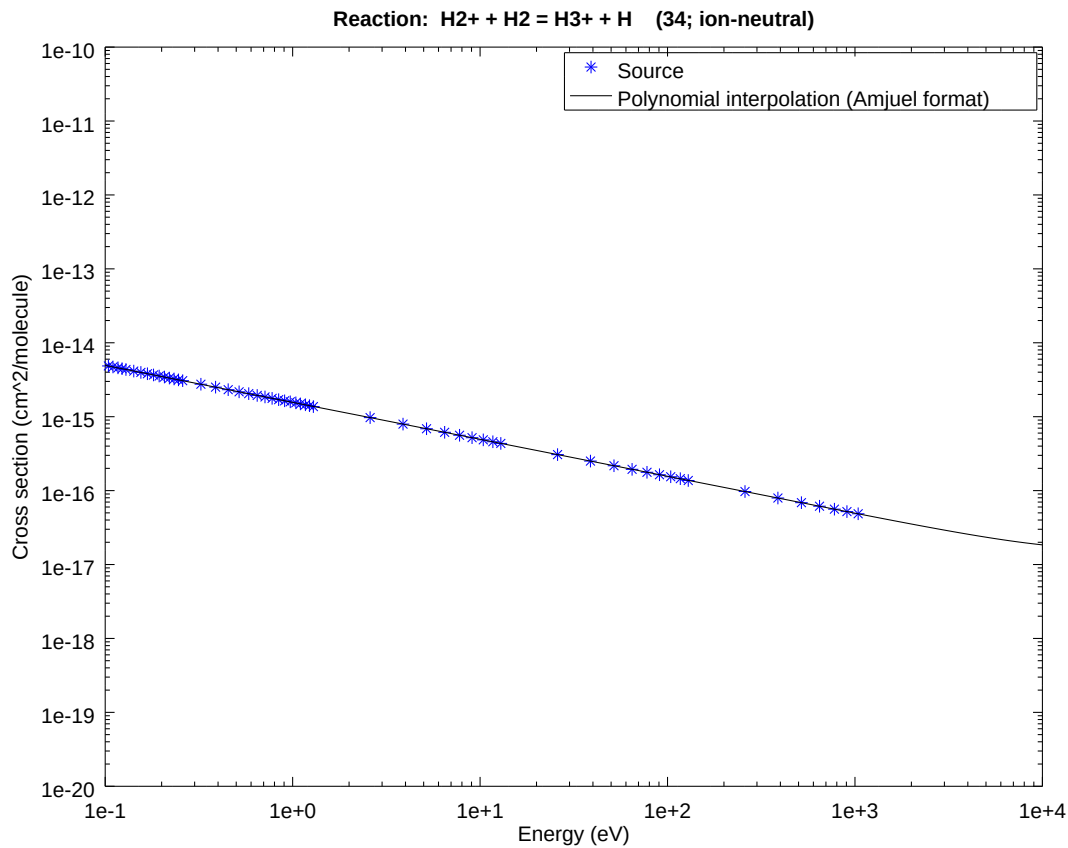
1.8 Reaction 33 $\text{H}_2^+ + \text{H} = \text{H}_2 + \text{H}^+$

a0	-3.543361362397E+01	a1	-4.992788117844E-01	a2	4.583388077472E-04
a3	-1.407924001098E-04	a4	-4.505298269726E-05	a5	1.169063046928E-05
a6	6.572405609520E-07	a7	-2.867139255236E-07	a8	1.628565543476E-08



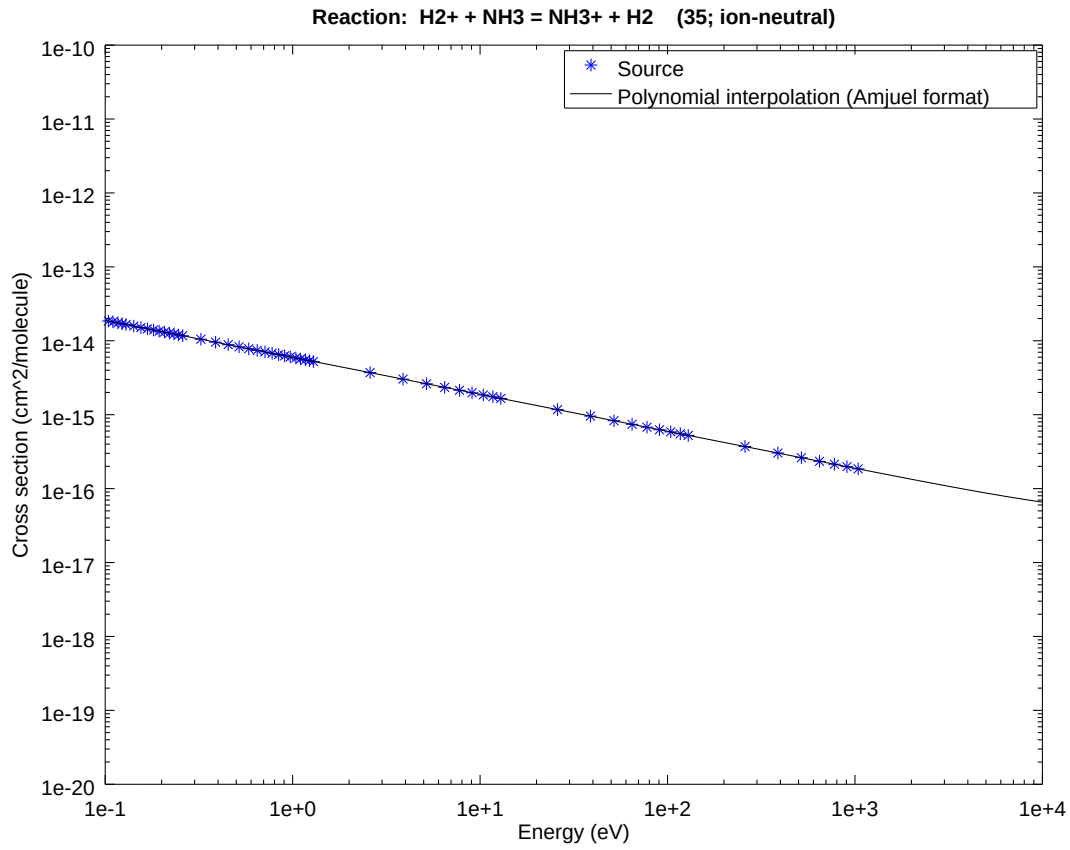
1.9 Reaction 34 $\text{H}_2^+ + \text{H}_2 = \text{H}_3^+ + \text{H}$

a0	-3.409202391649E+01	a1	-4.990685444208E-01	a2	1.437669809837E-04
a3	-2.595121452718E-04	a4	9.995622582894E-07	a5	1.869798439979E-05
a6	-1.485708040220E-06	a7	-3.285779522997E-07	a8	3.583715168109E-08



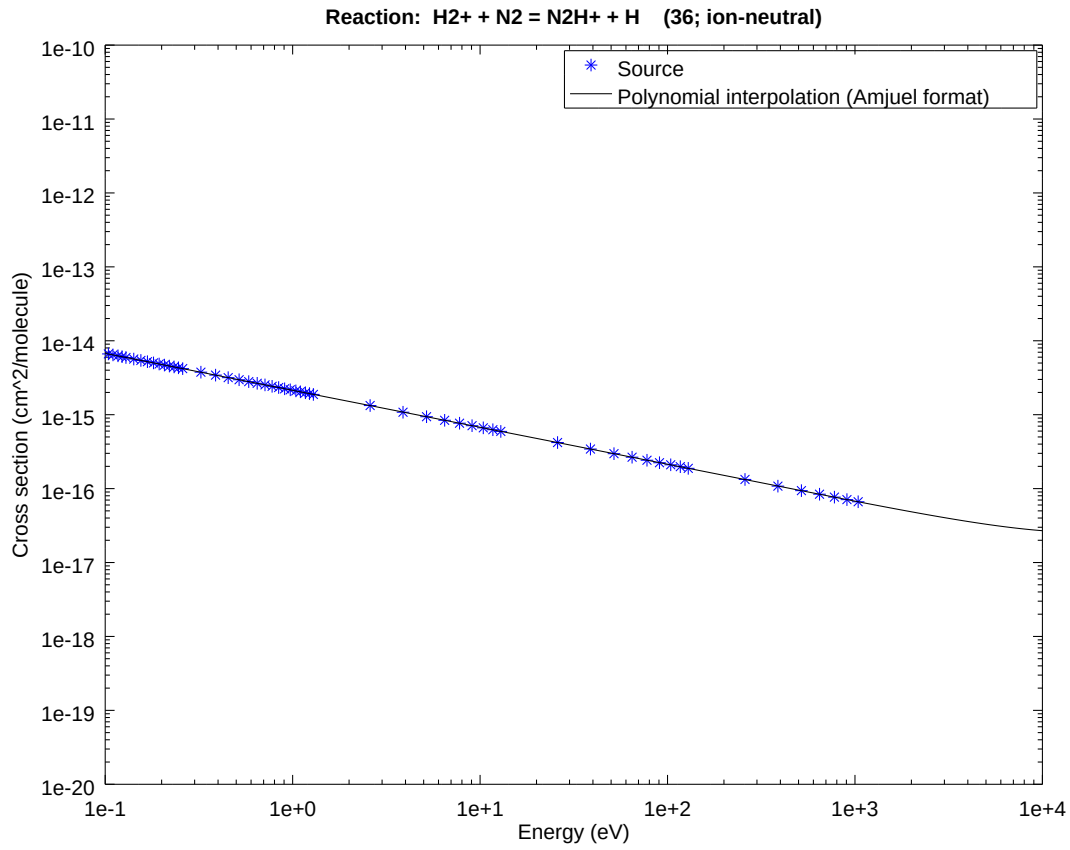
1.10 Reaction 35 $\text{H}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{H}_2$

a0	-3.275288968305E+01	a1	-4.994787050446E-01	a2	-2.188016275804E-04
a3	-1.033806506382E-04	a4	3.316204729448E-05	a5	4.858481757960E-06
a6	-1.522128427499E-06	a7	-1.004154955784E-08	a8	1.187729688822E-08



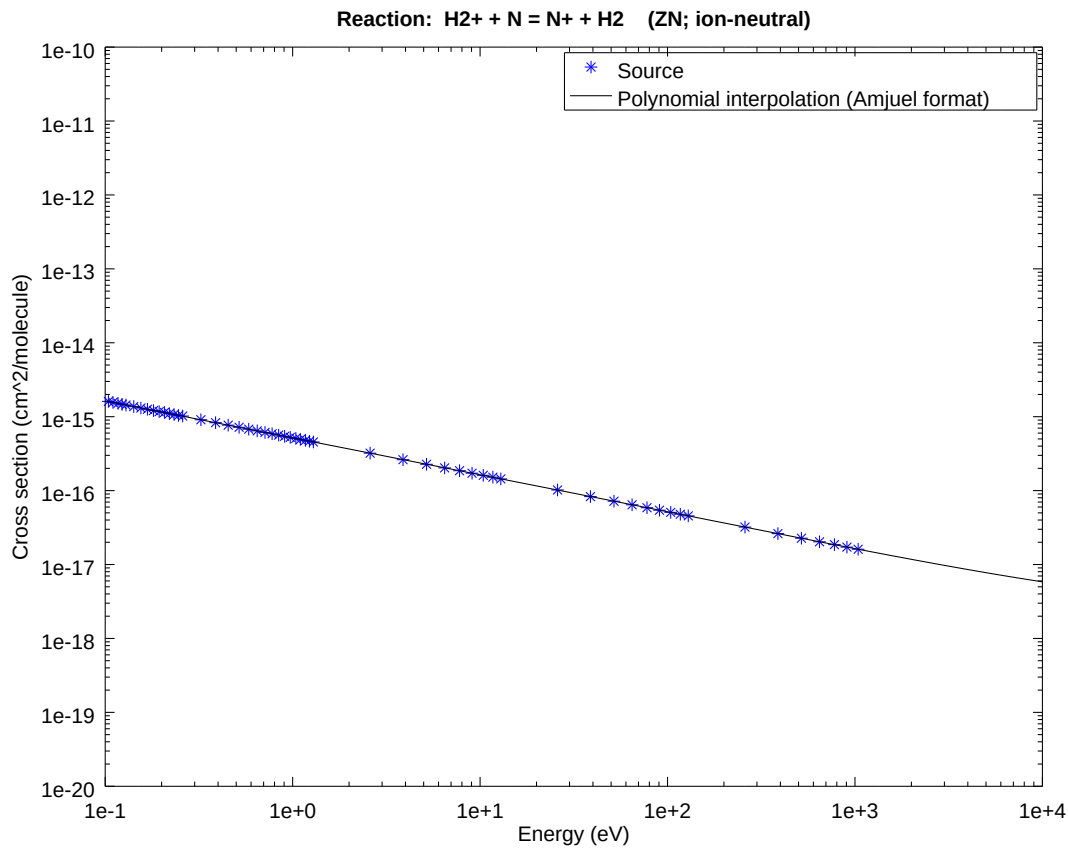
1.11 Reaction 36 $\text{H}_2^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{H}$

a0	-3.377920738035E+01	a1	-4.997687175351E-01	a2	-9.948208681396E-05
a3	-5.978514387256E-05	a4	1.295890511041E-05	a5	6.455654301945E-06
a6	-1.012966875438E-06	a7	-1.864498029942E-07	a8	2.762522101525E-08



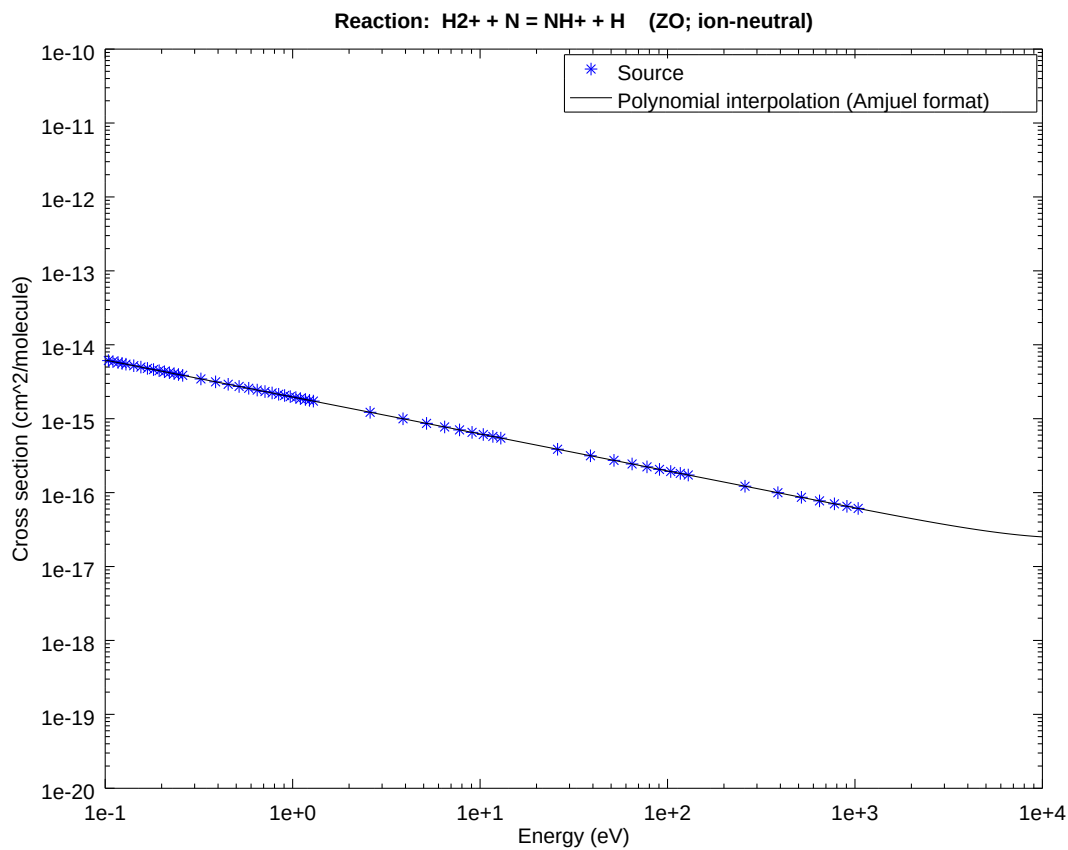
1.12 Reaction $\text{ZN H2}^+ + \text{N} = \text{N}^+ + \text{H2}$

a0	-3.519826157435E+01	a1	-4.998014584796E-01	a2	1.530815192373E-04
a3	5.036422855746E-05	a4	-5.083768450143E-06	a5	-5.745853671067E-06
a6	4.837645055788E-09	a7	1.667353837445E-07	a8	-8.813920612450E-09



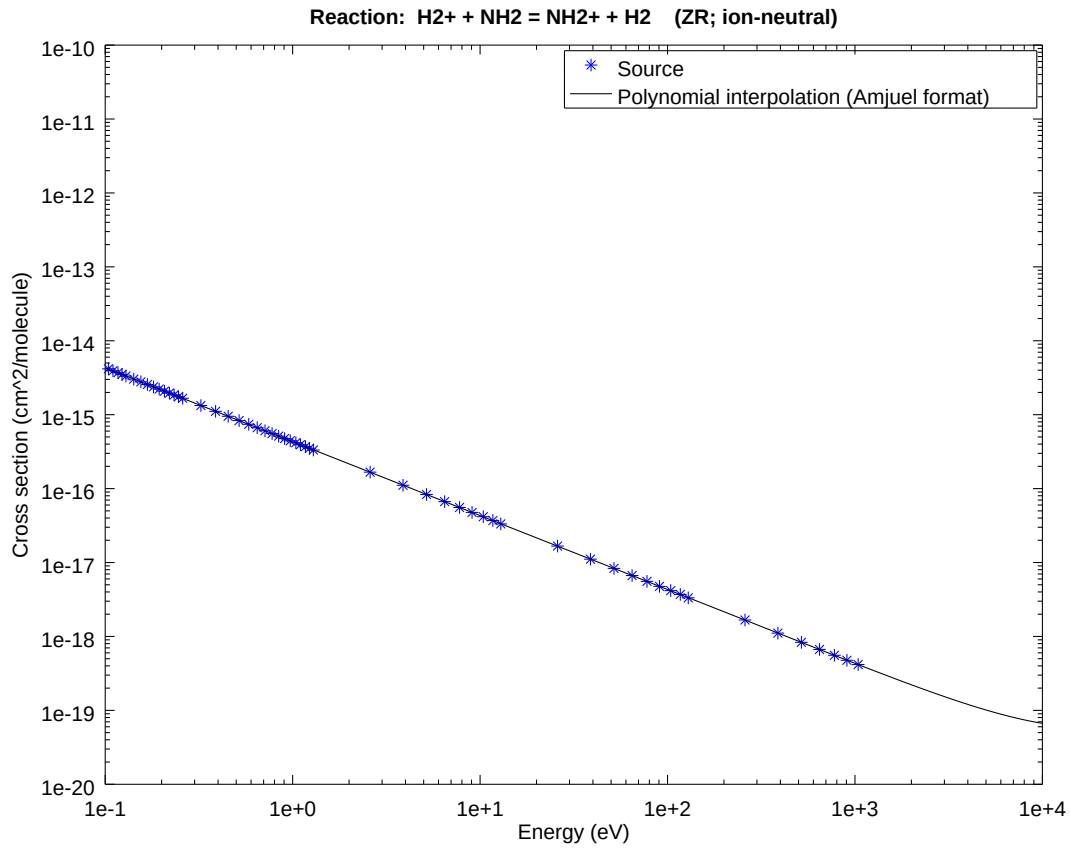
1.13 Reaction ZO H2+ + N = NH+ + H

a0	-3.386357554600E+01	a1	-4.994103923210E-01	a2	2.706129685180E-04
a3	-2.095646572375E-04	a4	-2.749472868472E-05	a5	2.142732365011E-05
a6	-3.458818819641E-07	a7	-6.382759584083E-07	a8	5.716798847223E-08



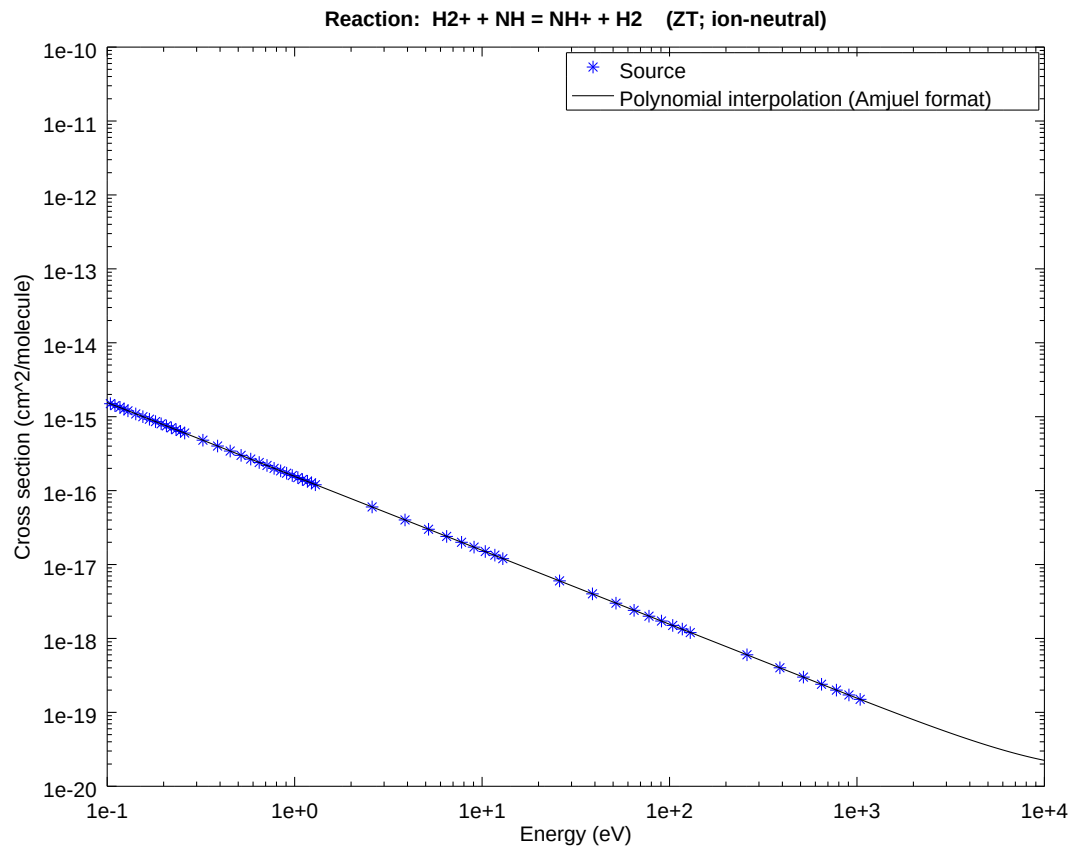
1.14 Reaction ZR $\text{H}_2^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}_2$

a0	-3.537889371688E+01	a1	-9.988111947000E-01	a2	-2.580324837846E-04
a3	-2.911740540974E-04	a4	5.386582495464E-05	a5	1.997041706219E-05
a6	-3.537831334797E-06	a7	-3.296643009103E-07	a8	5.745822253520E-08



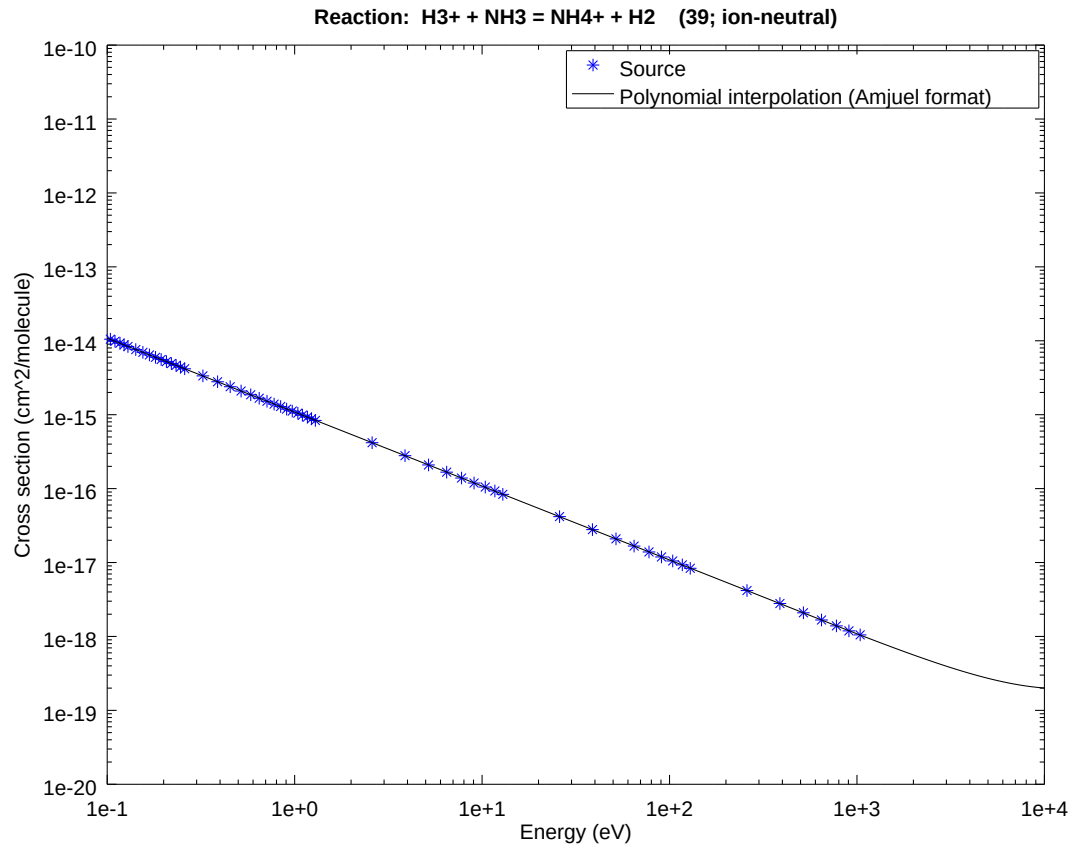
1.15 Reaction ZT $\text{H}_2^+ + \text{NH} = \text{NH}^+ + \text{H}_2$

a0	-3.639877685839E+01	a1	-9.991932271353E-01	a2	-3.786106055822E-04
a3	-2.000209526831E-04	a4	6.490058121546E-05	a5	1.204772346475E-05
a6	-3.491148657210E-06	a7	-1.039499856348E-07	a8	3.854441560333E-08



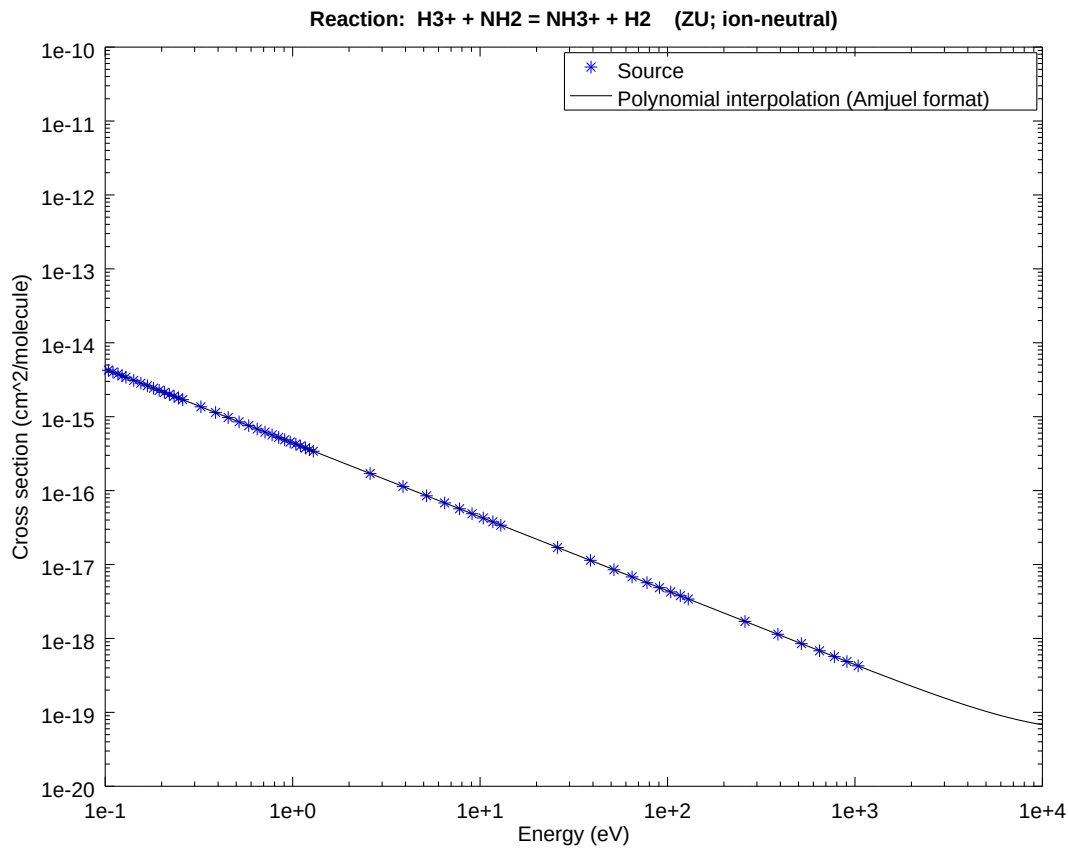
1.16 Reaction 39 $\text{H}_3^+ + \text{NH}_3 = \text{NH}_4^+ + \text{H}_2$

a0	-3.445855504336E+01	a1	-9.991800597753E-01	a2	-4.749471155324E-04
a3	-2.864635224663E-04	a4	7.700583560252E-05	a5	2.201247421731E-05
a6	-4.531304197853E-06	a7	-4.179190244699E-07	a8	7.678547276233E-08



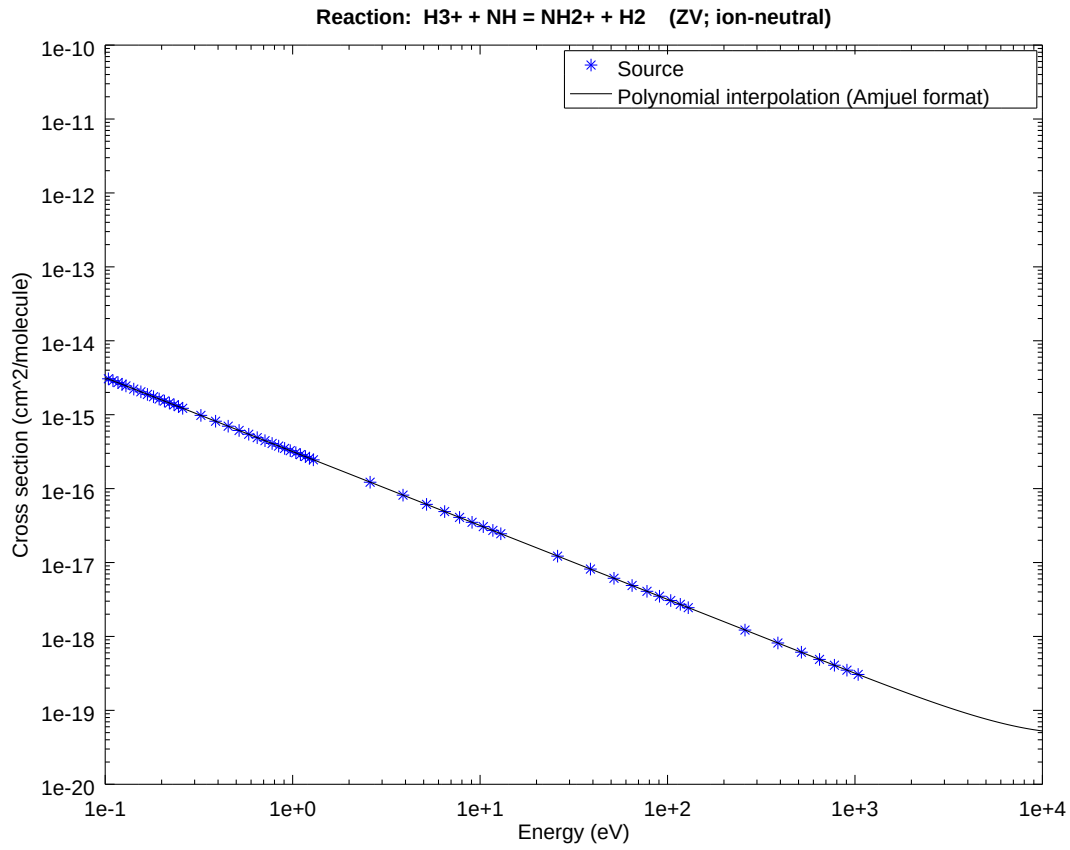
1.17 Reaction ZU $\text{H3}^+ + \text{NH2} = \text{NH3}^+ + \text{H2}$

a0	-3.535718875007E+01	a1	-9.988041211578E-01	a2	-1.327073962939E-04
a3	-3.024750238876E-04	a4	3.537215304242E-05	a5	2.306215109979E-05
a6	-2.938634729118E-06	a7	-4.885550353885E-07	a8	6.619909023077E-08



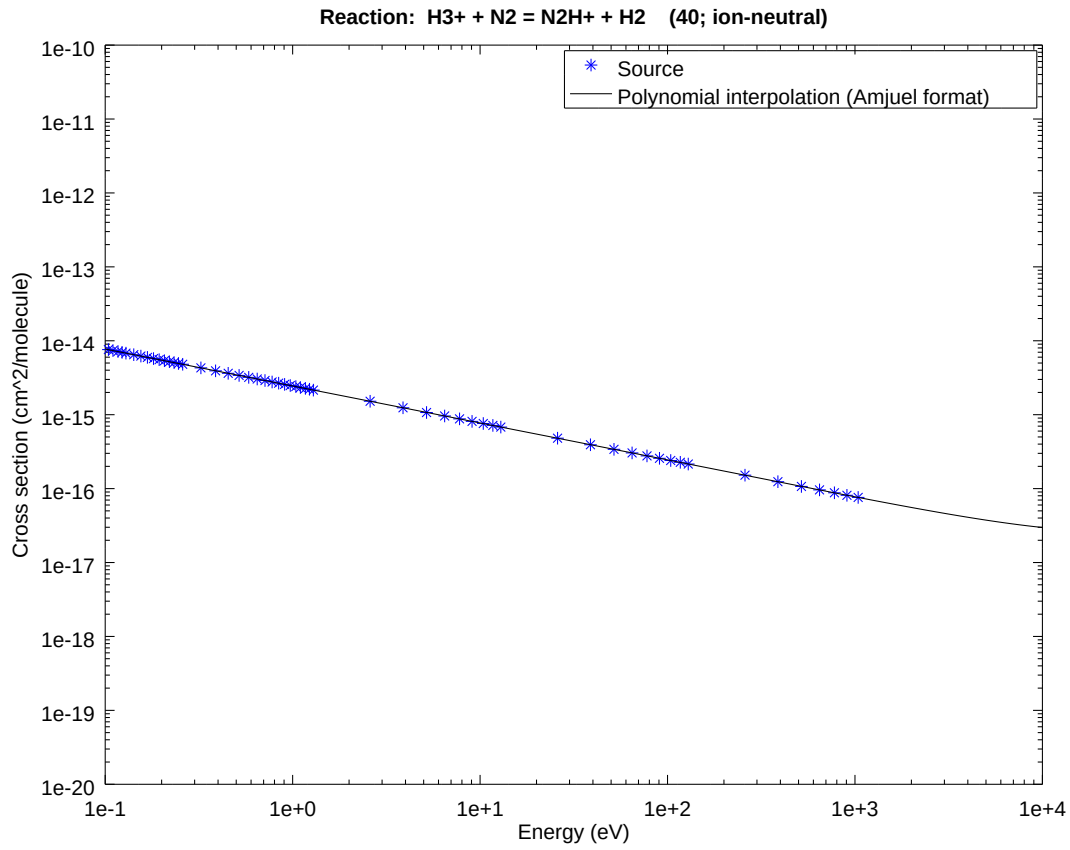
1.18 Reaction ZV $\text{H3}^+ + \text{NH} = \text{NH2}^+ + \text{H2}$

a0	-3.568779264120E+01	a1	-9.991262523277E-01	a2	-1.490267028808E-04
a3	-2.583679126152E-04	a4	3.122615471999E-05	a5	2.265333790401E-05
a6	-2.824905844315E-06	a7	-5.468812609031E-07	a8	7.313382420176E-08



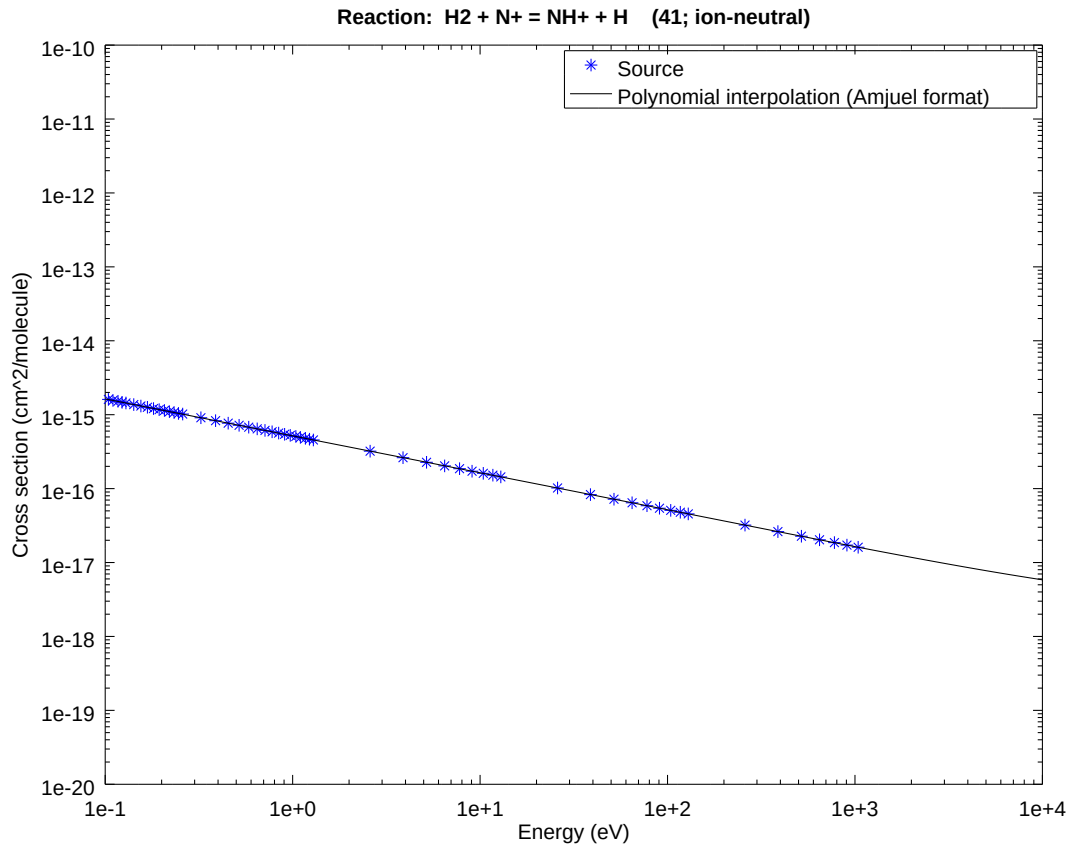
1.19 Reaction 40 $\text{H}_3^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{H}_2$

a0	-3.364424583043E+01	a1	-5.001053947537E-01	a2	-3.489472719550E-04
a3	5.217830017246E-05	a4	4.832027801732E-05	a5	-6.339876962418E-06
a6	-1.623185920202E-06	a7	2.249465107832E-07	a8	-2.791634048550E-10



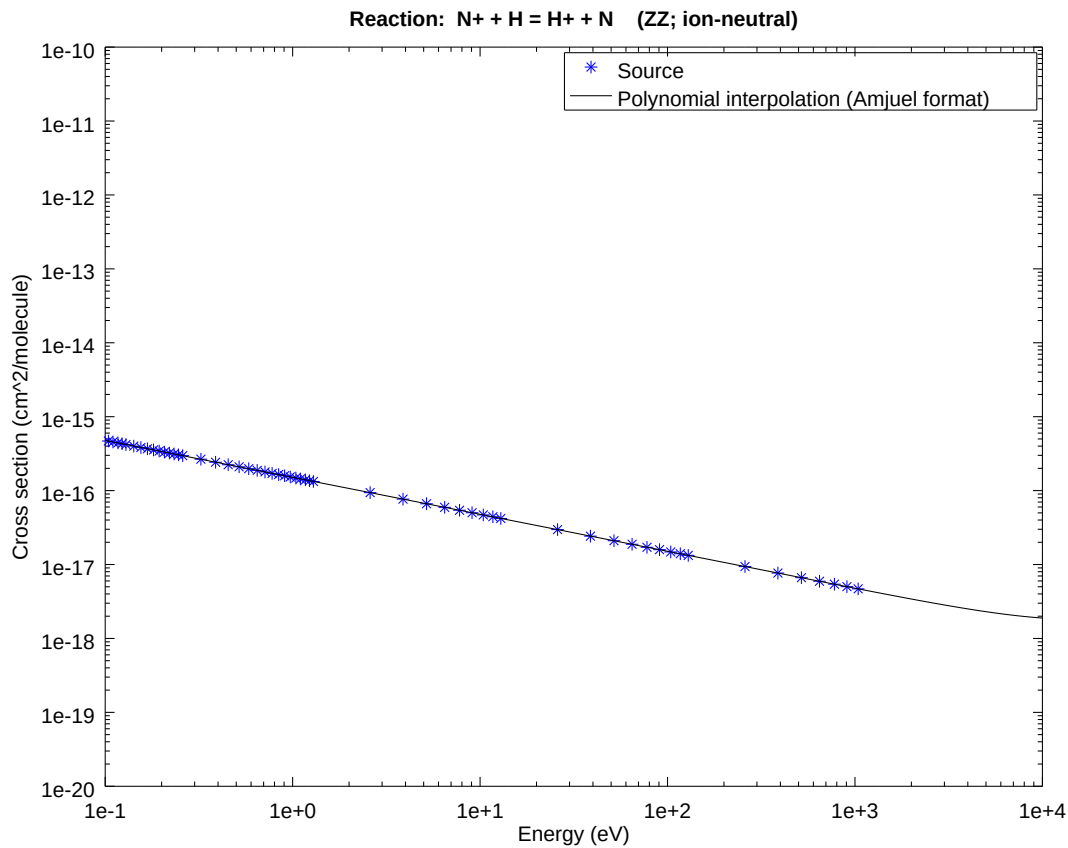
1.20 Reaction 41 $\text{H}_2 + \text{N}^+ = \text{NH}^+ + \text{H}$

a0	-3.519826157435E+01	a1	-4.998014584796E-01	a2	1.530815192373E-04
a3	5.036422855746E-05	a4	-5.083768450143E-06	a5	-5.745853671067E-06
a6	4.837645055788E-09	a7	1.667353837445E-07	a8	-8.813920612450E-09



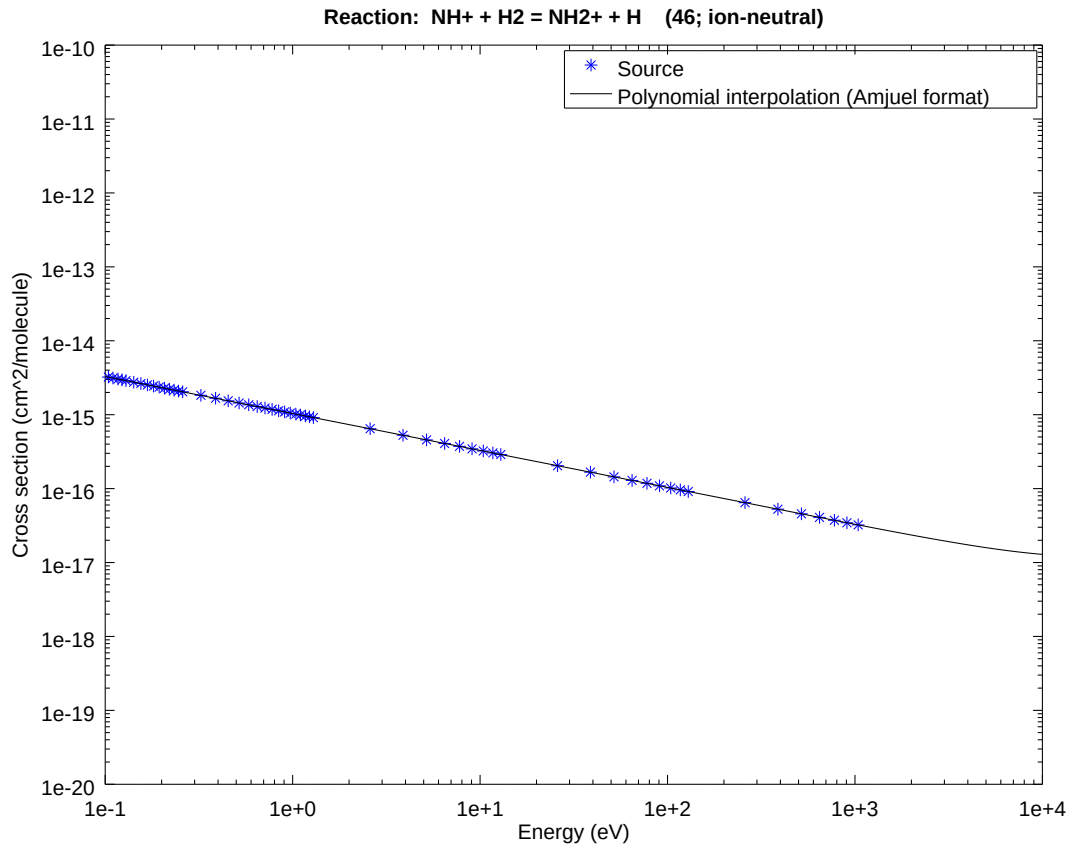
1.21 Reaction ZZ N+ + H = H+ + N

a0	-3.642877046760E+01	a1	-4.993284140026E-01	a2	1.110120593205E-04
a3	-2.158746434780E-04	a4	-8.302489080848E-06	a5	2.011967096965E-05
a6	-9.264918567094E-07	a7	-5.415644349055E-07	a8	5.239164855073E-08



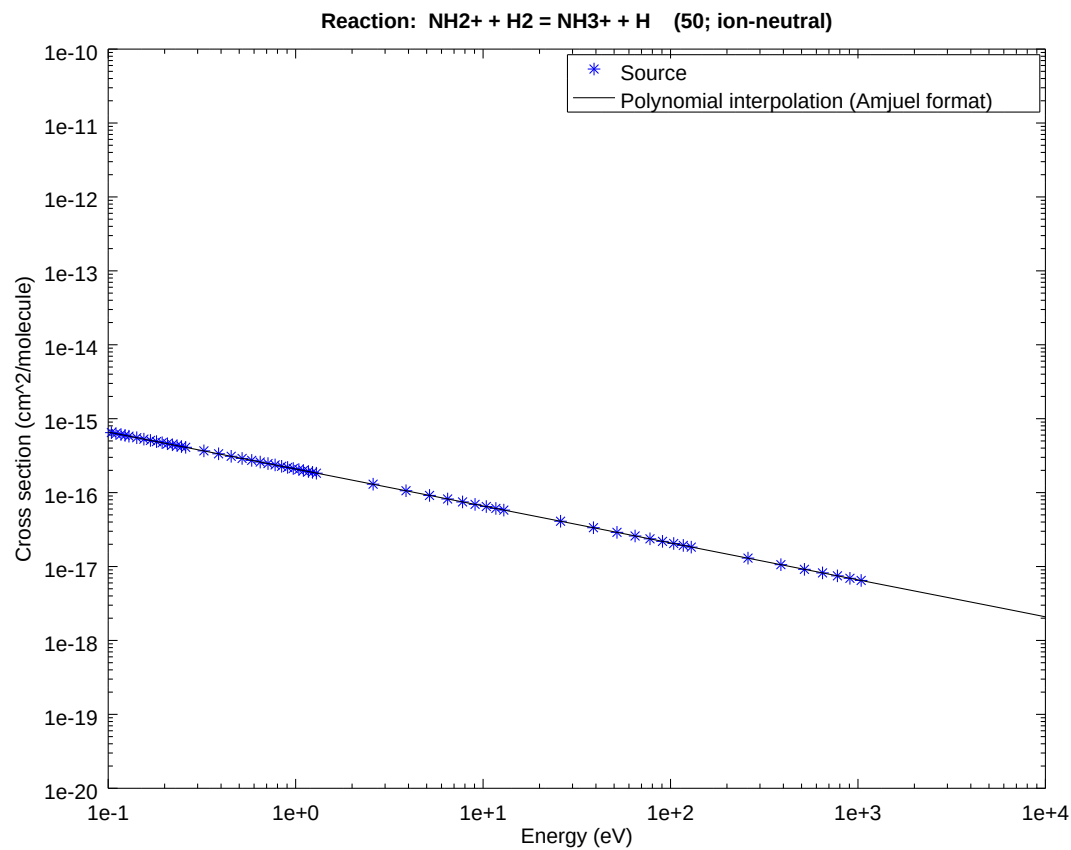
1.22 Reaction 46 $\text{NH}^+ + \text{H}_2 = \text{NH}_2^+ + \text{H}$

a0	-3.450120601495E+01	a1	-4.994452038861E-01	a2	7.638532994153E-05
a3	-1.448123543562E-04	a4	4.073634739478E-06	a5	1.176427290163E-05
a6	-1.088326346202E-06	a7	-2.720391880305E-07	a8	3.287797730196E-08



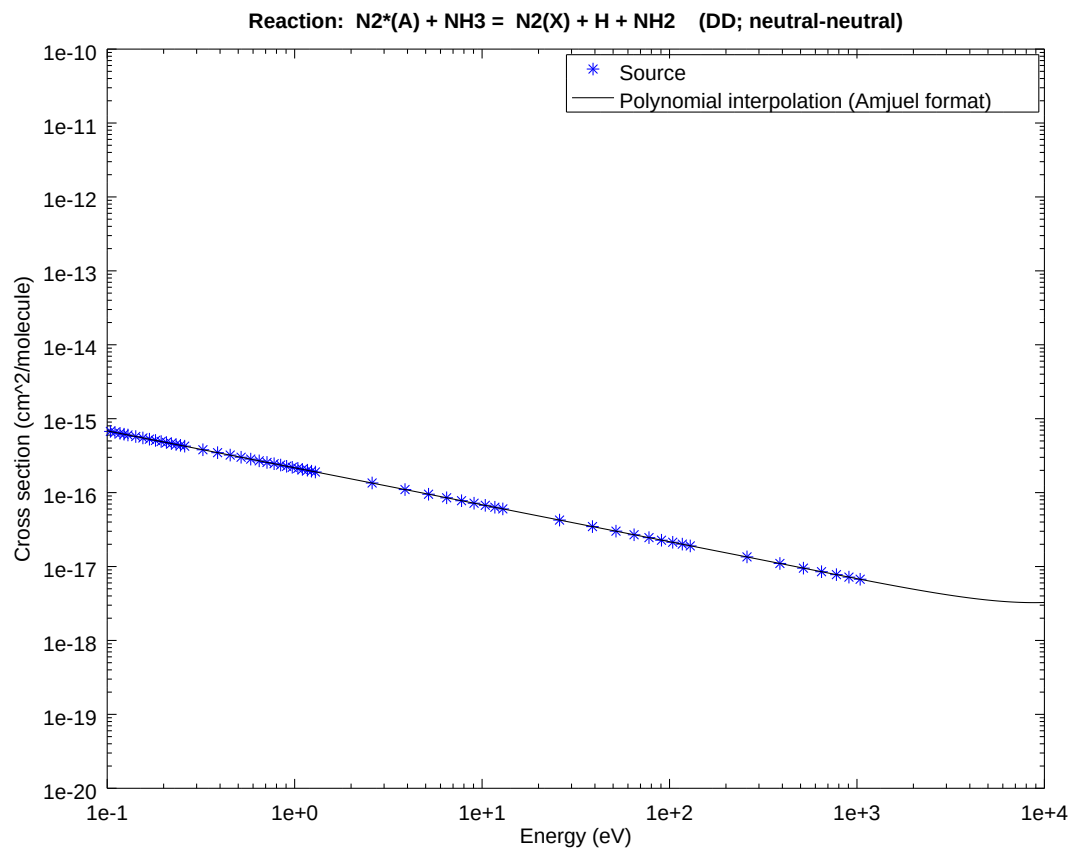
1.23 Reaction 50 $\text{NH}_2^+ + \text{H}_2 = \text{NH}_3^+ + \text{H}$

a0	-3.610585832408E+01	a1	-4.994904651358E-01	a2	-2.952907279827E-04
a3	-6.005065614576E-05	a4	5.388044003298E-05	a5	-4.149821036921E-06
a6	-1.914629649127E-06	a7	3.669195429192E-07	a8	-1.813946931009E-08



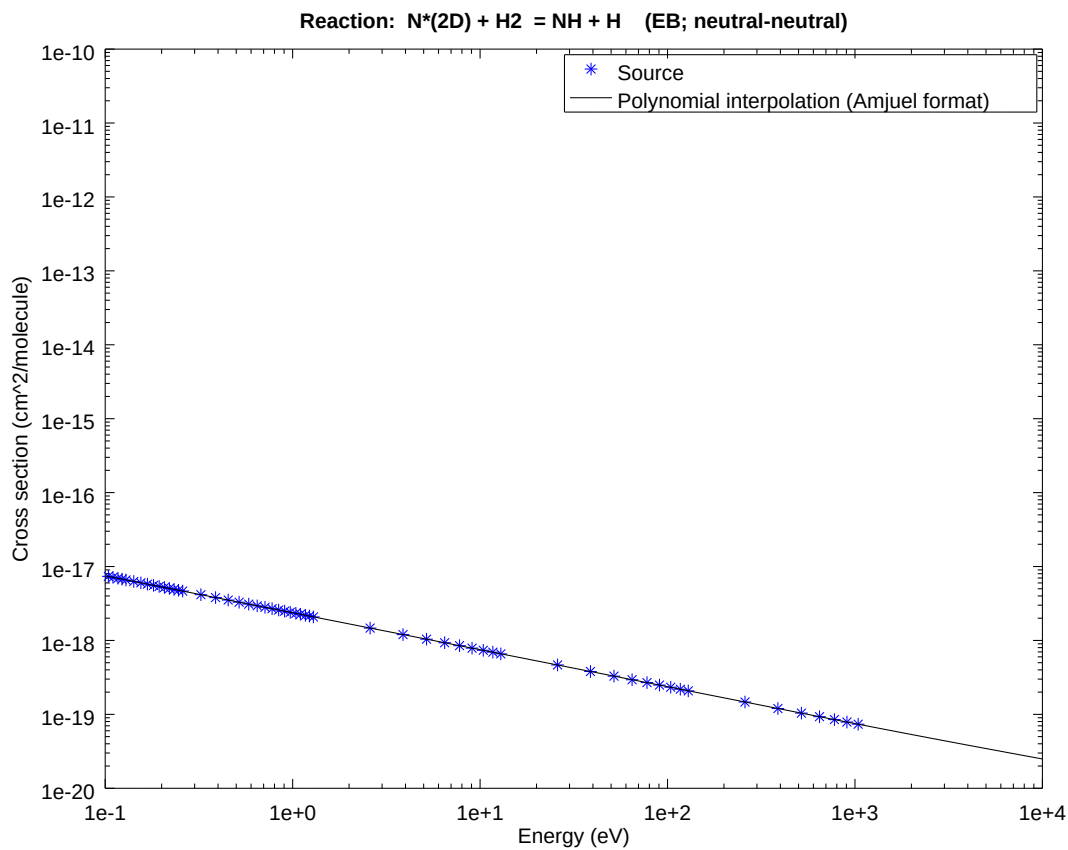
1.24 Reaction DD N2*(A) + NH3 = N2(X) + H + NH2

a0	-3.606850050331E+01	a1	-4.985692234760E-01	a2	-2.554692077144E-04
a3	-4.069952082798E-04	a4	5.028150025855E-05	a5	3.019730055286E-05
a6	-3.697880512623E-06	a7	-6.049288920032E-07	a8	7.786467574166E-08



1.25 Reaction EB $\text{N}^*(2\text{D}) + \text{H}_2 = \text{NH} + \text{H}$

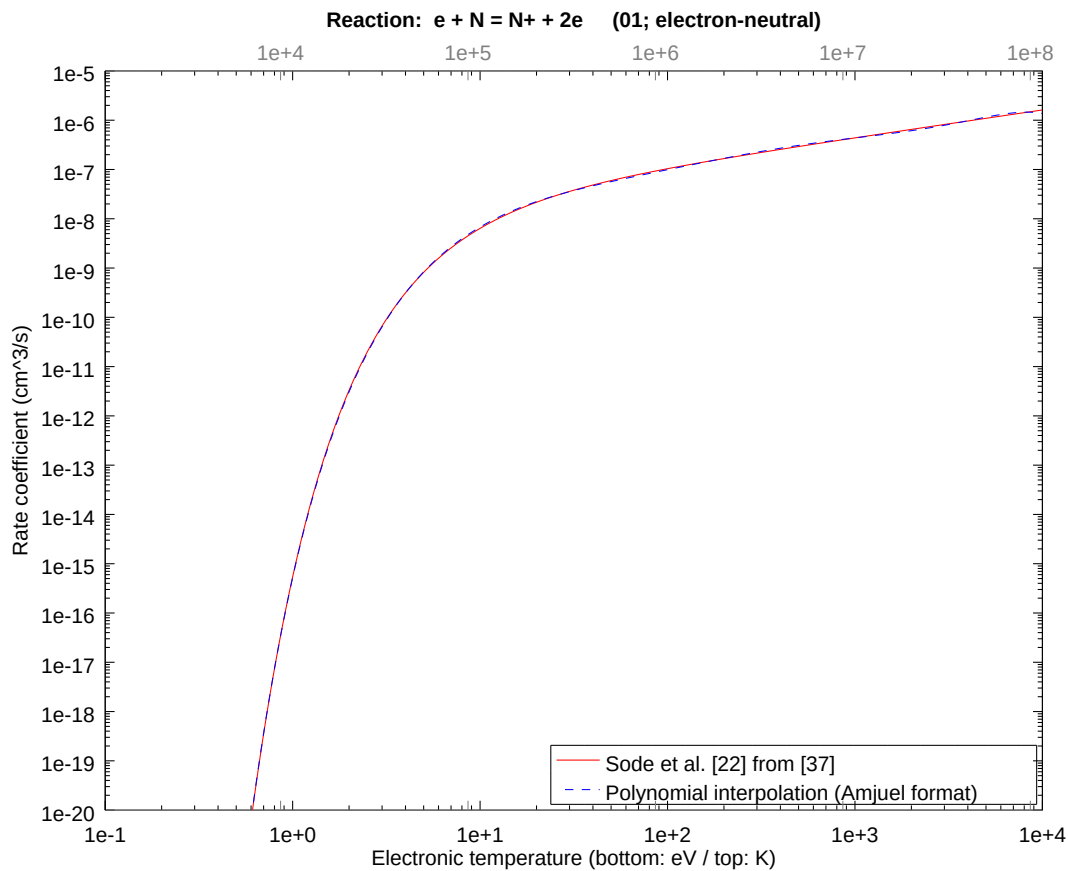
a0	-4.058426808527E+01	a1	-5.002334190407E-01	a2	-7.888387115786E-05
a3	9.295612376230E-05	a4	1.599541494452E-05	a5	-9.072793765241E-06
a6	-4.451575724006E-07	a7	3.001864279963E-07	a8	-1.820461730557E-08



2 H.2 : Fits for rate coefficient

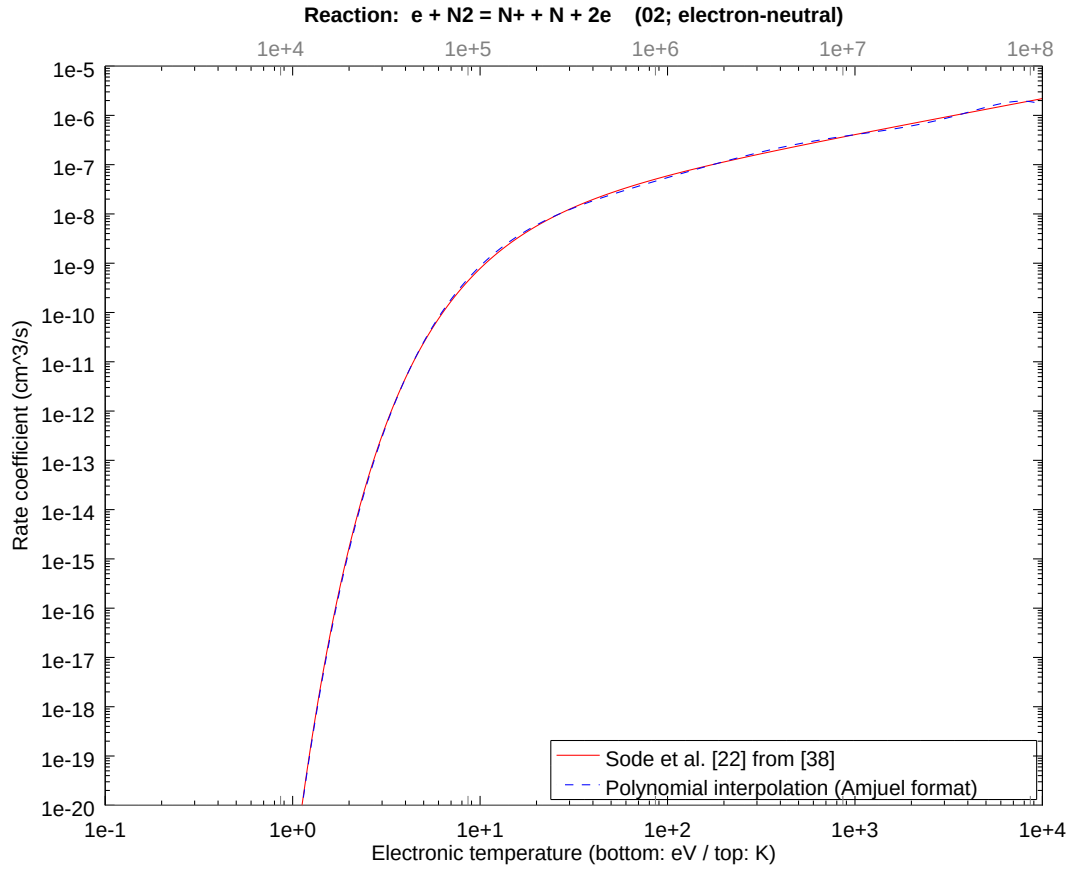
2.1 Reaction 01 $e + N = N^+ + 2e$

b0	-3.518518351008E+01	b1	1.707576198208E+01	b2	-8.227861906601E+00
b3	2.853001756746E+00	b4	-7.513126739699E-01	b5	1.408230464943E-01
b6	-1.675815431763E-02	b7	1.103977300446E-03	b8	-3.033294428047E-05



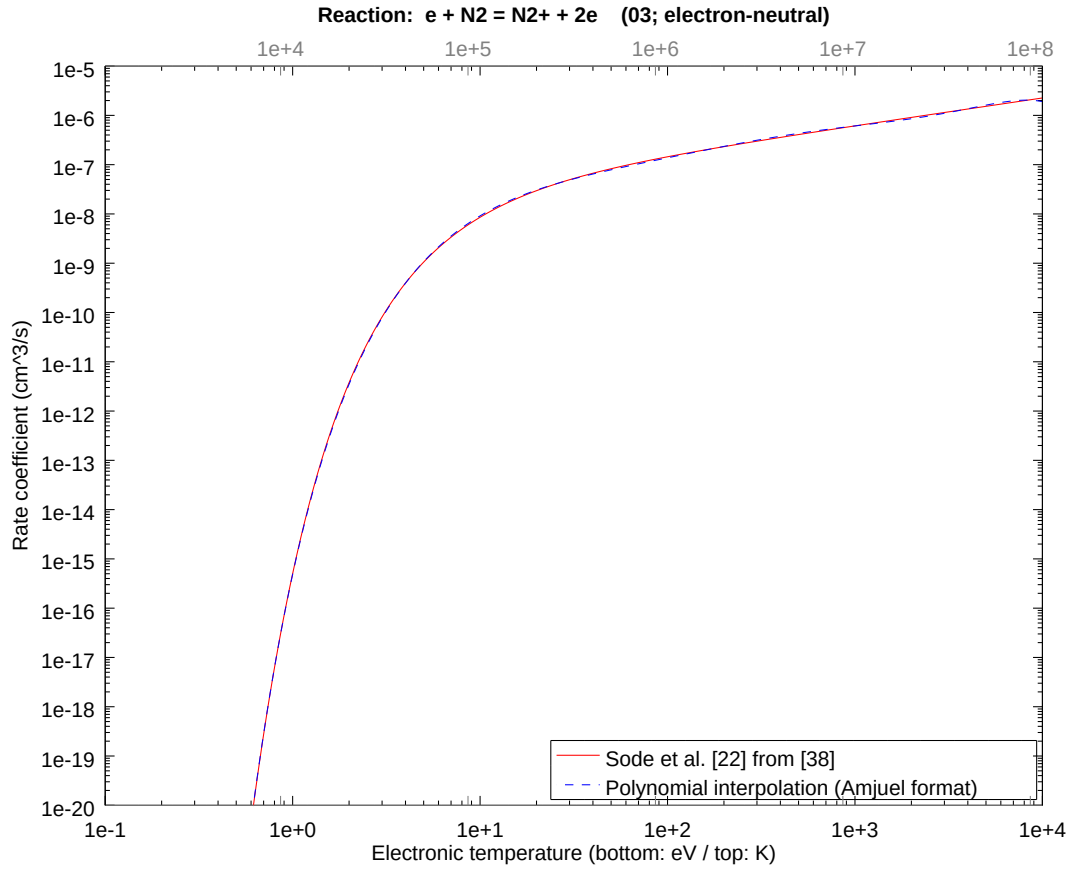
2.2 Reaction 02 $e + N_2 = N + N + 2e$

b0	-4.942549999176E+01	b1	3.017277842063E+01	b2	-1.467285577702E+01
b3	5.087796049996E+00	b4	-1.339825902980E+00	b5	2.511316153268E-01
b6	-2.988503990258E-02	b7	1.968737430747E-03	b8	-5.409314373186E-05



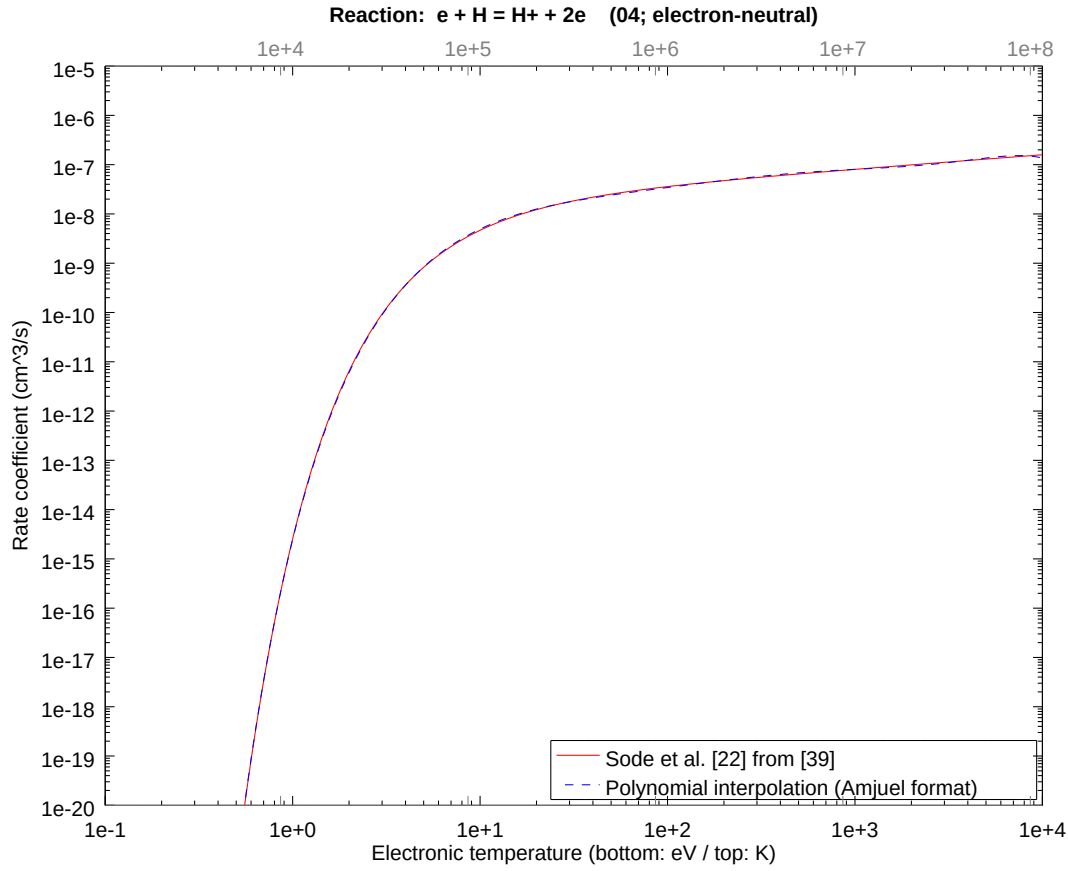
2.3 Reaction 03 $e + N_2 = N_2^+ + 2e$

b0	-3.526103439612E+01	b1	1.748221230697E+01	b2	-8.430348304062E+00
b3	2.923213684733E+00	b4	-7.698023616246E-01	b5	1.442886796913E-01
b6	-1.717056988007E-02	b7	1.131146009520E-03	b8	-3.107943330538E-05



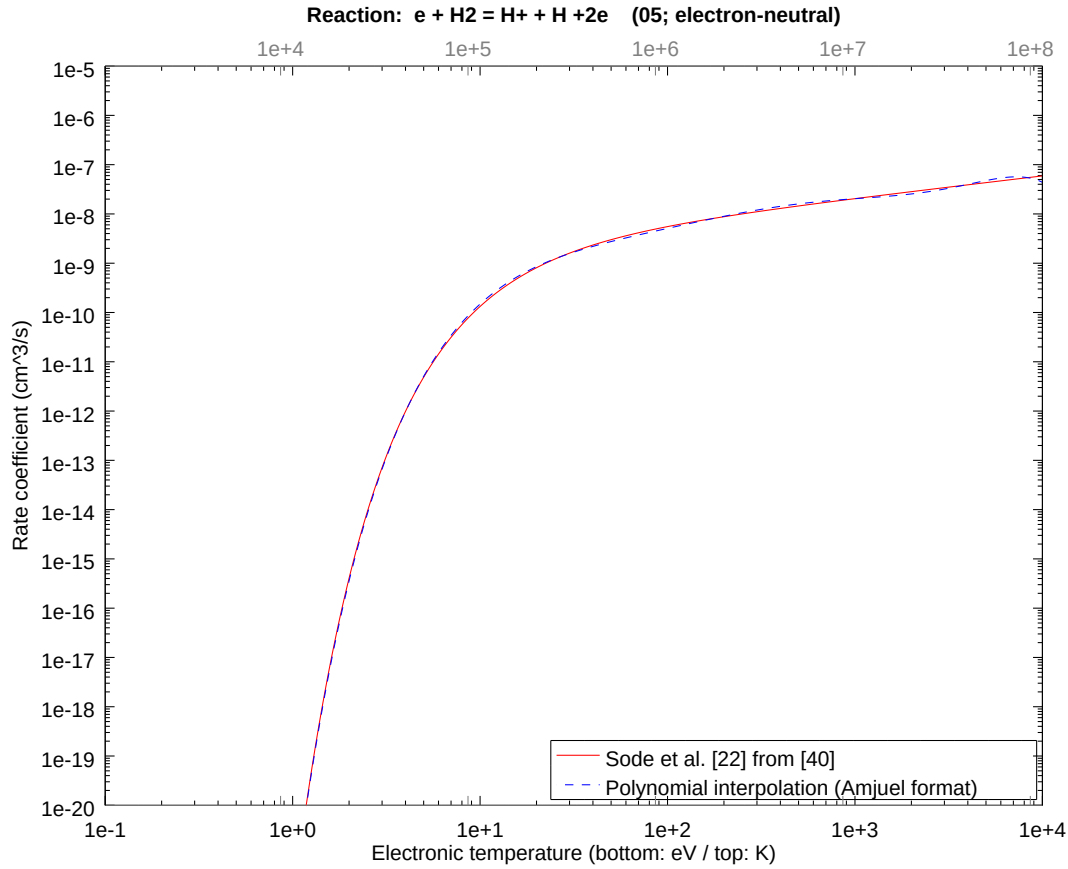
2.4 Reaction 04 $e + H = H^+ + 2e$

b0	-3.363465760376E+01	b1	1.543770966904E+01	b2	-7.546322324902E+00
b3	2.616678682058E+00	b4	-6.890790911320E-01	b5	1.291582323189E-01
b6	-1.537002388796E-02	b7	1.012531401609E-03	b8	-2.782037146492E-05



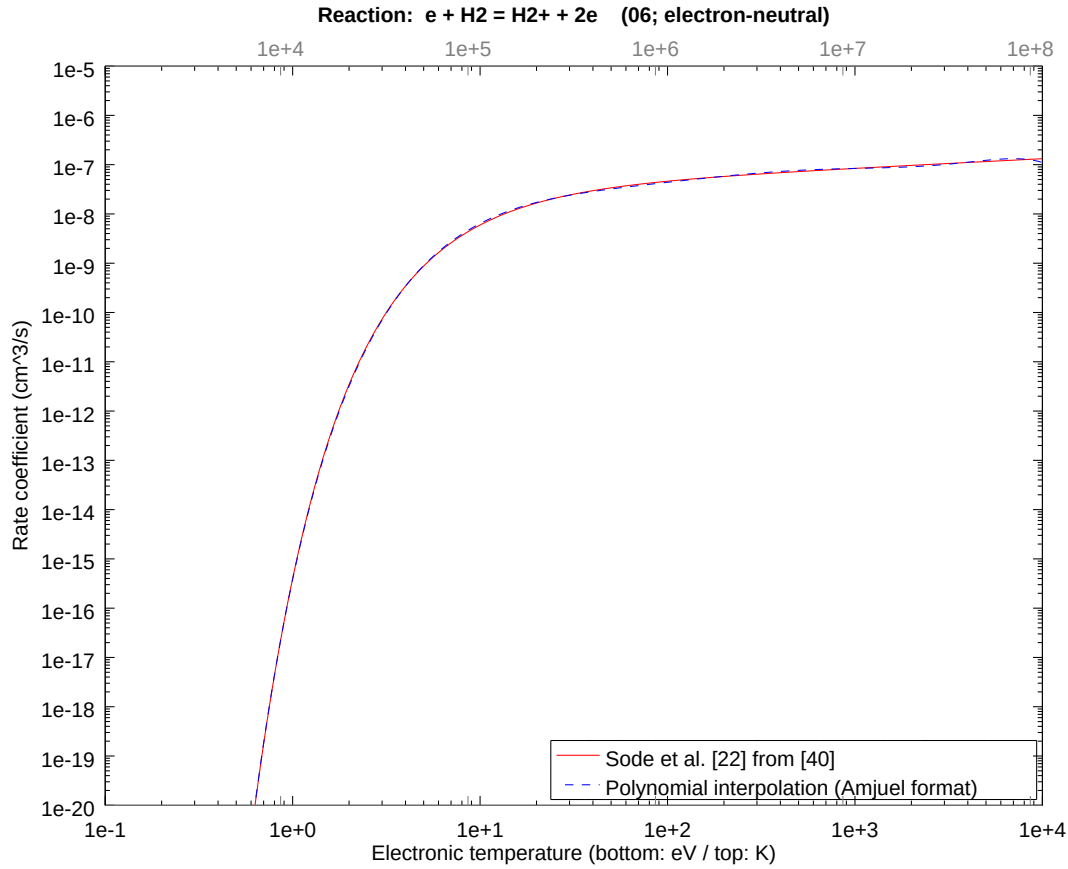
2.5 Reaction 05 $e + H_2 = H + H + 2e$

b0	-5.078252697142E+01	b1	3.013078713947E+01	b2	-1.478644570730E+01
b3	5.127183229110E+00	b4	-1.350198166786E+00	b5	2.530757510227E-01
b6	-3.011639497419E-02	b7	1.983978413886E-03	b8	-5.451190586779E-05



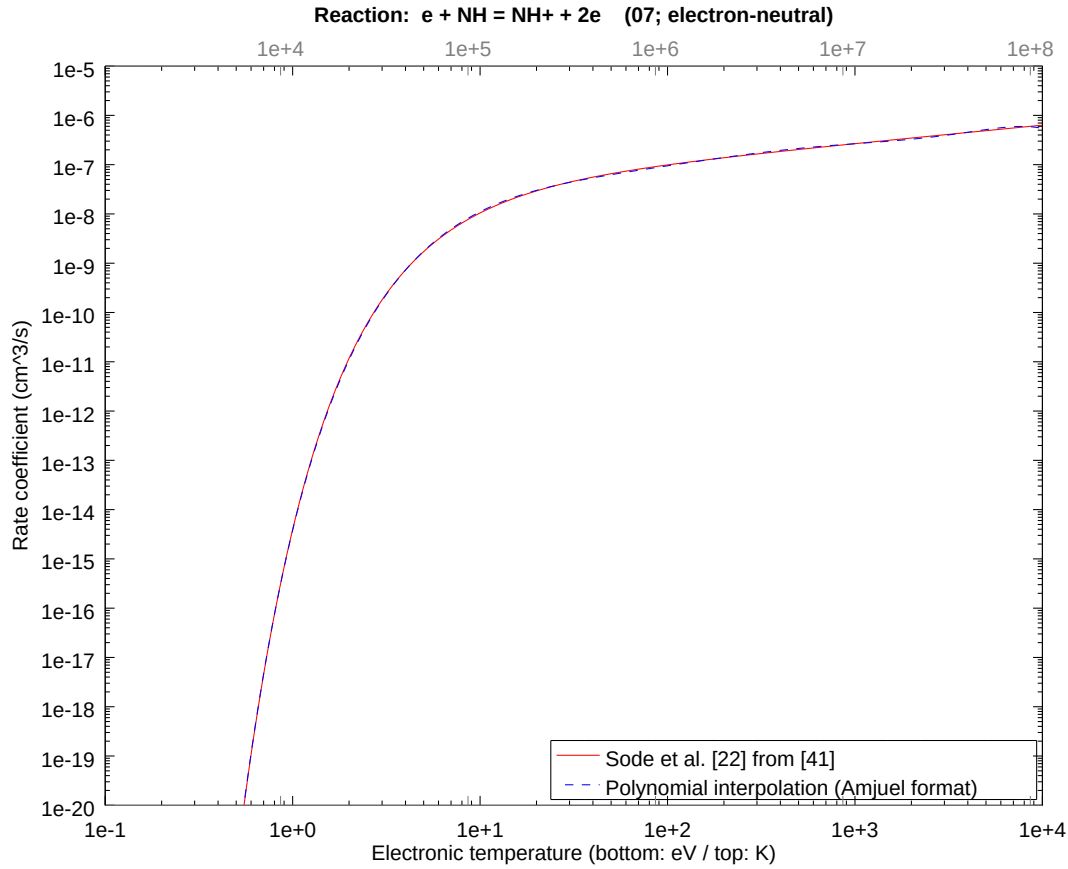
2.6 Reaction 06 $e + H_2 = H_2^+ + 2e$

b0	-3.549202289051E+01	b1	1.790528611163E+01	b2	-8.825443713743E+00
b3	3.060212568611E+00	b4	-8.058798009509E-01	b5	1.510508908075E-01
b6	-1.797528317264E-02	b7	1.184158124788E-03	b8	-3.253599725642E-05



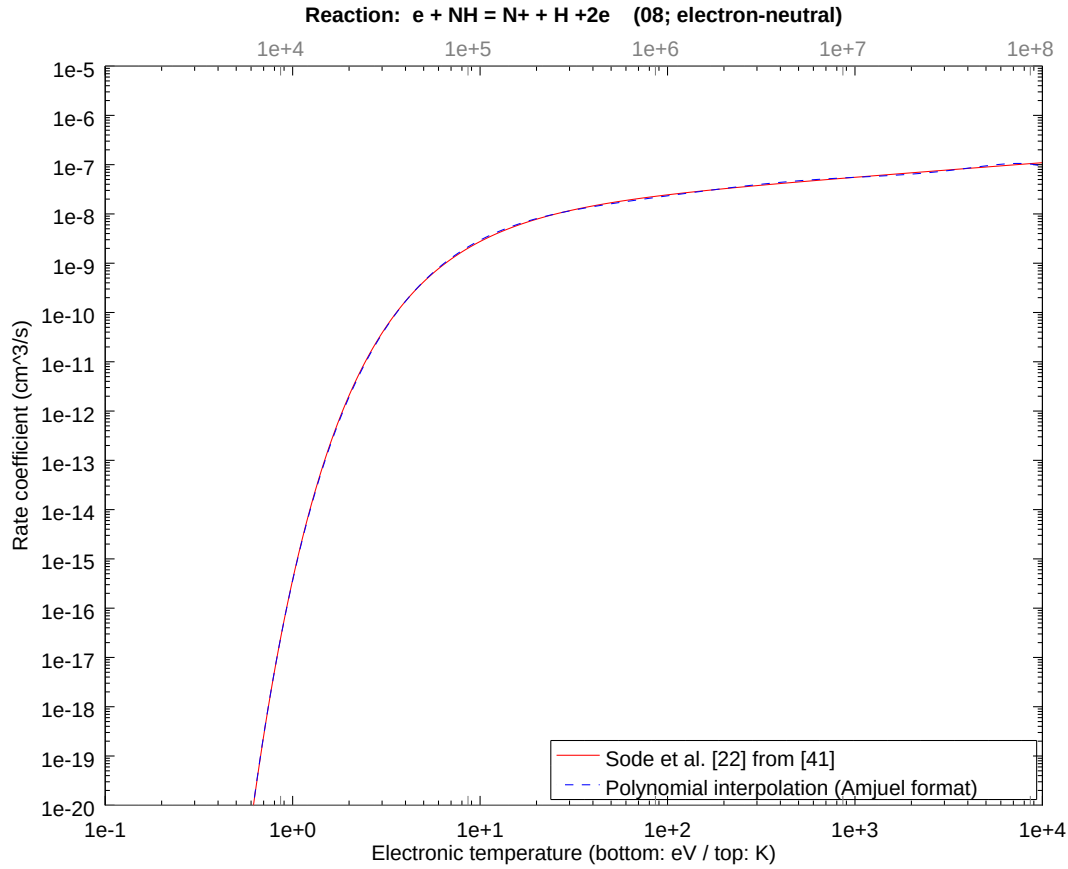
2.7 Reaction 07 $e + \text{NH} = \text{NH}^+ + 2e$

b0	-3.319843294396E+01	b1	1.572589154276E+01	b2	-7.650034869943E+00
b3	2.652640889075E+00	b4	-6.985494189552E-01	b5	1.309333127369E-01
b6	-1.558126112726E-02	b7	1.026447081867E-03	b8	-2.820271950207E-05



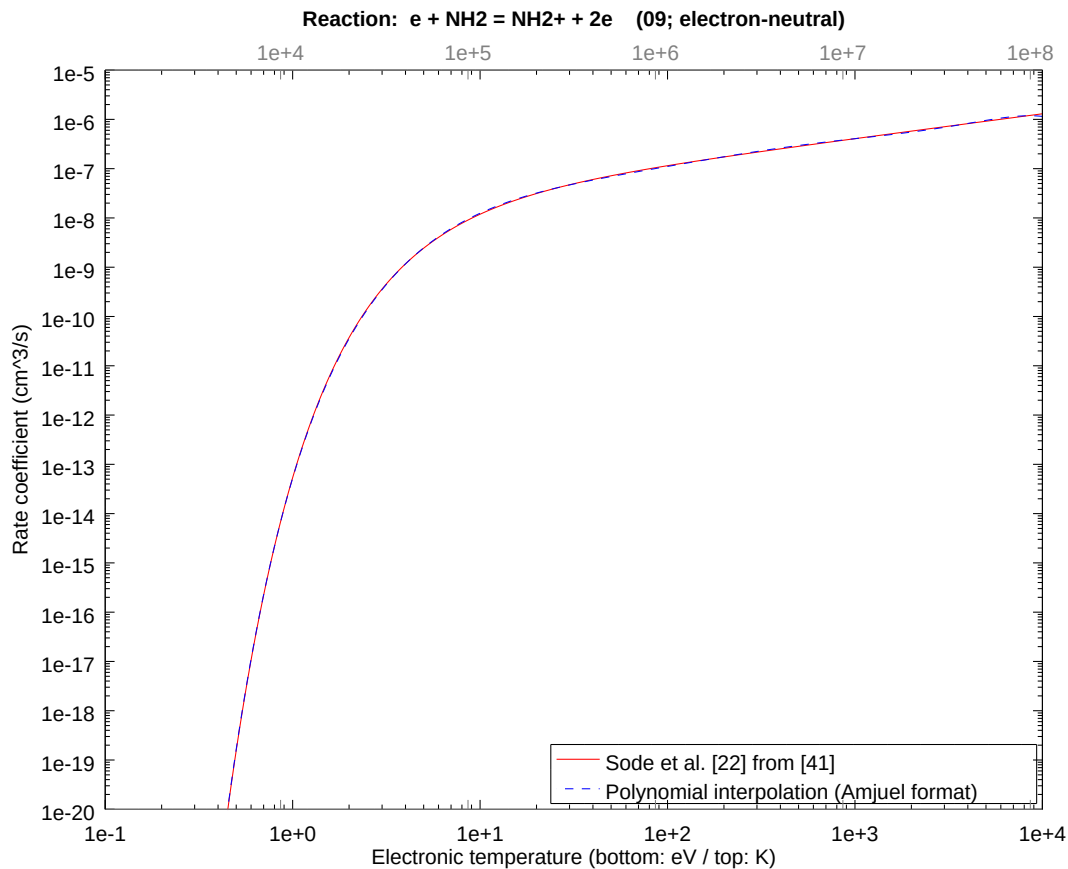
2.8 Reaction 08 $e + \text{NH} = \text{N}^+ + \text{H} + 2e$

b0	-3.554735633352E+01	b1	1.696437674301E+01	b2	-8.306880988537E+00
b3	2.880401533522E+00	b4	-7.585281618352E-01	b5	1.421754887175E-01
b6	-1.691909697615E-02	b7	1.114579723499E-03	b8	-3.062425707068E-05



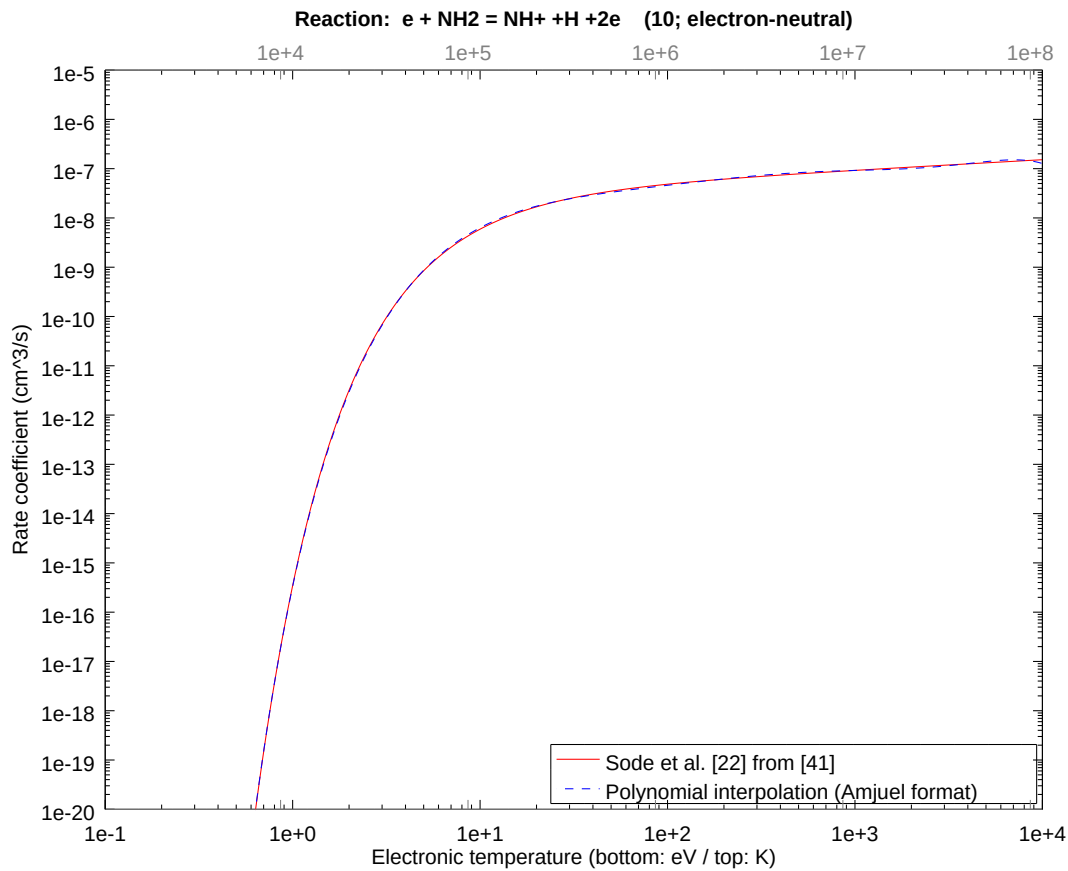
2.9 Reaction 09 $e + \text{NH}_2 = \text{NH}_2^+ + 2e$

b0	-3.058208344880E+01	b1	1.279264397226E+01	b2	-6.123978850051E+00
b3	2.123482700099E+00	b4	-5.592003095574E-01	b5	1.048142723007E-01
b6	-1.247305603473E-02	b7	8.216877866463E-04	b8	-2.257674124117E-05



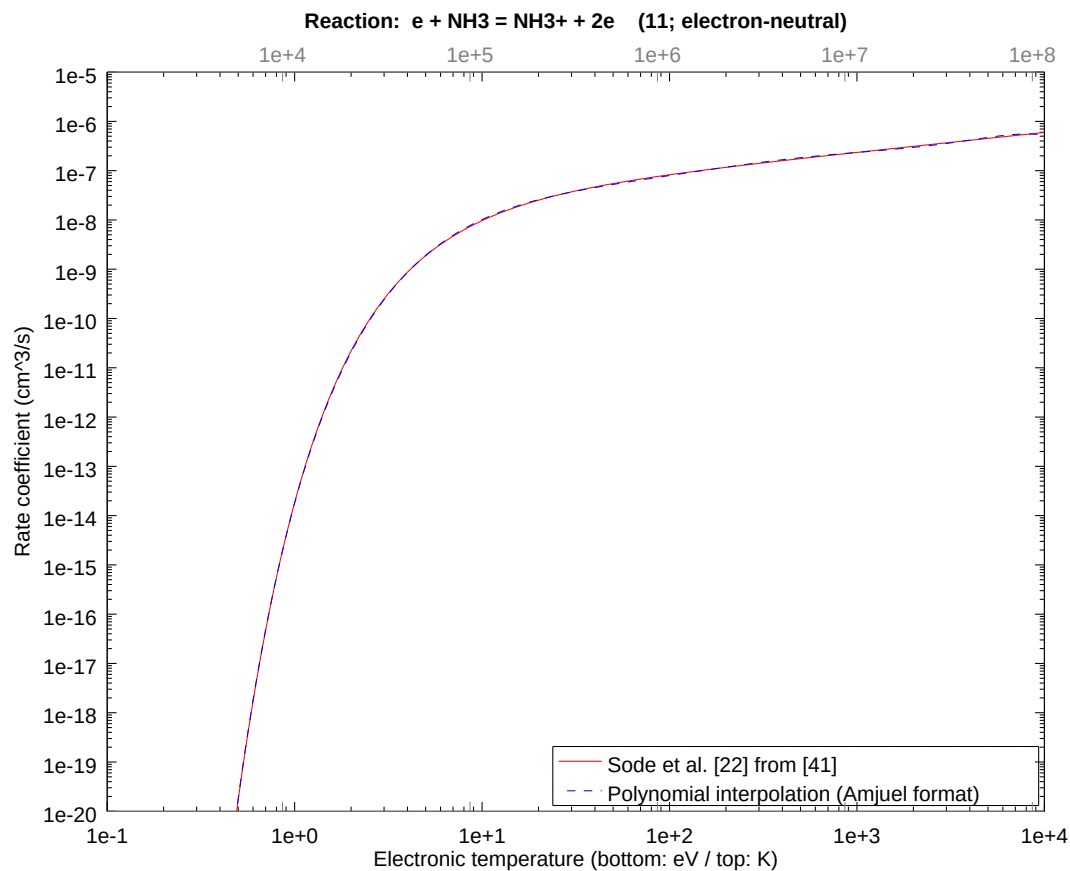
2.10 Reaction 10 $e + \text{NH}_2 = \text{NH}^+ + \text{H} + 2e$

b0	-3.563666632219E+01	b1	1.802442033722E+01	b2	-8.874830639953E+00
b3	3.077337429095E+00	b4	-8.103894808667E-01	b5	1.518961671970E-01
b6	-1.807587233421E-02	b7	1.190784639196E-03	b8	-3.271806775030E-05



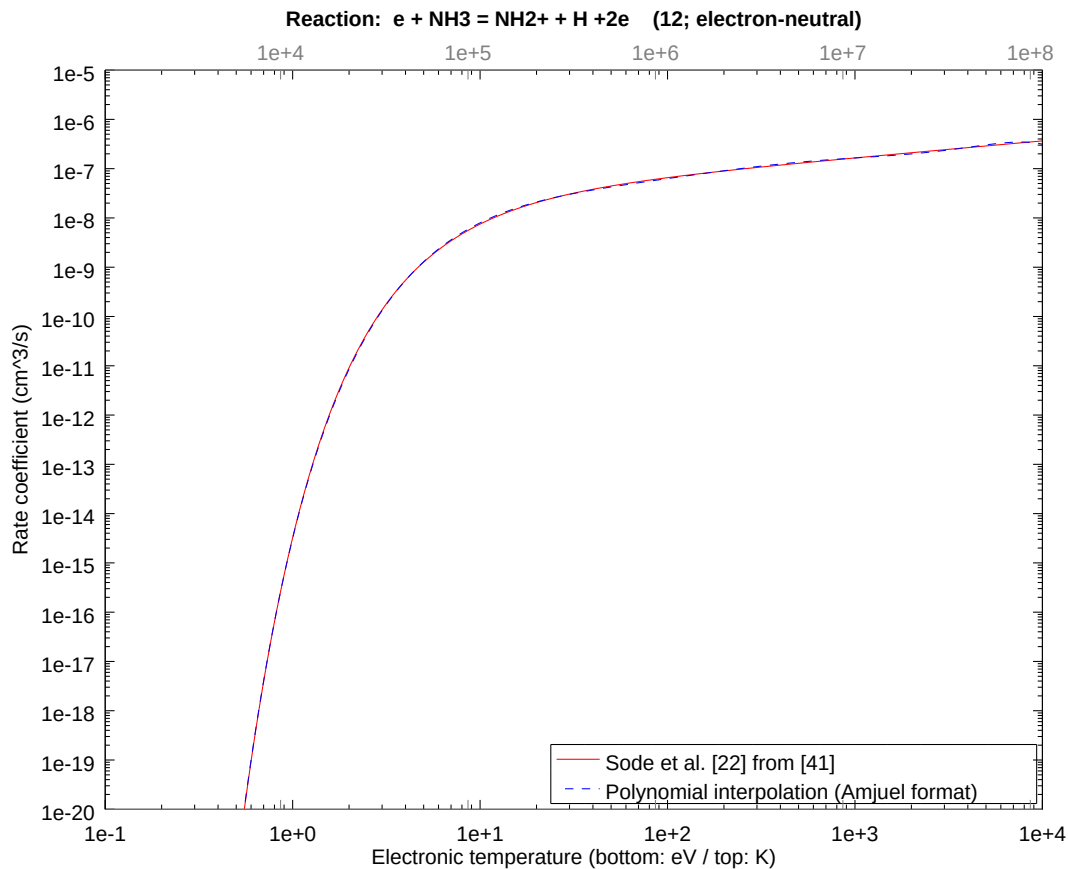
2.11 Reaction 11 $e + \text{NH}_3 = \text{NH}_3^+ + 2e$

b0	-3.165130180136E+01	b1	1.389216810181E+01	b2	-6.721560657193E+00
b3	2.330693511964E+00	b4	-6.137674365384E-01	b5	1.150421166139E-01
b6	-1.369018488974E-02	b7	9.018686109884E-04	b8	-2.477979421712E-05



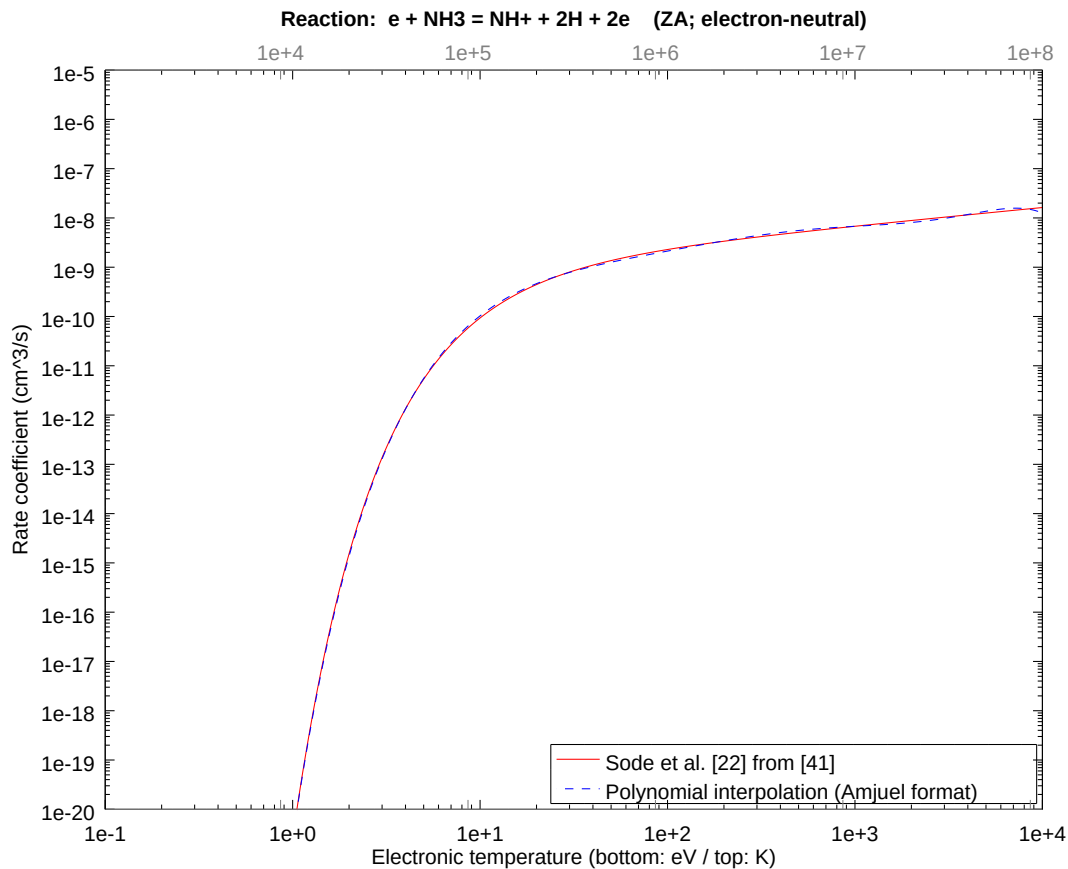
2.12 Reaction 12 $e + \text{NH}_3 = \text{NH}_2^+ + \text{H} + 2e$

b0	-3.339021332416E+01	b1	1.561658416230E+01	b2	-7.610525328975E+00
b3	2.638941000688E+00	b4	-6.949416750226E-01	b5	1.302570916252E-01
b6	-1.550078979800E-02	b7	1.021145870340E-03	b8	-2.805706310696E-05



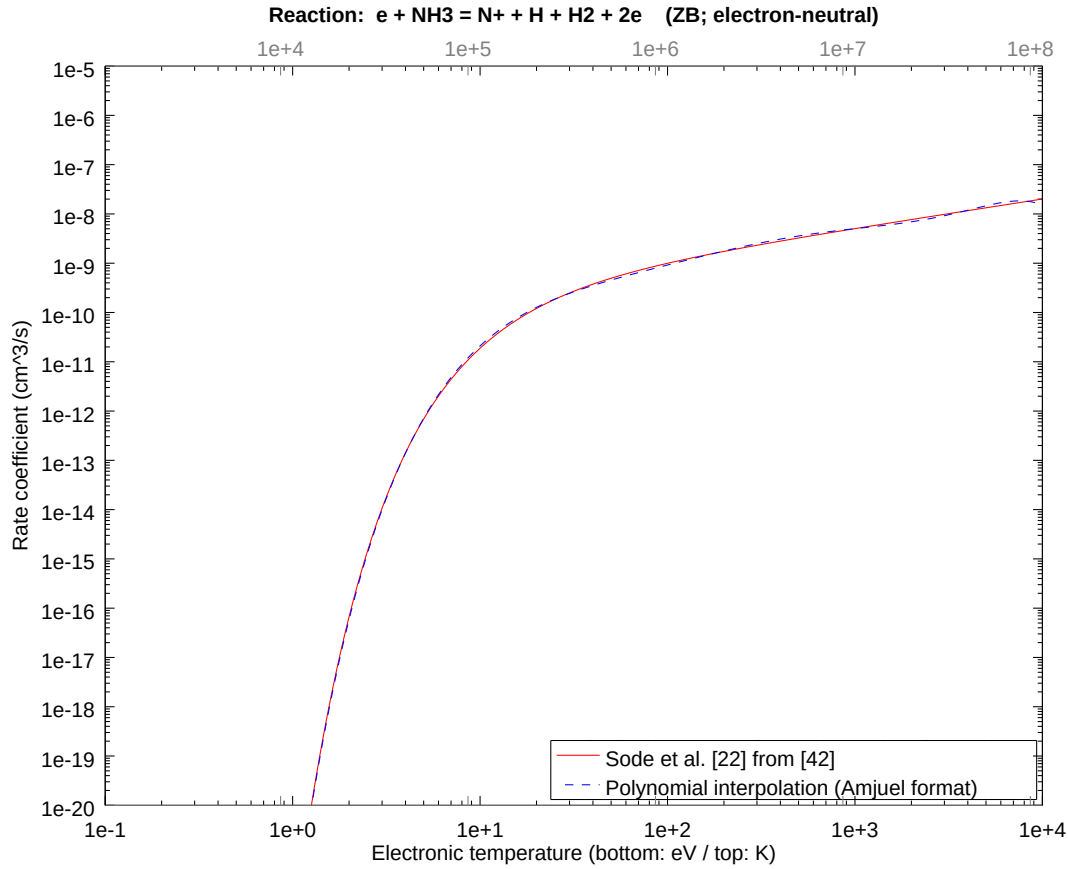
2.13 Reaction ZA $e + \text{NH}_3 = \text{NH}^+ + 2\text{H} + 2e$

b0	-4.744940094576E+01	b1	2.620437918686E+01	b2	-1.287023297035E+01
b3	4.462738642305E+00	b4	-1.175222586054E+00	b5	2.202790271093E-01
b6	-2.621353550525E-02	b7	1.726869654839E-03	b8	-4.744757070522E-05



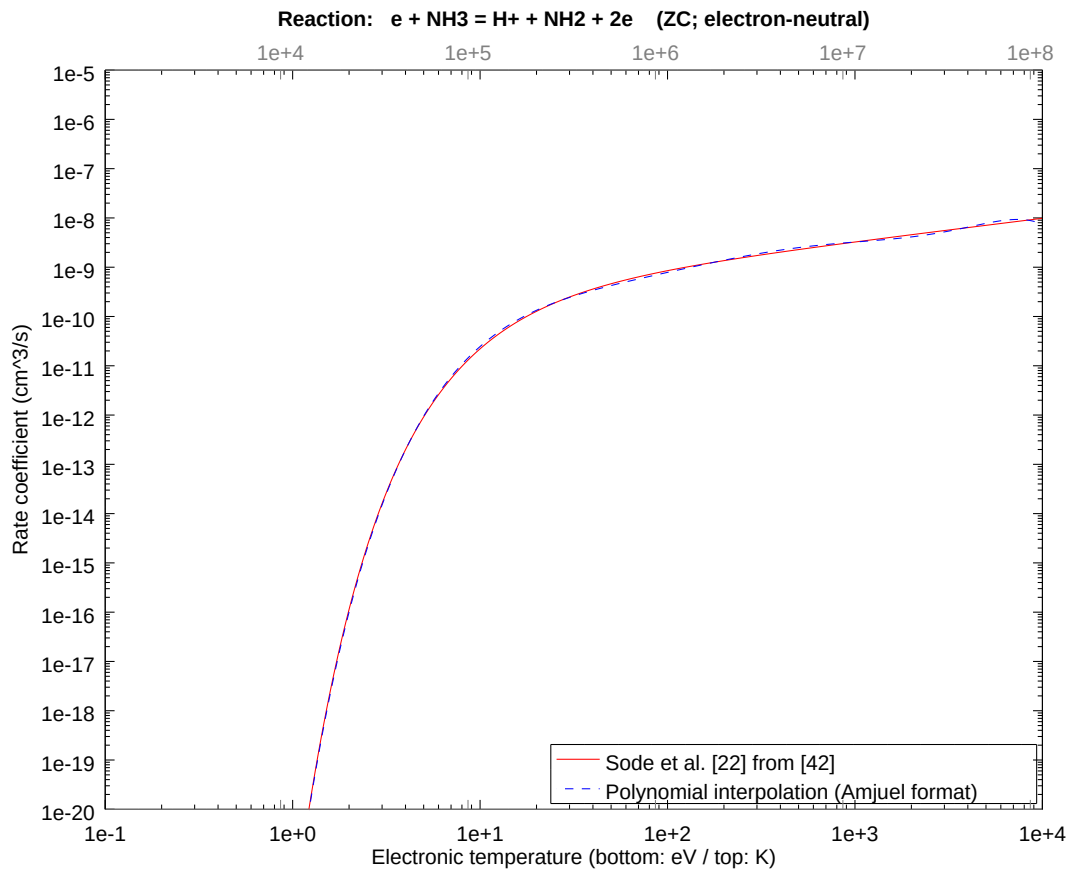
2.14 Reaction ZB $e + \text{NH}_3 = \text{N} + \text{H} + \text{H}_2 + 2e$

b0	-5.220926834260E+01	b1	2.933892541899E+01	b2	-1.432220860093E+01
b3	4.966209540554E+00	b4	-1.307807175578E+00	b5	2.451301529612E-01
b6	-2.917085685543E-02	b7	1.921689178447E-03	b8	-5.280044322530E-05



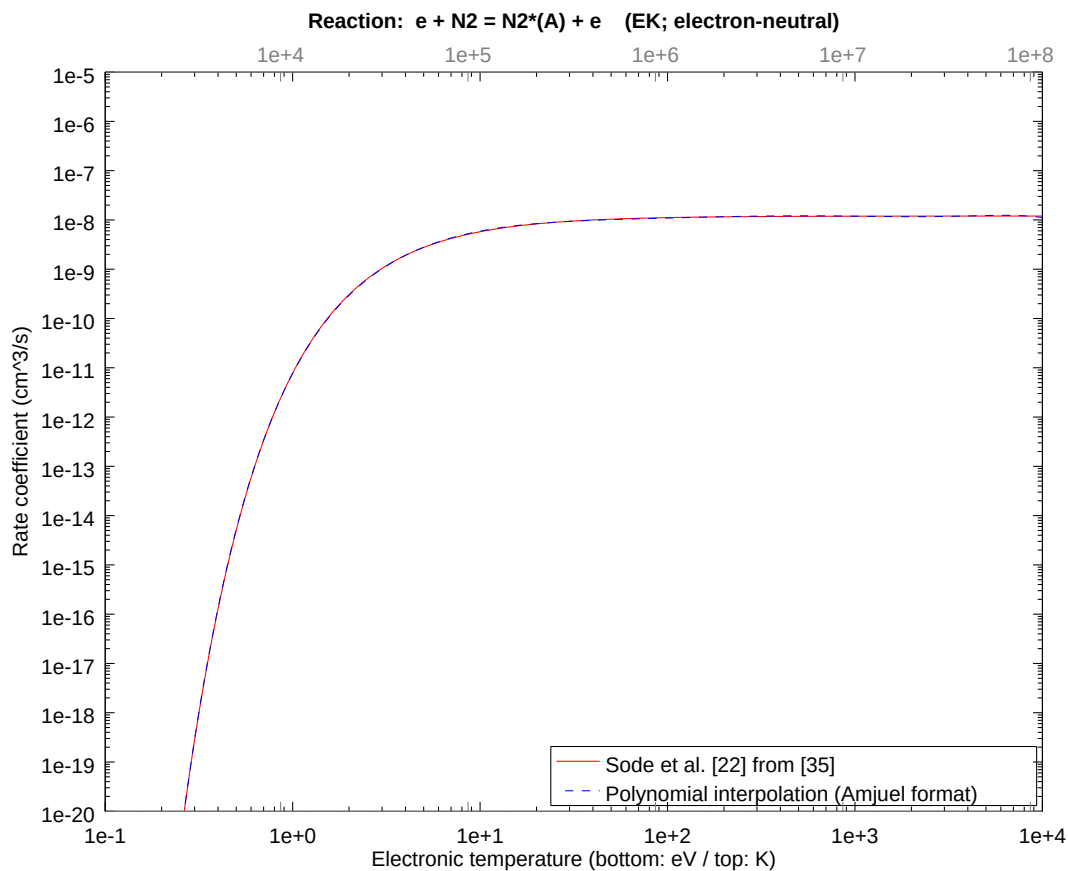
2.15 Reaction ZC $e + \text{NH}_3 = \text{H} + \text{NH}_2 + 2e$

b0	-5.136820819564E+01	b1	2.877282140387E+01	b2	-1.409996743298E+01
b3	4.889147668373E+00	b4	-1.287513615957E+00	b5	2.413264092084E-01
b6	-2.871820562836E-02	b7	1.891869863609E-03	b8	-5.198112600285E-05



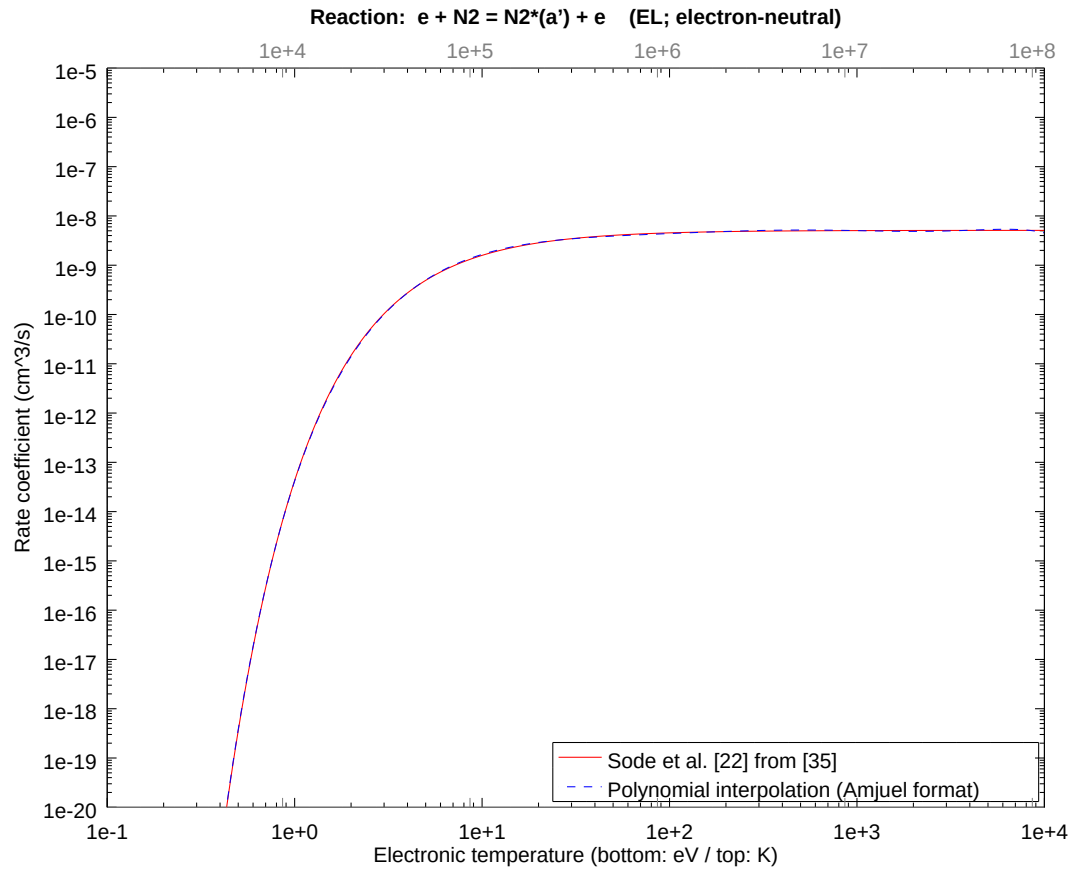
2.16 Reaction EK $e + N_2 = N_2^*(A) + e$

b0	-2.559242769964E+01	b1	7.276452157772E+00	b2	-3.625000383821E+00
b3	1.256964759575E+00	b4	-3.310105058187E-01	b5	6.204328699087E-02
b6	-7.383244459270E-03	b7	4.863861575793E-04	b8	-1.336397425082E-05



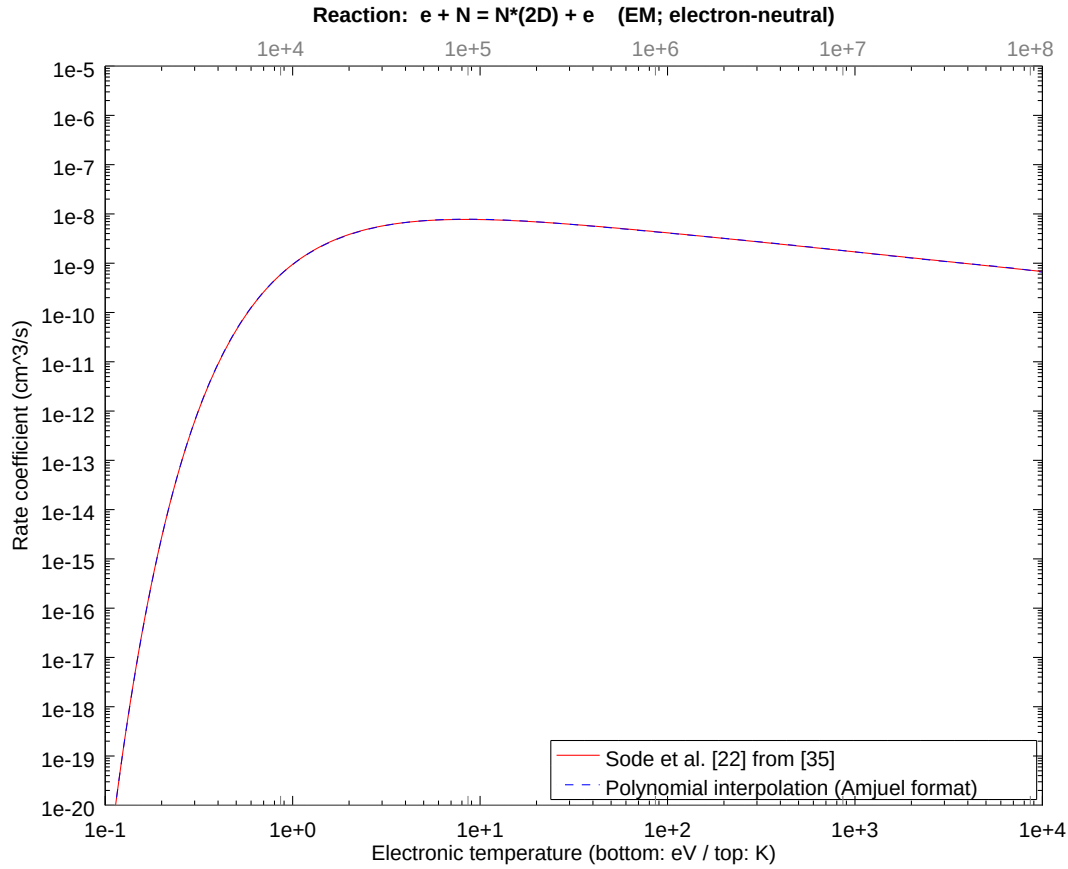
2.17 Reaction EL $e + N_2 = N_2^*(a') + e$

b0	-3.080643141587E+01	b1	1.158879097062E+01	b2	-5.773331673960E+00
b3	2.001896190658E+00	b4	-5.271815821553E-01	b5	9.881280993505E-02
b6	-1.175887298758E-02	b7	7.746395343463E-04	b8	-2.128404073461E-05



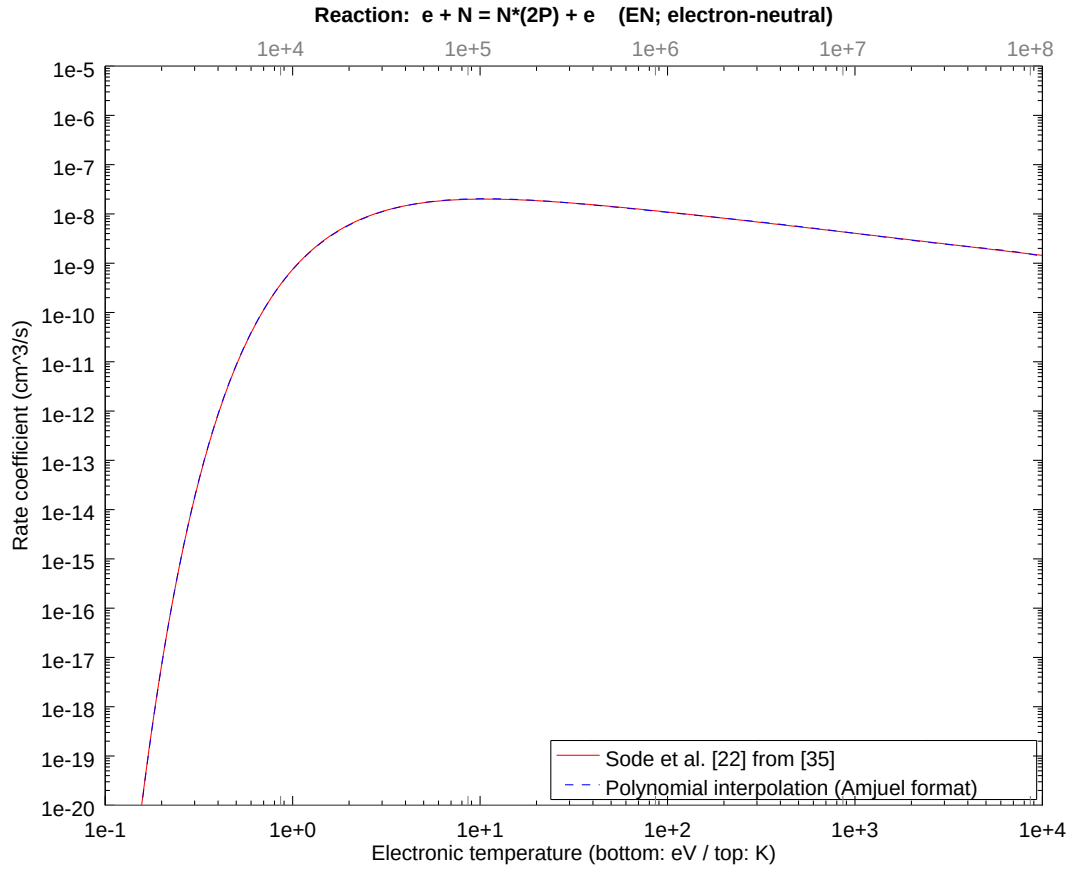
2.18 Reaction EM $e + N = N^*(2D) + e$

b0	-2.078384988604E+01	b1	2.920996557021E+00	b2	-1.654462028038E+00
b3	5.736828262365E-01	b4	-1.510742771788E-01	b5	2.831675904897E-02
b6	-3.369736912610E-03	b7	2.219882326827E-04	b8	-6.099361544993E-06



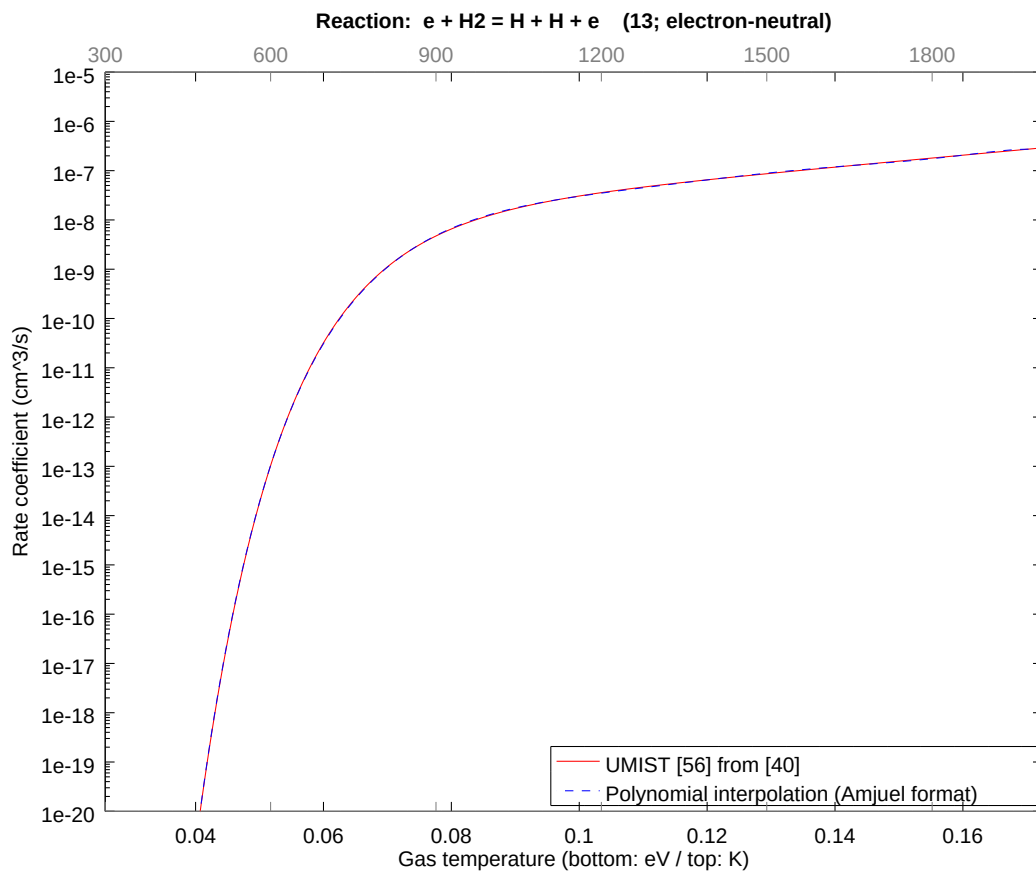
2.19 Reaction $\text{EN } e + \text{N} = \text{N}^*(2\text{P}) + e$

b0	-2.102160644758E+01	b1	4.308442827971E+00	b2	-2.370572458084E+00
b3	8.219933032642E-01	b4	-2.164646359577E-01	b5	4.057326669703E-02
b6	-4.828279755381E-03	b7	3.180726916050E-04	b8	-8.739383706258E-06



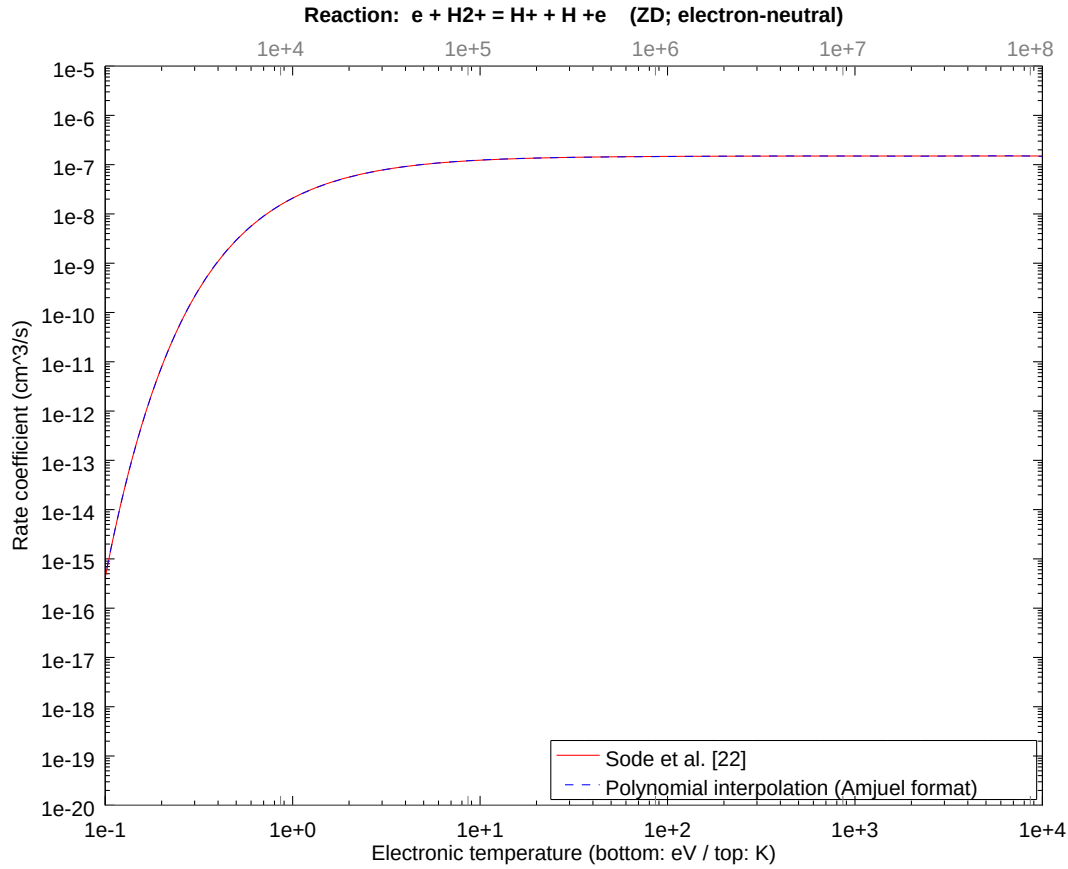
2.20 Reaction 13 $e + H_2 = H + H + e$

b0	-2.708571324418E+01	b1	9.068035062033E+00	b2	-4.343171611769E+00
b3	1.505989815931E+00	b4	-3.965890427172E-01	b5	7.433506599401E-02
b6	-8.845984646340E-03	b7	5.827471250443E-04	b8	-1.601159377683E-05



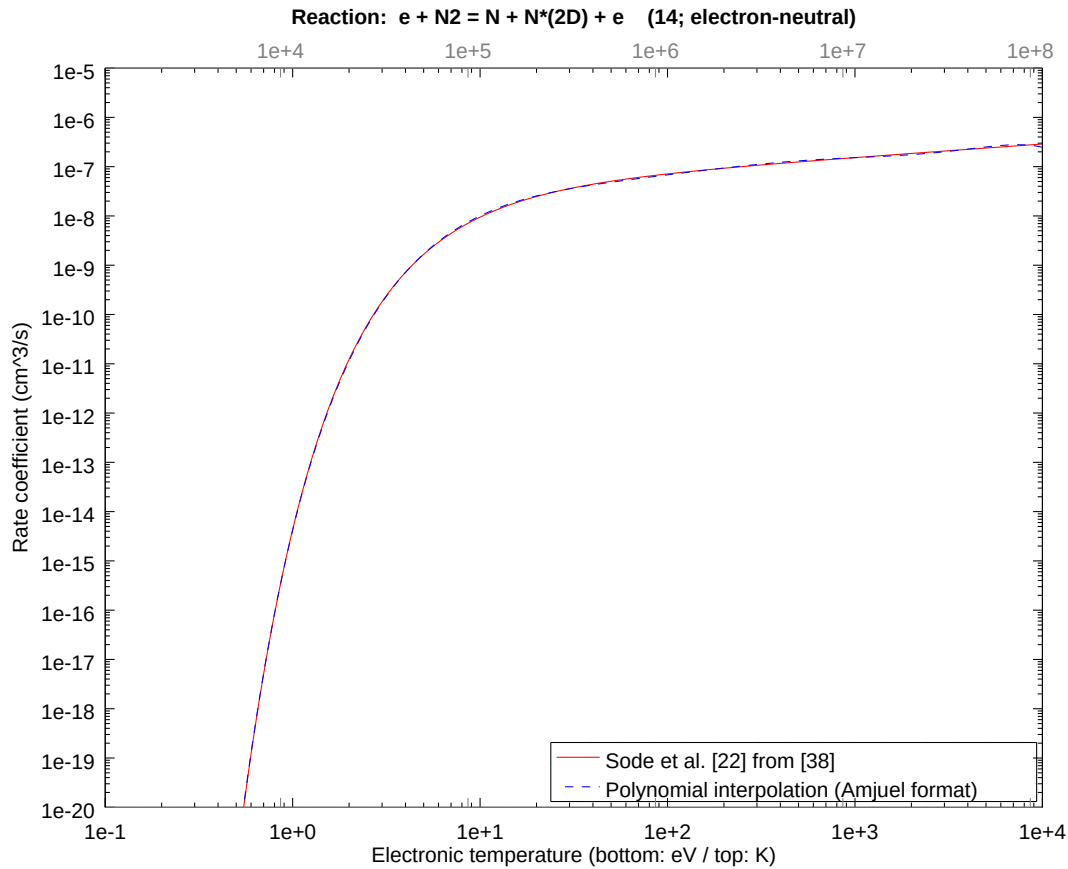
2.21 Reaction ZD $e + H_2^+ = H^+ + H + e$

b0	-1.768640642427E+01	b1	1.952944243980E+00	b2	-9.729224463388E-01
b3	3.373597515480E-01	b4	-8.884069434098E-02	b5	1.665194487357E-02
b6	-1.981606482938E-03	b7	1.305423338462E-04	b8	-3.586788729444E-06



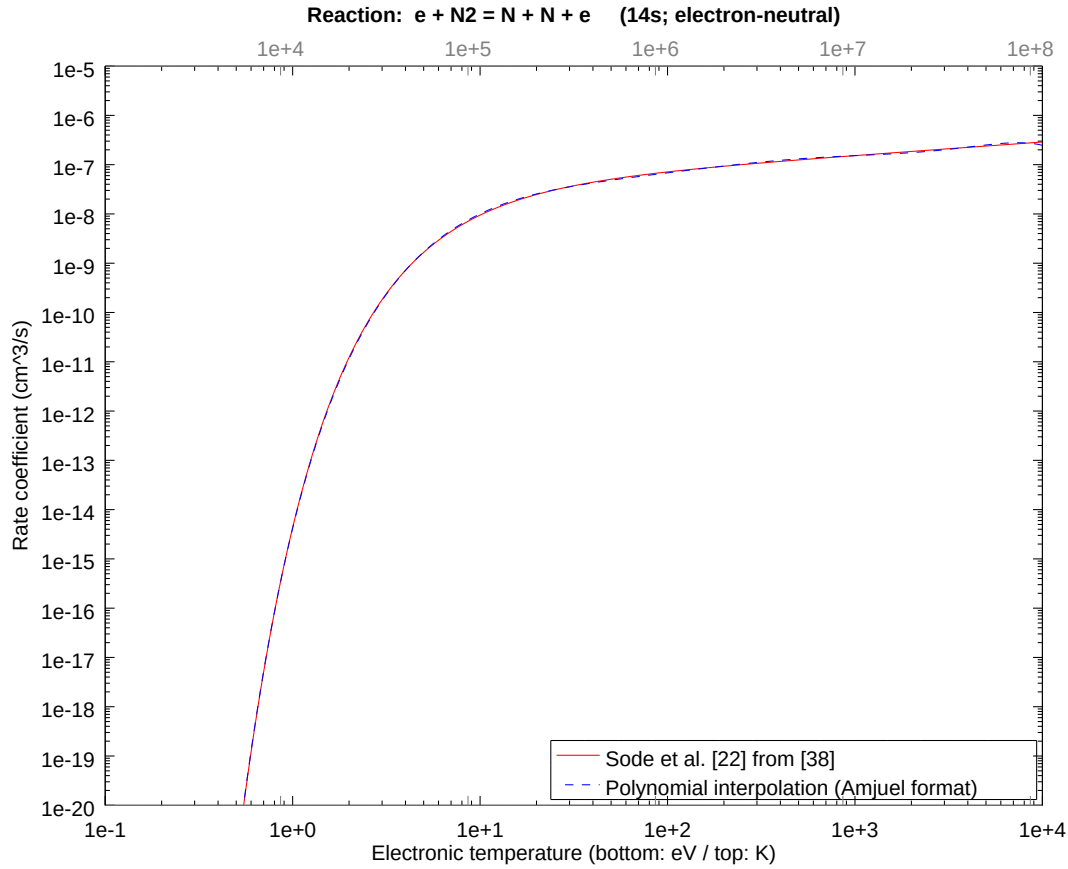
2.22 Reaction 14 $e + N_2 = N + N^*(2D) + e$

b0	-3.310497821898E+01	b1	1.566554523300E+01	b2	-7.669789640427E+00
b3	2.659490833269E+00	b4	-7.003532909215E-01	b5	1.312714232927E-01
b6	-1.562149679189E-02	b7	1.029097687630E-03	b8	-2.827554769962E-05



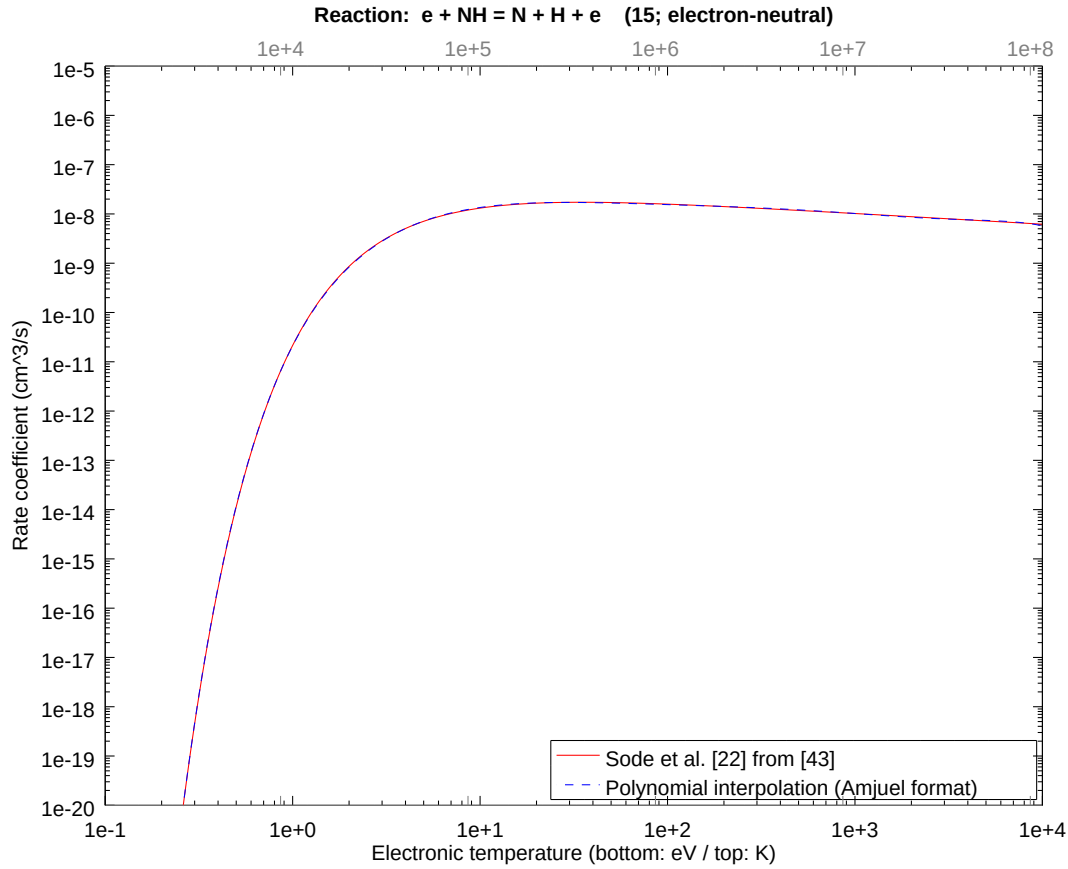
2.23 Reaction 14s $e + N_2 = N + N + e$

b0	-3.310497821898E+01	b1	1.566554523300E+01	b2	-7.669789640427E+00
b3	2.659490833269E+00	b4	-7.003532909215E-01	b5	1.312714232927E-01
b6	-1.562149679189E-02	b7	1.029097687630E-03	b8	-2.827554769962E-05



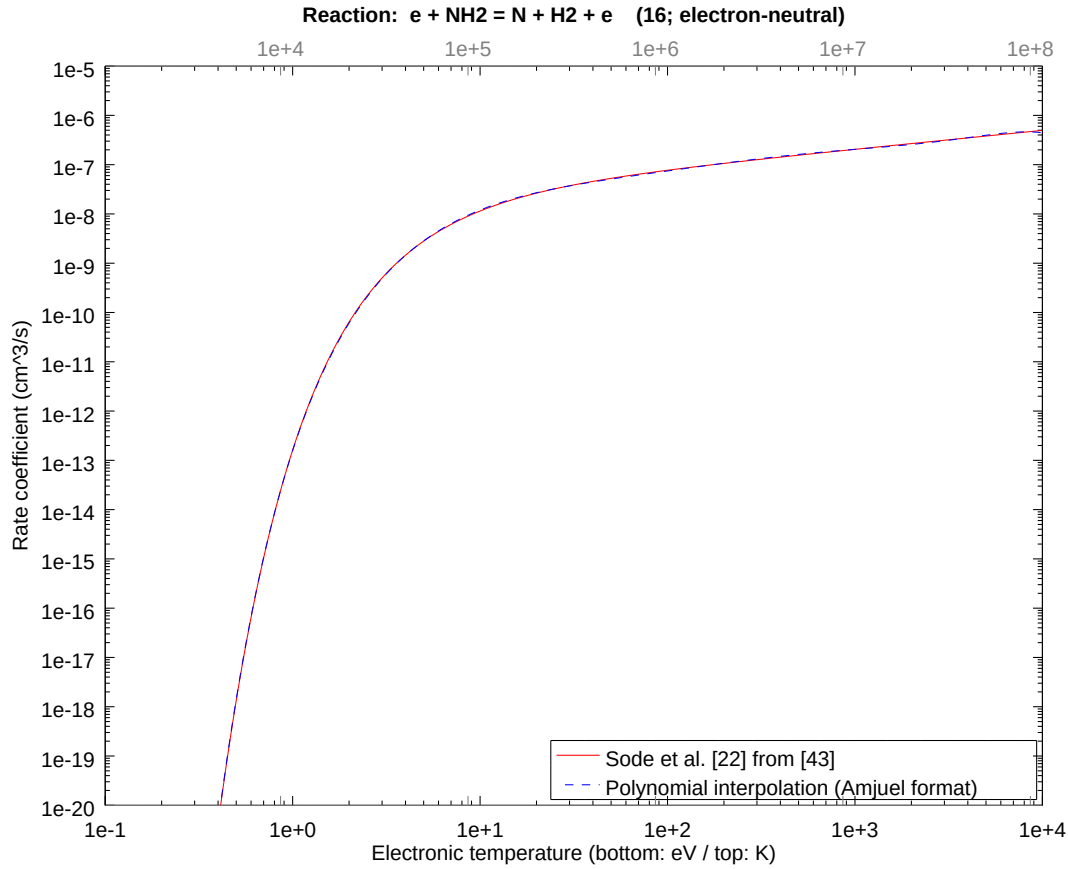
2.24 Reaction 15 $e + \text{NH} = \text{N} + \text{H} + e$

b0	-2.457785758960E+01	b1	7.403421947312E+00	b2	-3.797854625556E+00
b3	1.316901771271E+00	b4	-3.467943855239E-01	b5	6.500175435420E-02
b6	-7.735306524768E-03	b7	5.095789580089E-04	b8	-1.400122097940E-05



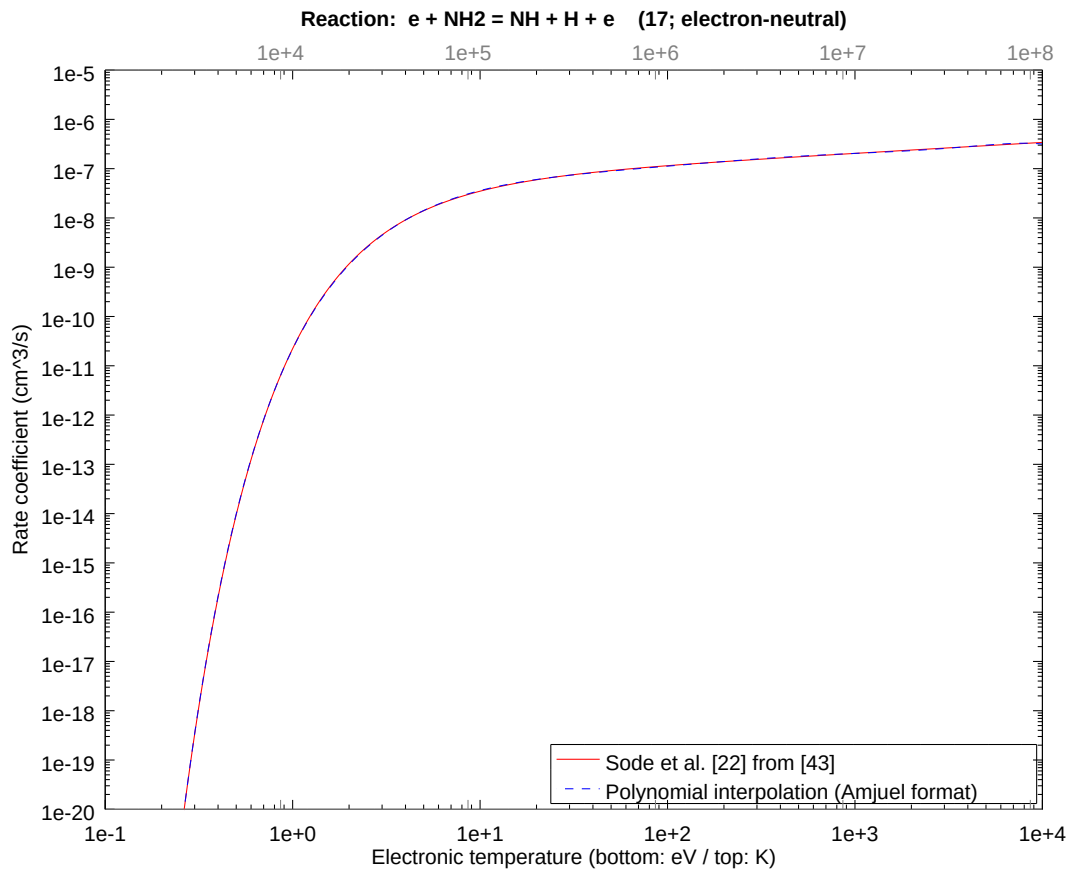
2.25 Reaction 16 $e + \text{NH}_2 = \text{N} + \text{H}_2 + e$

b0	-2.947714258173E+01	b1	1.172095540666E+01	b2	-5.649864358434E+00
b3	1.959084039446E+00	b4	-5.159073823659E-01	b5	9.669961896125E-02
b6	-1.150740008366E-02	b7	7.580732483252E-04	b8	-2.082886449991E-05



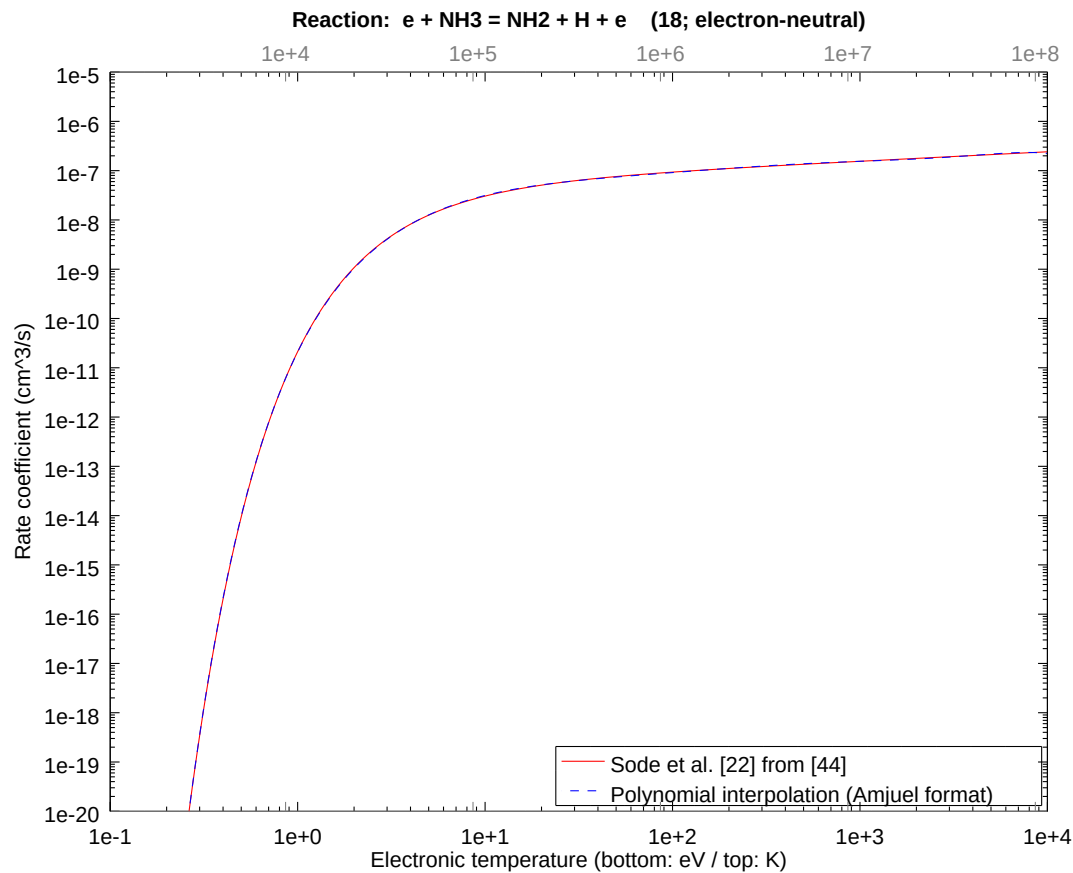
2.26 Reaction 17 $e + \text{NH}_2 = \text{NH} + \text{H} + e$

b0	-2.454118936625E+01	b1	7.764114566845E+00	b2	-3.758345084588E+00
b3	1.303201882883E+00	b4	-3.431866415913E-01	b5	6.432553324258E-02
b6	-7.654835195510E-03	b7	5.042777464821E-04	b8	-1.385556458430E-05



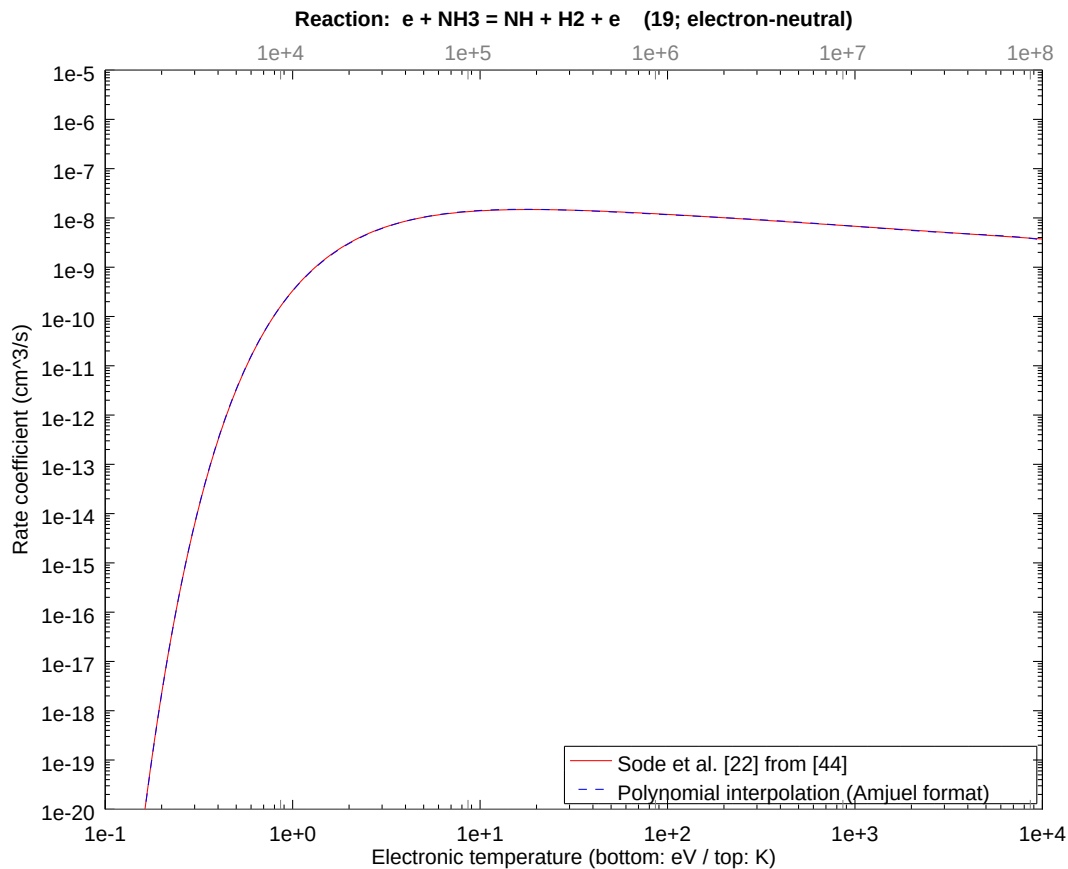
2.27 Reaction 18 $e + \text{NH}_3 = \text{NH}_2 + \text{H} + e$

b0	-2.459014390392E+01	b1	7.714287721729E+00	b2	-3.748467699346E+00
b3	1.299776910786E+00	b4	-3.422847056081E-01	b5	6.415647796467E-02
b6	-7.634717363196E-03	b7	5.029524436004E-04	b8	-1.381915048552E-05



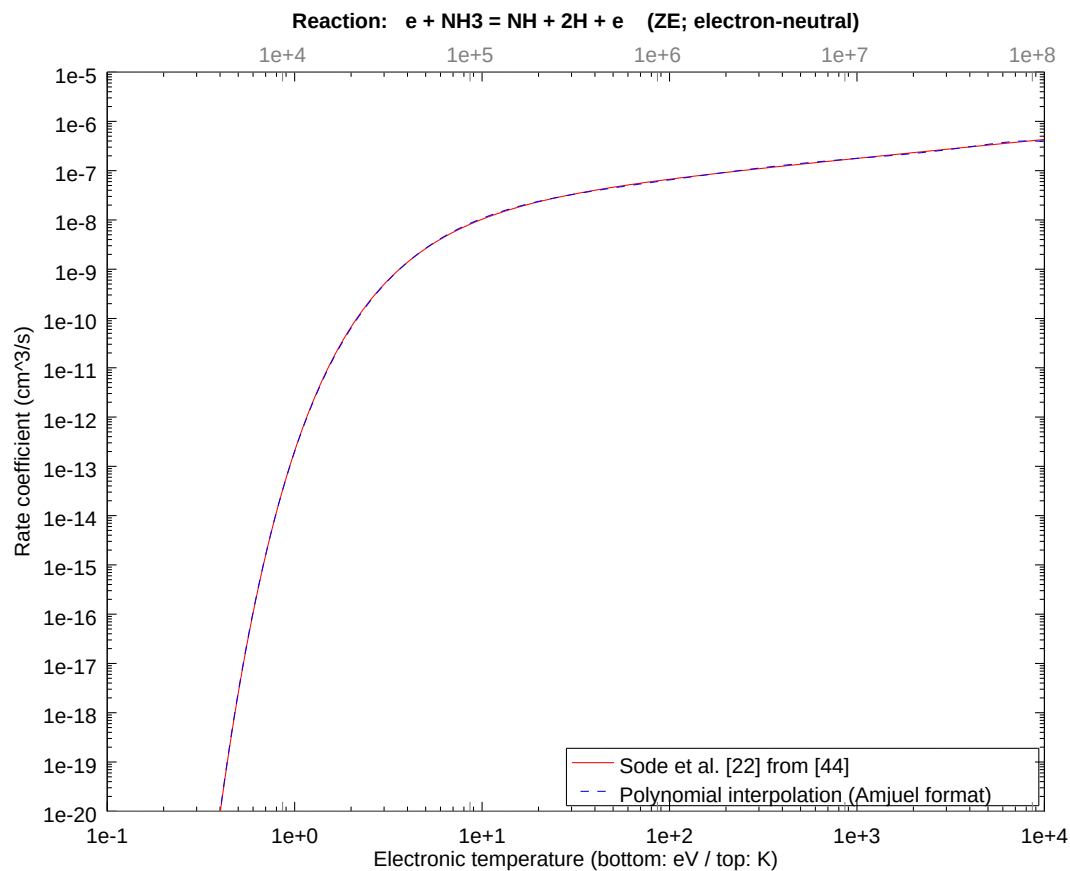
2.28 Reaction 19 $e + \text{NH}_3 = \text{NH} + \text{H}_2 + e$

b0	-2.182891305431E+01	b1	4.508356250529E+00	b2	-2.375511150705E+00
b3	8.237057893126E-01	b4	-2.169156039493E-01	b5	4.065779433598E-02
b6	-4.838338671538E-03	b7	3.187353430458E-04	b8	-8.757590755646E-06



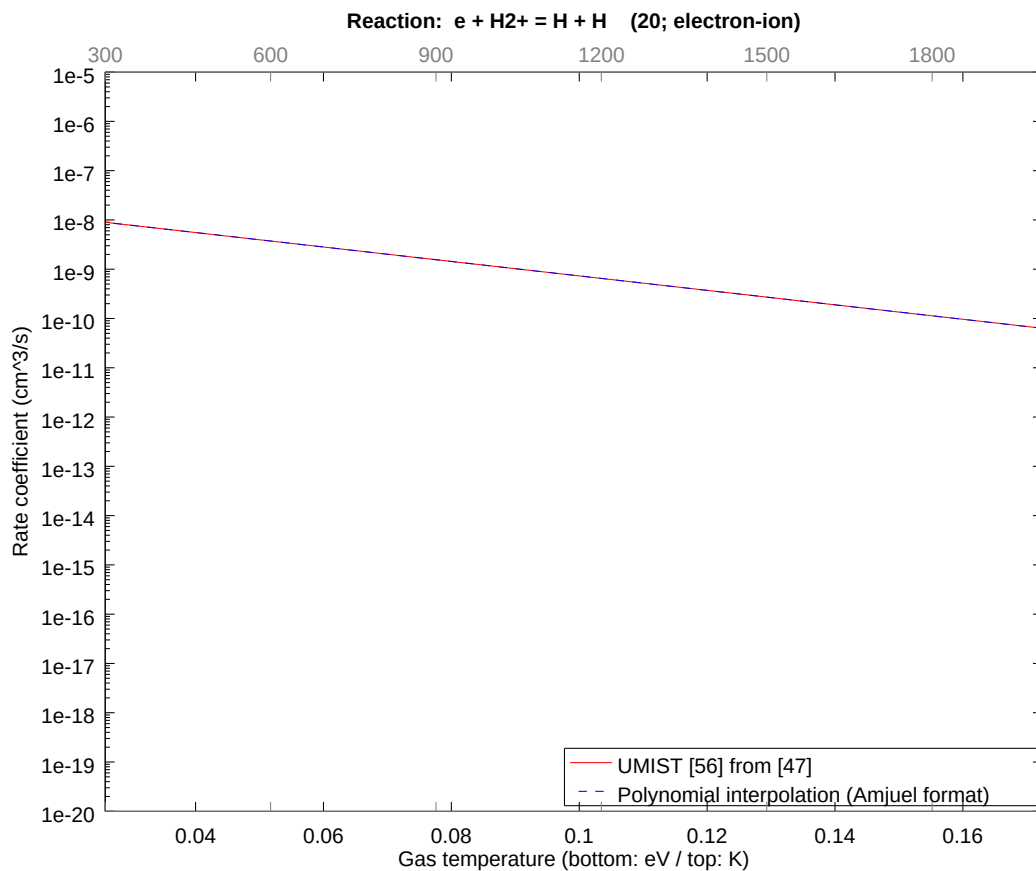
2.29 Reaction ZE $e + \text{NH}_3 = \text{NH} + 2\text{H} + e$

b0	-2.923951508276E+01	b1	1.134424534945E+01	b2	-5.462194038836E+00
b3	1.894009569605E+00	b4	-4.987705986859E-01	b5	9.348756868107E-02
b6	-1.112516126969E-02	b7	7.328924935732E-04	b8	-2.013699662317E-05



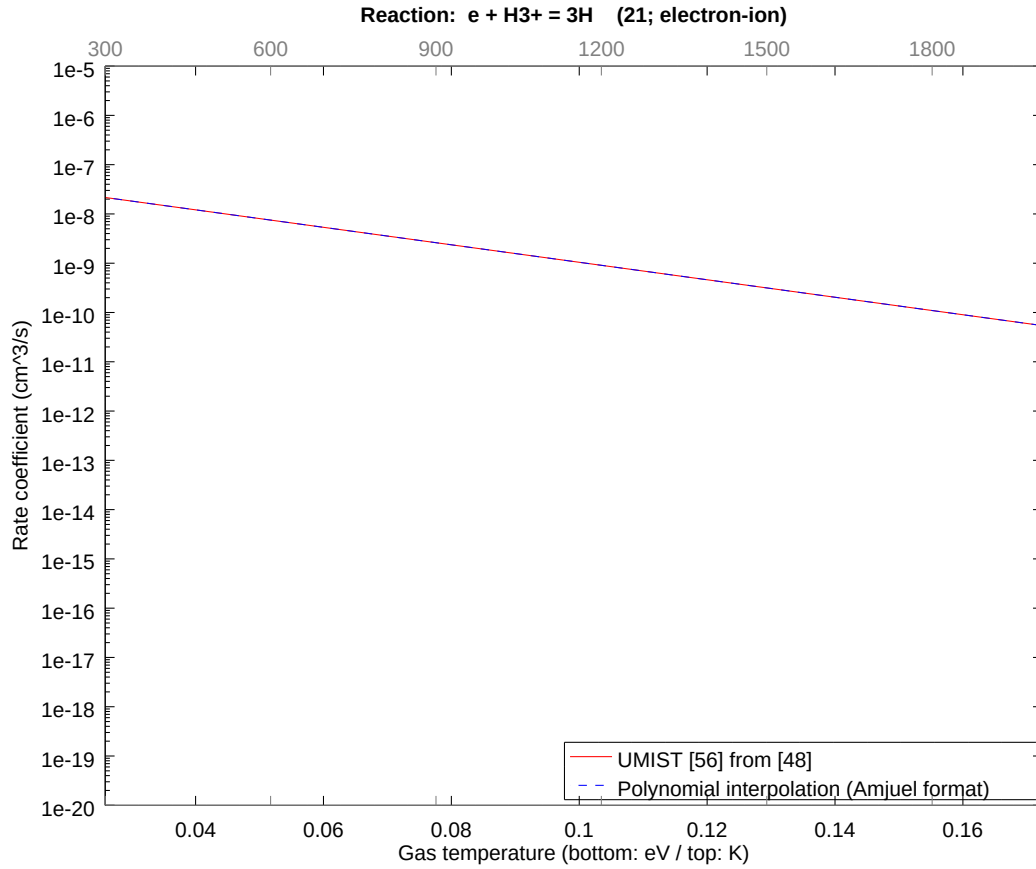
2.30 Reaction 20 $e + H_2^+ = H + H$

b0	-1.952226532449E+01	b1	-4.300000000000E-01	b2	1.176699630083E-14
b3	-4.306129821113E-15	b4	-1.976279697331E-15	b5	1.188464274130E-15
b6	-2.107638125050E-16	b7	1.540936682618E-17	b8	-3.878633051697E-19



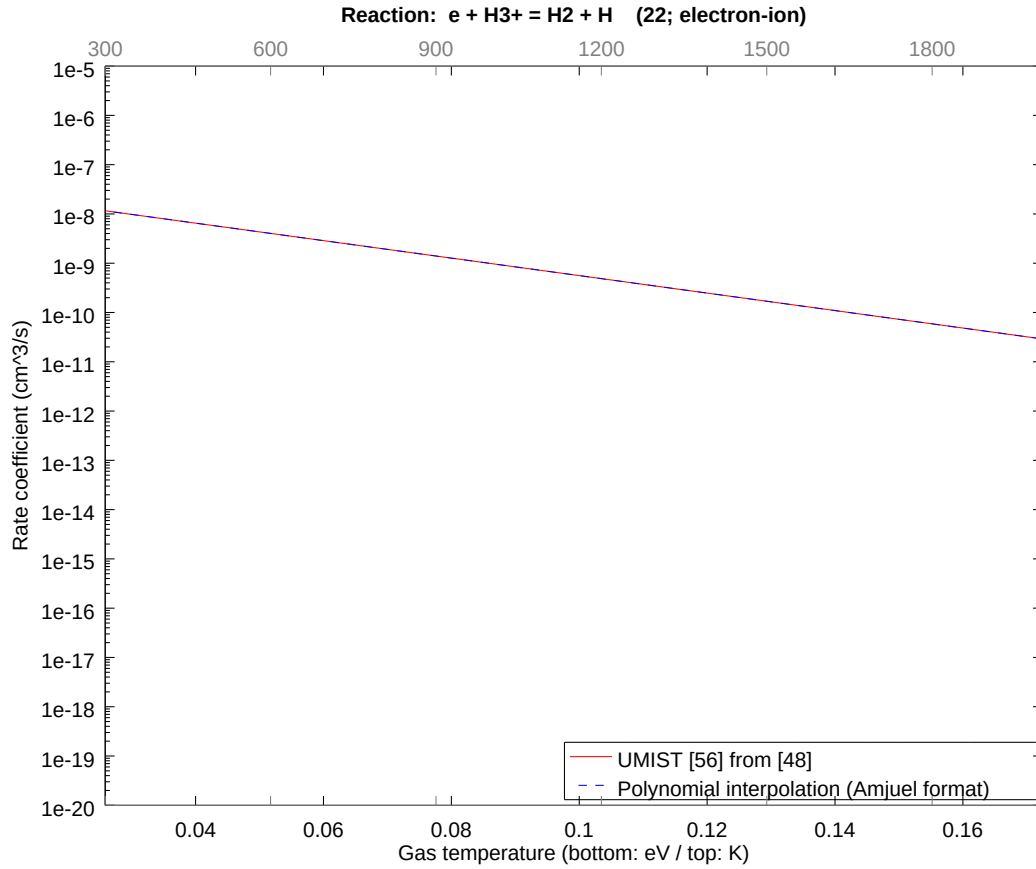
2.31 Reaction 21 $e + H_3^+ = 3H$

b0	-1.884873396354E+01	b1	-5.200000000000E-01	b2	1.568932840110E-14
b3	-3.526376612459E-15	b4	-2.691319284541E-15	b5	1.154103018397E-15
b6	-1.604149159172E-16	b7	8.482332459835E-18	b8	-1.050069624873E-19



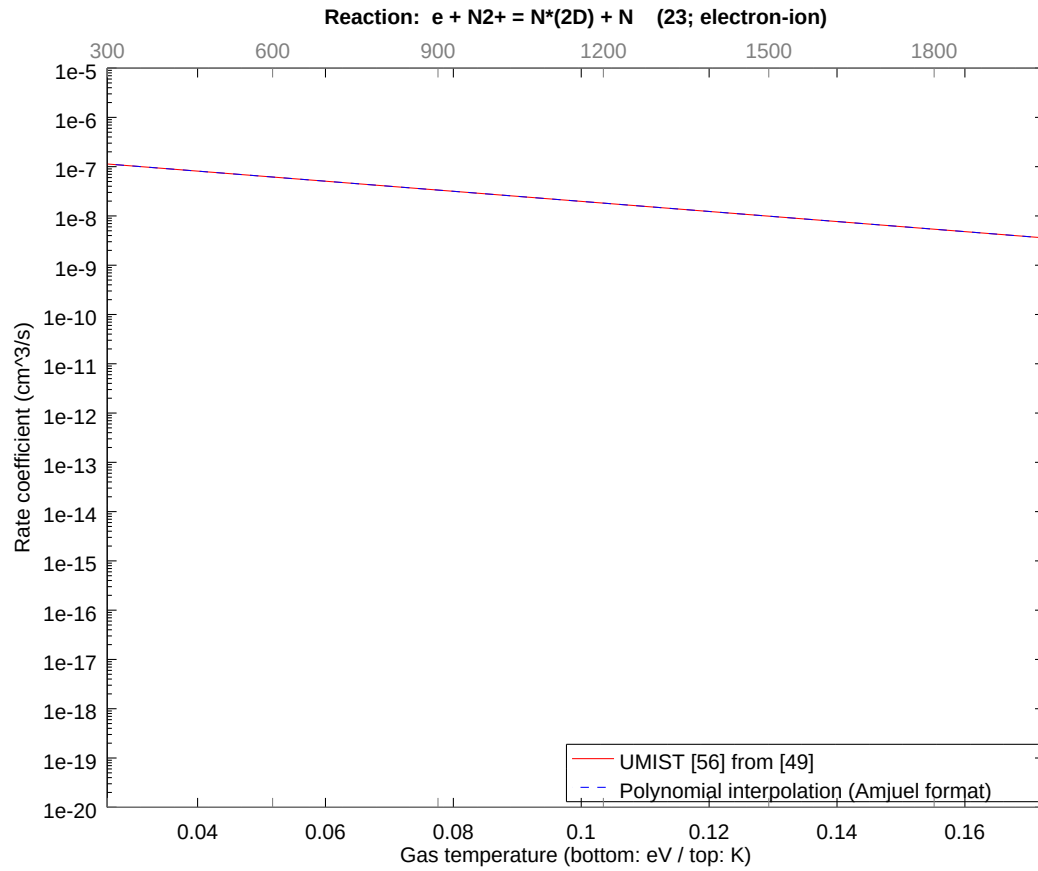
2.32 Reaction 22 $e + H_3^+ = H_2 + H$

b0	-1.947105509154E+01	b1	-5.200000000000E-01	b2	1.176699630083E-14
b3	3.446824533805E-15	b4	-2.138917803282E-15	b5	-2.721988437046E-16
b6	2.082826560054E-16	b7	-2.858807325302E-17	b8	1.225739194722E-18



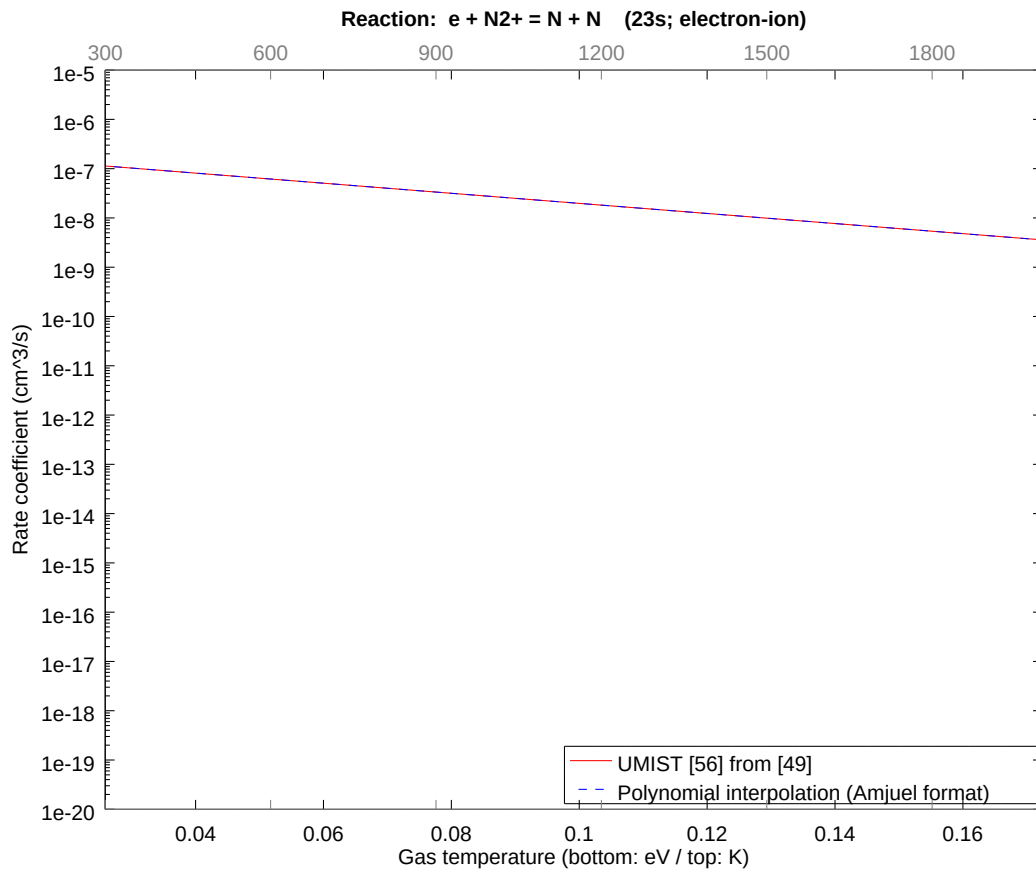
2.33 Reaction 23 $e + N_2^+ = N^*(2D) + N$

b0	-1.668392429045E+01	b1	-3.000000000000E-01	b2	1.961166050138E-14
b3	-5.930872513862E-15	b4	-2.419718156833E-15	b5	1.337121931740E-15
b6	-2.261907631853E-16	b7	1.639312063691E-17	b8	-4.303619611140E-19



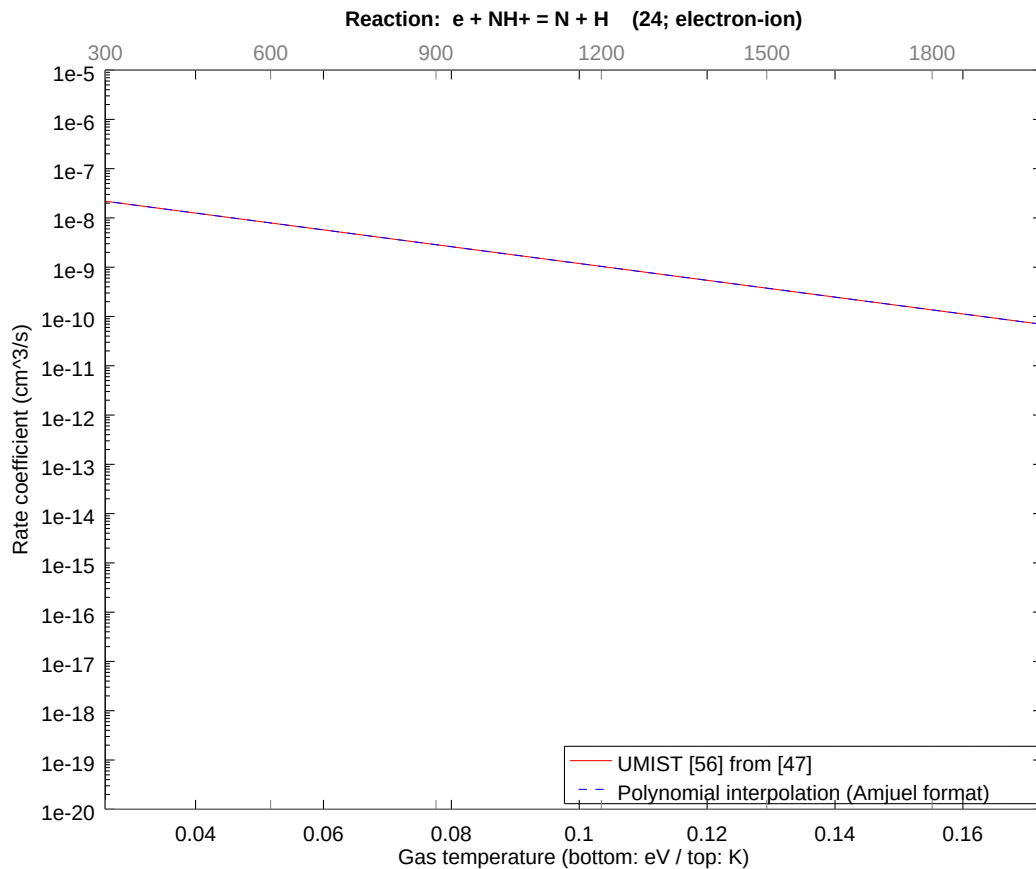
2.34 Reaction 23s $e + N_2^+ = N + N$

b0	-1.668392429045E+01	b1	-3.000000000000E-01	b2	1.961166050138E-14
b3	-5.930872513862E-15	b4	-2.419718156833E-15	b5	1.337121931740E-15
b6	-2.261907631853E-16	b7	1.639312063691E-17	b8	-4.303619611140E-19



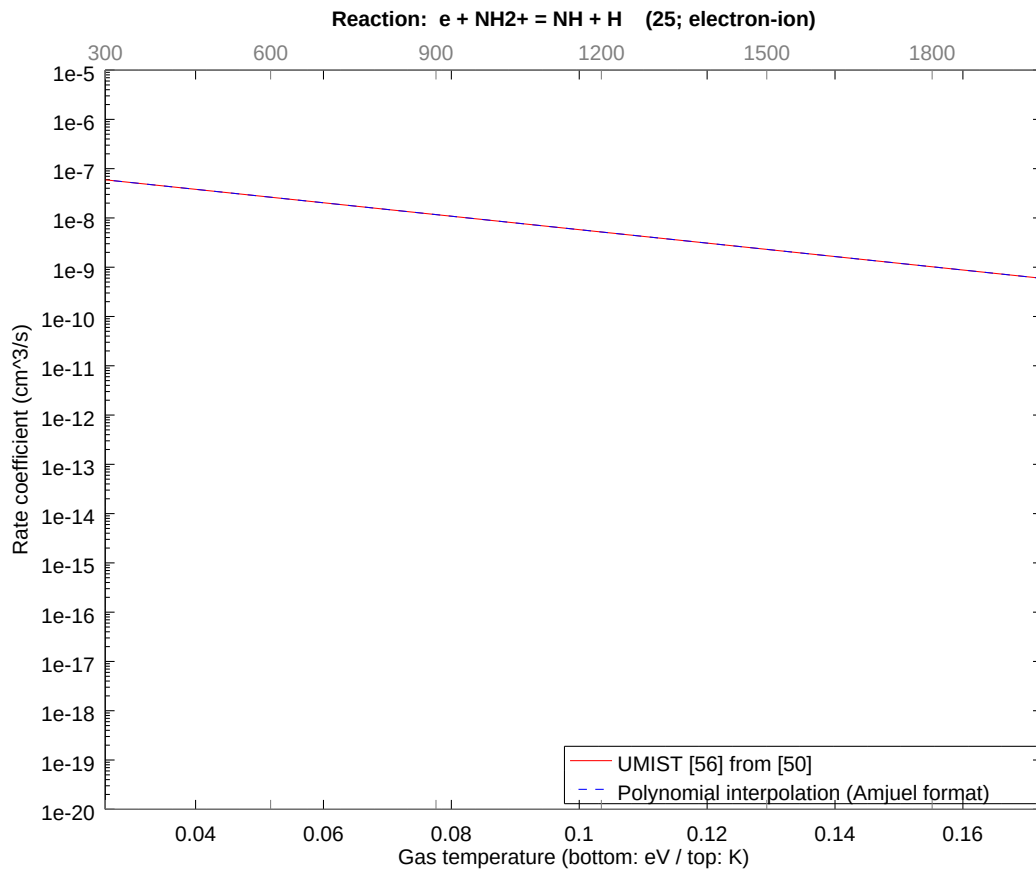
2.35 Reaction 24 $e + \text{NH}^+ = \text{N} + \text{H}$

b0	-1.878949387217E+01	b1	-5.000000000000E-01	b2	3.726215495262E-14
b3	-8.098253227630E-15	b4	-4.407151722654E-15	b5	2.057457077627E-15
b6	-3.302568924012E-16	b7	2.375034314407E-17	b8	-6.443341406356E-19



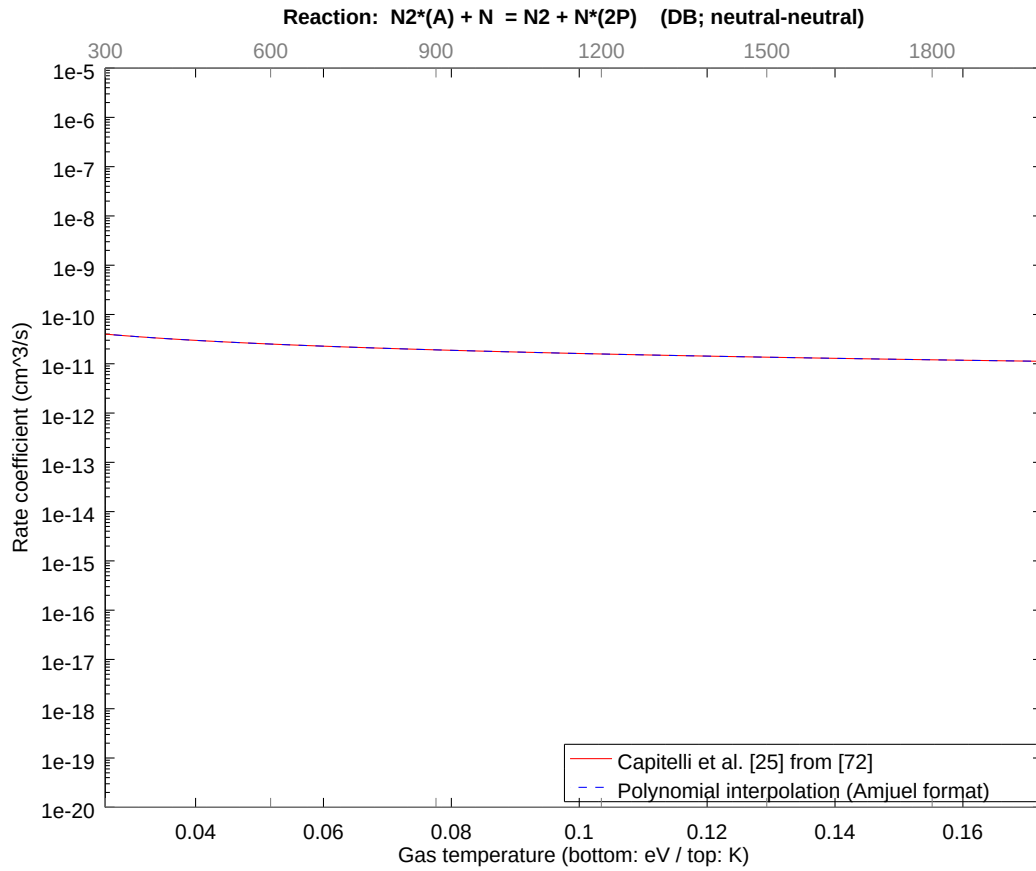
2.36 Reaction 25 $e + \text{NH}_2^+ = \text{NH} + \text{H}$

b0	-1.756023554439E+01	b1	-4.000000000000E-01	b2	3.137865680221E-14
b3	-4.837623409228E-15	b4	-5.267214245188E-15	b5	1.760403301084E-15
b6	-1.625073412217E-16	b7	-2.977237729308E-19	b8	4.512501894627E-19



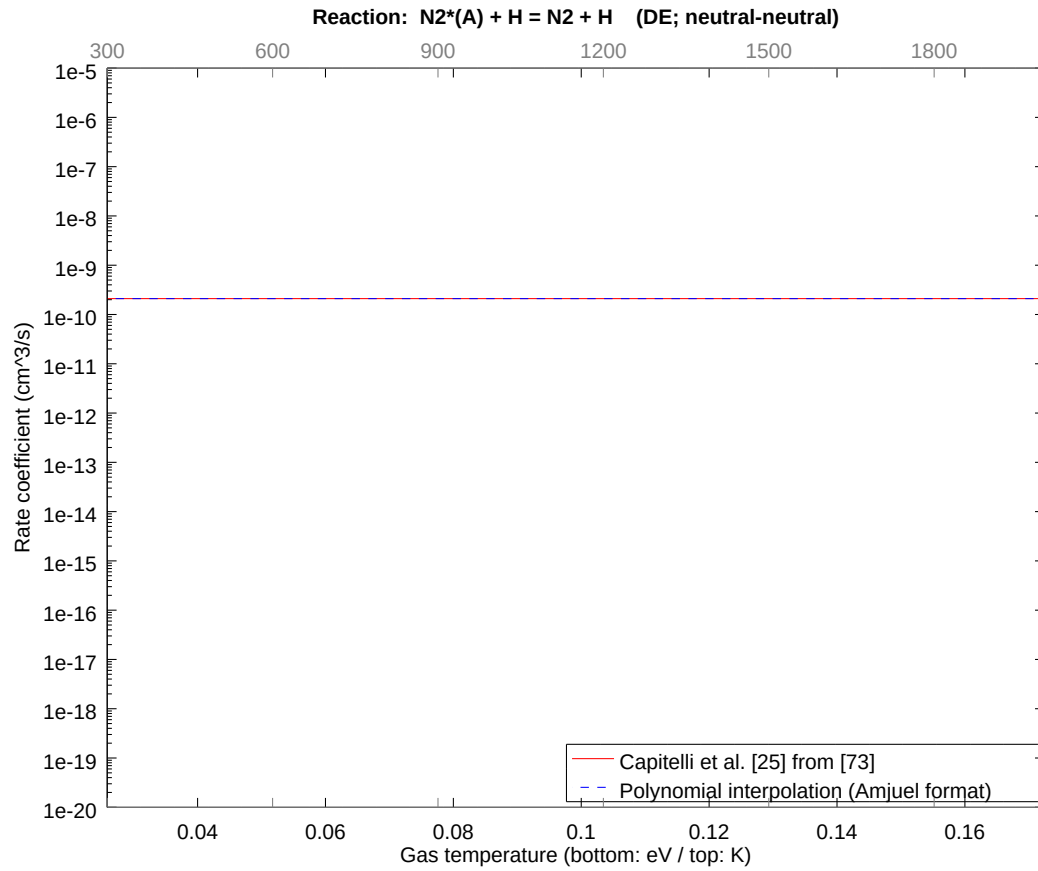
2.37 Reaction DB N2*(A) + N = N2 + N*(2P)

b0 -2.639089538605E+01	b1 -6.700000063335E-01	b2 -8.624699809858E-09
b3 -6.633570679721E-09	b4 -3.151551870313E-09	b5 -9.470599538949E-10
b6 -1.758156215834E-10	b7 -1.843916759489E-11	b8 -8.367108693434E-13



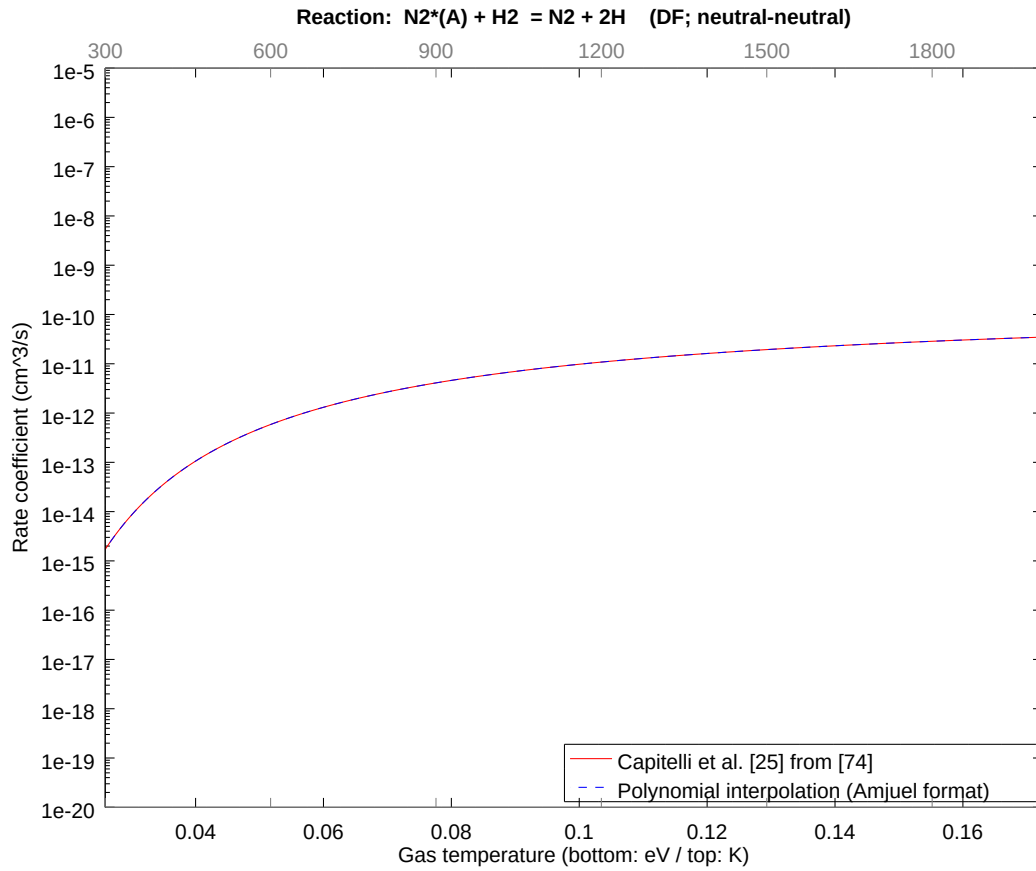
2.38 Reaction DE $\text{N}_2^*(\text{A}) + \text{H} = \text{N}_2 + \text{H}$

b0	-2.228391358521E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



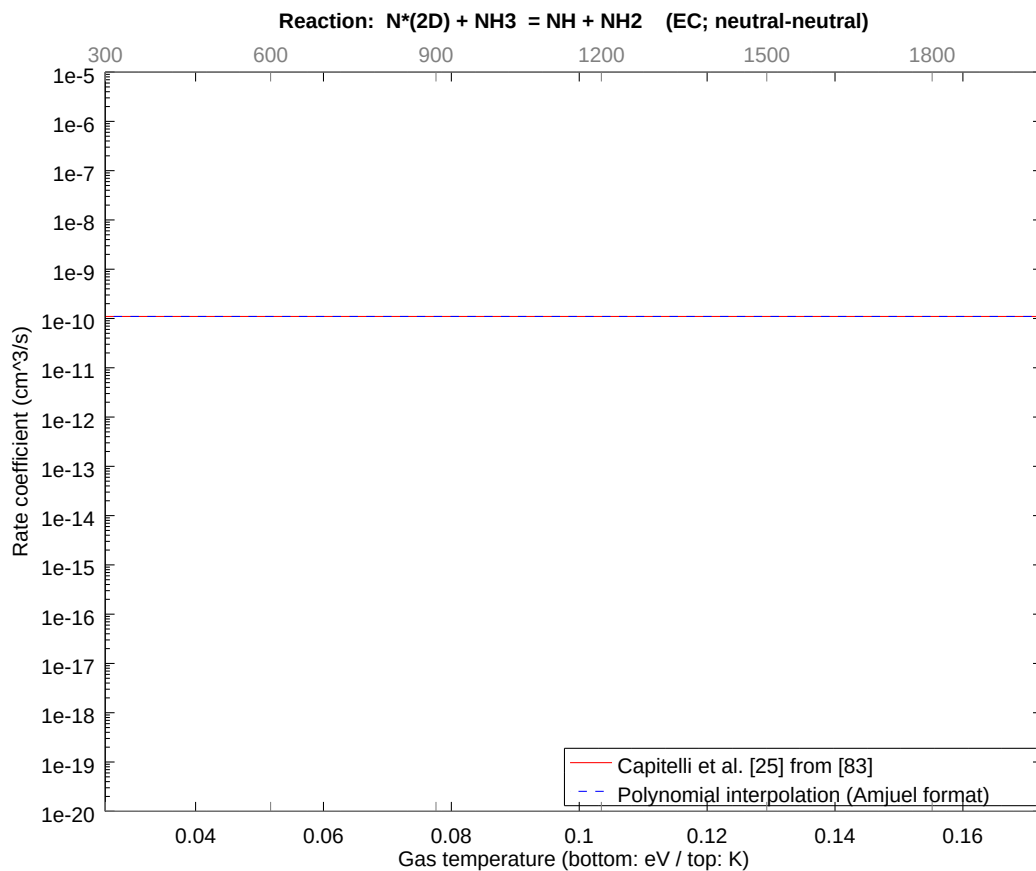
2.39 Reaction DF $\text{N}_2^*(\text{A}) + \text{H}_2 = \text{N}_2 + 2\text{H}$

b0	-2.268826321125E+01	b1	1.143045937131E-01	b2	-4.372098380053E-01
b3	-2.007521826120E-01	b4	-1.512801491418E-01	b5	-4.722928220100E-02
b6	-1.184186593315E-02	b7	-1.507809868440E-03	b8	-1.121565090195E-04



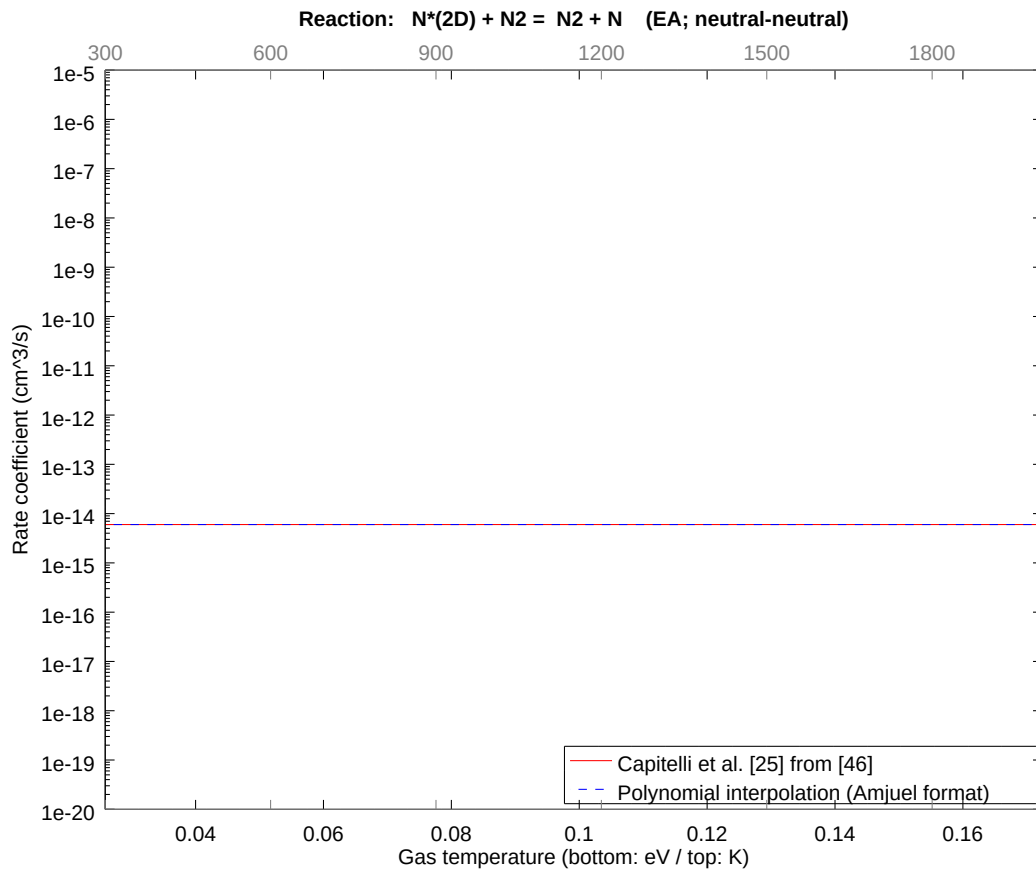
2.40 Reaction EC N*(2D) + NH3 = NH + NH2

b0	-2.293054075014E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



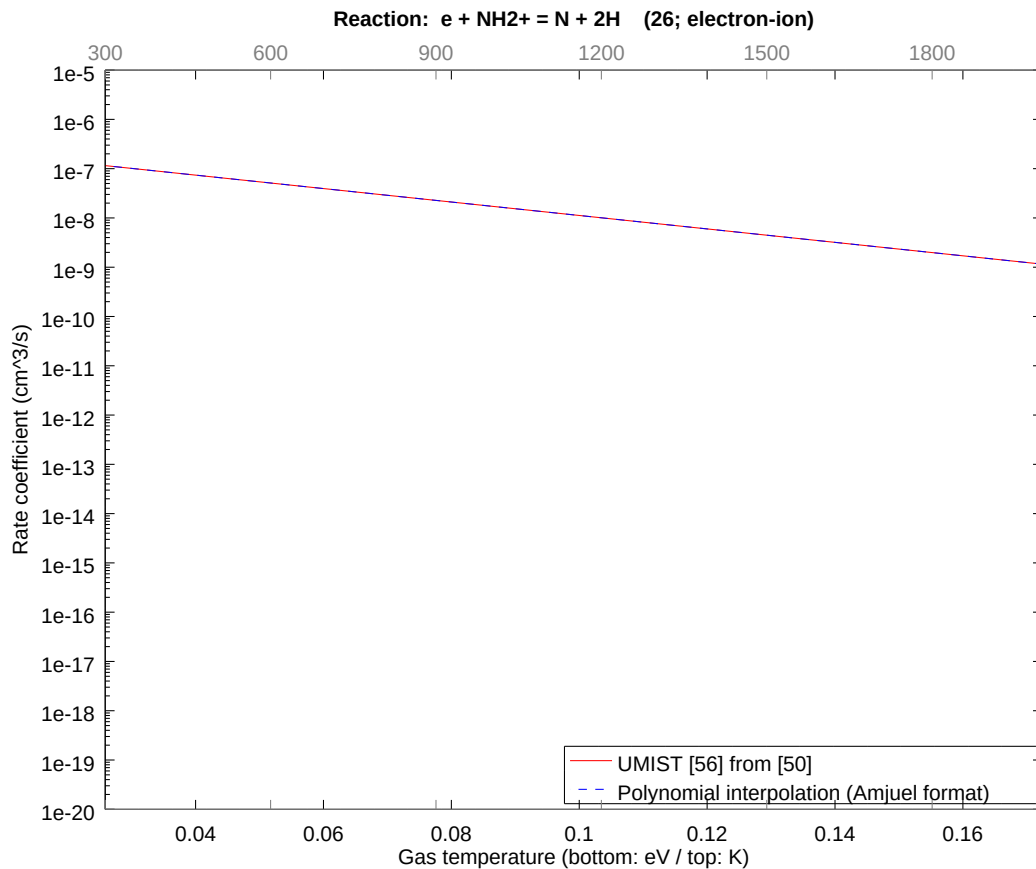
2.41 Reaction EA $\text{N}^*(2\text{D}) + \text{N}_2 = \text{N}_2 + \text{N}$

b0	-3.274701692568E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



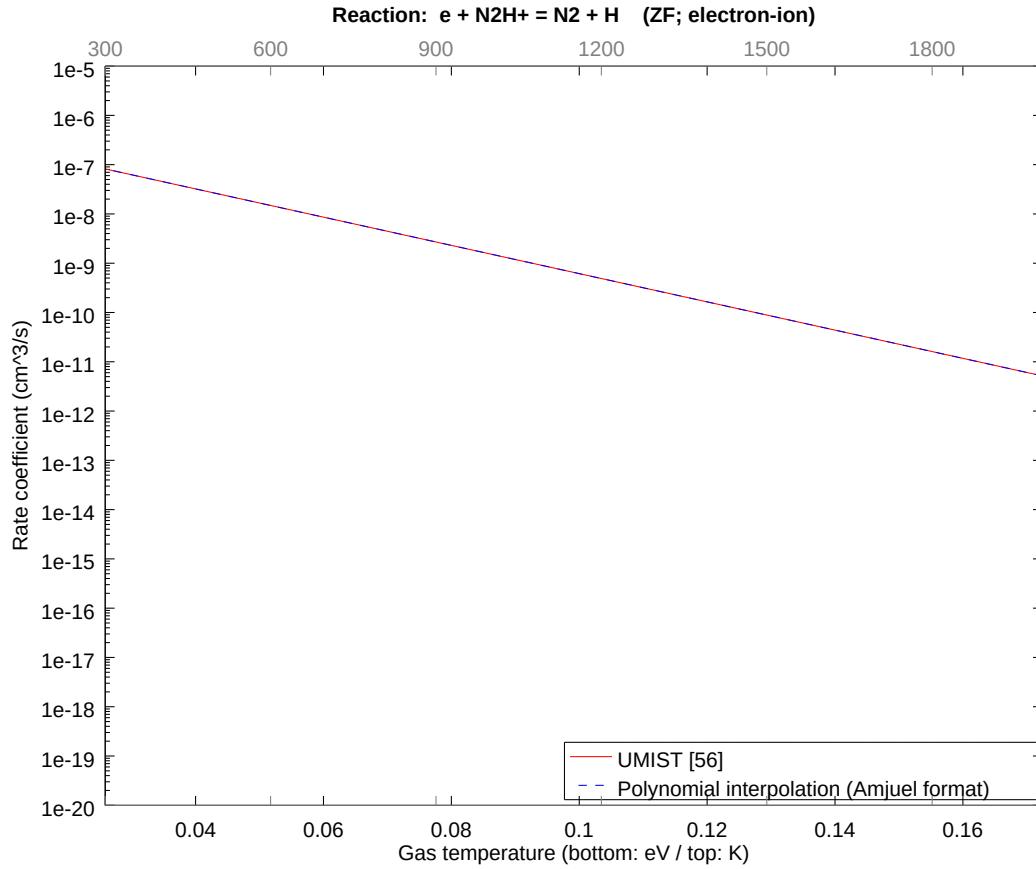
2.42 Reaction 26 $e + \text{NH}_2^+ = \text{N} + 2\text{H}$

b0	-1.689694132698E+01	b1	-4.000000000000E-01	b2	-3.922332100276E-15
b3	3.373615195767E-15	b4	-5.494869089446E-16	b5	-2.931721287741E-16
b6	1.188405030278E-16	b7	-1.507005324713E-17	b8	6.434779337134E-19



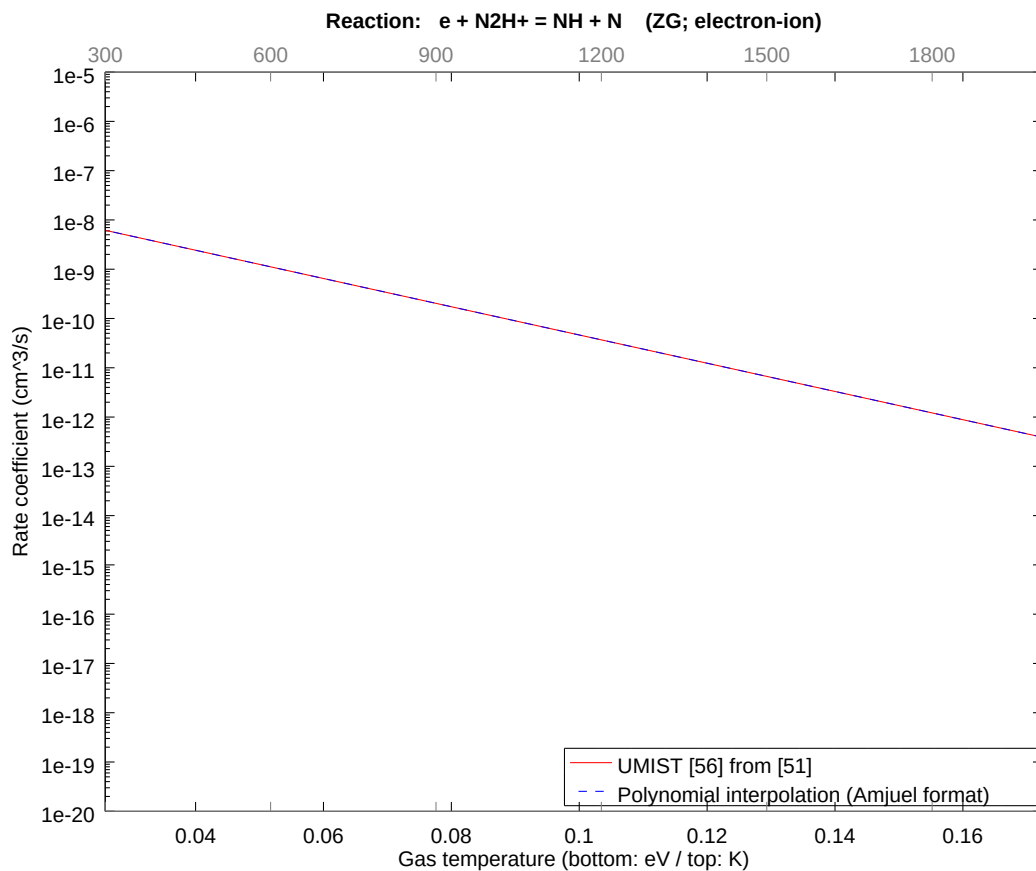
2.43 Reaction ZF $e + N_2H^+ = N_2 + H$

b0	-1.825208158533E+01	b1	-8.400000000000E-01	b2	1.961166050138E-15
b3	6.667678312880E-16	b4	4.608871172109E-18	b5	-8.196062952573E-17
b6	1.886554408333E-17	b7	-1.936160340987E-18	b8	8.166550190105E-20



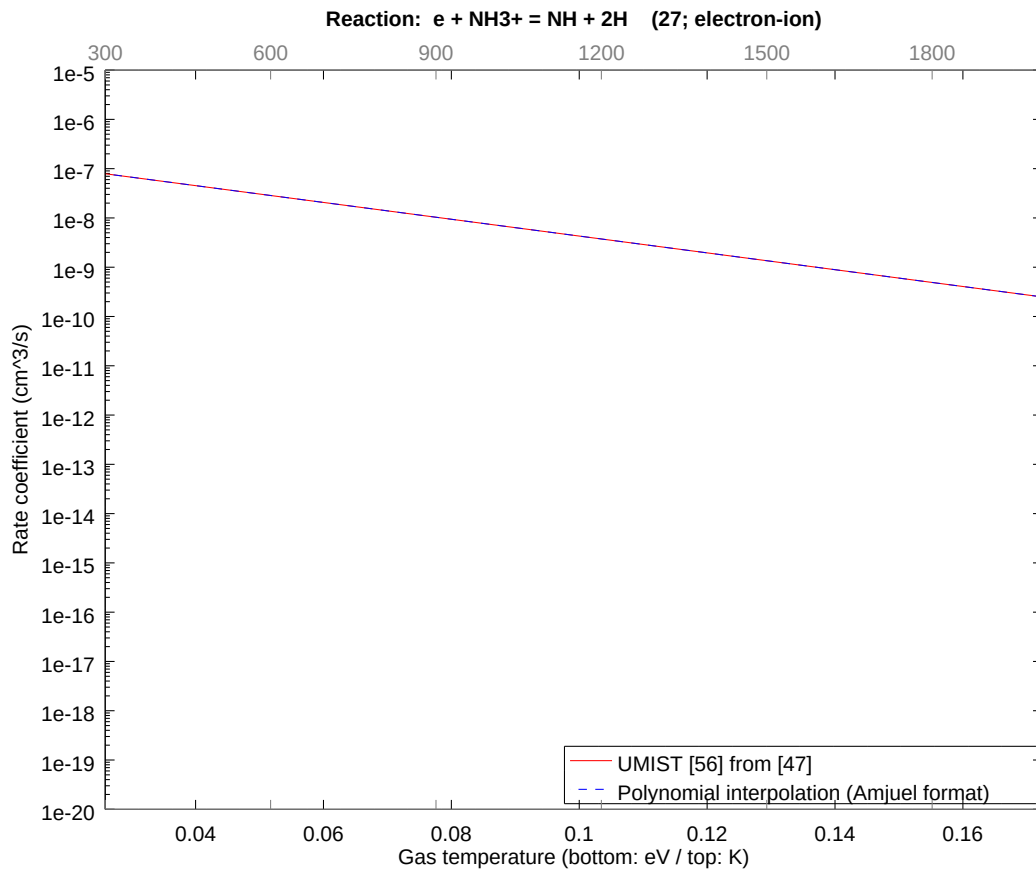
2.44 Reaction ZG $e + N_2H^+ = NH + N$

b0	-2.083843485145E+01	b1	-8.400000000000E-01	b2	3.530098890249E-14
b3	-8.211238604995E-15	b4	-4.923772150637E-15	b5	2.244719695150E-15
b6	-3.385173756315E-16	b7	2.160520453501E-17	b8	-4.775349462097E-19



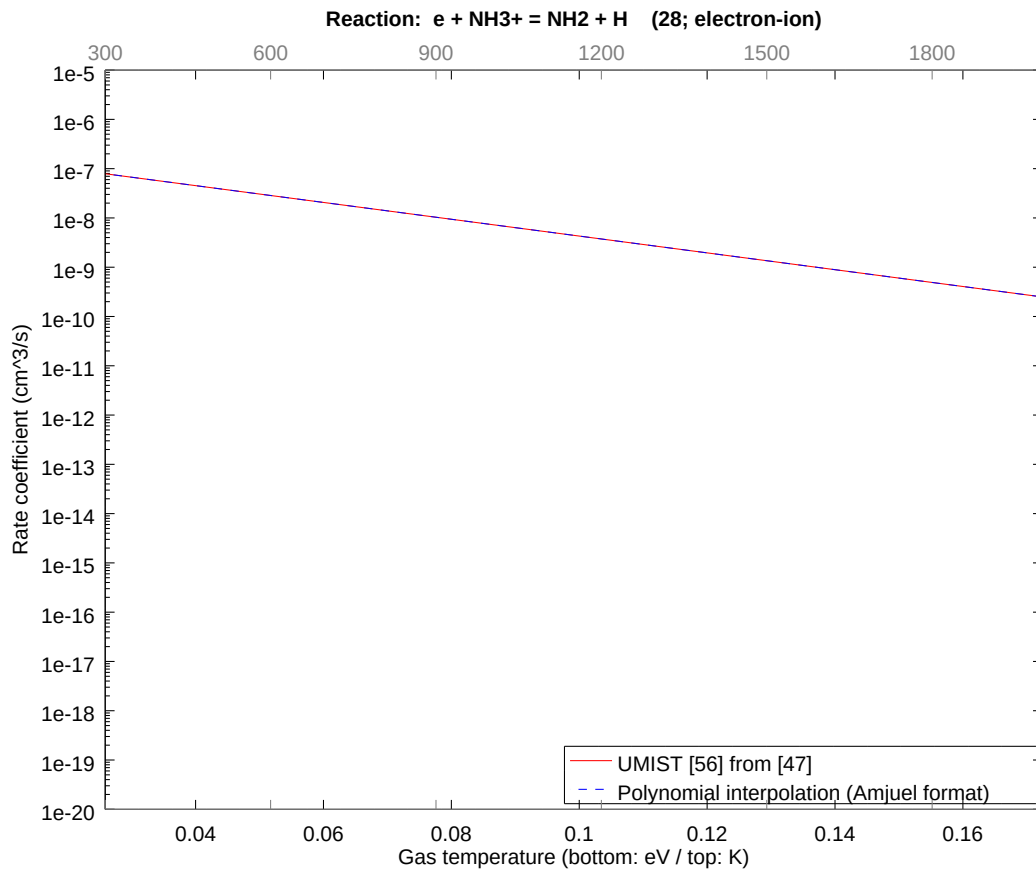
2.45 Reaction 27 $e + \text{NH}_3^+ = \text{NH} + 2\text{H}$

b0	-1.750726887094E+01	b1	-5.000000000000E-01	b2	2.745632470193E-14
b3	-1.004763624926E-14	b4	-2.722575161826E-15	b5	1.938931165252E-15
b6	-3.680508755642E-16	b7	2.957064295090E-17	b8	-8.674063015304E-19



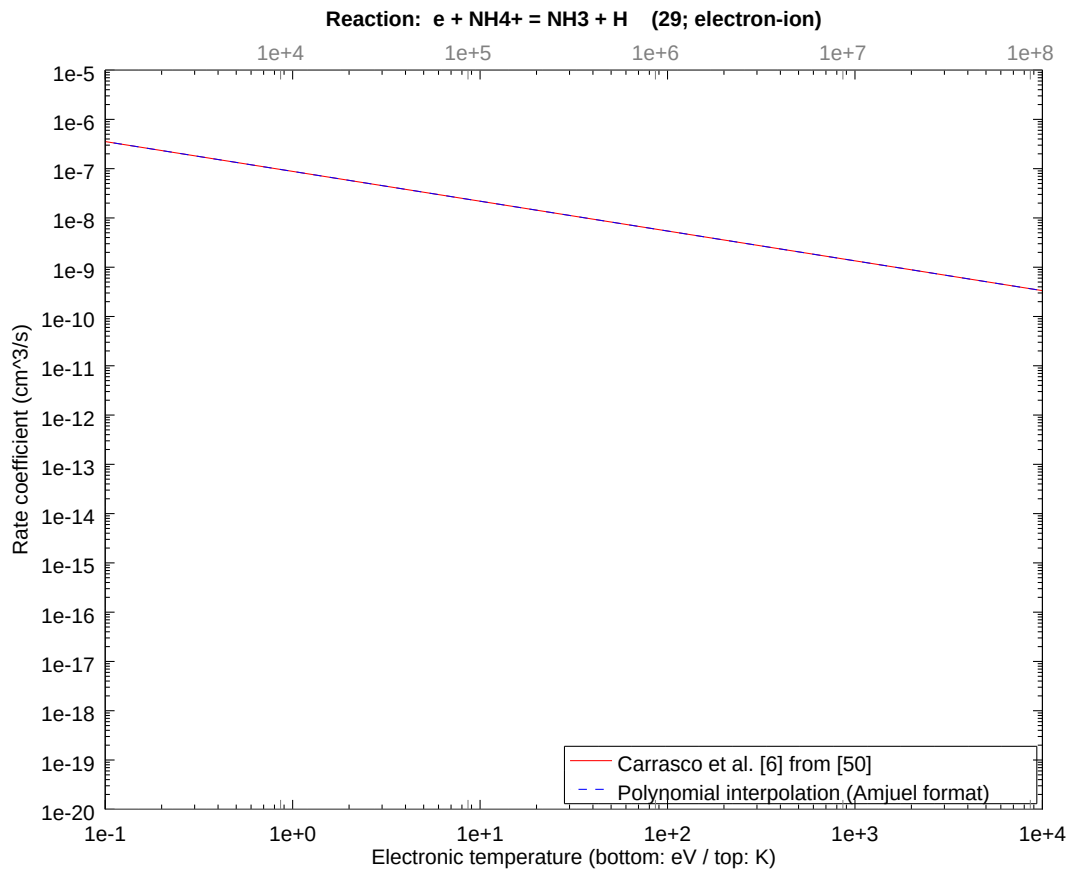
2.46 Reaction 28 $e + \text{NH}_3^+ = \text{NH}_2 + \text{H}$

b0	-1.750726887094E+01	b1	-5.000000000000E-01	b2	2.745632470193E-14
b3	-1.004763624926E-14	b4	-2.722575161826E-15	b5	1.938931165252E-15
b6	-3.680508755642E-16	b7	2.957064295090E-17	b8	-8.674063015304E-19



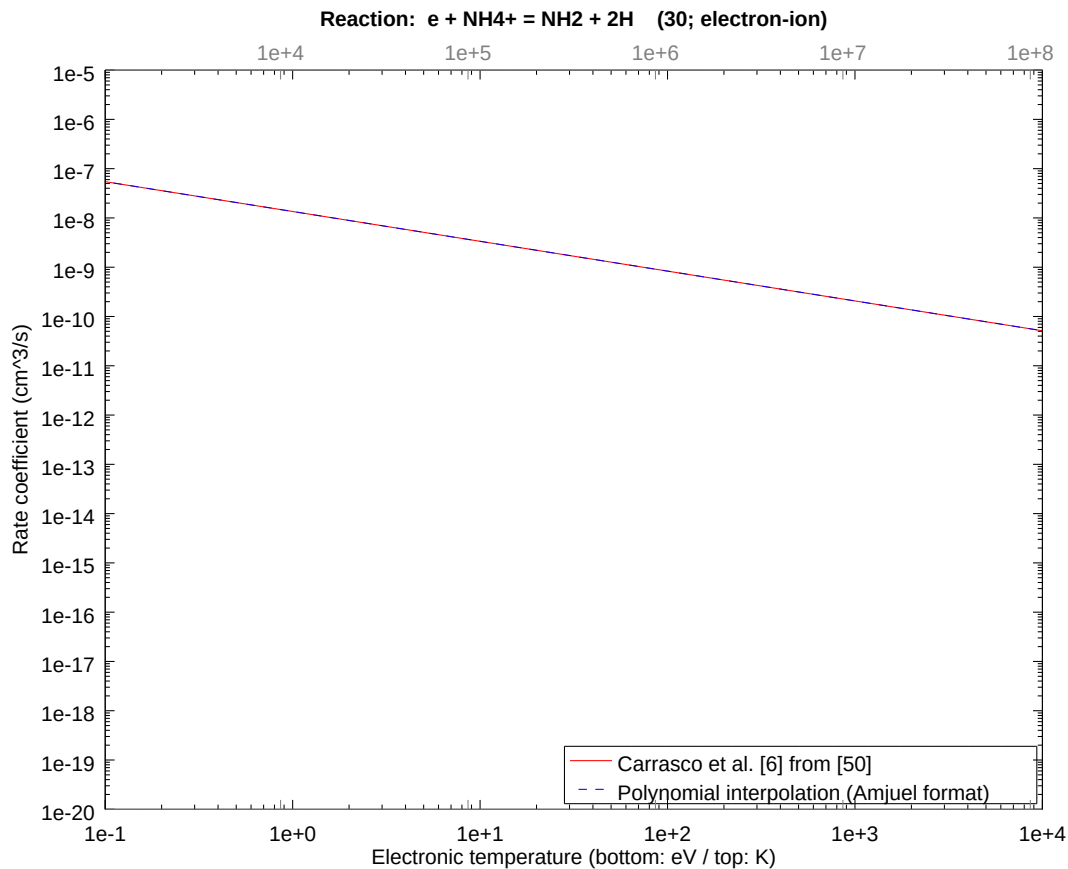
2.47 Reaction 29 $e + \text{NH}_4^+ = \text{NH}_3 + \text{H}$

b0	-1.624544842816E+01	b1	-6.050000000000E-01	b2	0.000000000000E+00
b3	8.306736808840E-16	b4	-4.686048483672E-17	b5	-2.950510937902E-16
b6	1.050373201078E-16	b7	-1.304766889864E-17	b8	5.502683070338E-19



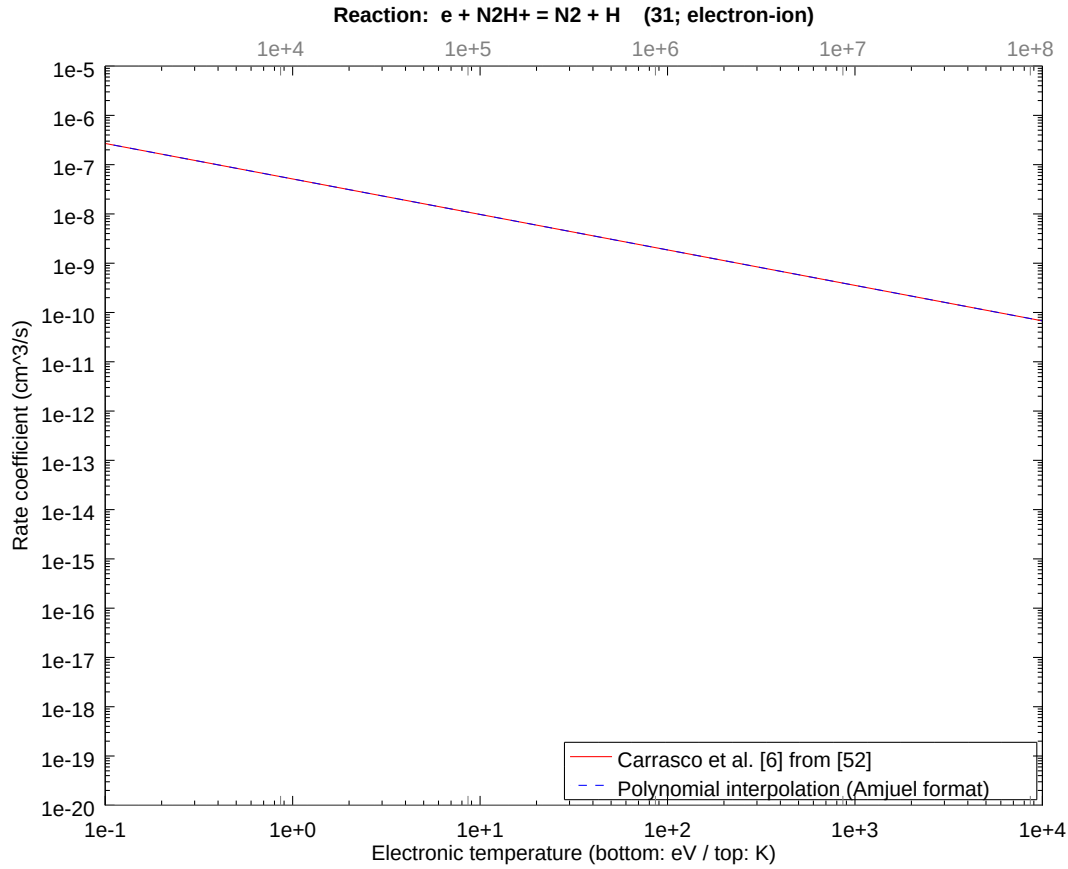
2.48 Reaction 30 $e + \text{NH}_4^+ = \text{NH}_2 + 2\text{H}$

b0	-1.811912501986E+01	b1	-6.050000000000E-01	b2	2.941749075207E-14
b3	-9.380868417975E-15	b4	-3.748190362625E-15	b5	2.200689446104E-15
b6	-3.784562783185E-16	b7	2.695309824768E-17	b8	-6.649703260841E-19



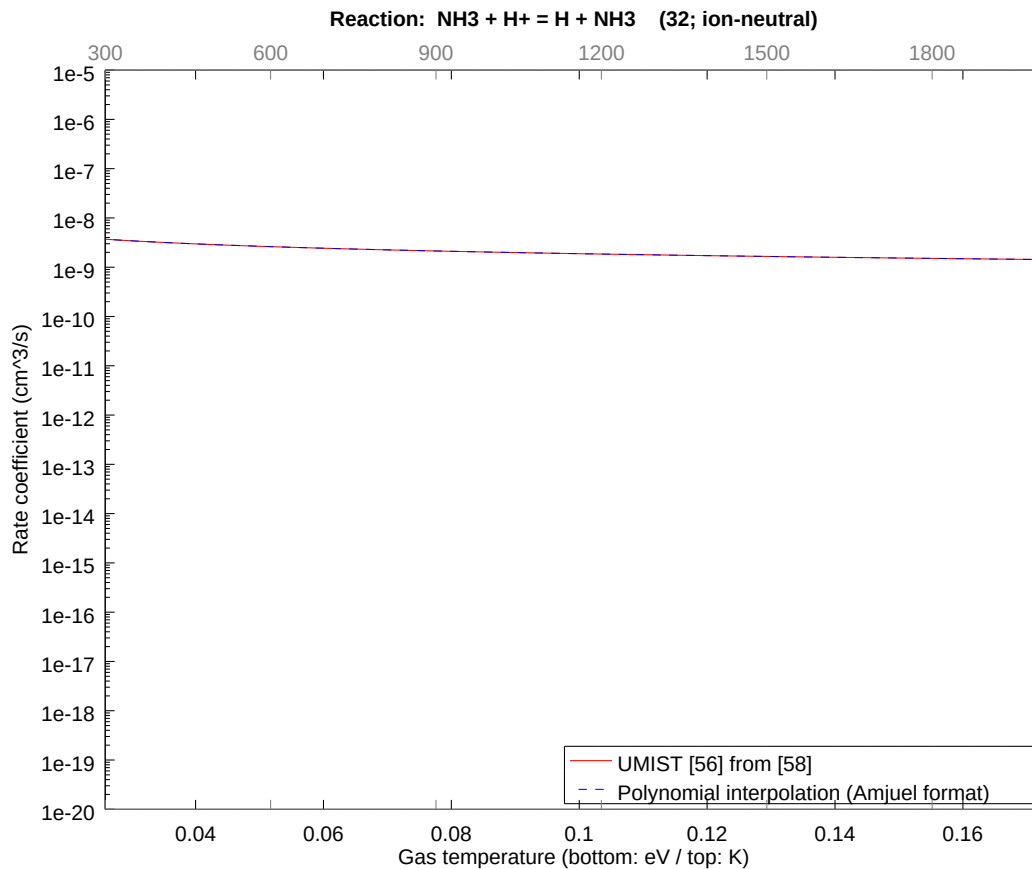
2.49 Reaction 31 $e + N_2H^+ = N_2 + H$

b0	-1.678575516040E+01	b1	-7.200000000000E-01	b2	2.941749075207E-14
b3	-1.766359676538E-15	b4	-4.297937615358E-15	b5	1.044560914165E-15
b6	-3.956040344312E-17	b7	-8.226431945192E-18	b8	6.016644534124E-19



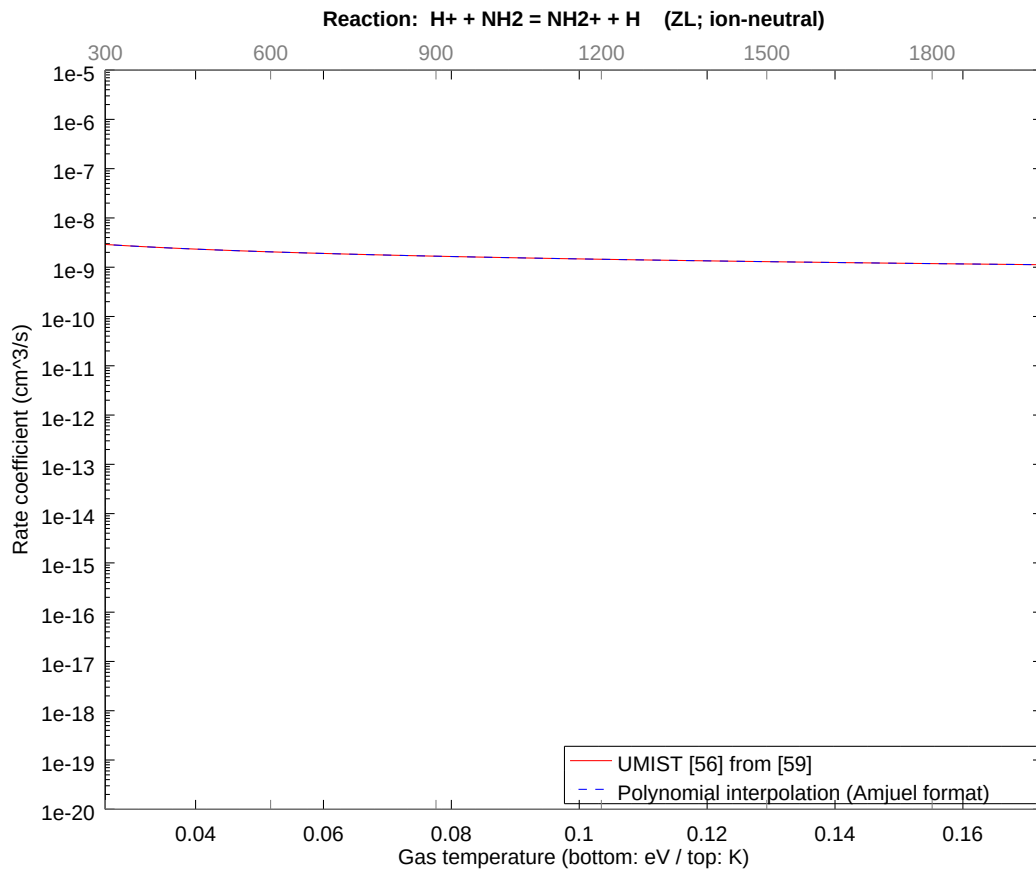
2.50 Reaction 32 $\text{NH}_3 + \text{H}^+ = \text{H} + \text{NH}_3$

b0	-2.124236116983E+01	b1	-5.000000049540E-01	b2	-6.581921337736E-09
b3	-4.941457323182E-09	b4	-2.292305524667E-09	b5	-6.727088691196E-10
b6	-1.219481205389E-10	b7	-1.248479911283E-11	b8	-5.526669688591E-13



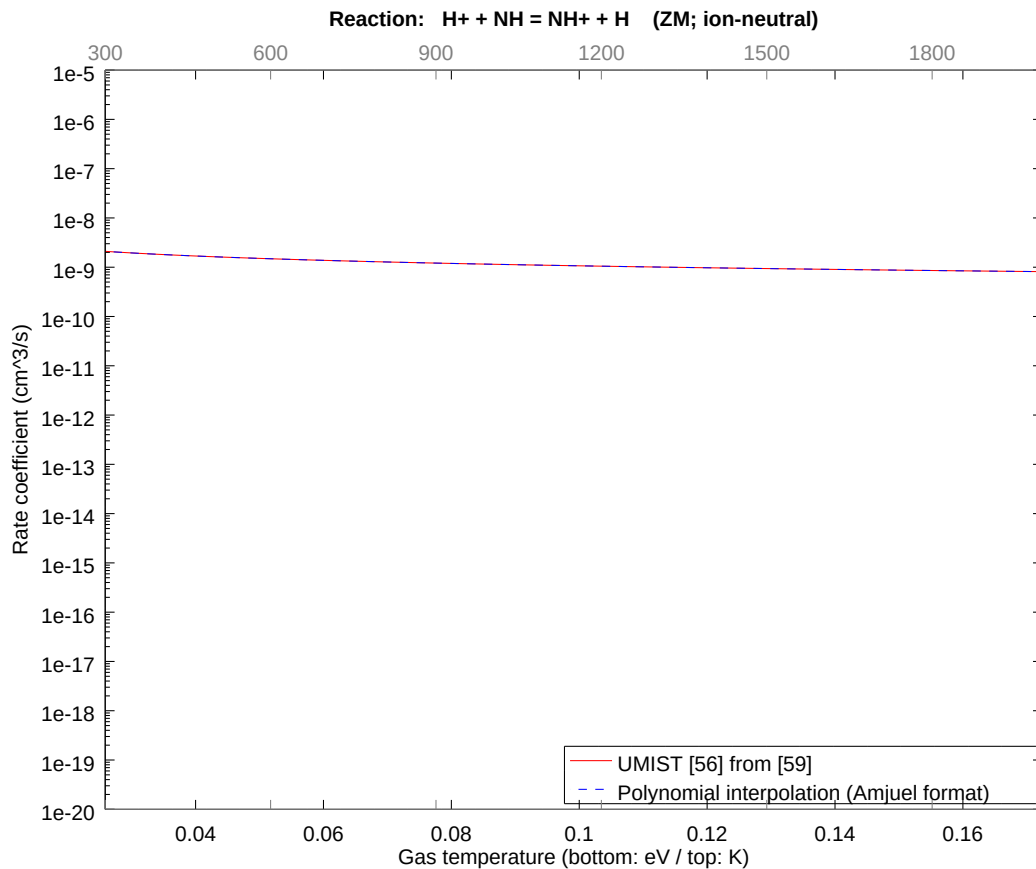
2.51 Reaction ZL $\text{H}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}$

b0	-2.148598325252E+01	b1	-5.000000052634E-01	b2	-7.274686331173E-09
b3	-5.686468493740E-09	b4	-2.749065636477E-09	b5	-8.416173449789E-10
b6	-1.593528795054E-10	b7	-1.706404308800E-11	b8	-7.914454258785E-13



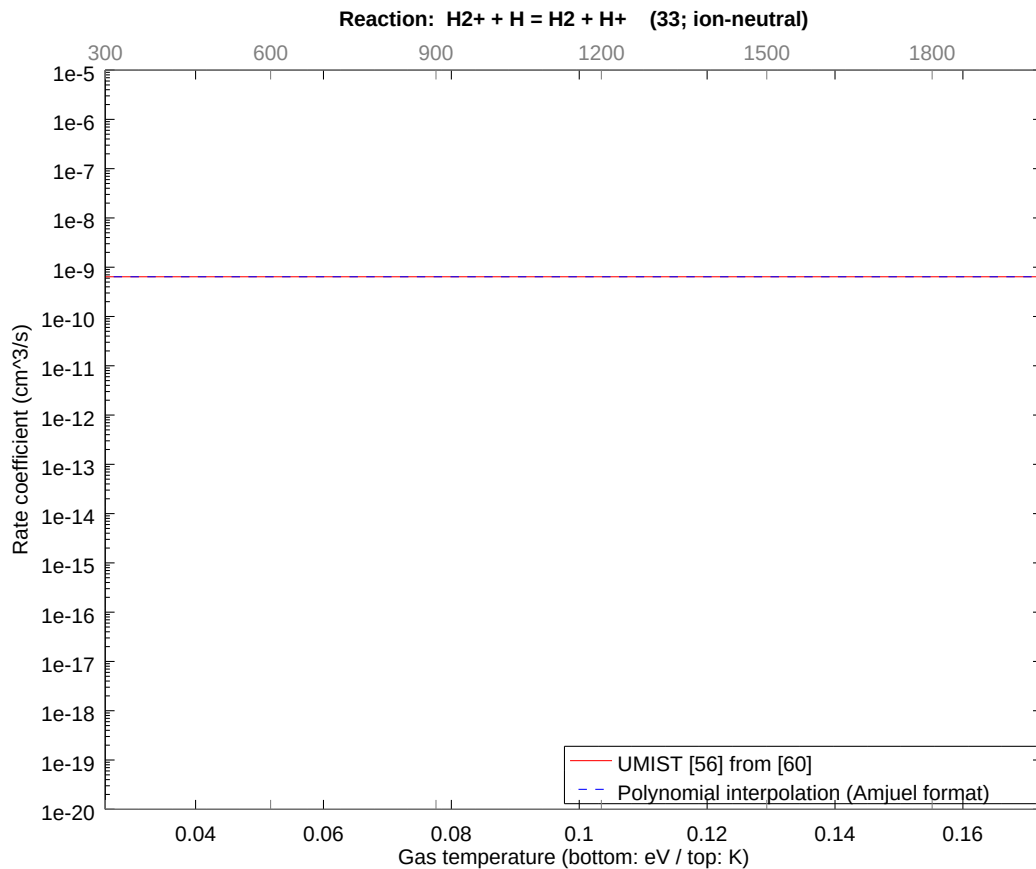
2.52 Reaction ZM $\text{H}^+ + \text{NH} = \text{NH}^+ + \text{H}$

b0	-2.180875664609E+01	b1	-5.000000092849E-01	b2	-1.261517857558E-08
b3	-9.699489564652E-09	b4	-4.615551314815E-09	b5	-1.391879349947E-09
b6	-2.597688881768E-10	b7	-2.743376818213E-11	b8	-1.255322335567E-12



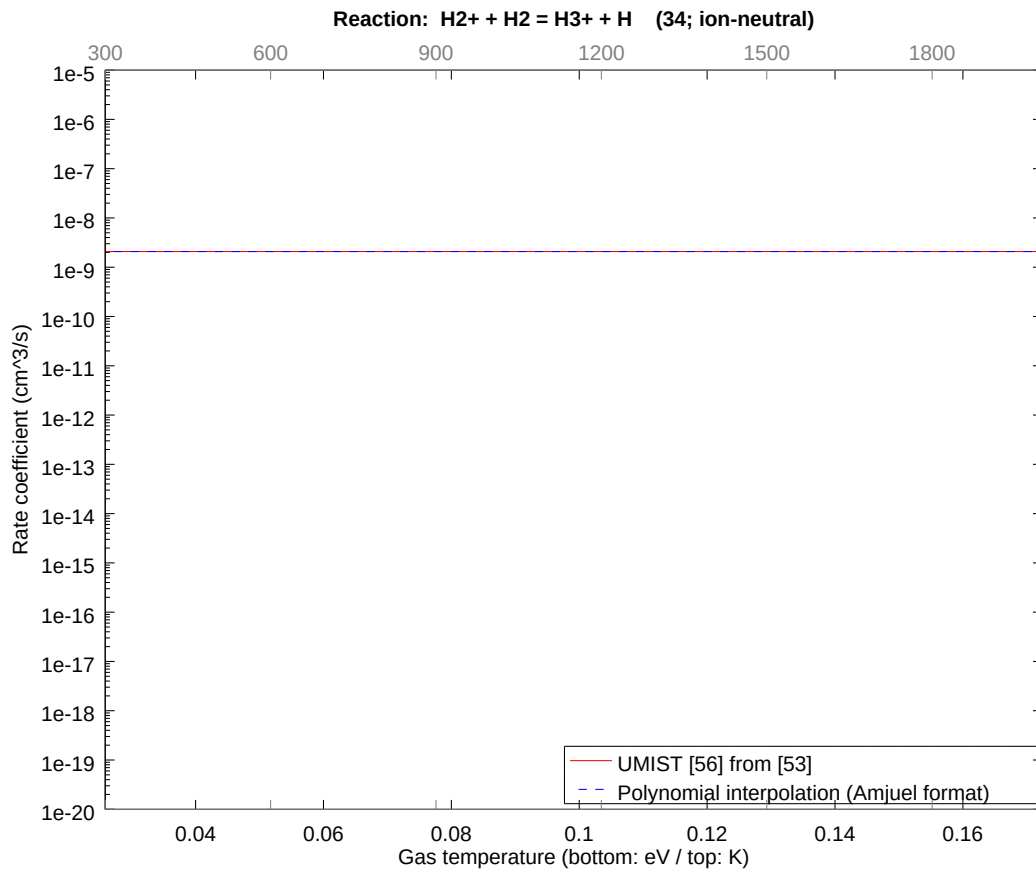
2.53 Reaction 33 $\text{H}_2^+ + \text{H} = \text{H}_2 + \text{H}^+$

b0	-2.116955293957E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



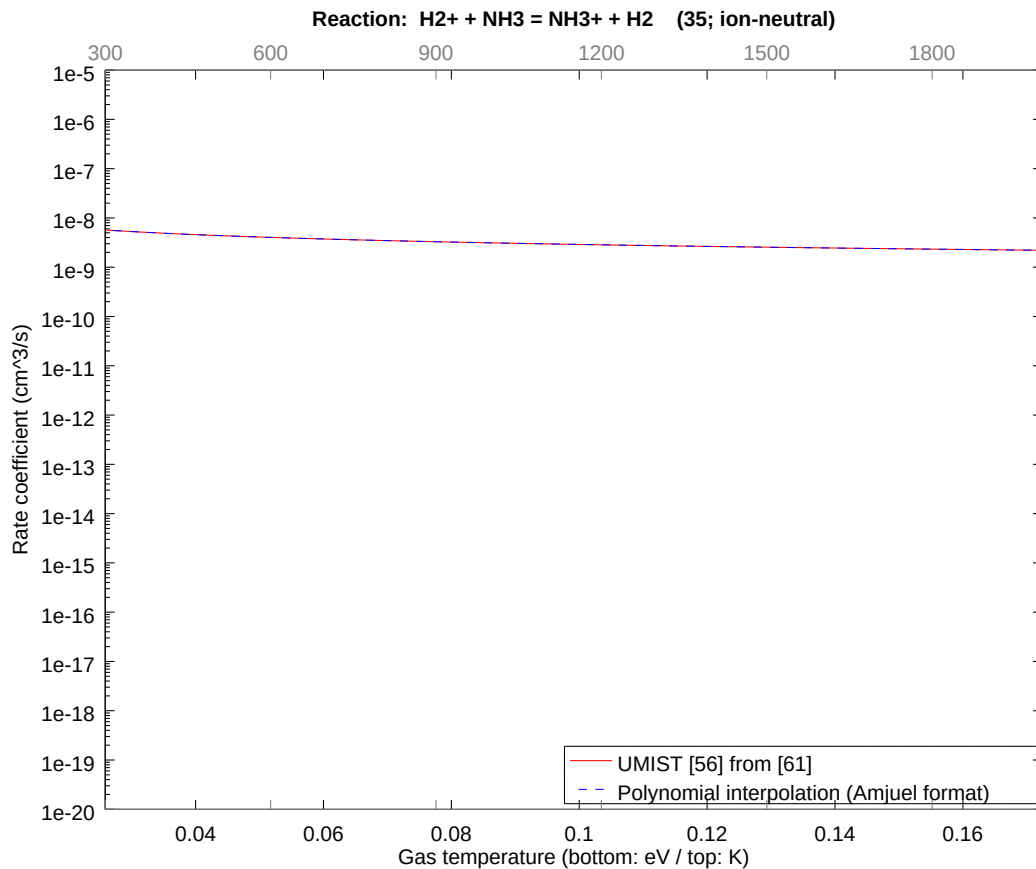
2.54 Reaction 34 $\text{H}_2^+ + \text{H}_2 = \text{H}_3^+ + \text{H}$

b0	-1.999089794323E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



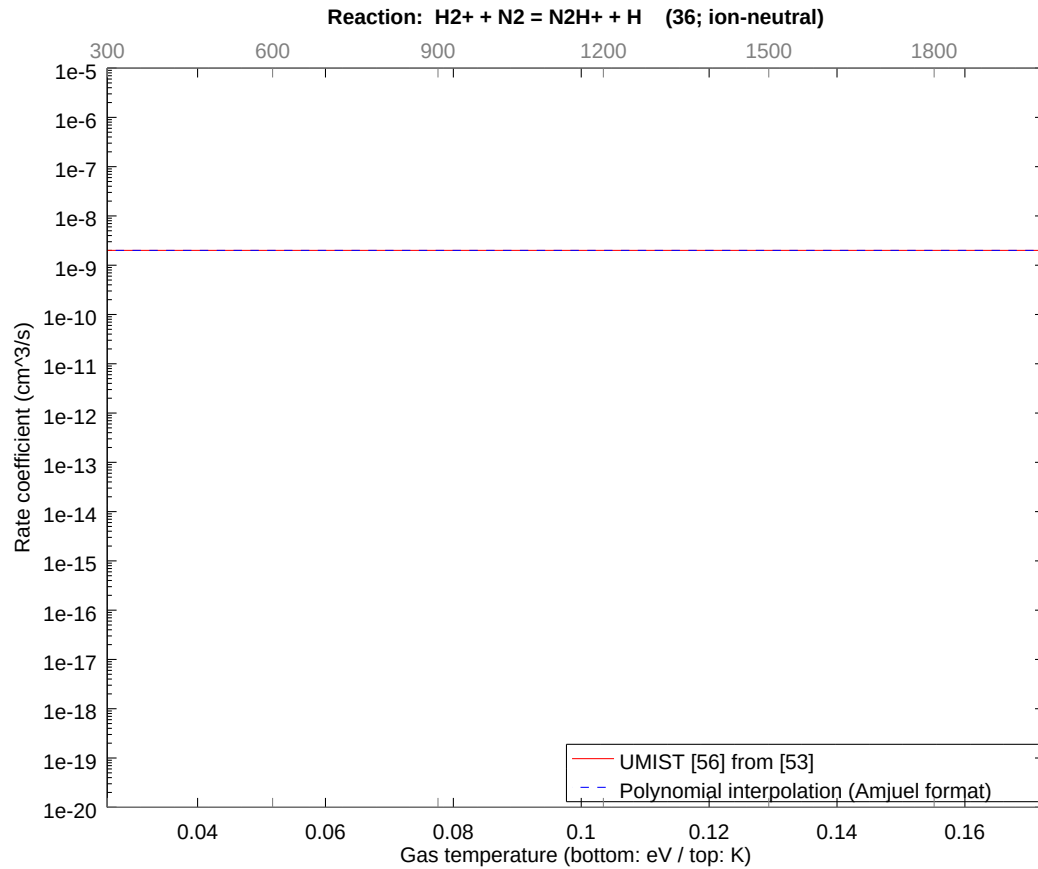
2.55 Reaction 35 $\text{H}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{H}_2$

b0	-2.081022781504E+01	b1	-5.000000063682E-01	b2	-8.716351898605E-09
b3	-6.744375376163E-09	b4	-3.226162467434E-09	b5	-9.769045577405E-10
b6	-1.828822359175E-10	b7	-1.935531934856E-11	b8	-8.868789870933E-13



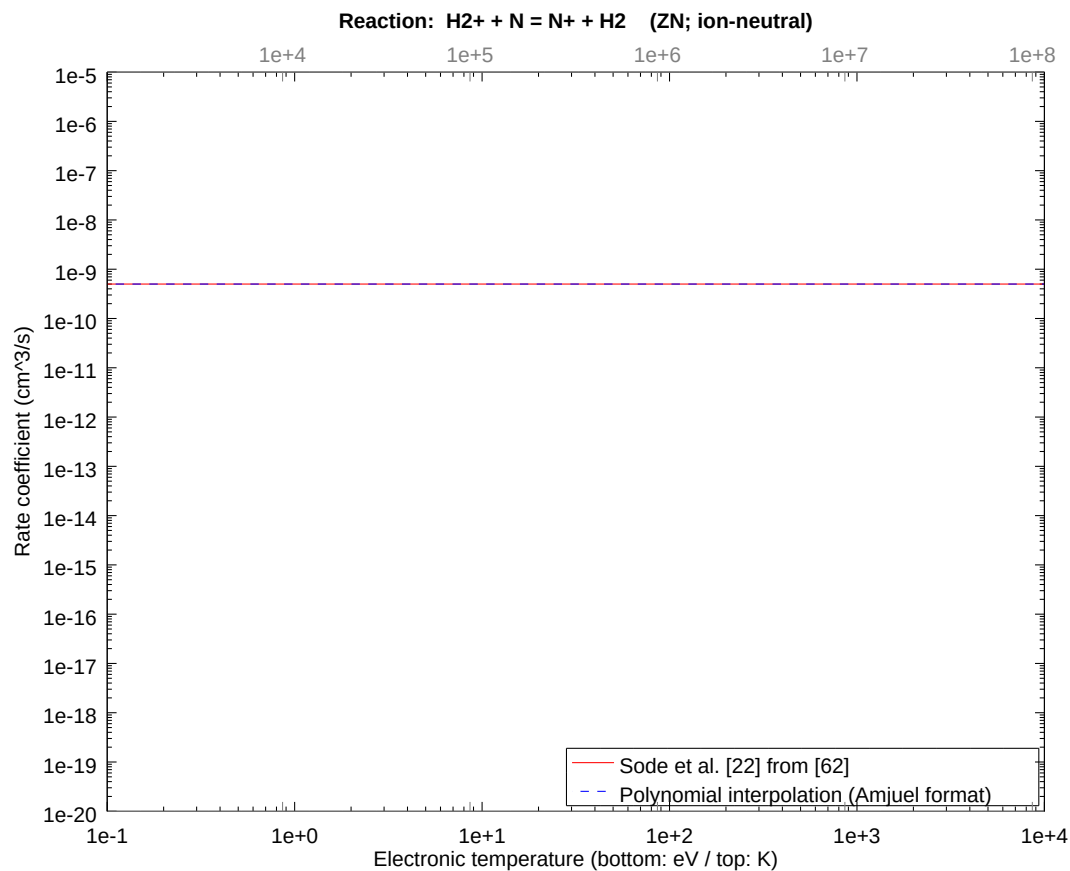
2.56 Reaction 36 $\text{H}_2^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{H}$

b0	-2.003011865639E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



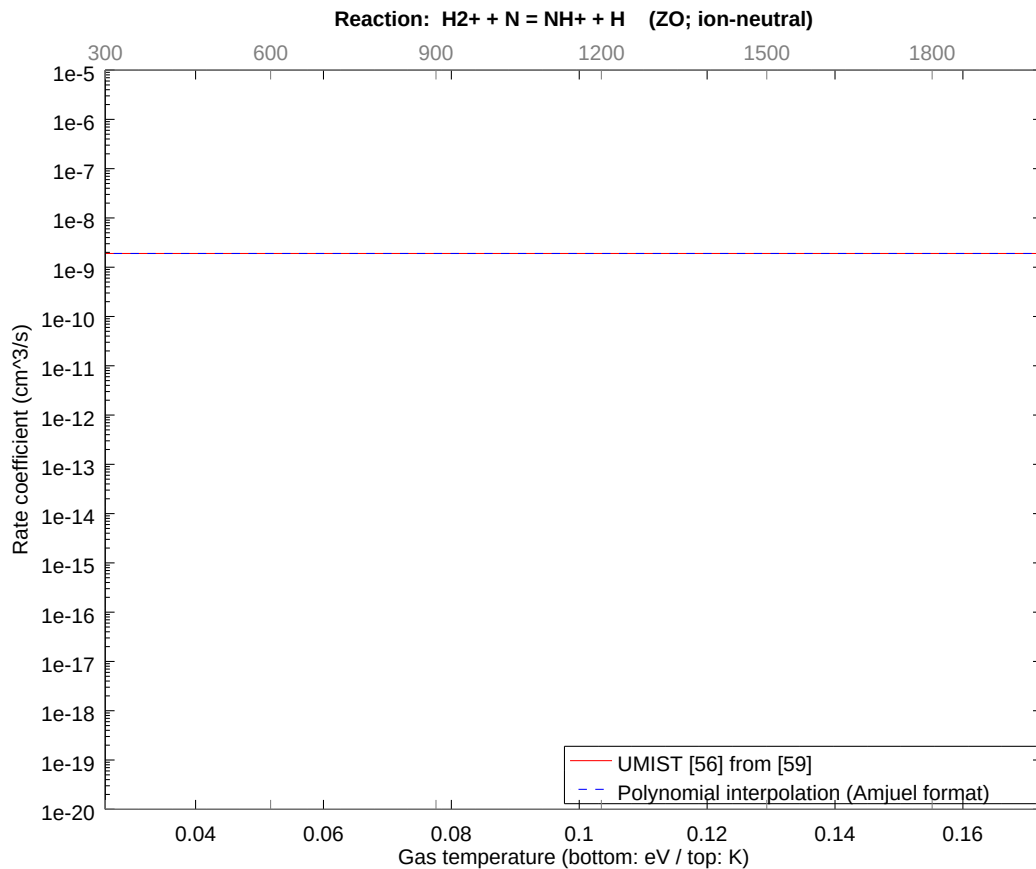
2.57 Reaction $\text{ZN H2}^+ + \text{N} = \text{N}^+ + \text{H2}$

b0	-2.141641301751E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



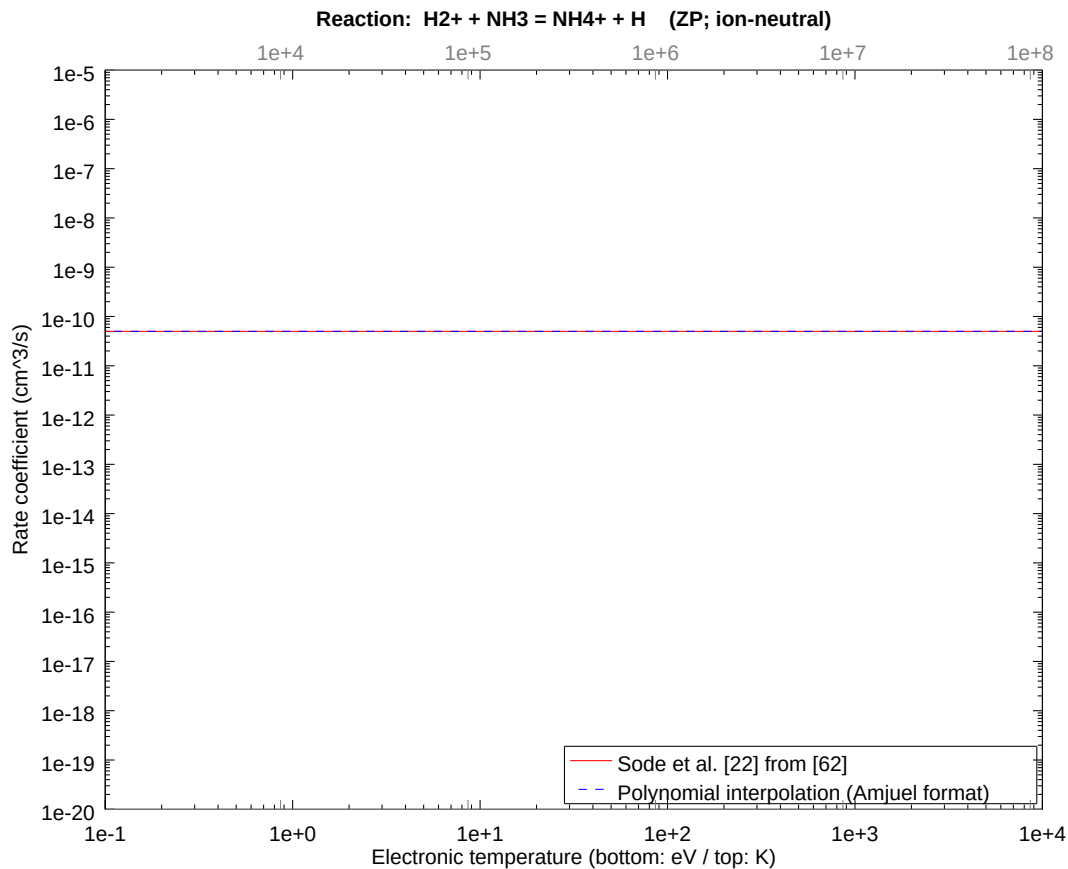
2.58 Reaction ZO H2+ + N = NH+ + H

b0	-2.008141195077E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



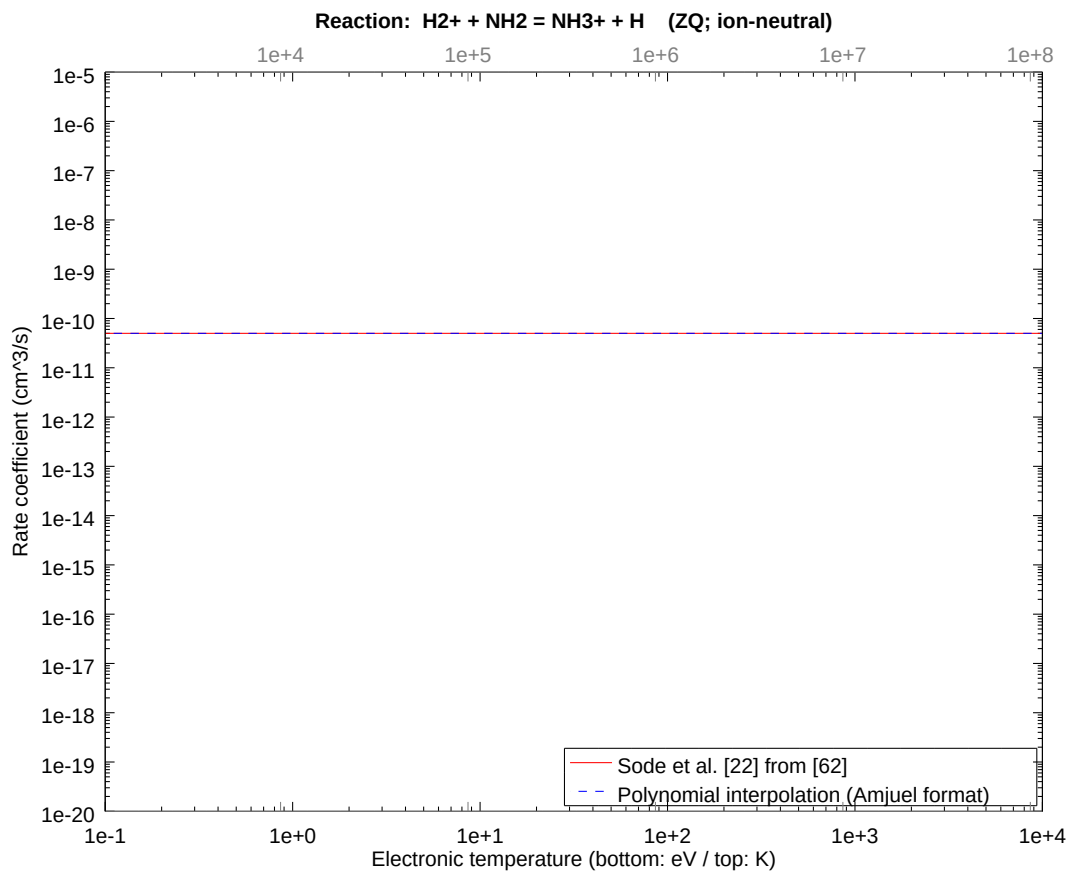
2.59 Reaction ZP $\text{H}_2^+ + \text{NH}_3 = \text{NH}_4^+ + \text{H}$

b0	-2.371899811050E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



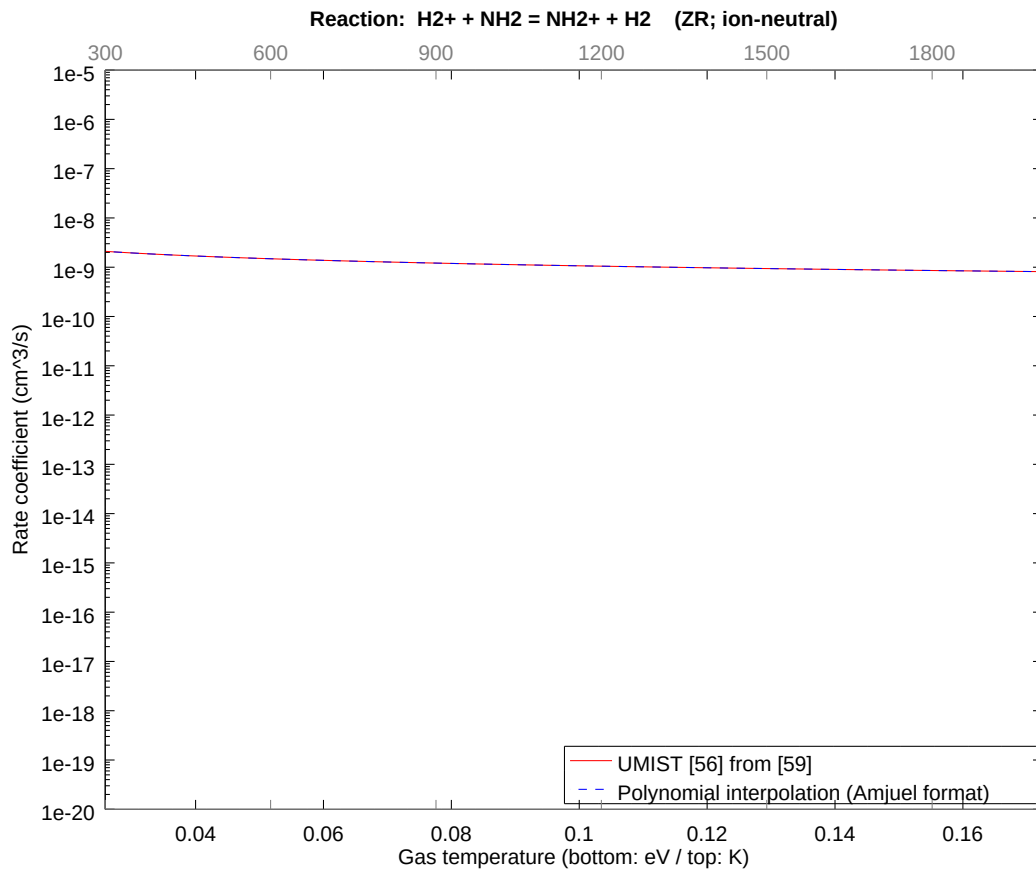
2.60 Reaction ZQ $\text{H}_2^+ + \text{NH}_2 = \text{NH}_3^+ + \text{H}$

b0	-2.371899811050E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



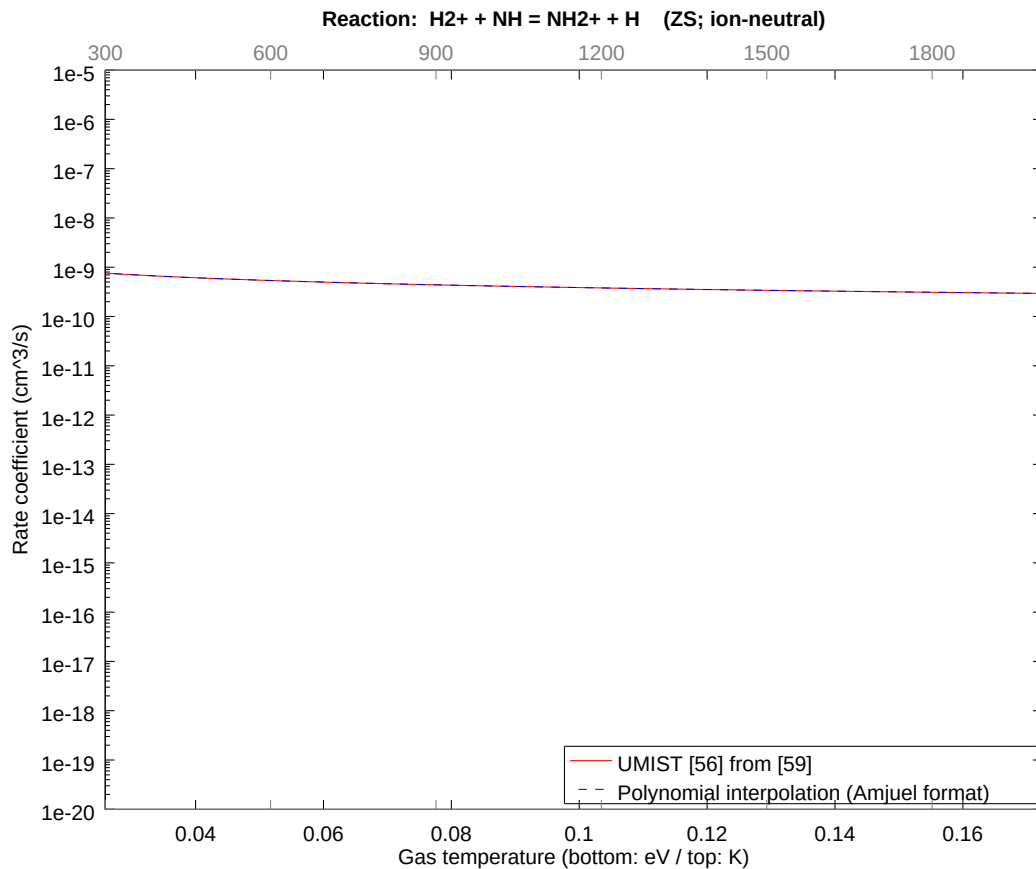
2.61 Reaction ZR $\text{H}_2^+ + \text{NH}_2 = \text{NH}_2^+ + \text{H}_2$

b0	-2.180875664609E+01	b1	-5.000000092849E-01	b2	-1.261517857558E-08
b3	-9.699489564652E-09	b4	-4.615551314815E-09	b5	-1.391879349947E-09
b6	-2.597688881768E-10	b7	-2.743376818213E-11	b8	-1.255322335567E-12



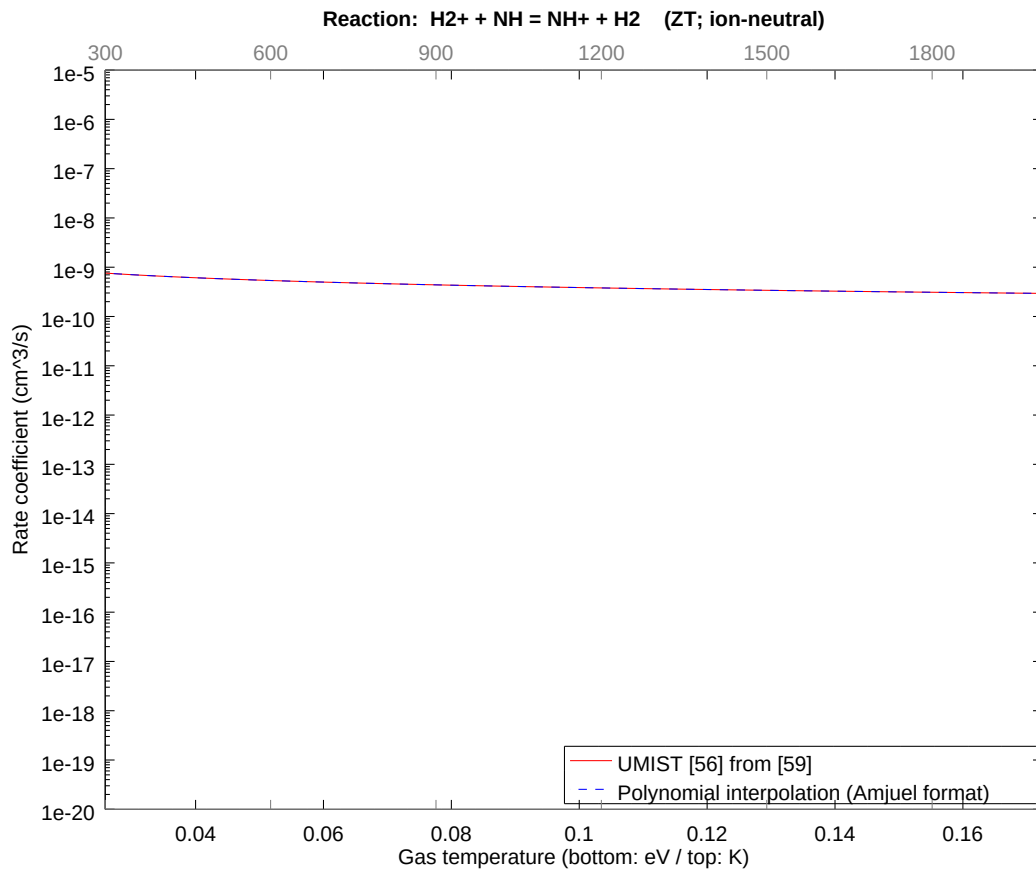
2.62 Reaction ZS H2+ + NH = NH2+ + H

b0	-2.282513083456E+01	b1	-5.000000030980E-01	b2	-4.172533795604E-09
b3	-3.178279948201E-09	b4	-1.497002555752E-09	b5	-4.463695735110E-10
b6	-8.227558117594E-11	b7	-8.571561358925E-12	b8	-3.865197085583E-13



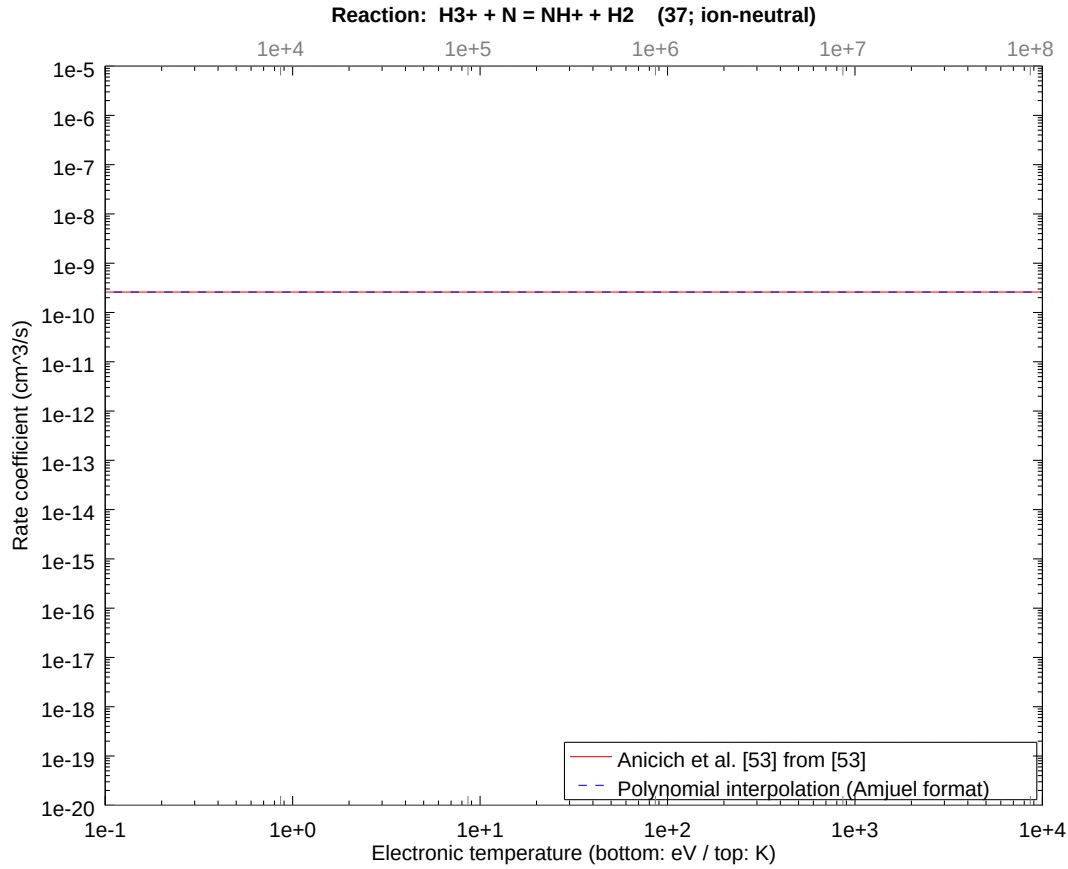
2.63 Reaction ZT $\text{H}_2^+ + \text{NH} = \text{NH}^+ + \text{H}_2$

b0	-2.282513083456E+01	b1	-5.000000030980E-01	b2	-4.172533795604E-09
b3	-3.178279948201E-09	b4	-1.497002555752E-09	b5	-4.463695735110E-10
b6	-8.227558117594E-11	b7	-8.571561358925E-12	b8	-3.865197085583E-13



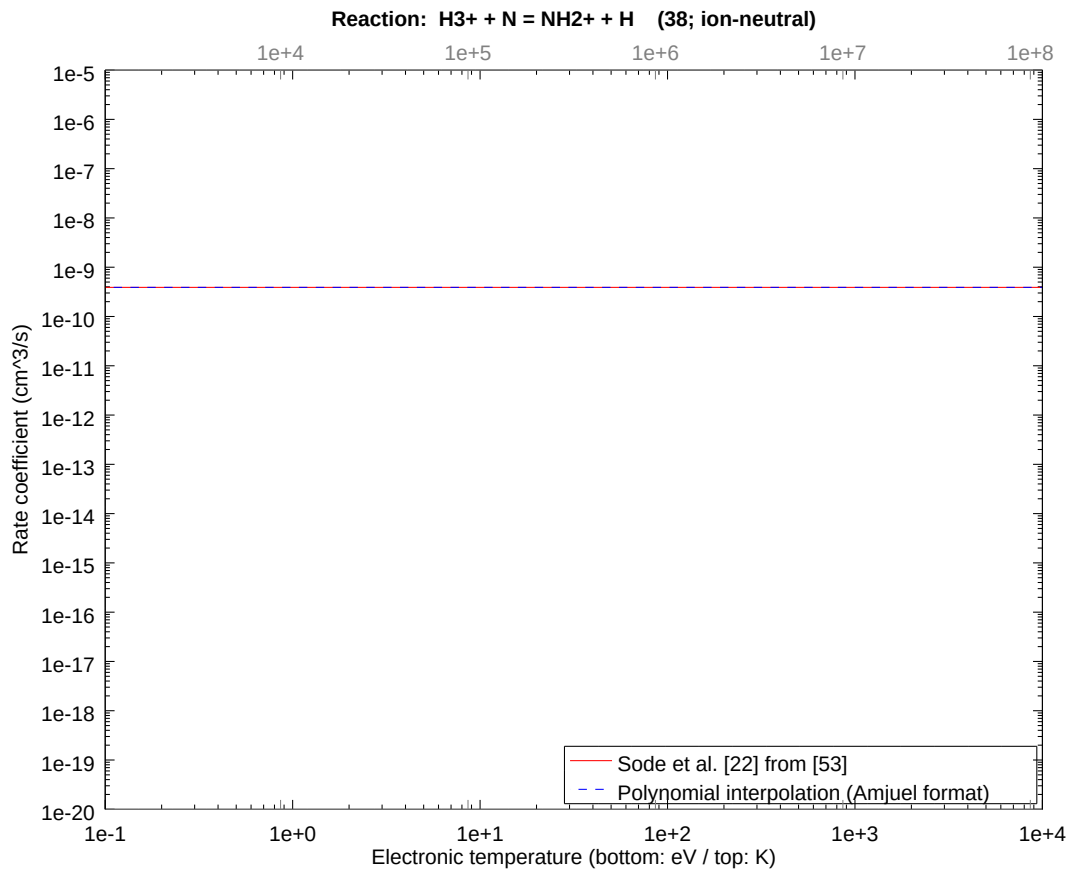
2.64 Reaction 37 $\text{H3}^+ + \text{N} = \text{NH}^+ + \text{H2}$

b0	-2.207033948491E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



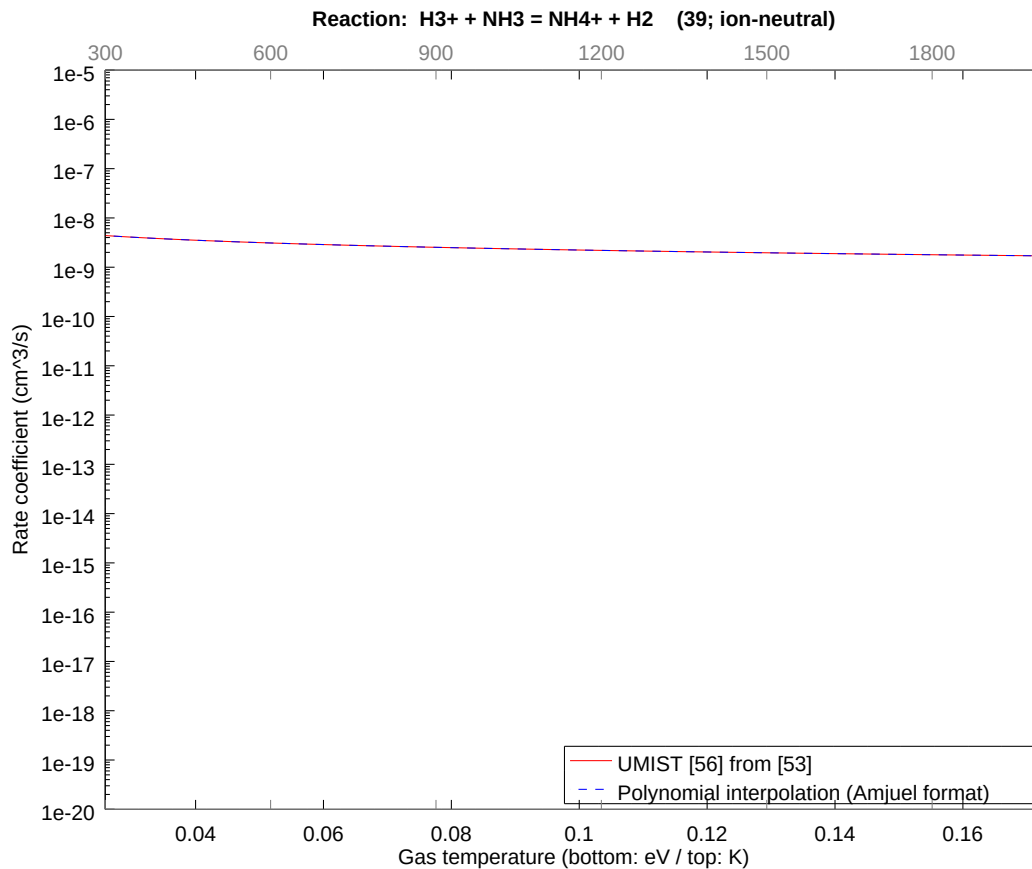
2.65 Reaction 38 $\text{H3}^+ + \text{N} = \text{NH2}^+ + \text{H}$ (38; ion-neutral)

b0	-2.166487437680E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



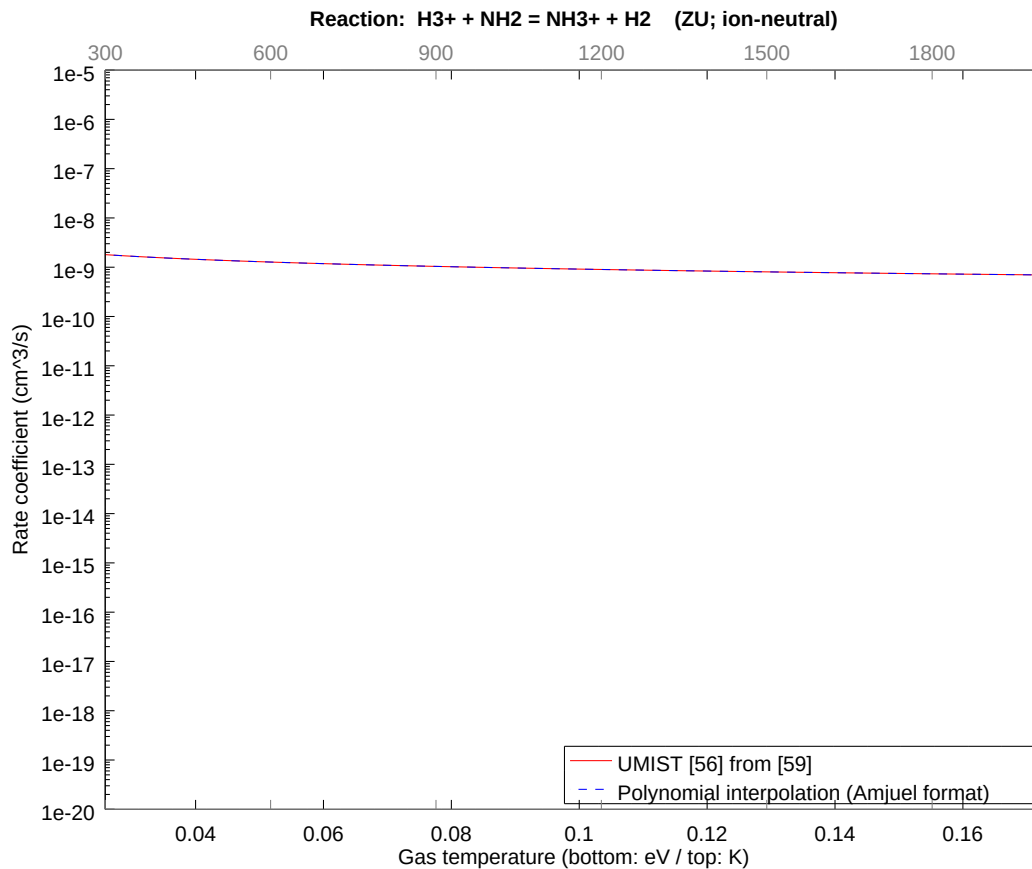
2.66 Reaction 39 $\text{H3}^+ + \text{NH3} = \text{NH4}^+ + \text{H2}$

b0	-2.107136476258E+01	b1	-5.000000055727E-01	b2	-7.437148651905E-09
b3	-5.609819632118E-09	b4	-2.615584427083E-09	b5	-7.718869172379E-10
b6	-1.408075560704E-10	b7	-1.451862542892E-11	b8	-6.479800800281E-13



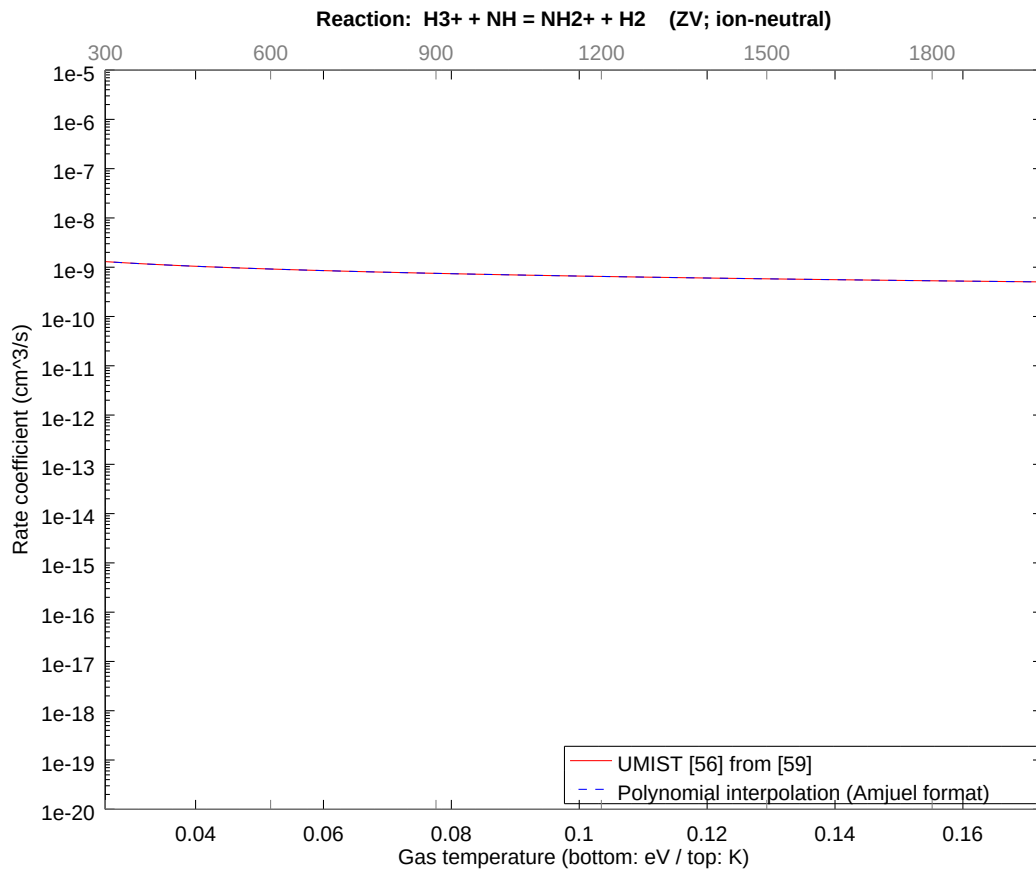
2.67 Reaction ZU $\text{H3}^+ + \text{NH2} = \text{NH3}^+ + \text{H2}$

b0	-2.196290732349E+01	b1	-5.000000017280E-01	b2	-2.428579288053E-09
b3	-1.921777919375E-09	b4	-9.364377031497E-10	b5	-2.877434523747E-10
b6	-5.445759602353E-11	b7	-5.805294394799E-12	b8	-2.669522116361E-13



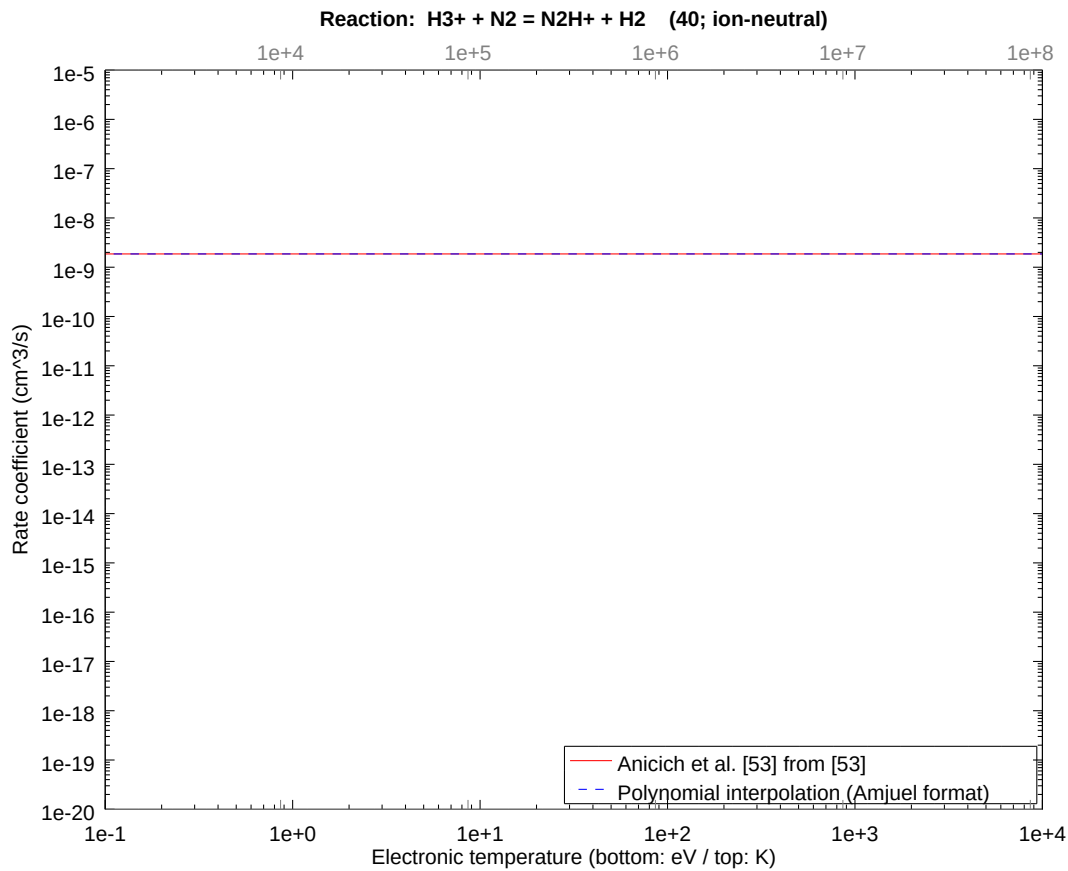
2.68 Reaction ZV $\text{H3}^+ + \text{NH} = \text{NH2}^+ + \text{H2}$

b0	-2.228832972476E+01	b1	-5.000000042028E-01	b2	-5.569379050783E-09
b3	-4.161520963593E-09	b4	-1.916979254164E-09	b5	-5.572768434265E-10
b6	-9.981865098488E-11	b7	-1.007015872766E-11	b8	-4.380051097582E-13



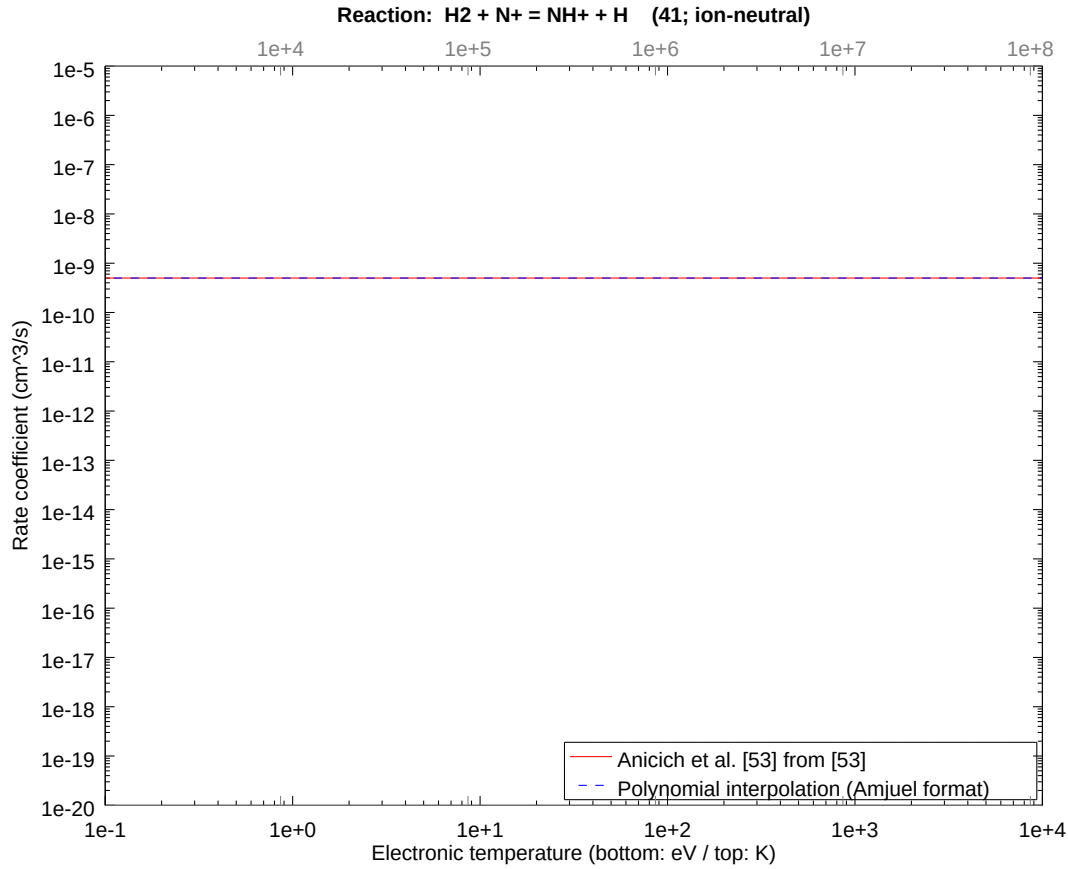
2.69 Reaction 40 $\text{H3}^+ + \text{N2} = \text{N2H}^+ + \text{H2}$

b0	-2.010268934922E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



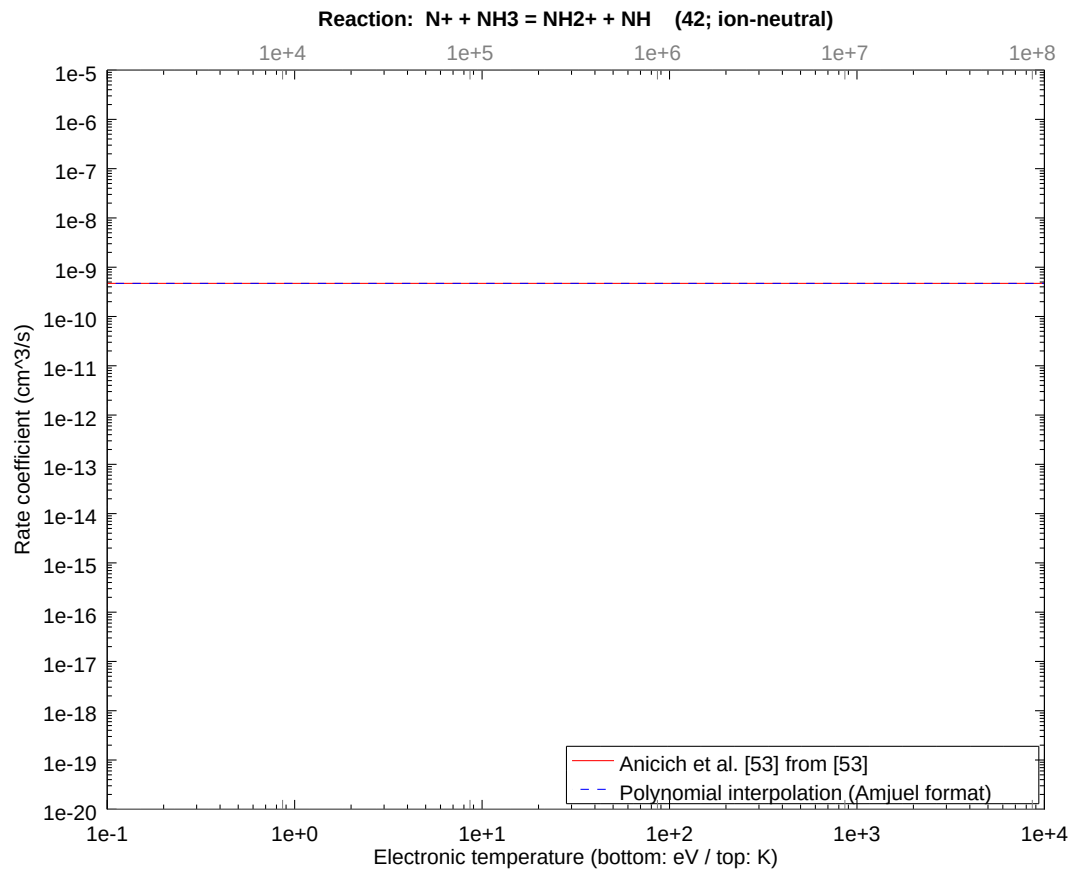
2.70 Reaction 41 $\text{H}_2 + \text{N}^+ = \text{NH}^+ + \text{H}$

b0	-2.141641301751E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



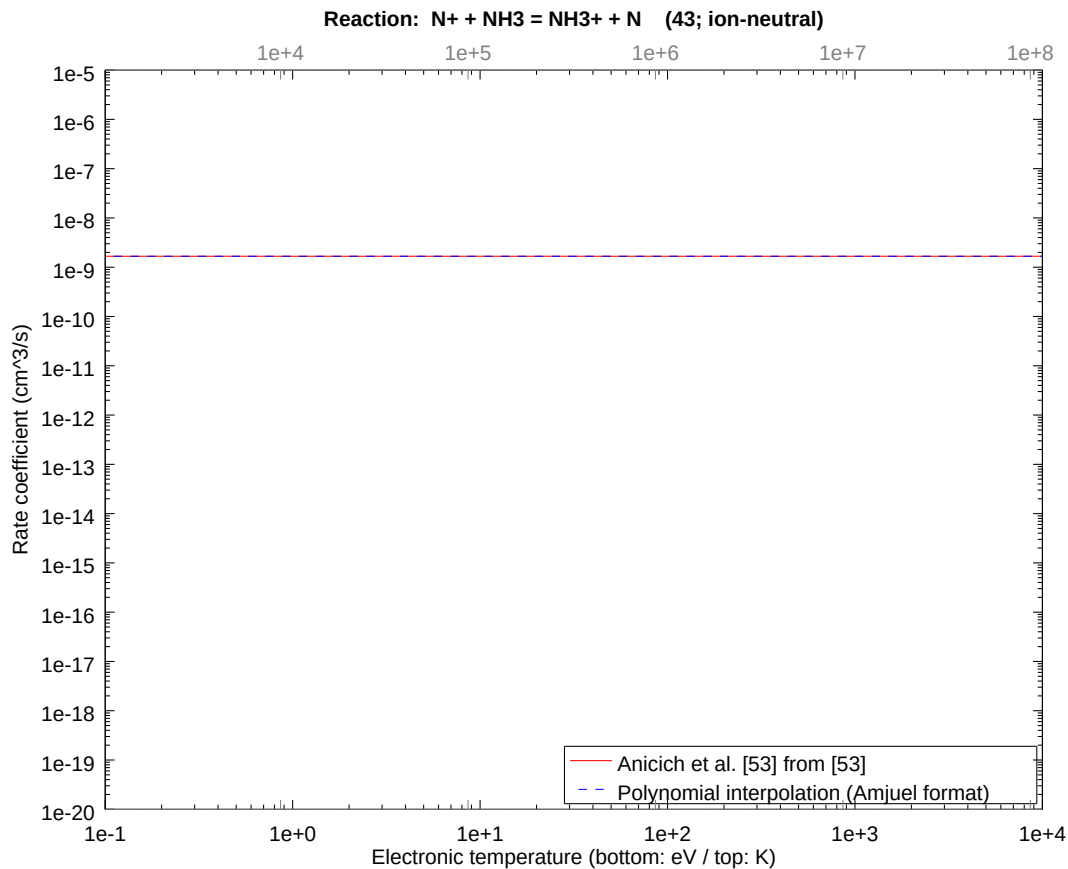
2.71 Reaction 42 $\text{N}^+ + \text{NH}_3 = \text{NH}_2^+ + \text{NH}$

b0	-2.147828842122E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



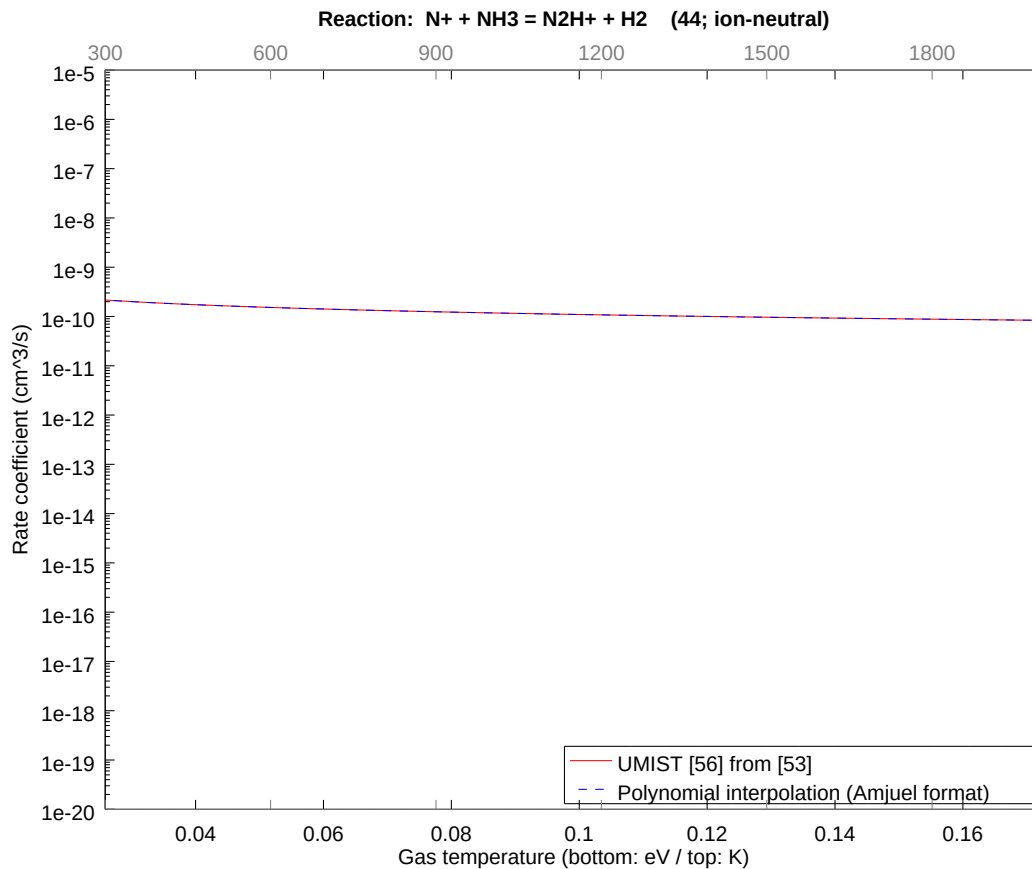
2.72 Reaction 43 $\text{N}^+ + \text{NH}_3 = \text{NH}_3^+ + \text{N}$

b0	-2.021044221052E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



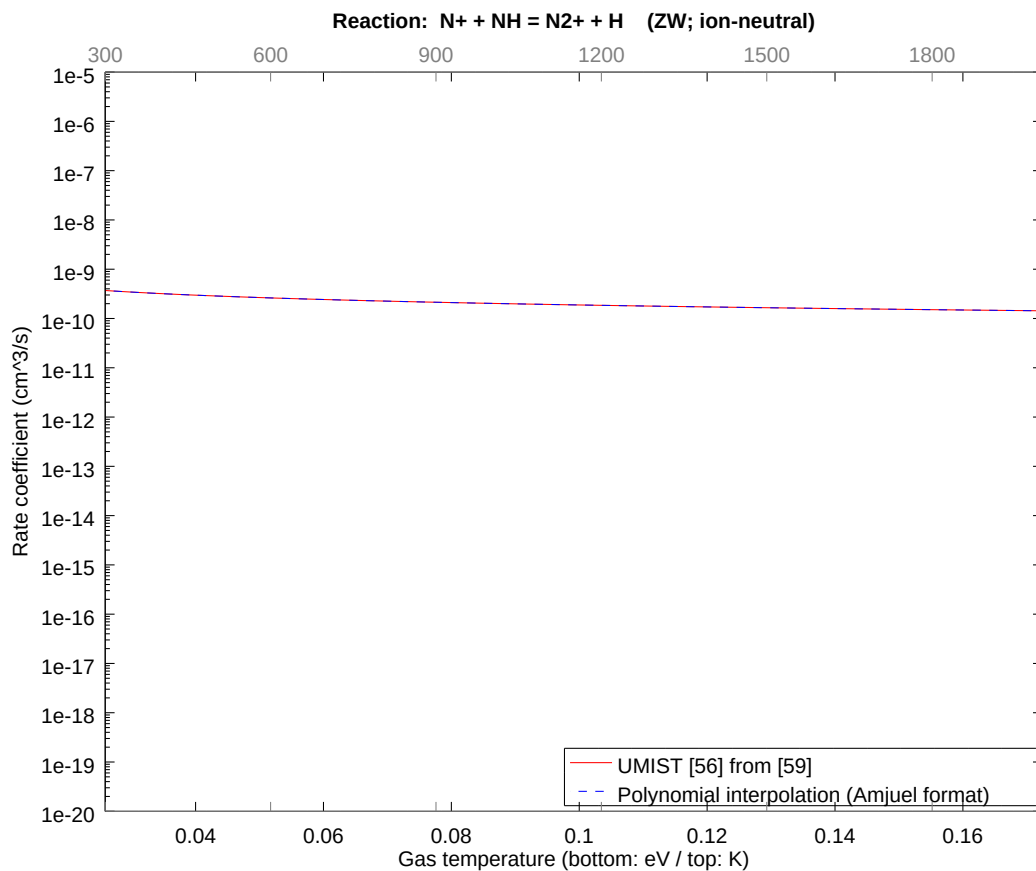
2.73 Reaction 44 $\text{N}^+ + \text{NH}_3 = \text{N}_2\text{H}^+ + \text{H}_2$

b0 -2.408317086045E+01	b1 -5.000000039818E-01	b2 -5.322487393845E-09
b3 -4.009817581528E-09	b4 -1.860909331983E-09	b5 -5.443952121280E-10
b6 -9.796847626584E-11	b7 -9.908533882743E-12	b8 -4.308943182685E-13



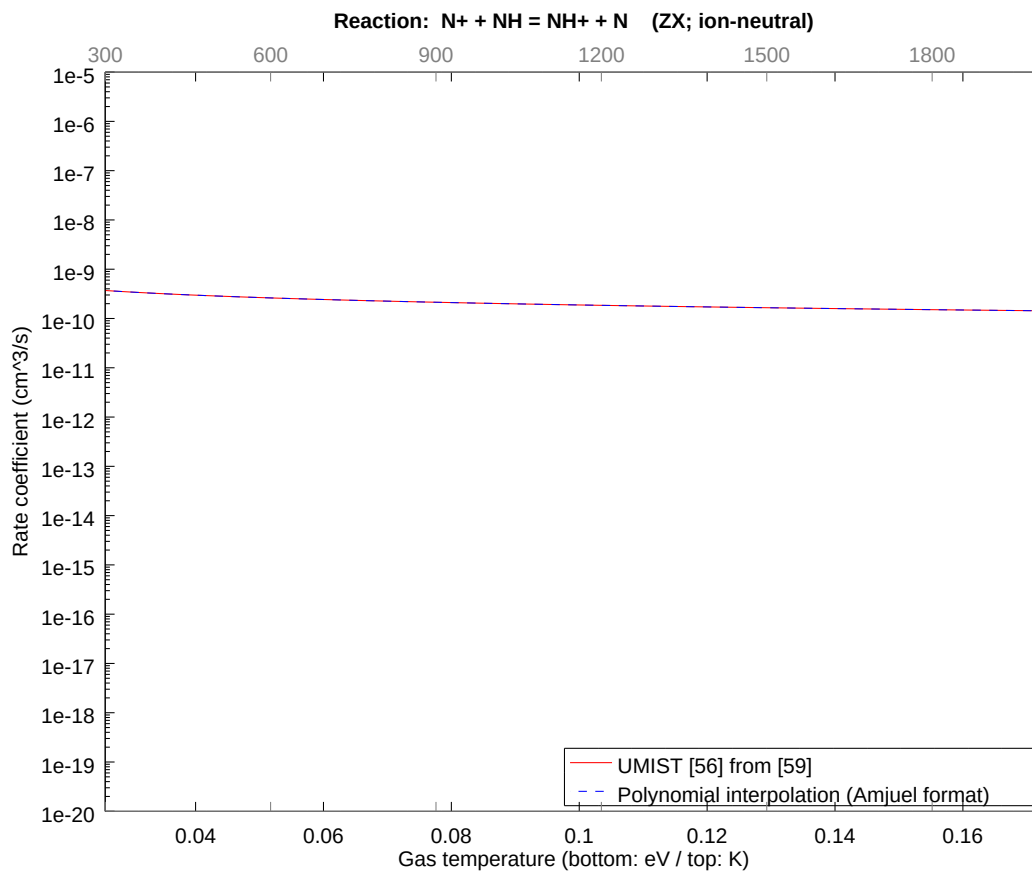
2.74 Reaction ZW $\text{N}^+ + \text{NH} = \text{N}_2^+ + \text{H}$

b0	-2.354494626224E+01	b1	-5.000000032306E-01	b2	-4.347944933015E-09
b3	-3.299966277679E-09	b4	-1.544289864373E-09	b5	-4.561831738007E-10
b6	-8.305450654833E-11	b7	-8.519342746733E-12	b8	-3.768628601479E-13



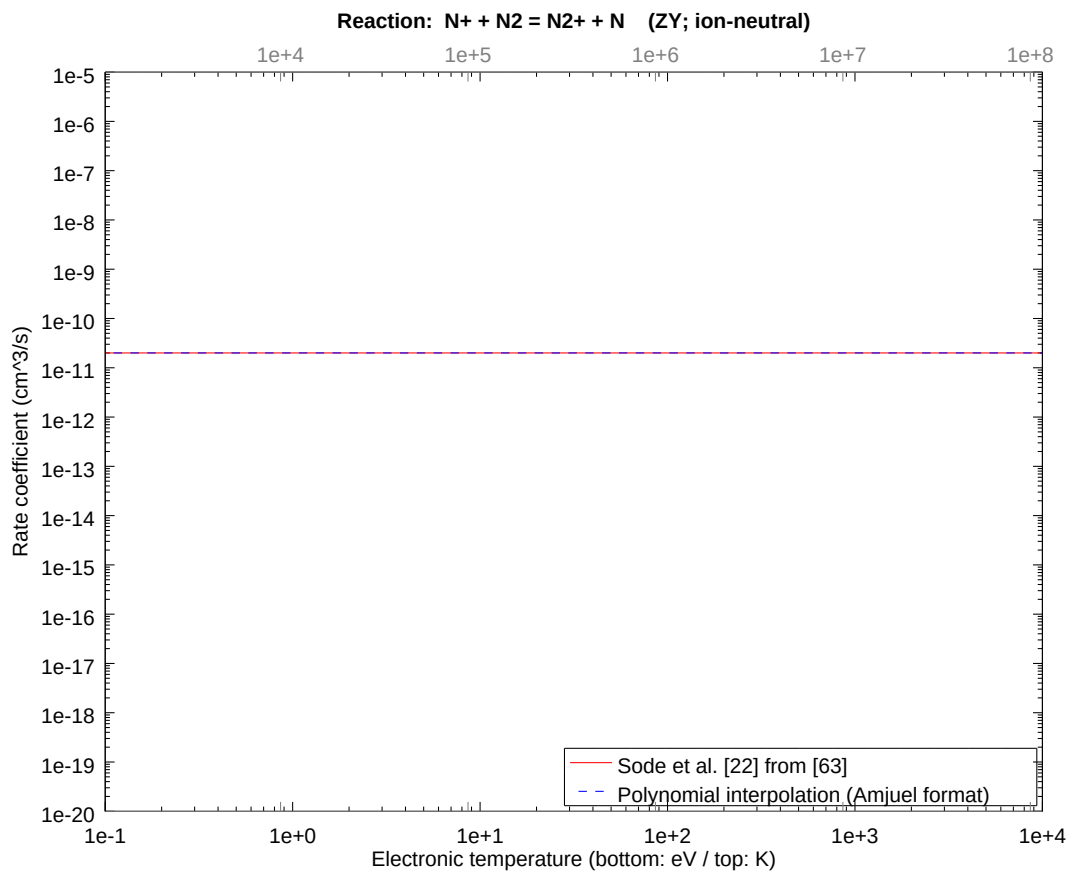
2.75 Reaction ZX $\text{N}^+ + \text{NH} = \text{NH}^+ + \text{N}$

b0	-2.354494626224E+01	b1	-5.000000032306E-01	b2	-4.347944933015E-09
b3	-3.299966277679E-09	b4	-1.544289864373E-09	b5	-4.561831738007E-10
b6	-8.305450654833E-11	b7	-8.519342746733E-12	b8	-3.768628601479E-13



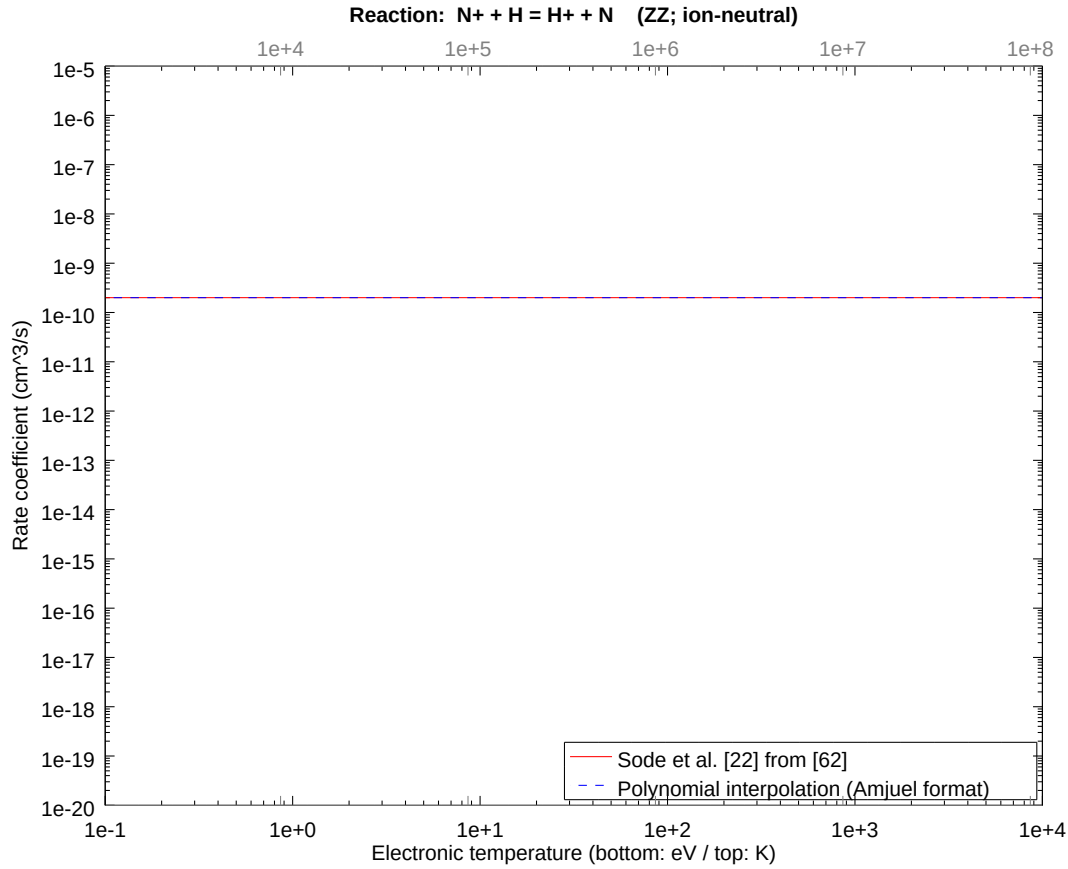
2.76 Reaction ZY $\text{N}^+ + \text{N}_2 = \text{N}_2^+ + \text{N}$

b0	-2.463528884237E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



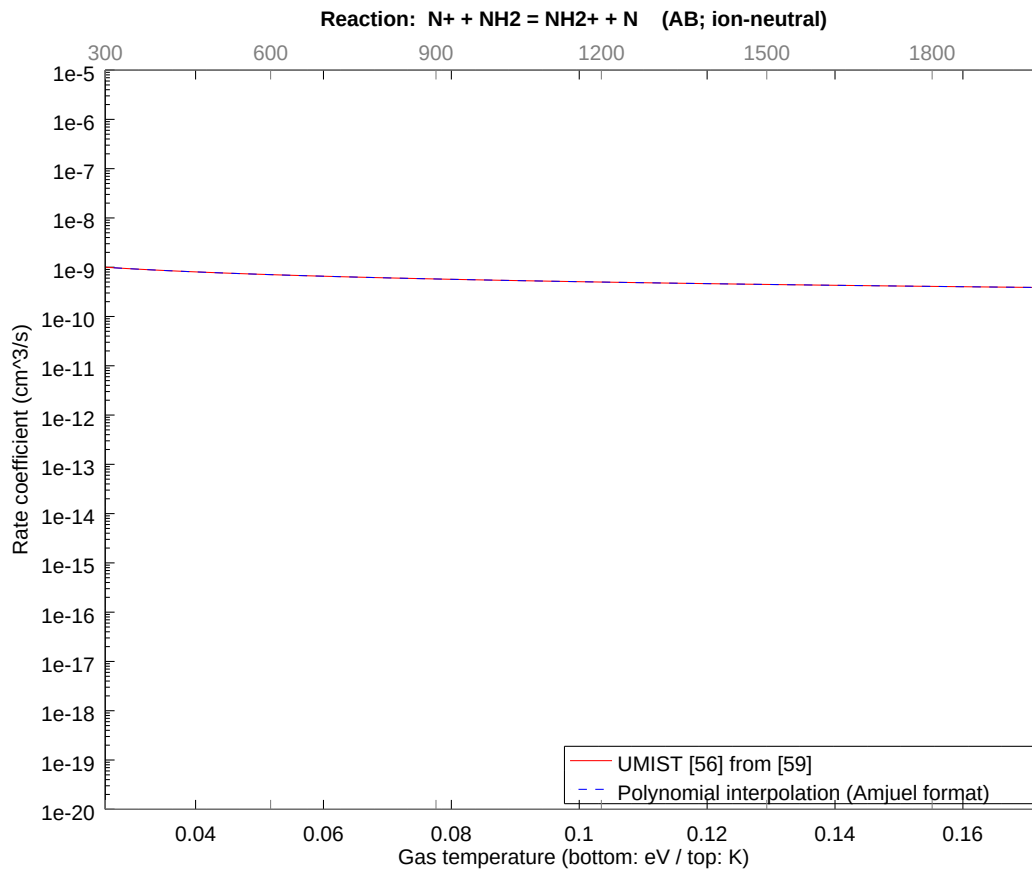
2.77 Reaction ZZ N+ + H = H+ + N

b0	-2.233270374938E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



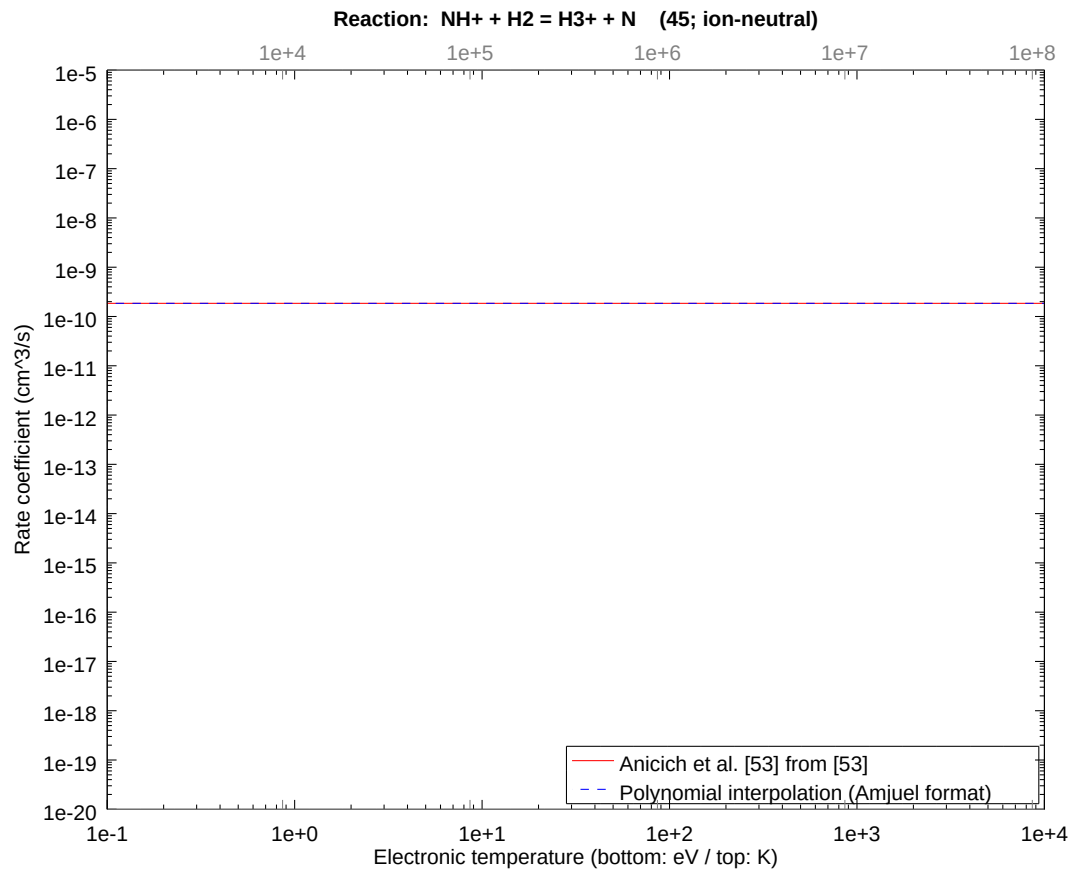
2.78 Reaction AB $\text{N}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{N}$

b0	-2.255069399034E+01	b1	-5.000000077382E-01	b2	-1.048331289759E-08
b3	-8.030169368750E-09	b4	-3.804180693869E-09	b5	-1.141479368811E-09
b6	-2.119003834735E-10	b7	-2.225565586486E-11	b8	-1.012819638805E-12



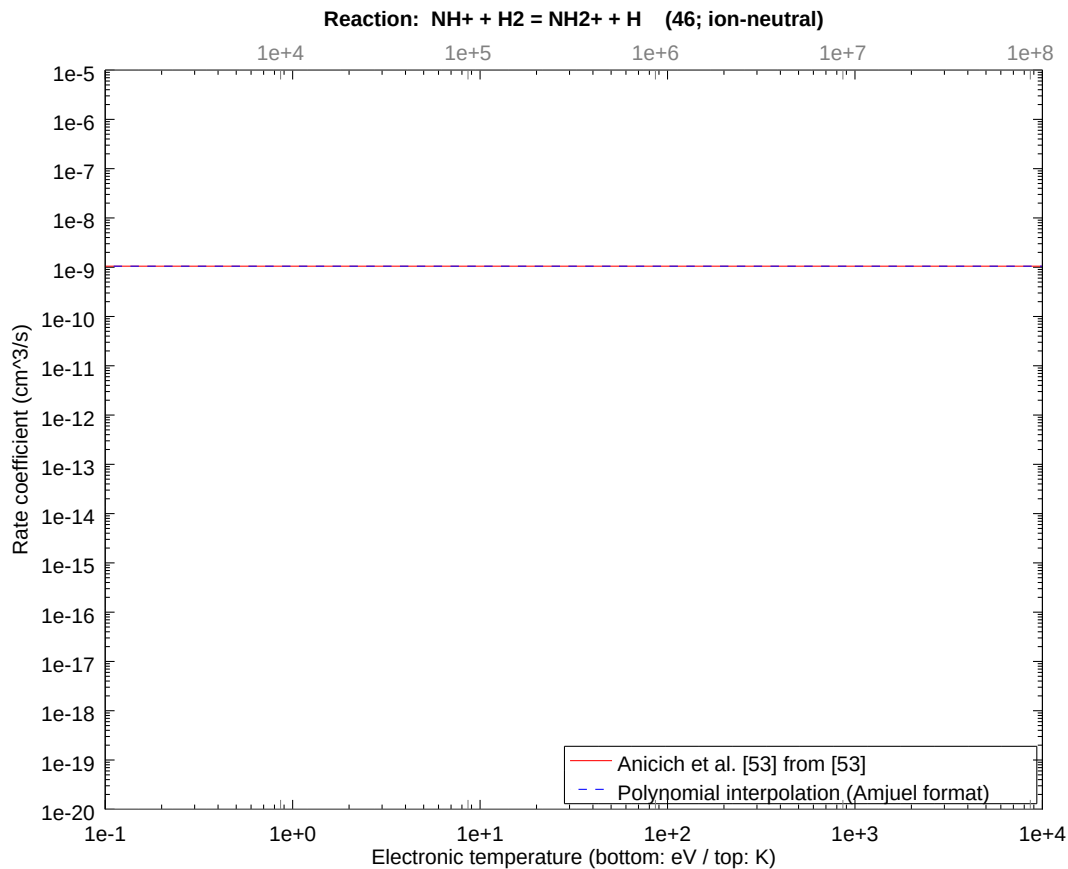
2.79 Reaction 45 $\text{NH}^+ + \text{H}_2 = \text{H}_3^+ + \text{N}$

b0	-2.241066529085E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



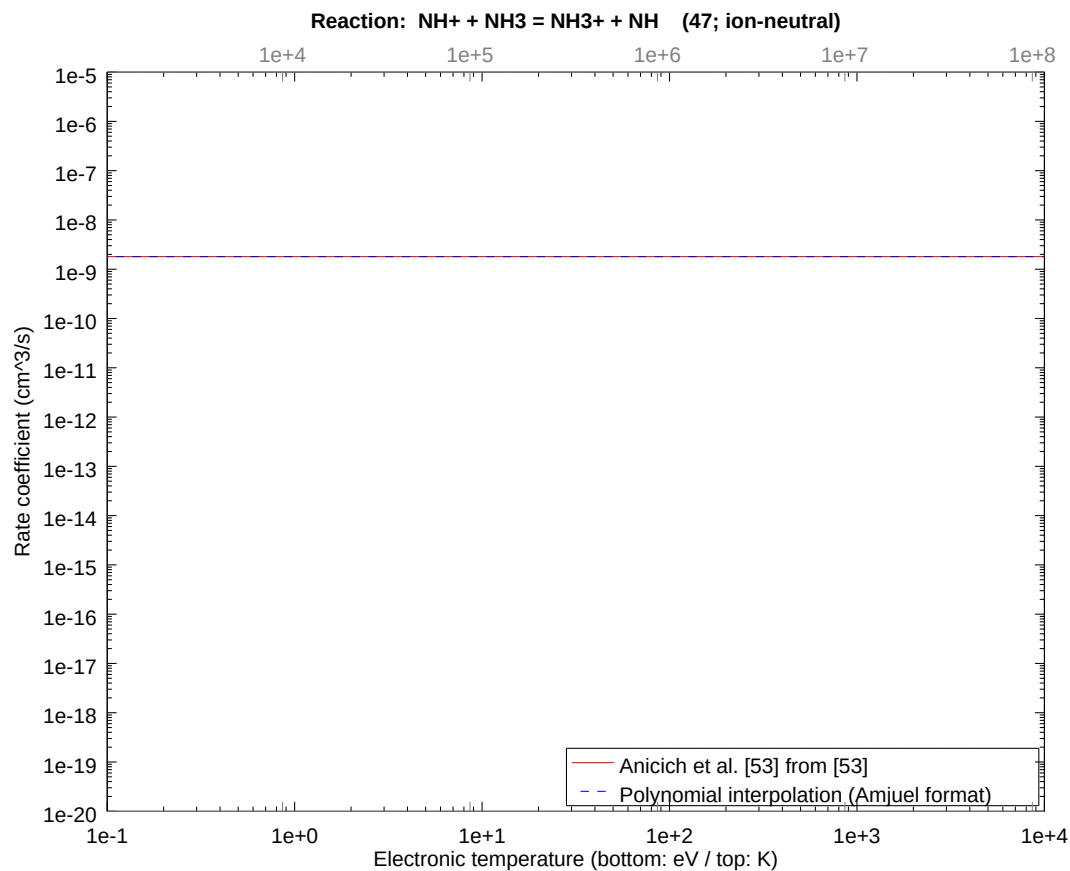
2.80 Reaction 46 $\text{NH}^+ + \text{H}_2 = \text{NH}_2^+ + \text{H}$

b0	-2.067447567278E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



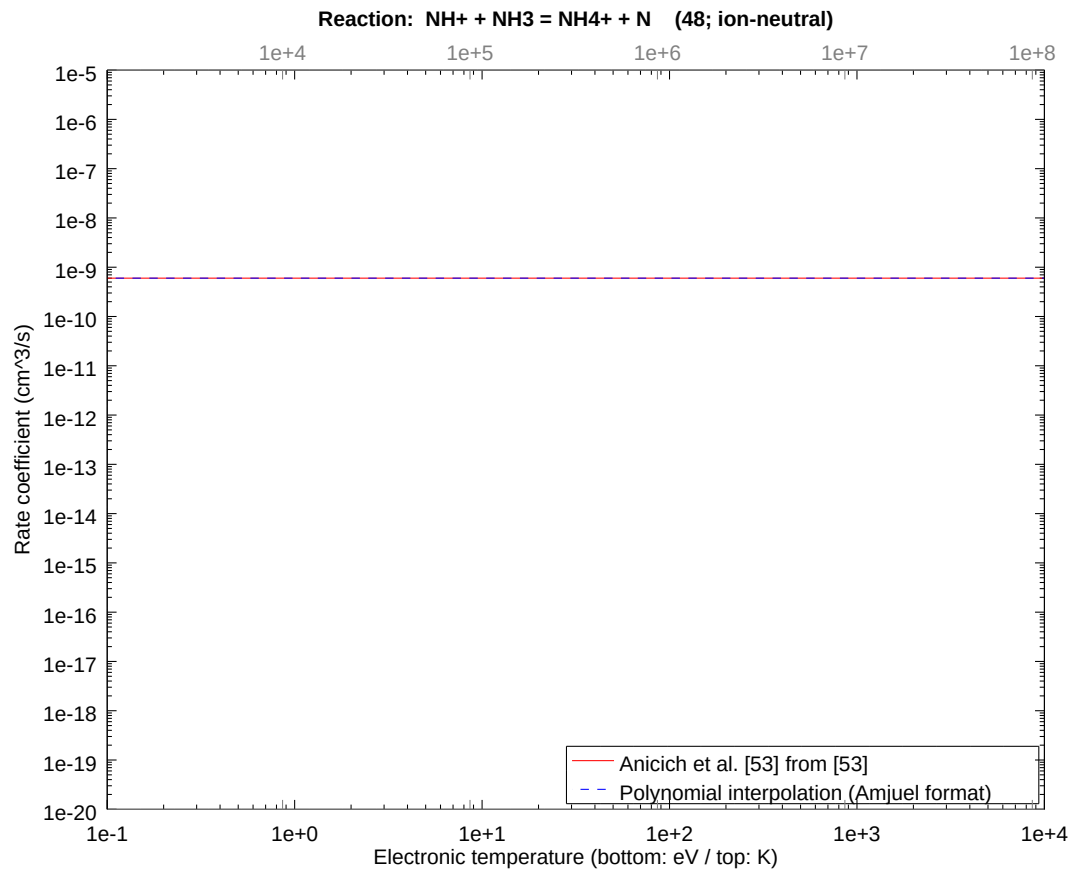
2.81 Reaction 47 $\text{NH}^+ + \text{NH}_3 = \text{NH}_3^+ + \text{NH}$

b0	-2.013547917204E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



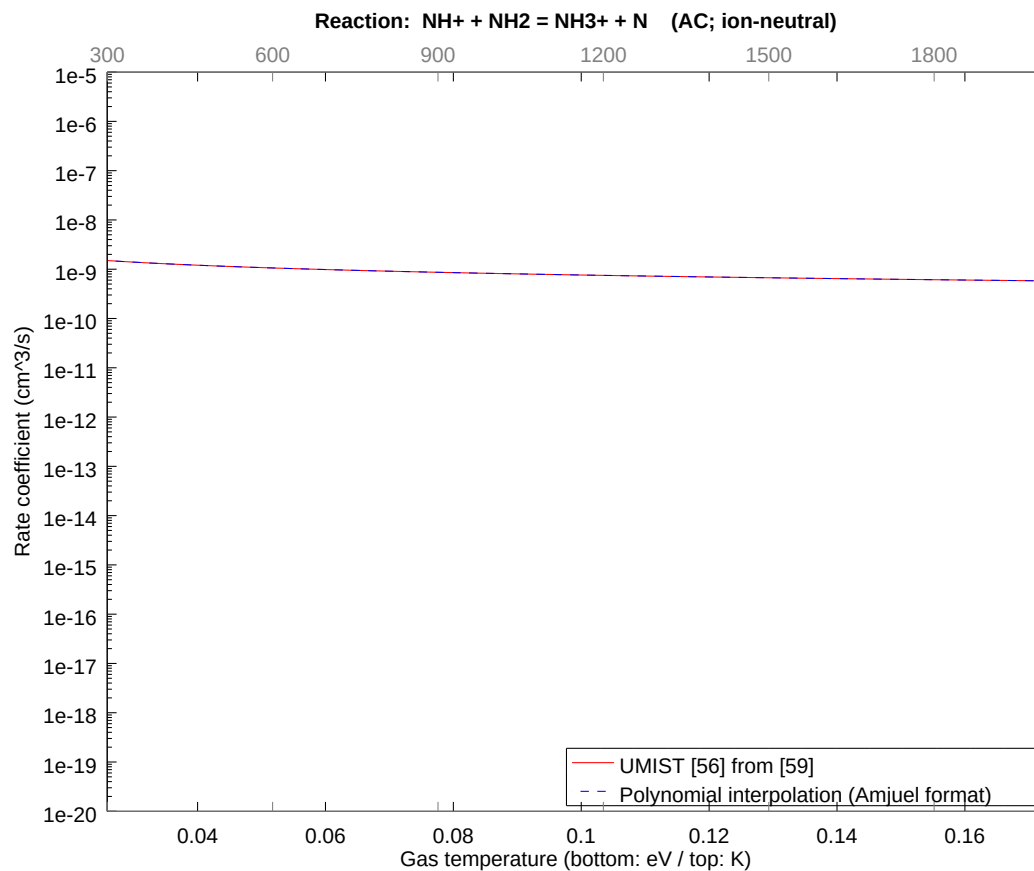
2.82 Reaction 48 $\text{NH}^+ + \text{NH}_3 = \text{NH}_4^+ + \text{N}$

b0	-2.123409146071E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



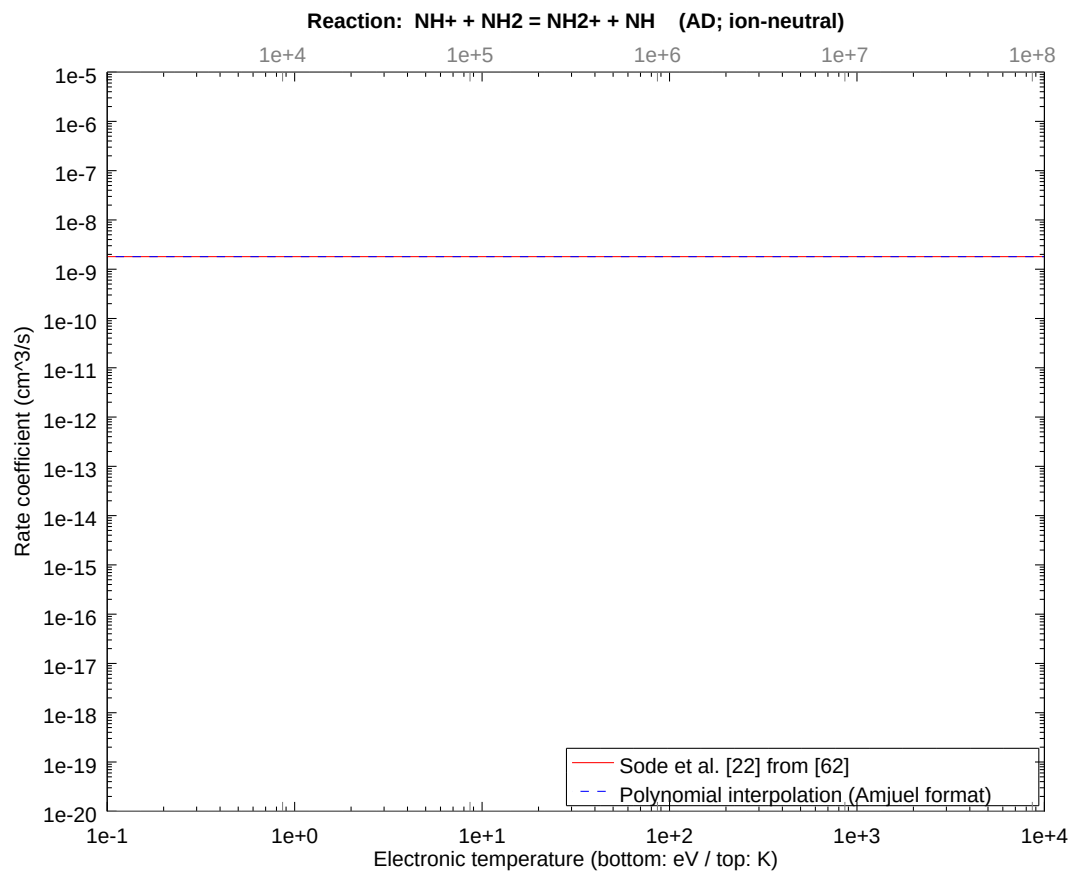
2.83 Reaction AC $\text{NH}^+ + \text{NH}_2 = \text{NH}_3^+ + \text{N}$

b0	-2.214522888071E+01	b1	-5.000000029212E-01	b2	-3.860018050693E-09
b3	-2.872424232405E-09	b4	-1.316010331162E-09	b5	-3.799779029463E-10
b6	-6.750141857656E-11	b7	-6.743736474333E-12	b8	-2.900439964921E-13



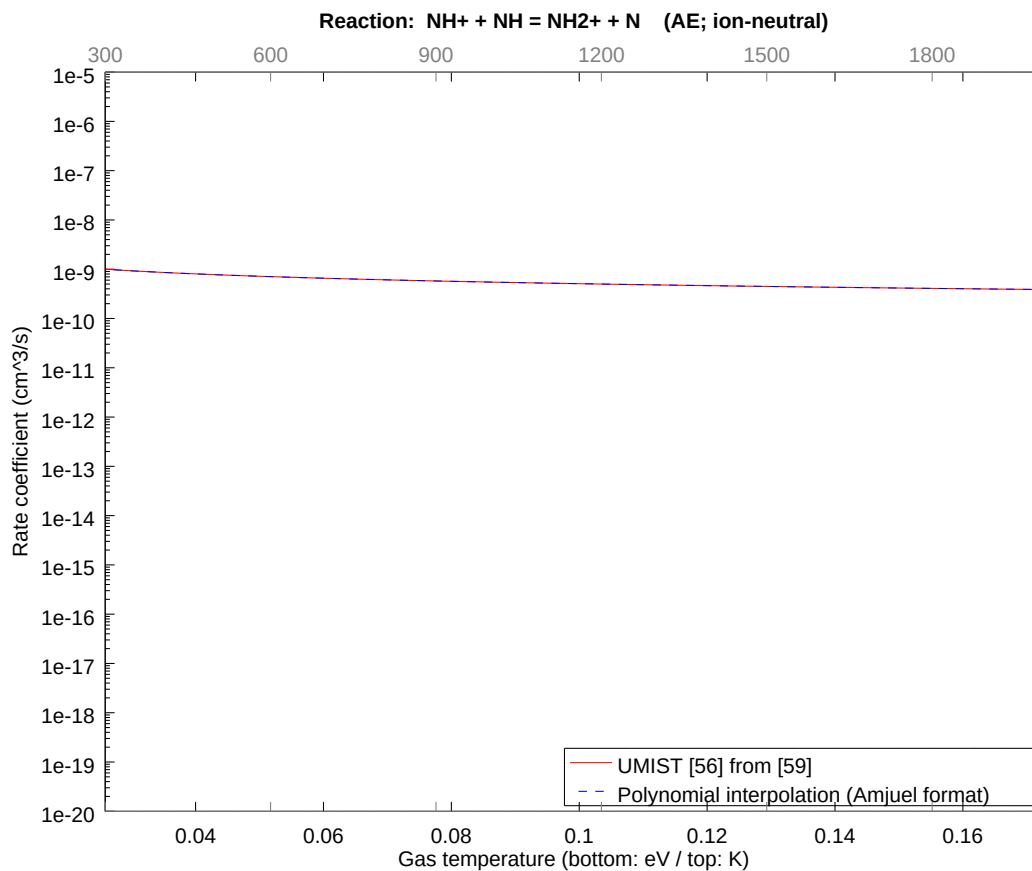
2.84 Reaction AD $\text{NH}^+ + \text{NH}_2 = \text{NH}_2^+ + \text{NH}$

b0	-2.013547917204E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



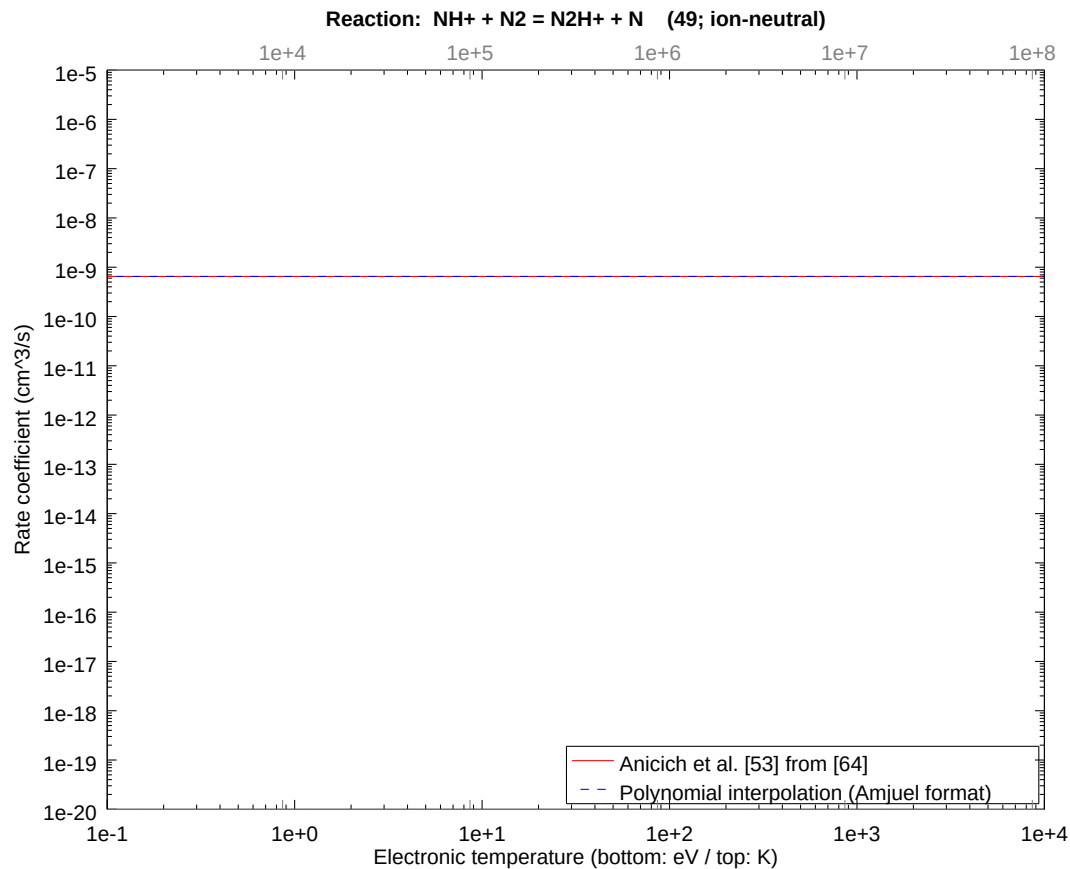
2.85 Reaction AE $\text{NH}^+ + \text{NH} = \text{NH}_2^+ + \text{N}$

b0 -2.255069399034E+01	b1 -5.000000077382E-01	b2 -1.048331289759E-08
b3 -8.030169368750E-09	b4 -3.804180693869E-09	b5 -1.141479368811E-09
b6 -2.119003834735E-10	b7 -2.225565586486E-11	b8 -1.012819638805E-12



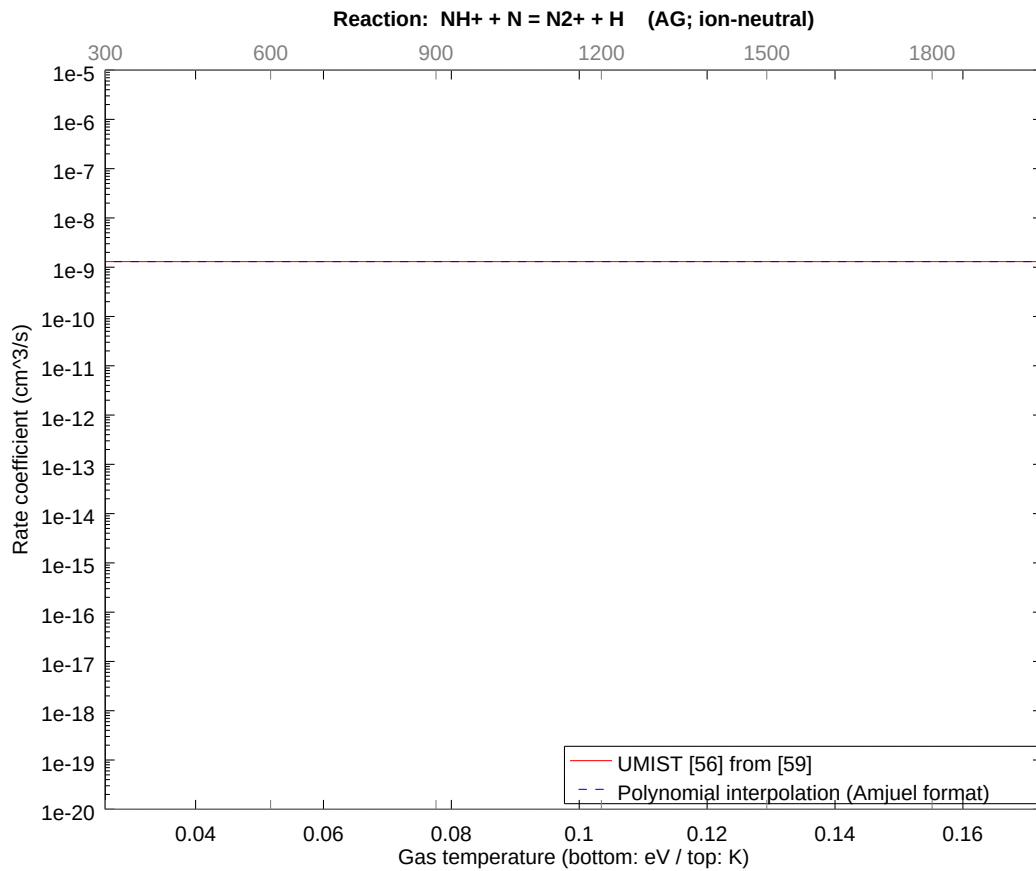
2.86 Reaction 49 $\text{NH}^+ + \text{N}_2 = \text{N}_2\text{H}^+ + \text{N}$

b0	-2.115404875304E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



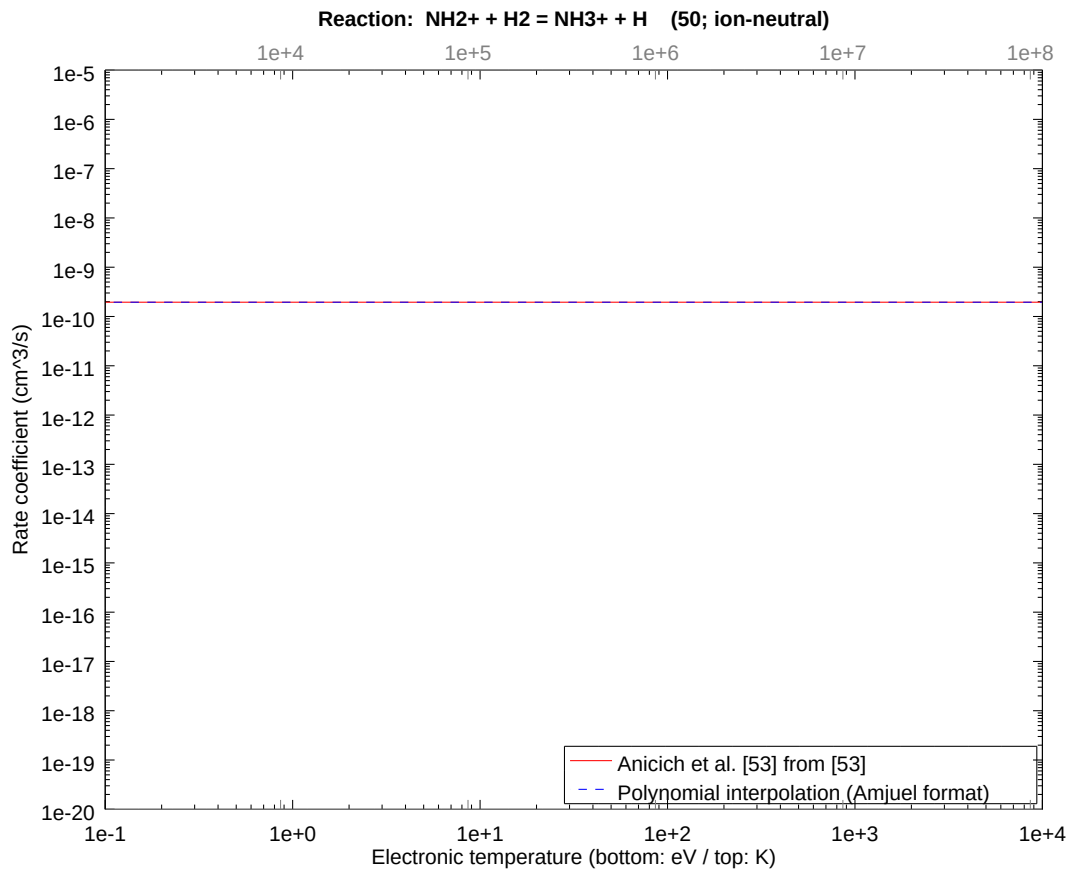
2.87 Reaction AG $\text{NH}^+ + \text{N} = \text{N}_2^+ + \text{H}$

b0	-2.046090157248E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



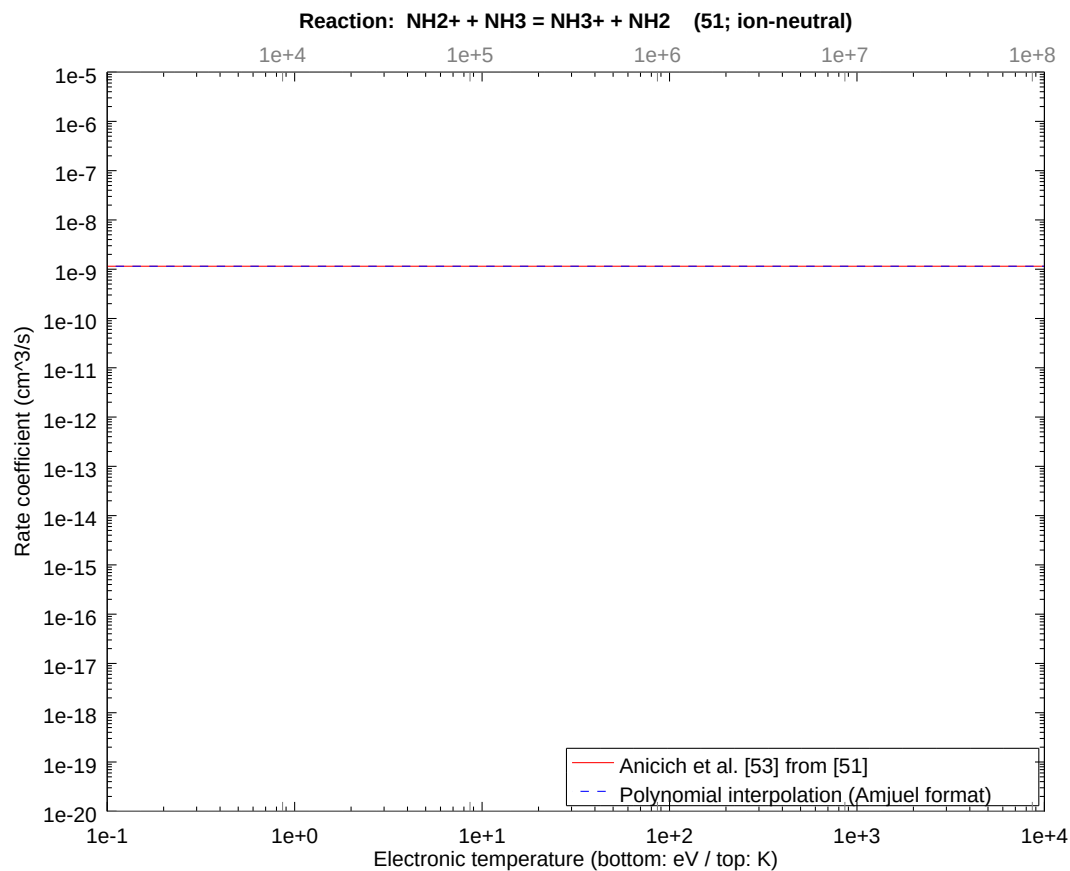
2.88 Reaction 50 $\text{NH}_2^+ + \text{H}_2 = \text{NH}_3^+ + \text{H}$

b0	-2.235802155736E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



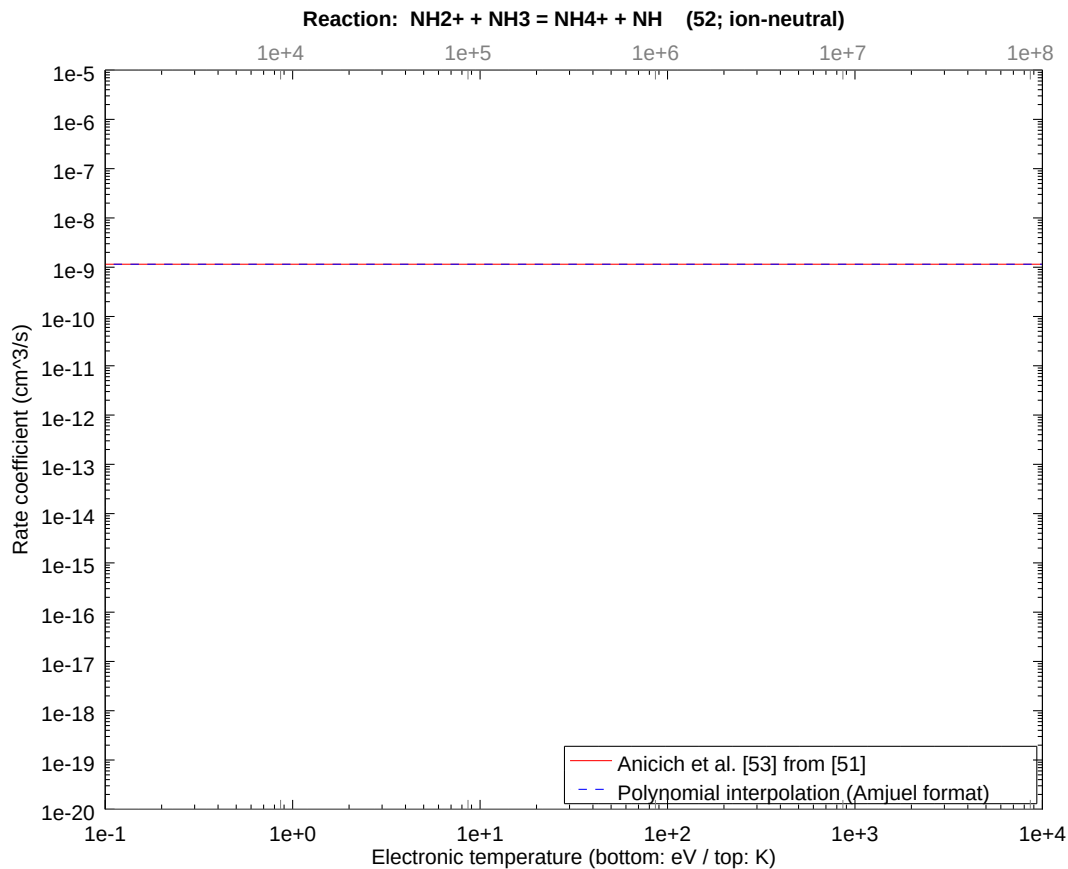
2.89 Reaction 51 $\text{NH}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{NH}_2$

b0	-2.058350389457E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



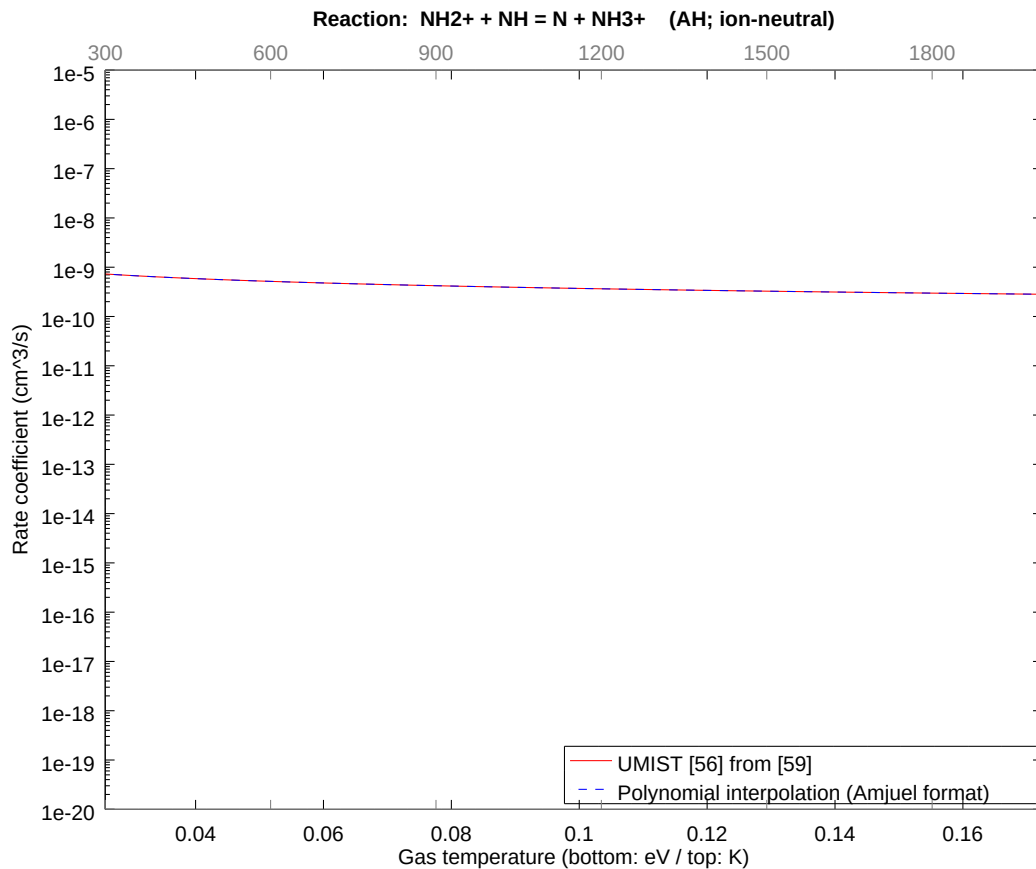
2.90 Reaction 52 $\text{NH}_2^+ + \text{NH}_3 = \text{NH}_4^+ + \text{NH}$

b0	-2.058350389457E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



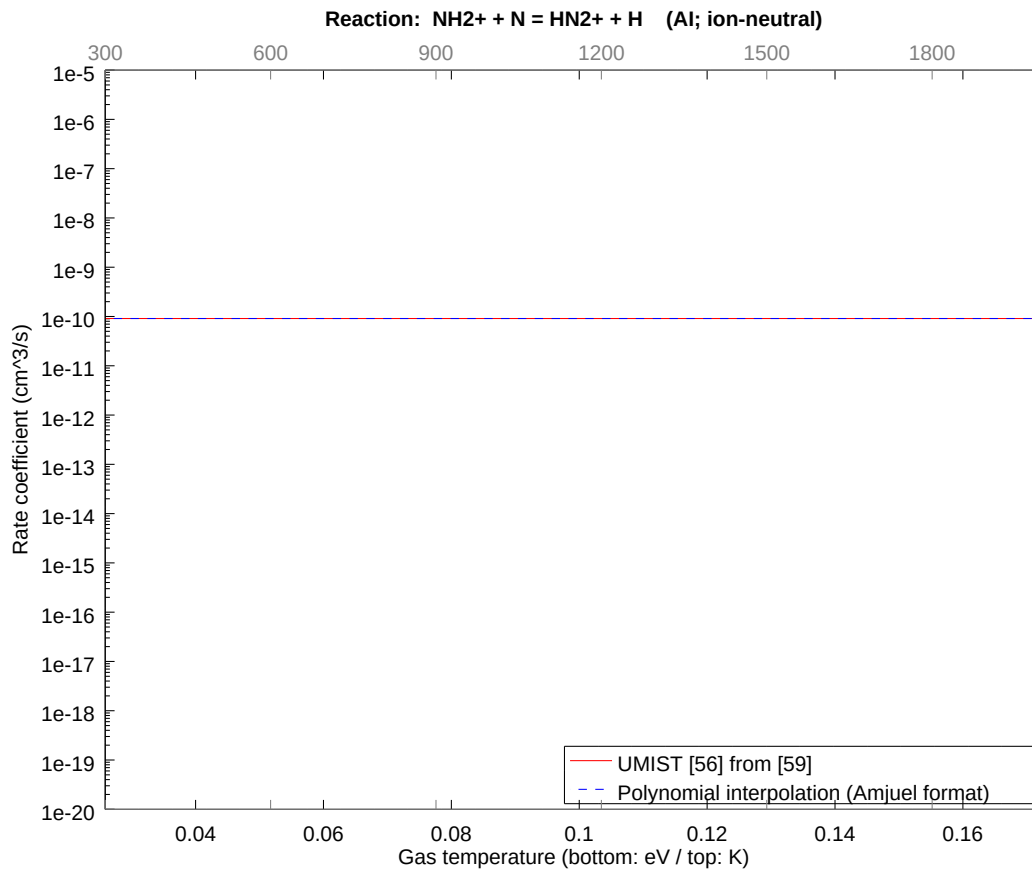
2.91 Reaction $\text{AH NH}_2^+ + \text{NH} = \text{N} + \text{NH}_3^+$

b0	-2.286540473346E+01	b1	-5.000000022583E-01	b2	-2.903991793554E-09
b3	-2.092602780058E-09	b4	-9.224932782956E-10	b5	-2.541818400592E-10
b6	-4.262482993526E-11	b7	-3.961221014971E-12	b8	-1.552398934101E-13



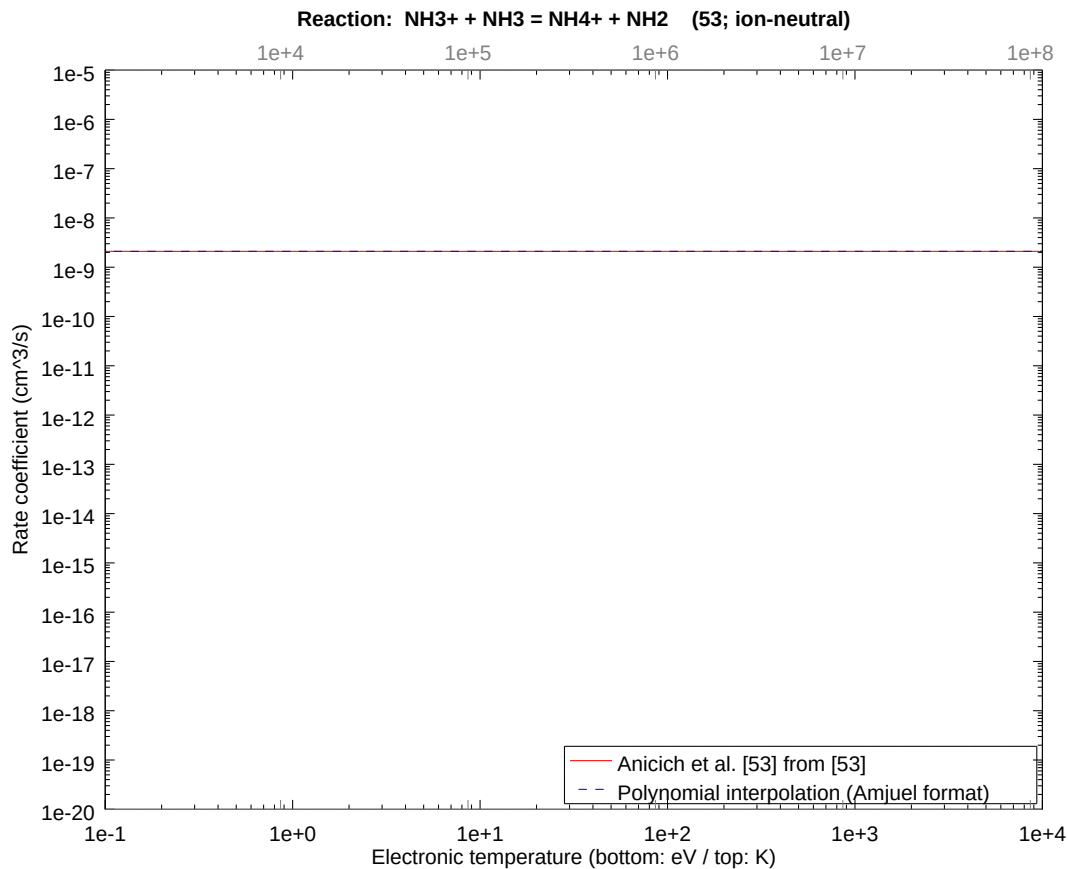
2.92 Reaction Al NH2+ + N = HN2+ + H

b0	-2.312016160941E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



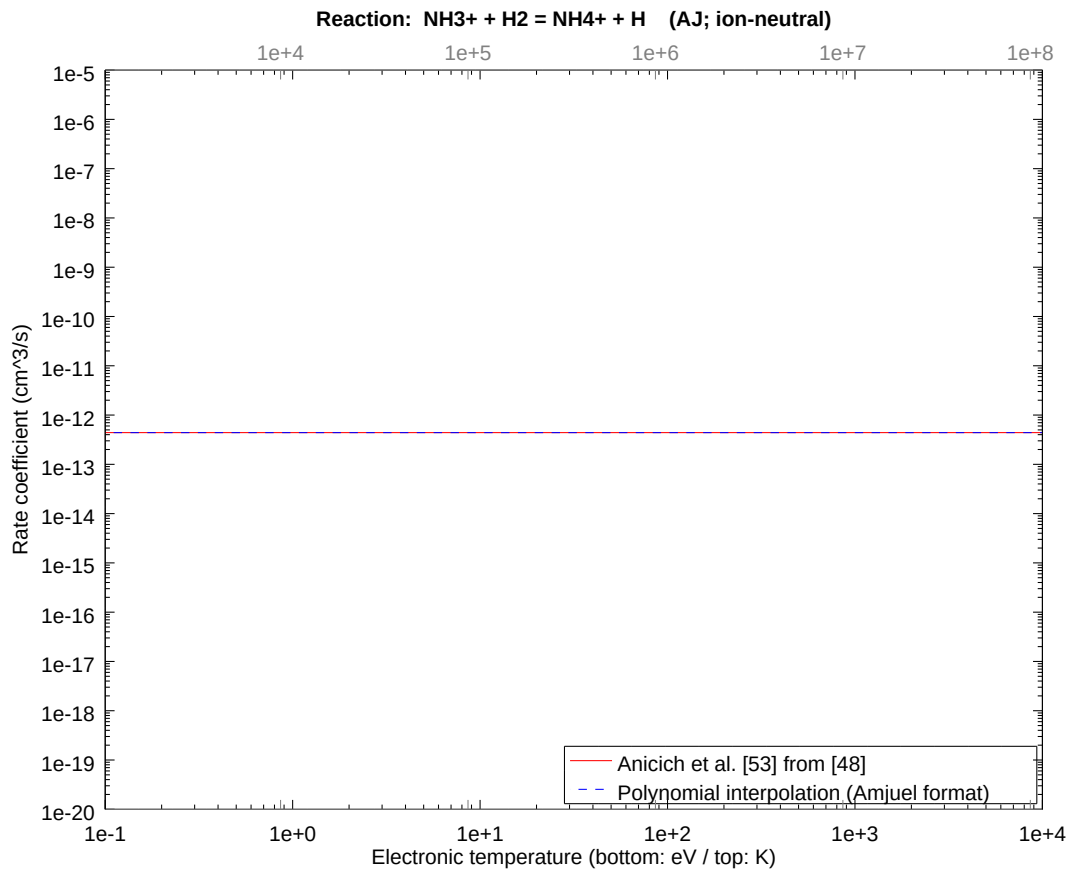
2.93 Reaction 53 $\text{NH}_3^+ + \text{NH}_3 = \text{NH}_4^+ + \text{NH}_2$

b0	-1.998132849222E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



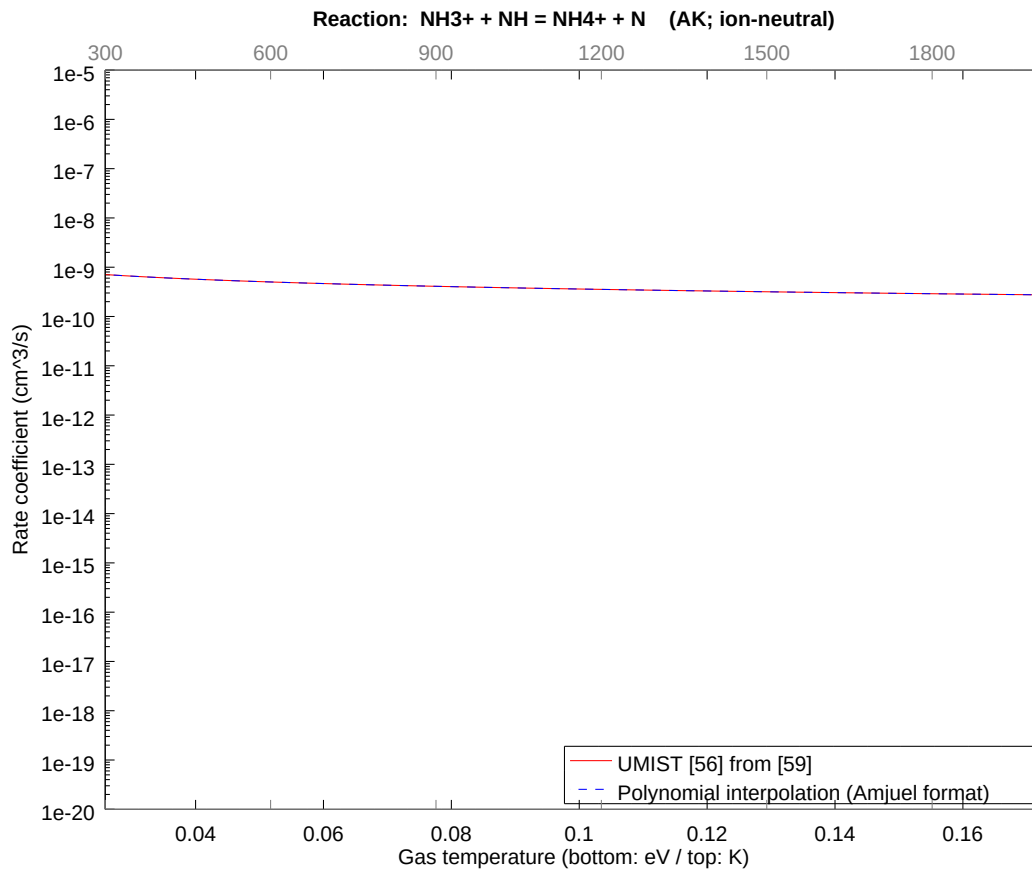
2.94 Reaction AJ $\text{NH}_3^+ + \text{H}_2 = \text{NH}_4^+ + \text{H}$

b0	-2.845200166800E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



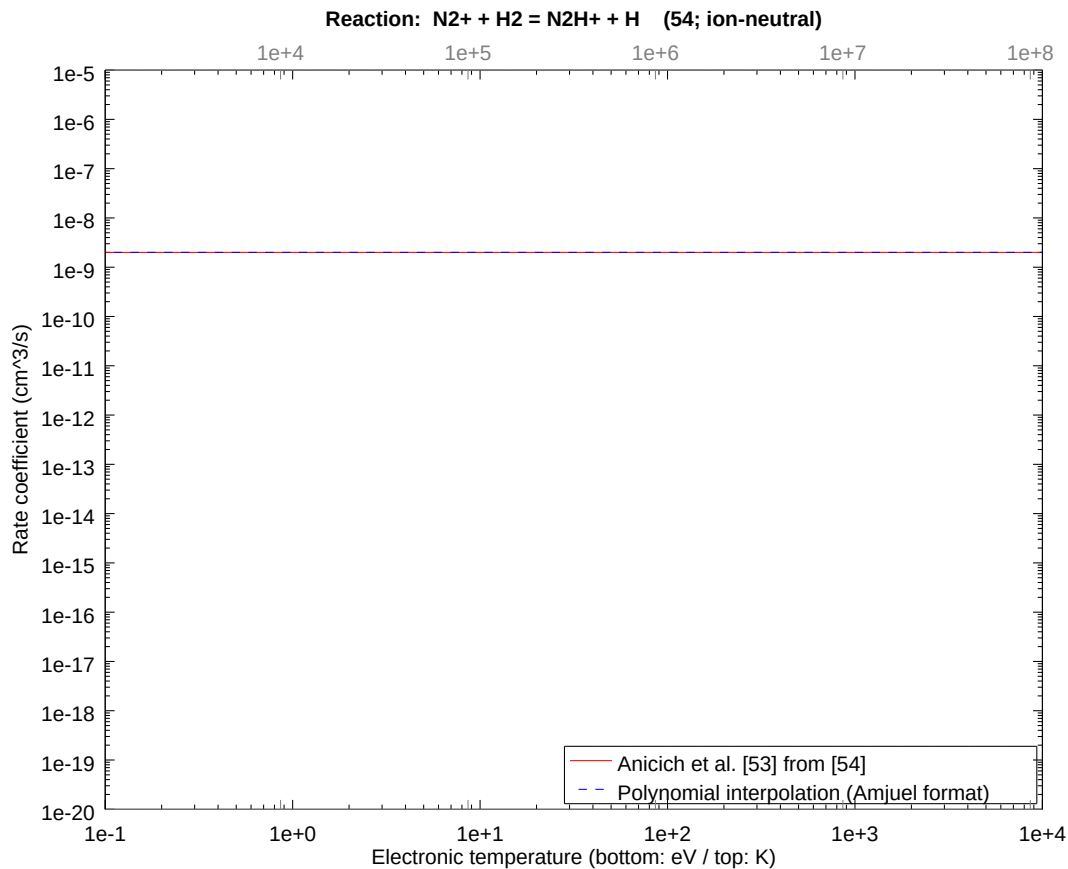
2.95 Reaction AK $\text{NH}_3^+ + \text{NH} = \text{NH}_4^+ + \text{N}$

b0 -2.289318429743E+01	b1 -5.000000018164E-01	b2 -2.303568193971E-09
b3 -1.639801354397E-09	b4 -7.151531646047E-10	b5 -1.951520773247E-10
b6 -3.242435734298E-11	b7 -2.983863276617E-12	b8 -1.155432594465E-13



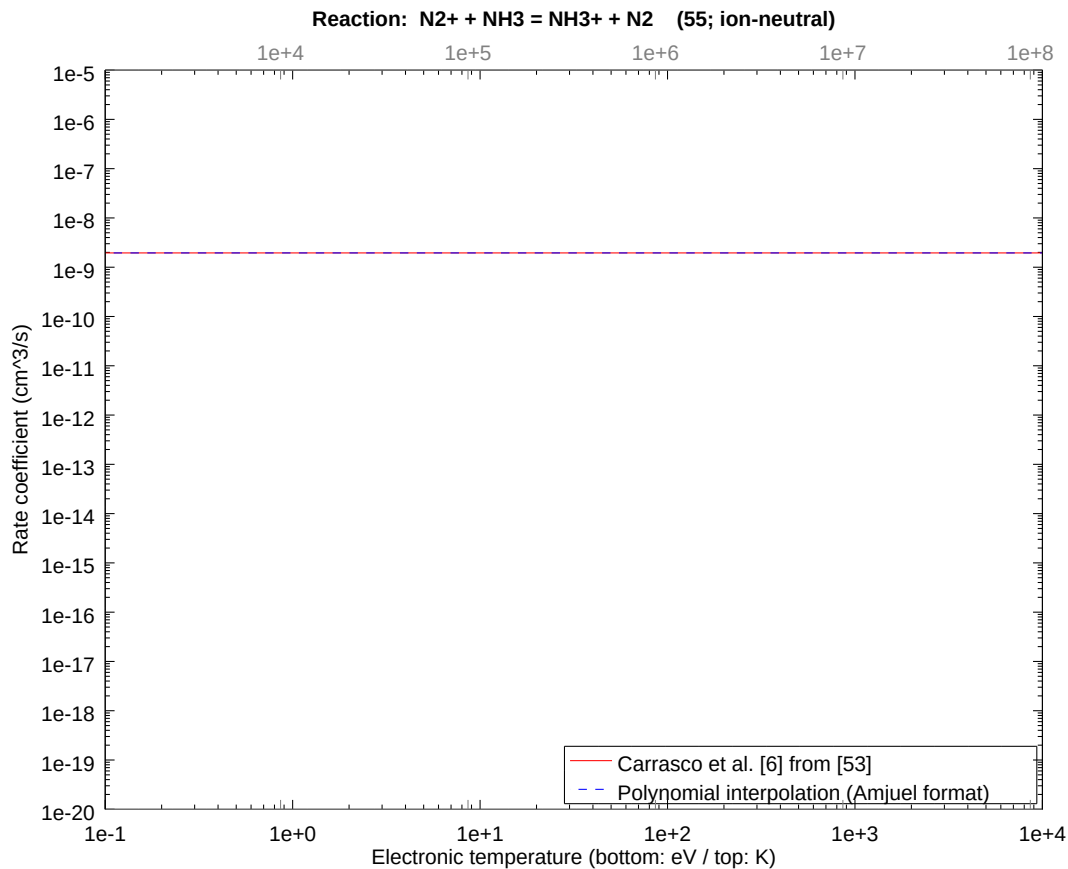
2.96 Reaction 54 $\text{N}_2^+ + \text{H}_2 = \text{N}_2\text{H}^+ + \text{H}$

b0	-2.003011865639E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



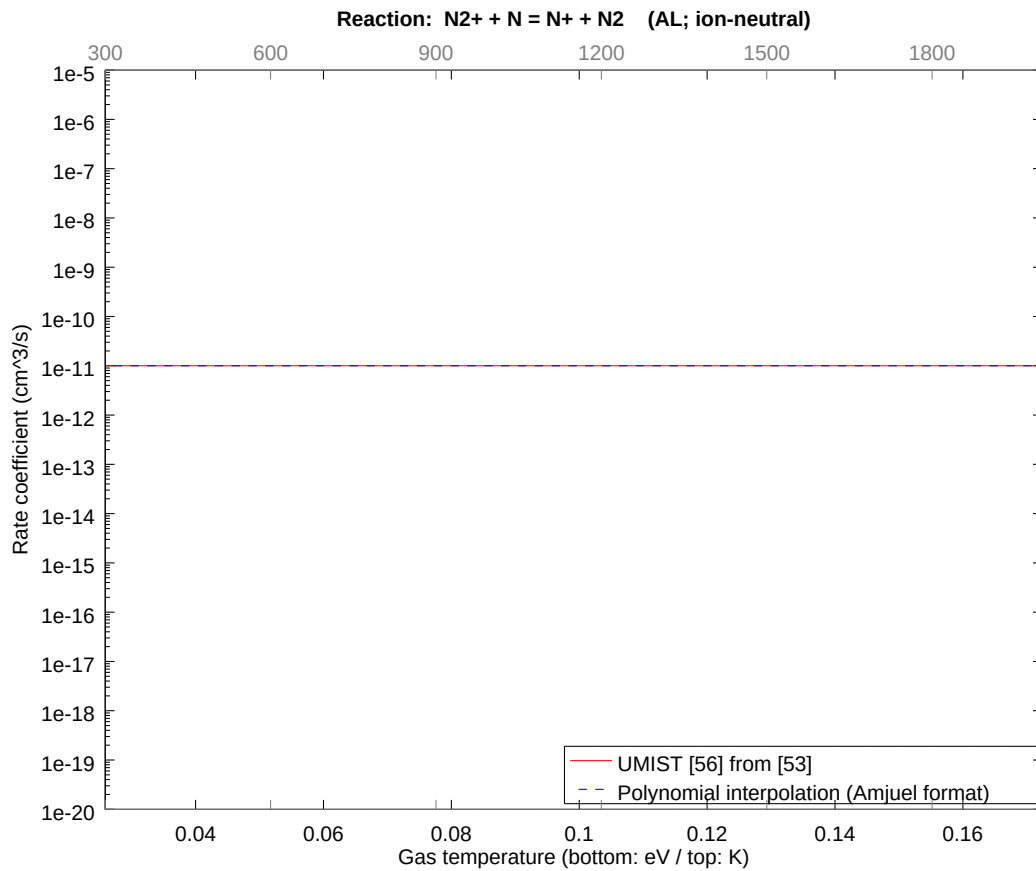
2.97 Reaction 55 $\text{N}_2^+ + \text{NH}_3 = \text{NH}_3^+ + \text{N}_2$

b0	-2.005543646437E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



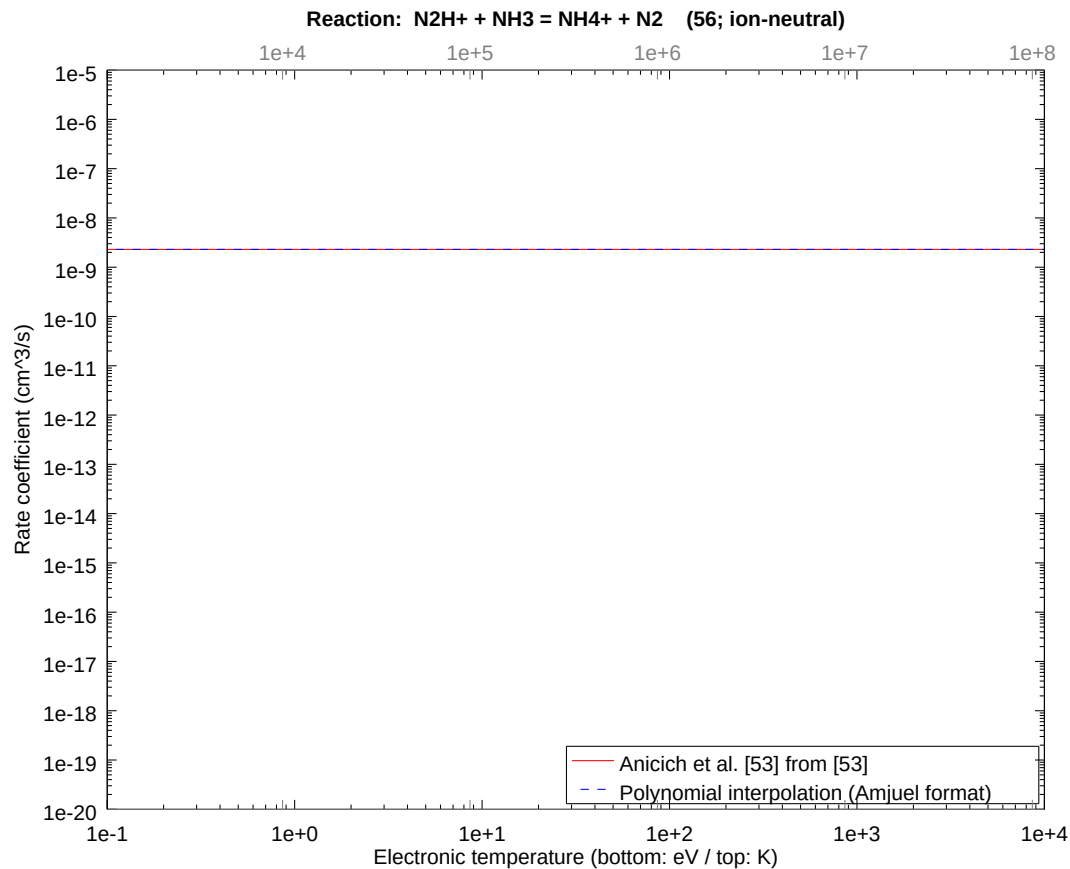
2.98 Reaction AL $\text{N}_2^+ + \text{N} = \text{N}^+ + \text{N}_2$

b0	-2.532843602293E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



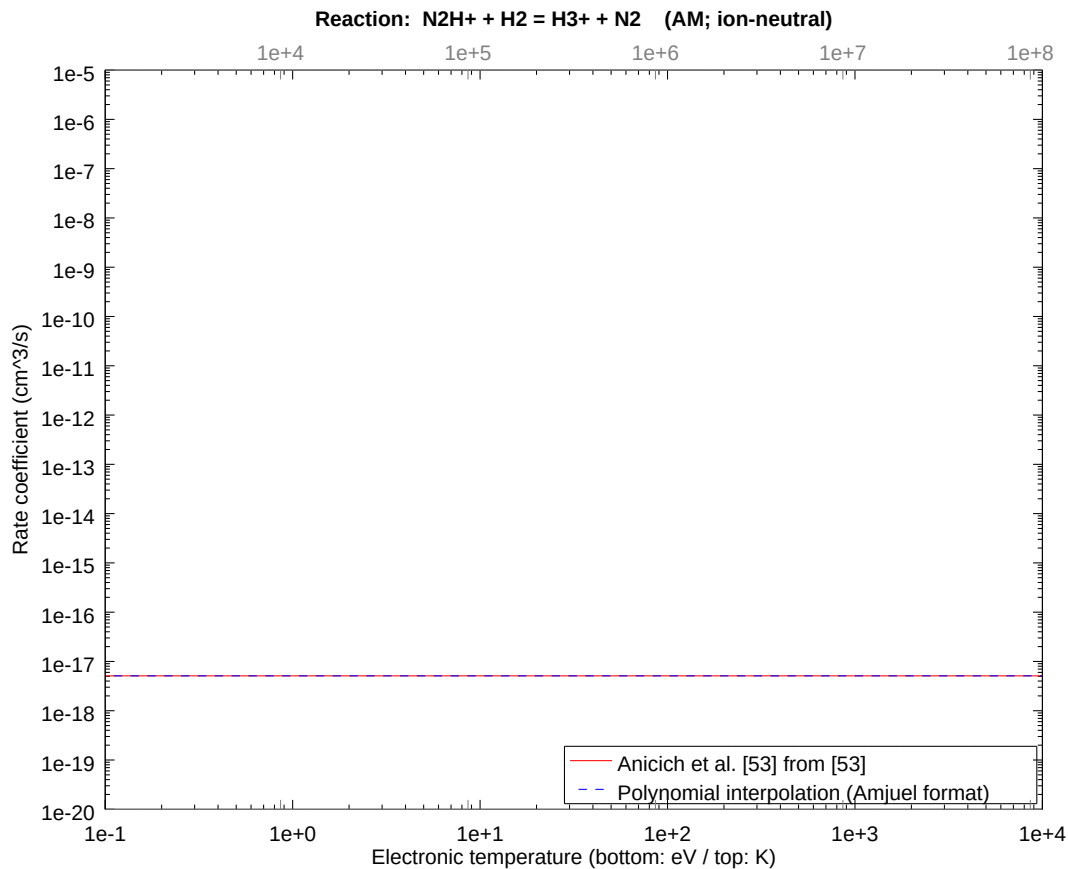
2.99 Reaction 56 $\text{N}_2\text{H}^+ + \text{NH}_3 = \text{NH}_4^+ + \text{N}_2$

b0	-1.989035671401E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



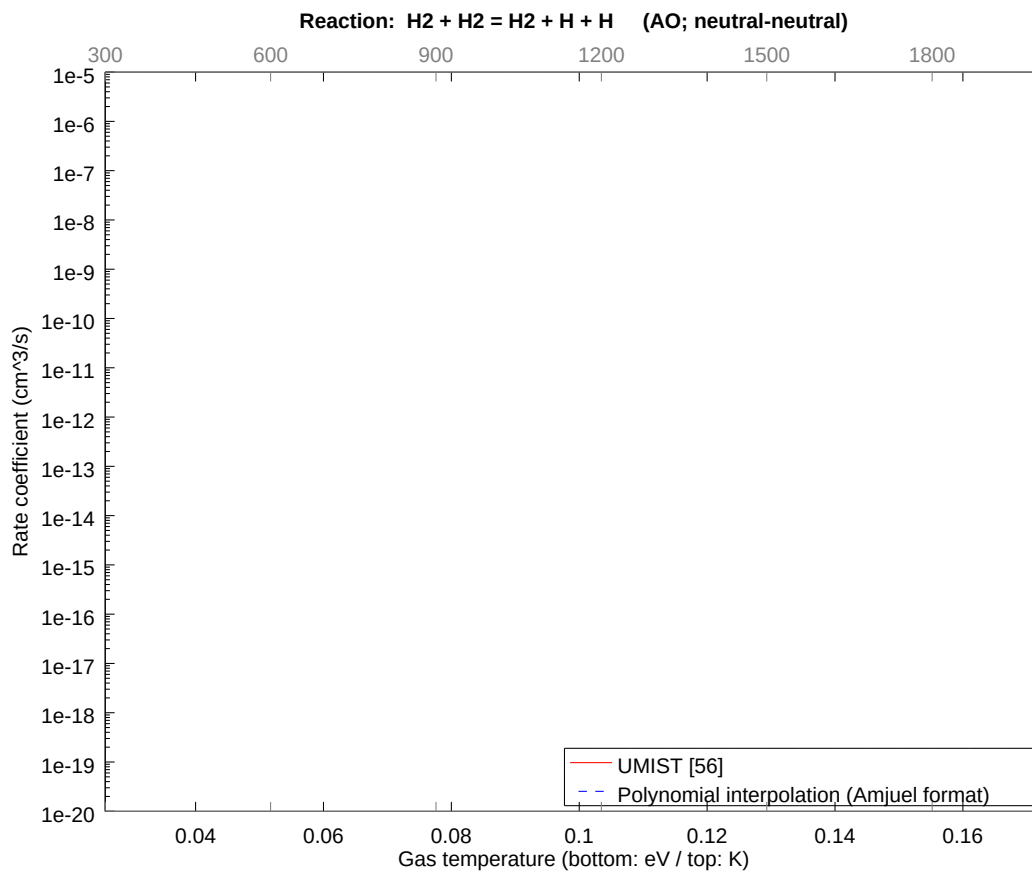
2.100 Reaction AM $\text{N}_2\text{H}^+ + \text{H}_2 = \text{H}_3^+ + \text{N}_2$

b0	-3.981729113416E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



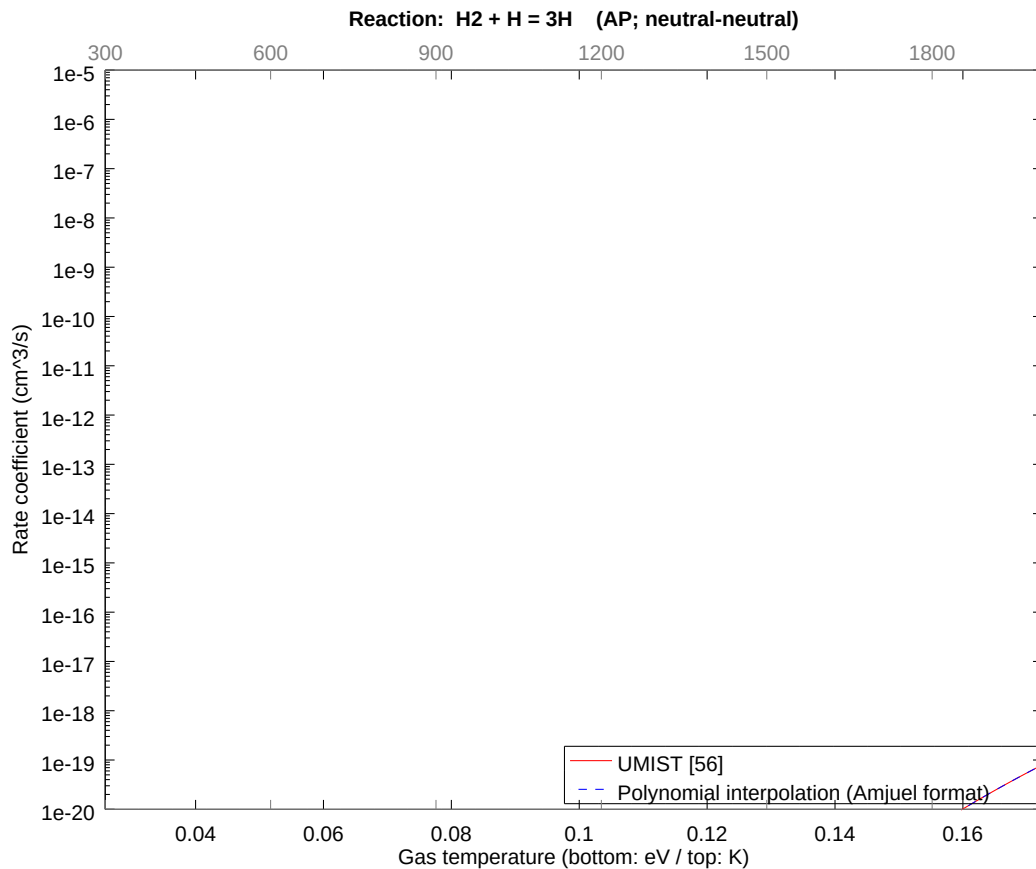
2.101 Reaction AO $\text{H}_2 + \text{H}_2 = \text{H}_2 + \text{H} + \text{H}$

b0	-2.696426662024E+01	b1	2.746576251943E+00	b2	-1.050552760935E+01
b3	-4.823787997126E+00	b4	-3.635045792807E+00	b5	-1.134852158008E+00
b6	-2.845431171969E-01	b7	-3.623051668352E-02	b8	-2.694960668238E-03



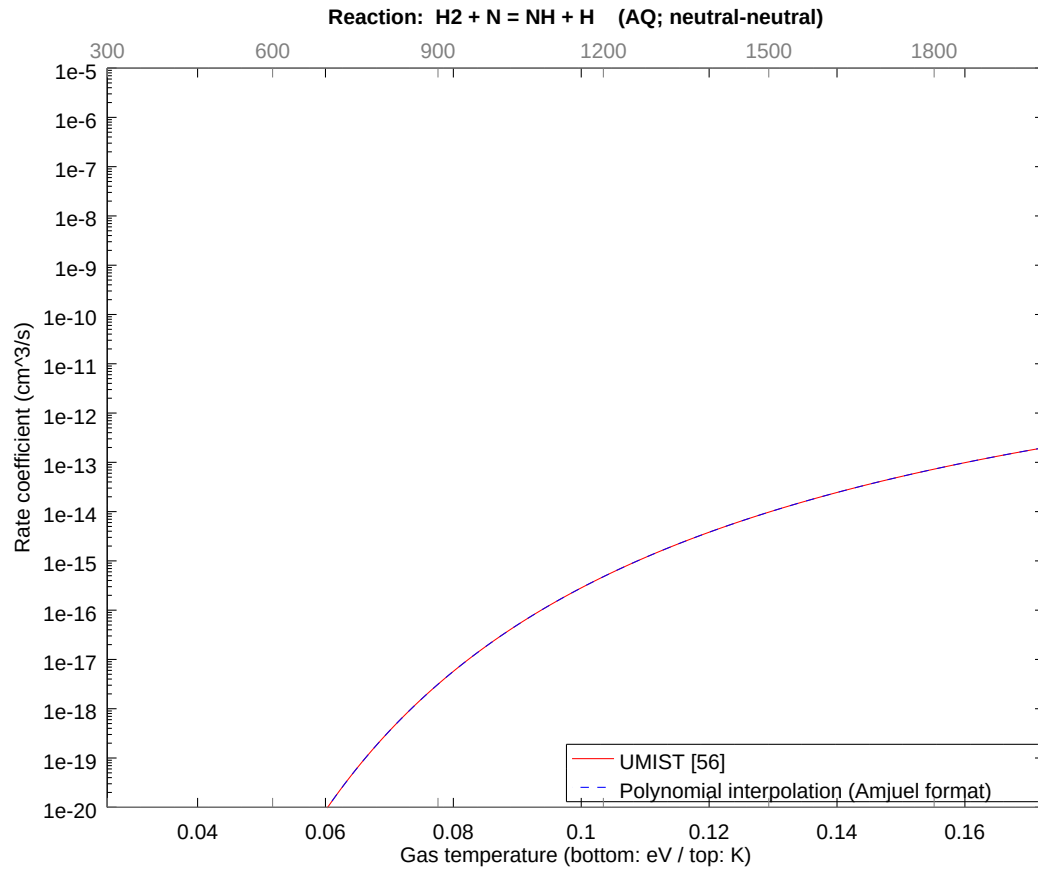
2.102 Reaction AP H2 + H = 3H

b0	-2.381915581932E+01	b1	7.962151505816E-01	b2	-6.870440167331E+00
b3	-3.154677044982E+00	b4	-2.377259434294E+00	b5	-7.421744189427E-01
b6	-1.860864617664E-01	b7	-2.369415477215E-02	b8	-1.762459413704E-03



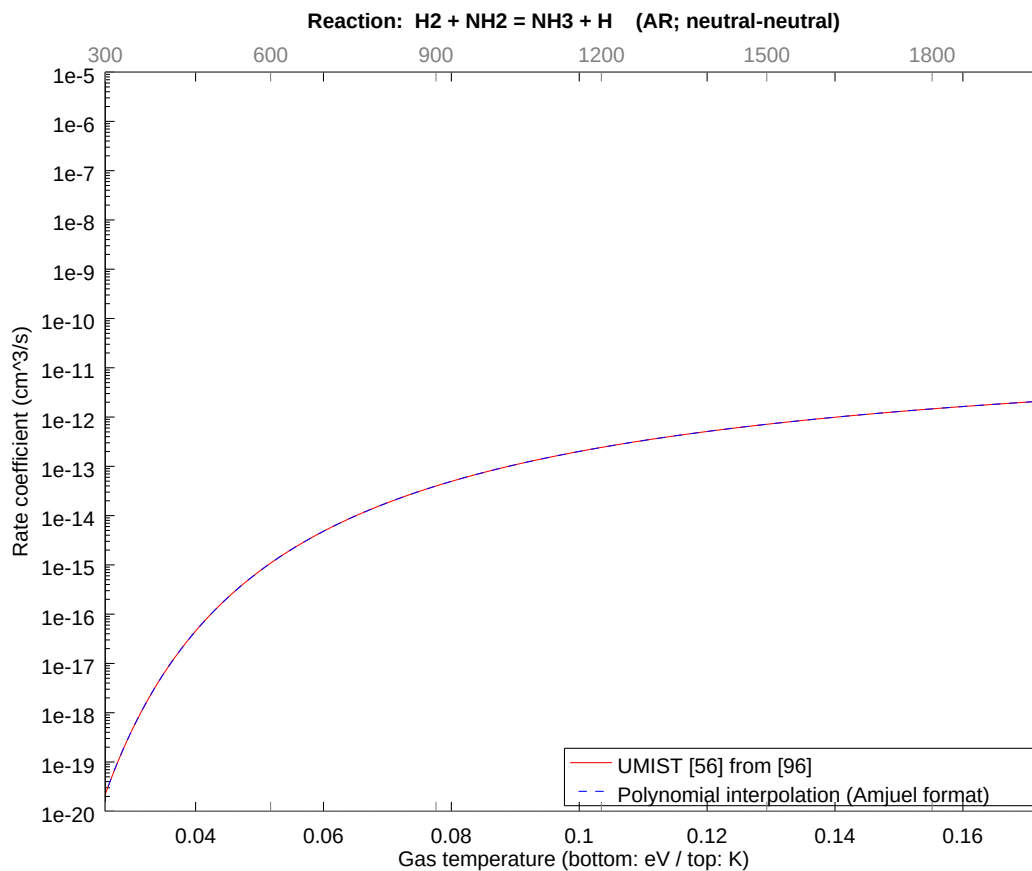
2.103 Reaction AQ H2 + N = NH + H

b0	-2.203677971440E+01	b1	5.909547855171E-01	b2	-2.260374813605E+00
b3	-1.037888746589E+00	b4	-7.821183532573E-01	b5	-2.441753836278E-01
b6	-6.122244587988E-02	b7	-7.795376915334E-03	b8	-5.798491468765E-04



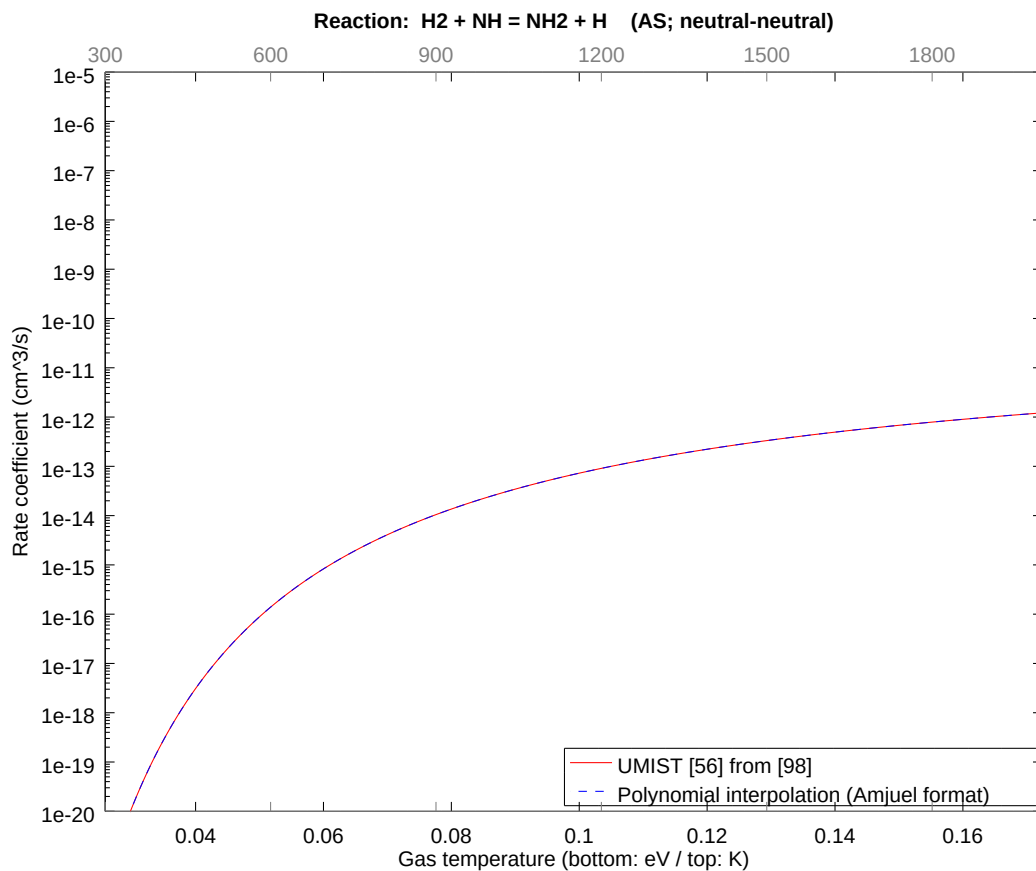
2.104 Reaction AR $\text{H}_2 + \text{NH}_2 = \text{NH}_3 + \text{H}$

b0	-2.430134589741E+01	b1	2.119533835672E-01	b2	-8.107119454755E-01
b3	-3.722518957301E-01	b4	-2.805166153577E-01	b5	-8.757658207336E-02
b6	-2.195820260768E-02	b7	-2.795910275707E-03	b8	-2.079702113924E-04



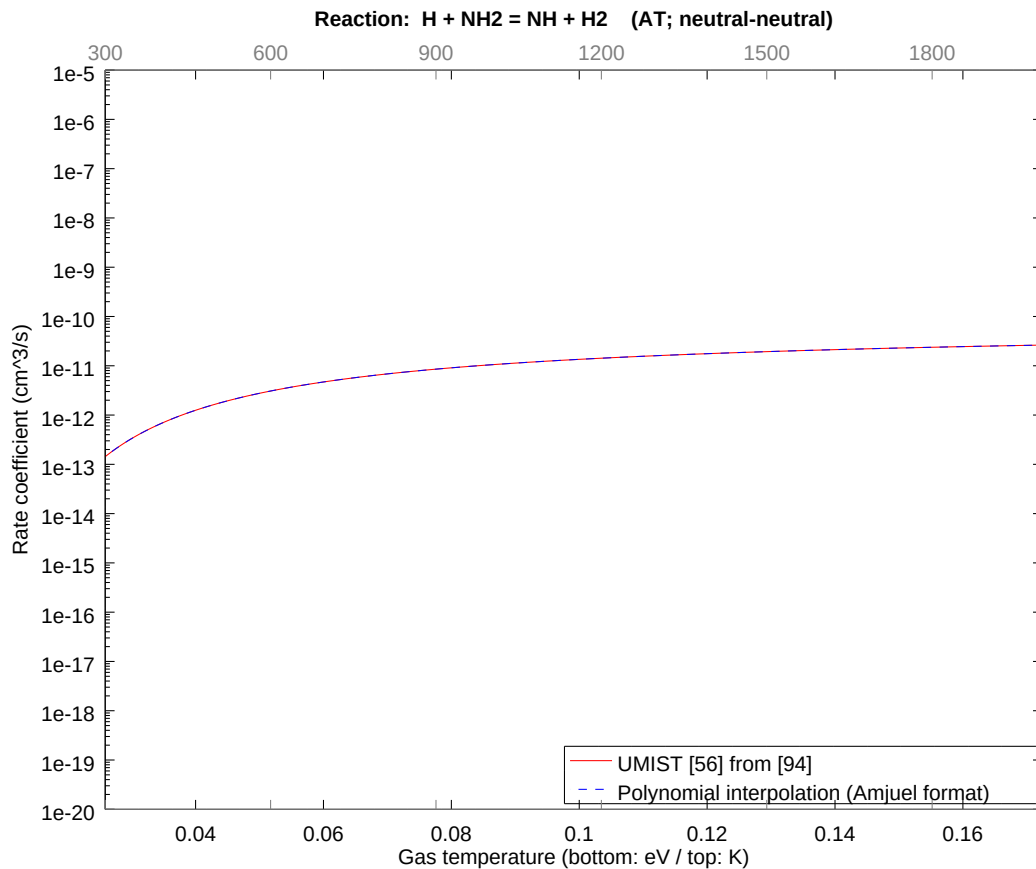
2.105 Reaction AS H2 + NH = NH2 + H

b0	-2.433392660574E+01	b1	2.541481071904E-01	b2	-9.721048352481E-01
b3	-4.463581308227E-01	b4	-3.363606022624E-01	b5	-1.050109343825E-01
b6	-2.632954285966E-02	b7	-3.352507521719E-03	b8	-2.493719857357E-04



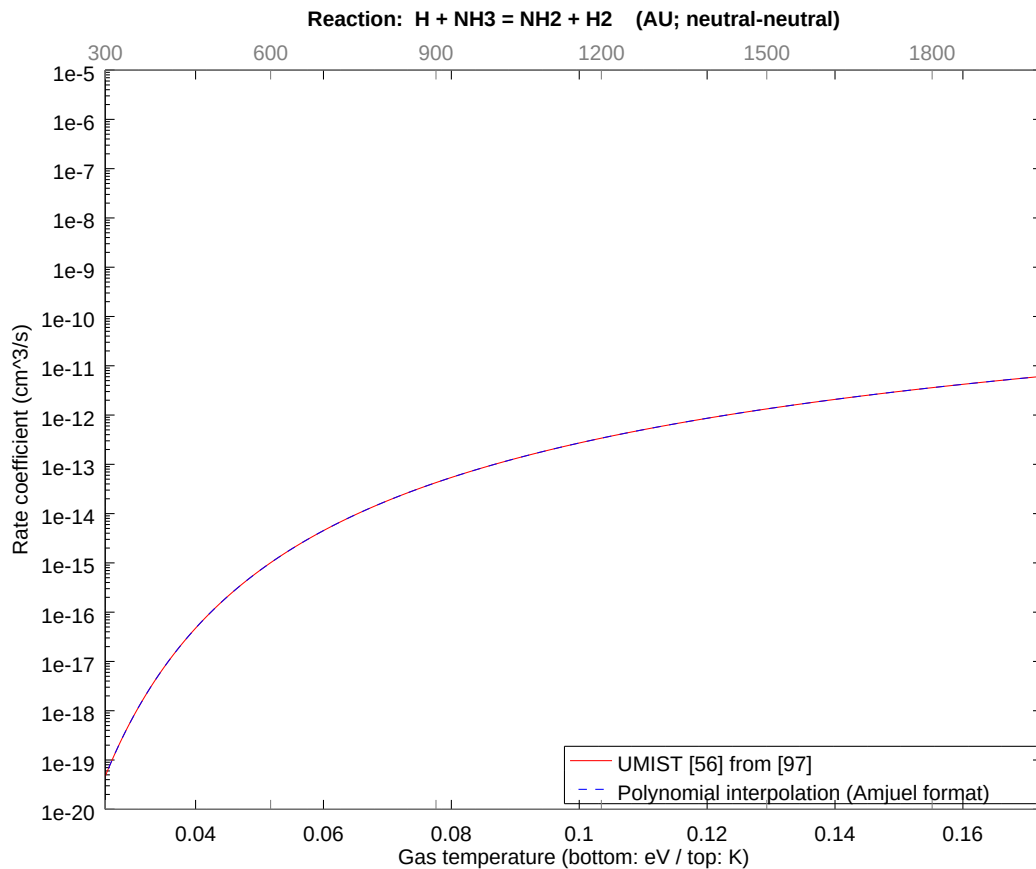
2.106 Reaction AT $\text{H} + \text{NH}_2 = \text{NH} + \text{H}_2$

b0	-2.362828906347E+01	b1	6.009155469604E-02	b2	-2.298474618058E-01
b3	-1.055382933279E-01	b4	-7.953013693836E-02	b5	-2.482910875909E-02
b6	-6.225438162482E-03	b7	-7.926771952545E-04	b8	-5.896227934612E-05



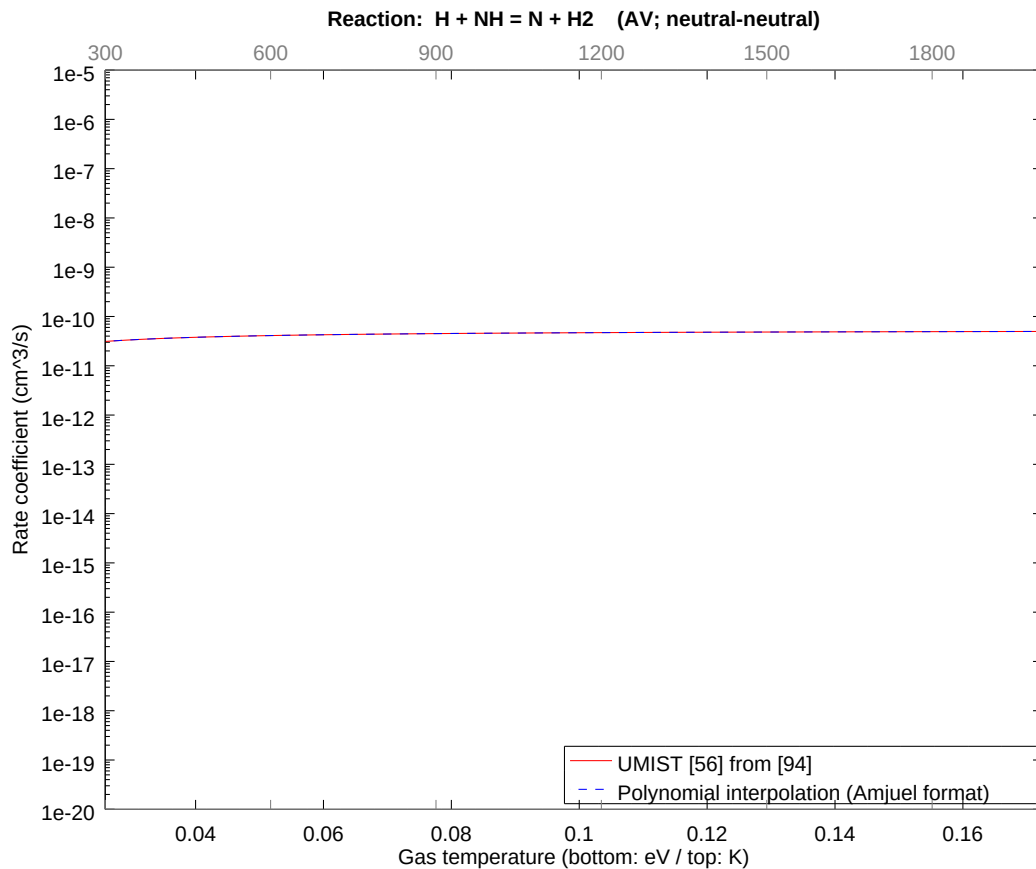
2.107 Reaction AU $\text{H} + \text{NH}_3 = \text{NH}_2 + \text{H}_2$

b0	-1.961475355238E+01	b1	2.562965700633E+00	b2	-6.233363002700E-01
b3	-2.862152455775E-01	b4	-2.156822654030E-01	b5	-6.733546099889E-02
b6	-1.688311717638E-02	b7	-2.149706042274E-03	b8	-1.599031358719E-04



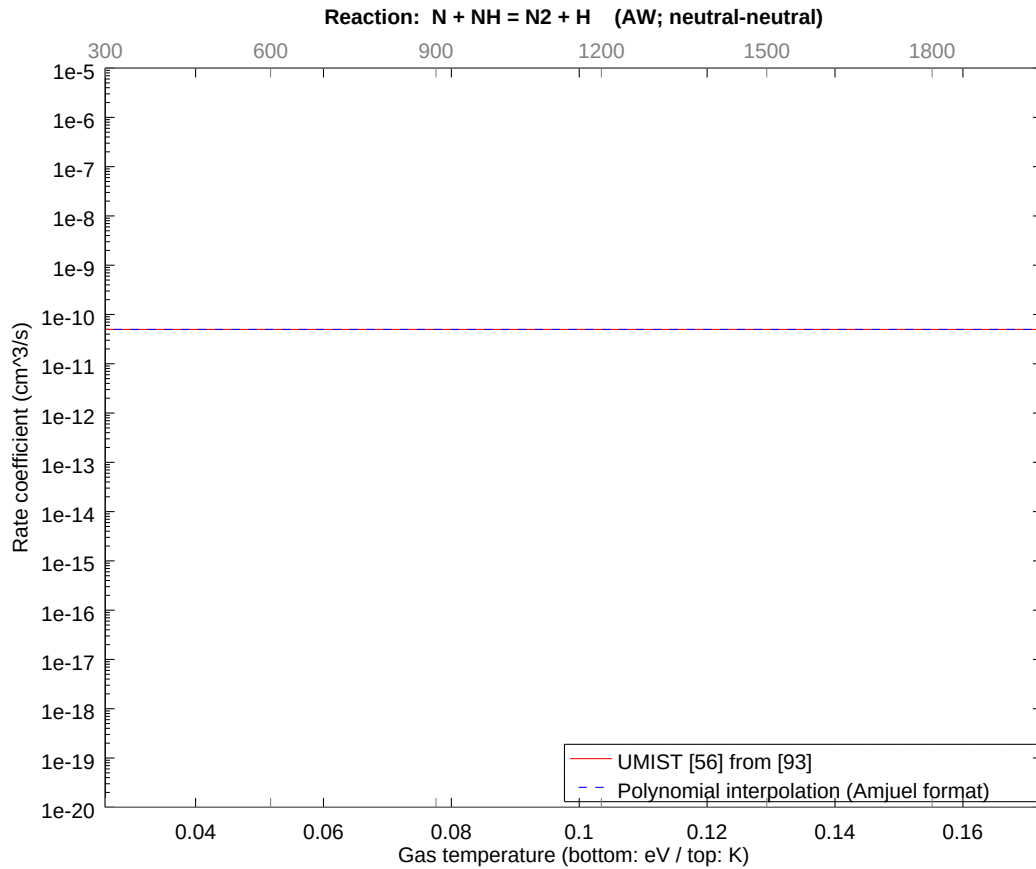
2.108 Reaction AV $\text{H} + \text{NH} = \text{N} + \text{H}_2$

b0	-2.365879916009E+01	b1	5.388639604441E-03	b2	-2.061132826434E-02
b3	-9.464036959063E-03	b4	-7.131781003725E-03	b5	-2.226524049453E-03
b6	-5.582595290652E-04	b7	-7.108247905394E-05	b8	-5.287378895012E-06



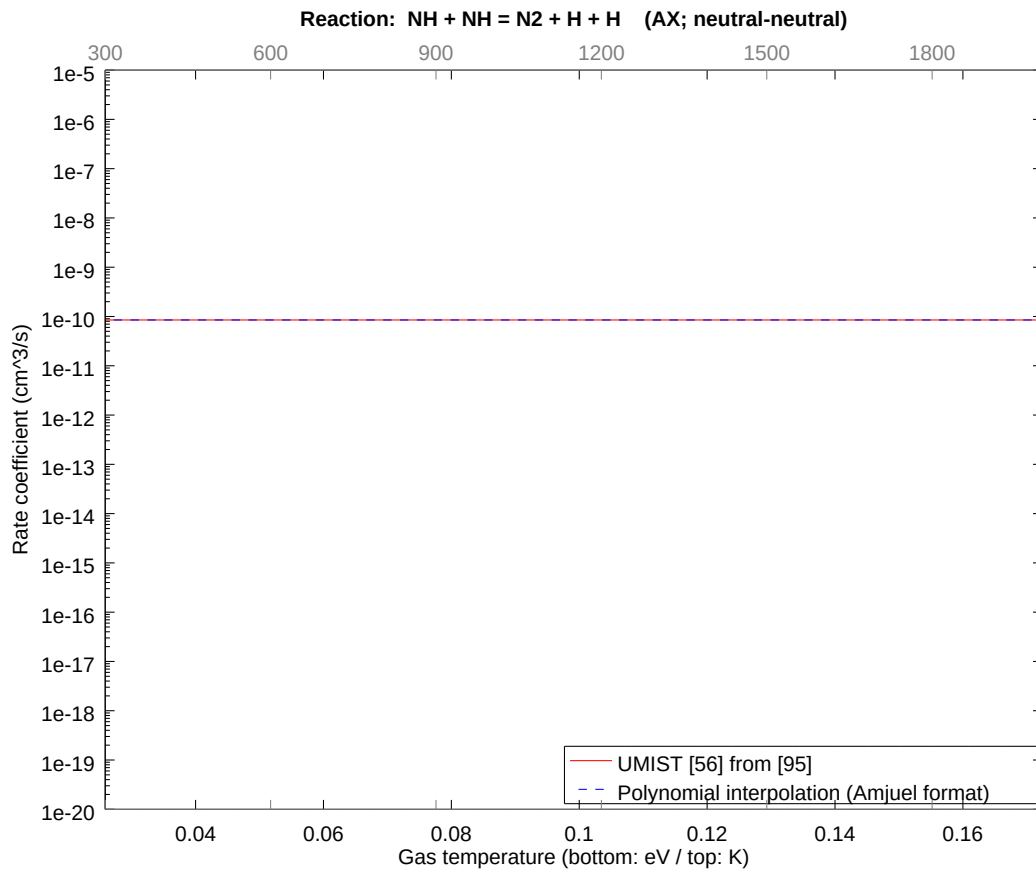
2.109 Reaction AW $\text{N} + \text{NH} = \text{N}_2 + \text{H}$

b0	-2.372300613190E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



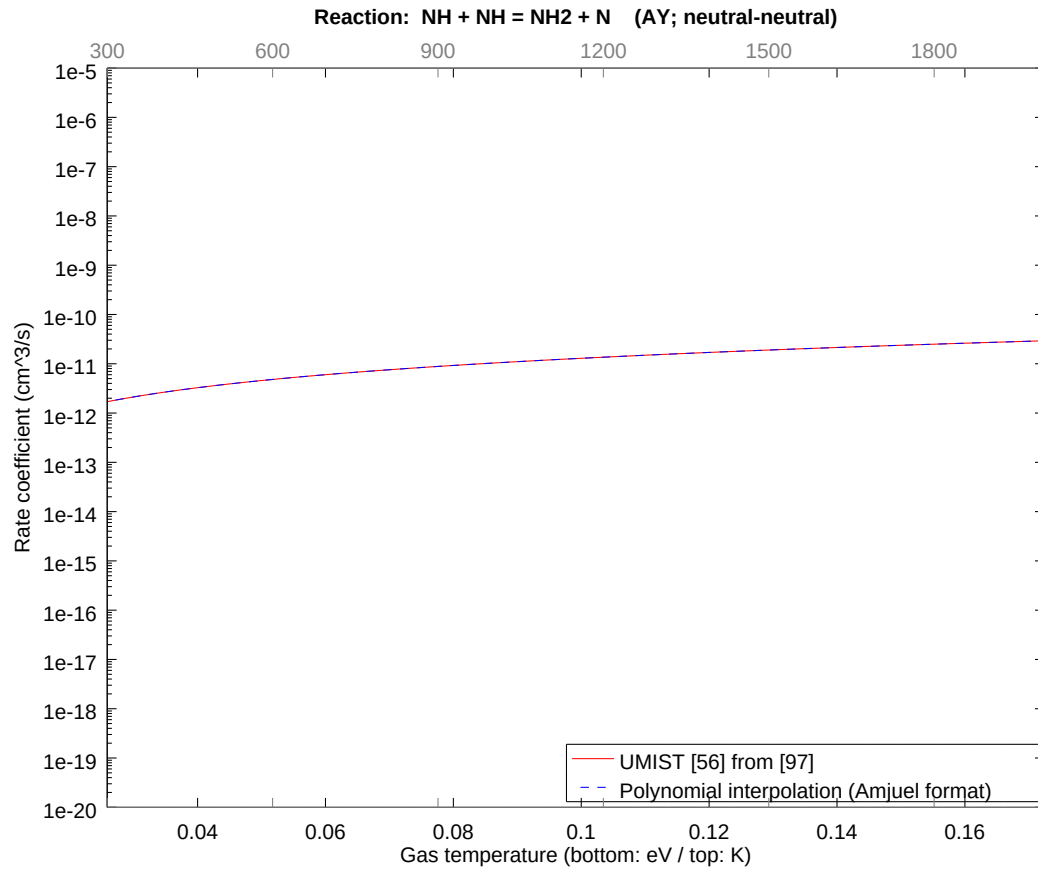
2.110 Reaction AX $\text{NH} + \text{NH} = \text{N}_2 + \text{H} + \text{H}$

b0	-2.318836985944E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



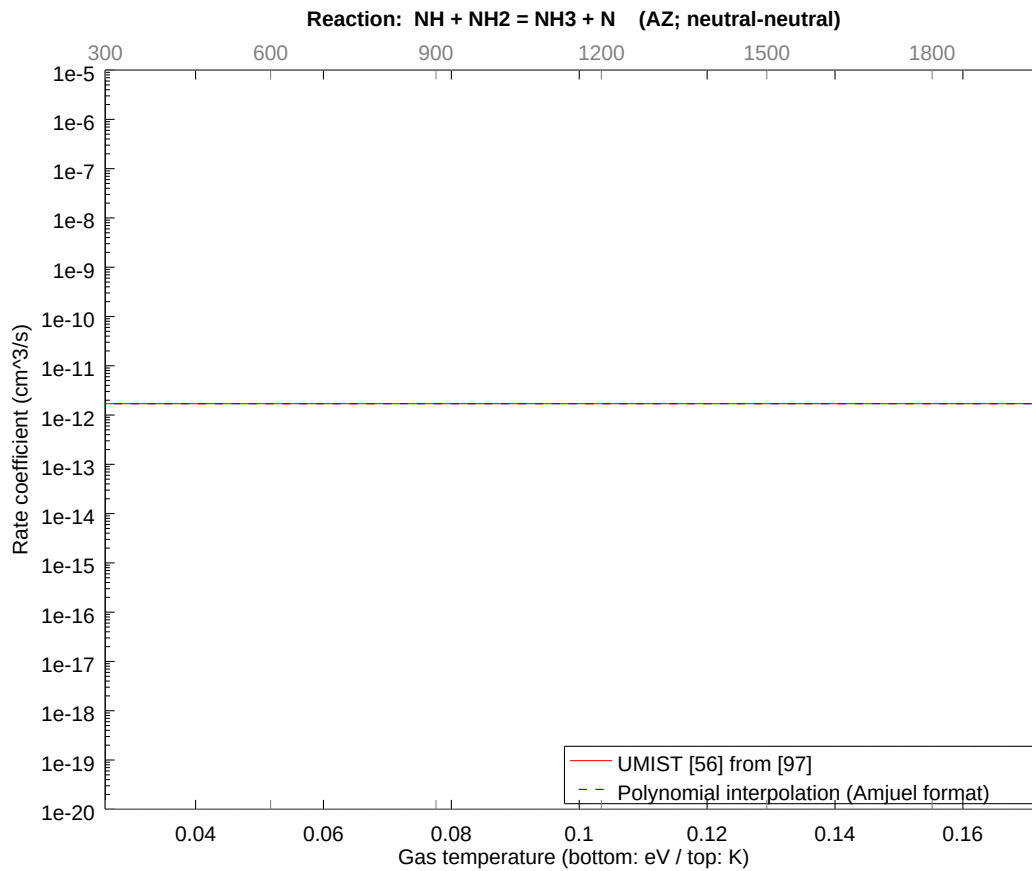
2.111 Reaction AY NH + NH = NH2 + N

b0	-2.161810841353E+01	b1	1.499999995506E+00	b2	-6.169272393228E-09
b3	-4.782398095465E-09	b4	-2.289625742652E-09	b5	-6.932775712898E-10
b6	-1.296647471885E-10	b7	-1.369812742491E-11	b8	-6.259358332704E-13



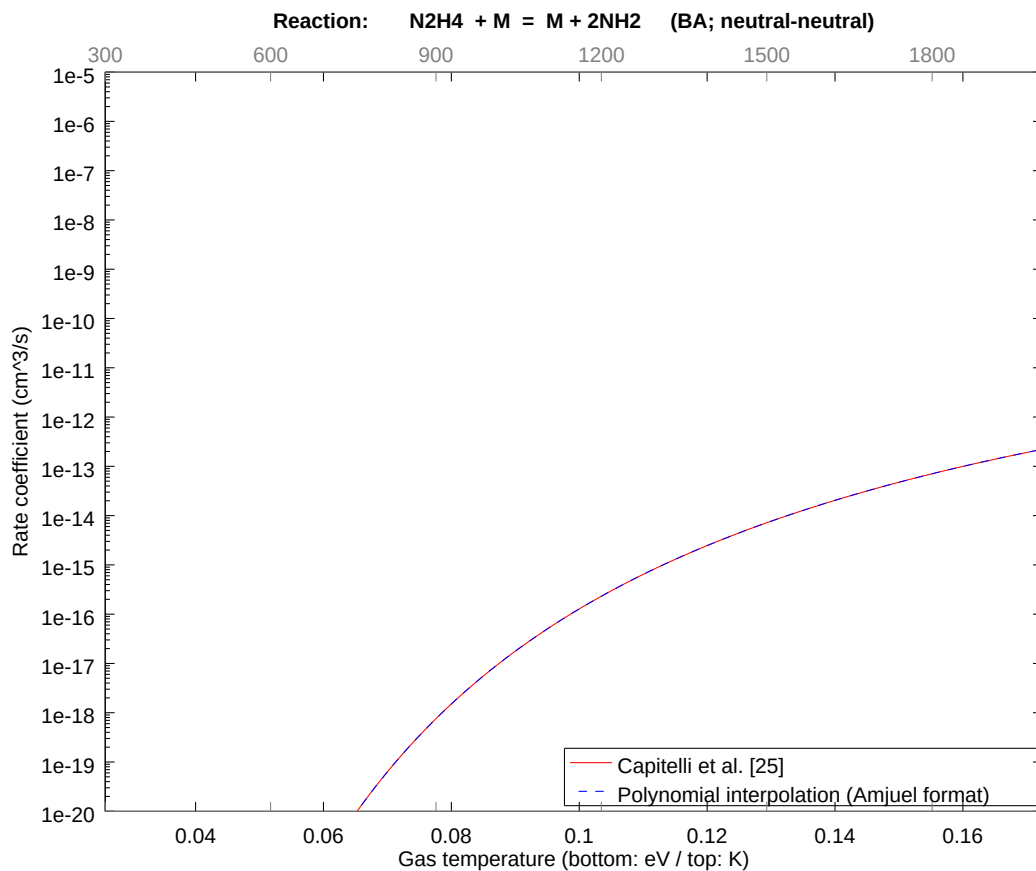
2.112 Reaction AZ $\text{NH} + \text{NH}_2 = \text{NH}_3 + \text{N}$

b0	-2.710039286487E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



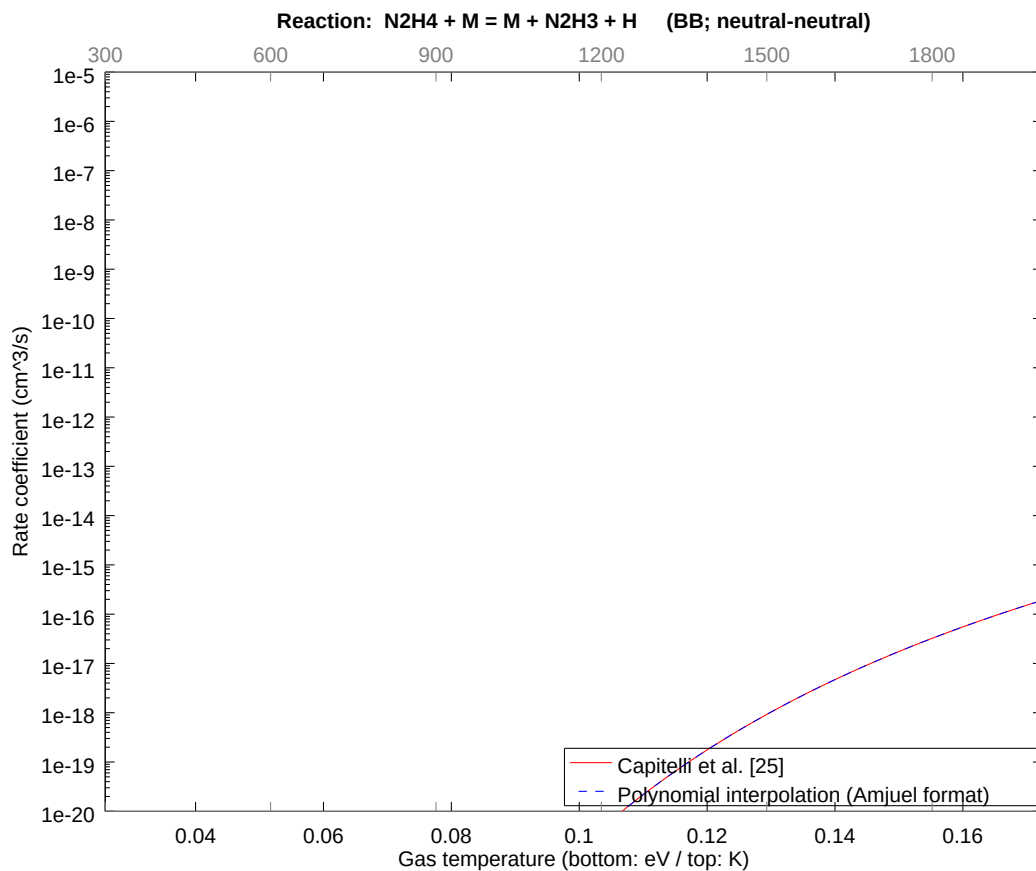
2.113 Reaction BA $\text{N}_2\text{H}_4 + \text{M} = \text{M} + 2\text{NH}_2$

b0	-2.092891758081E+01	b1	6.727642154328E-01	b2	-2.573292141534E+00
b3	-1.181569952382E+00	b4	-8.903917175497E-01	b5	-2.779780557347E-01
b6	-6.969783852452E-02	b7	-8.874537981438E-03	b8	-6.601211627768E-04



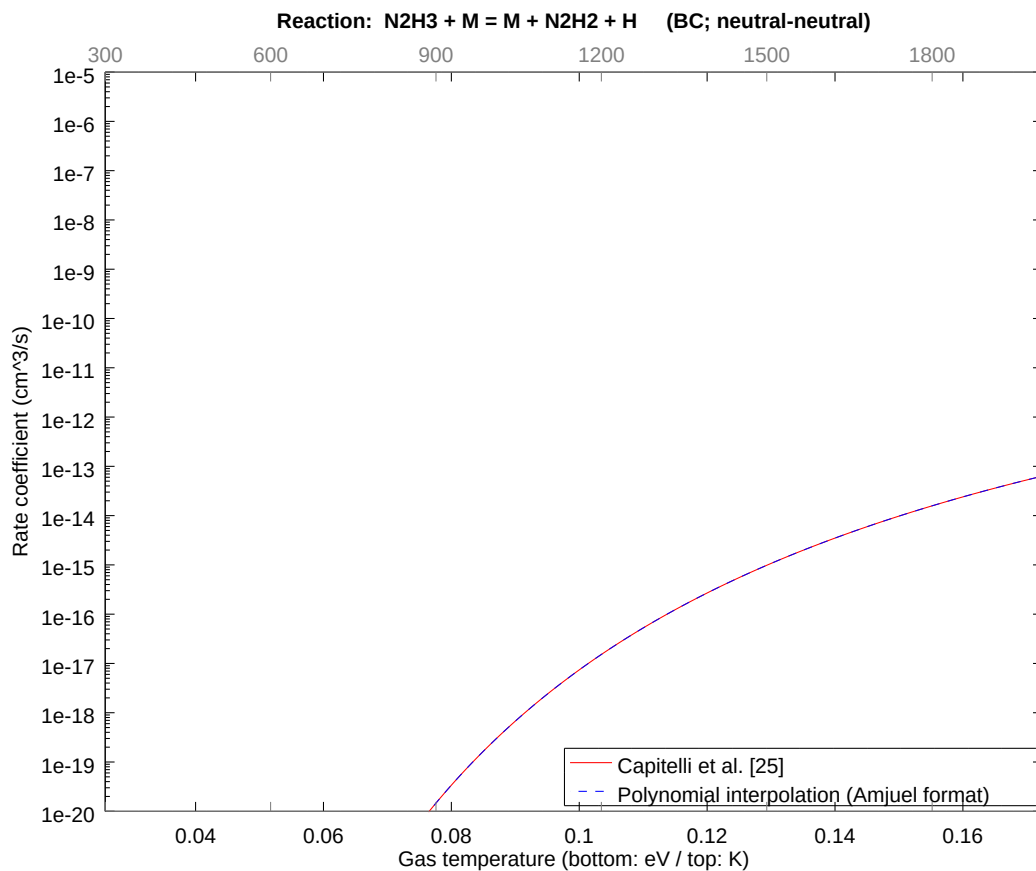
2.114 Reaction BB N2H4 + M = M + N2H3 + H

b0	-2.344346693000E+01	b1	1.045070638427E+00	b2	-3.997346999314E+00
b3	-1.835448457155E+00	b4	-1.383132759255E+00	b5	-4.318105702736E-01
b6	-1.082684867080E-01	b7	-1.378569003483E-02	b8	-1.025430930953E-03



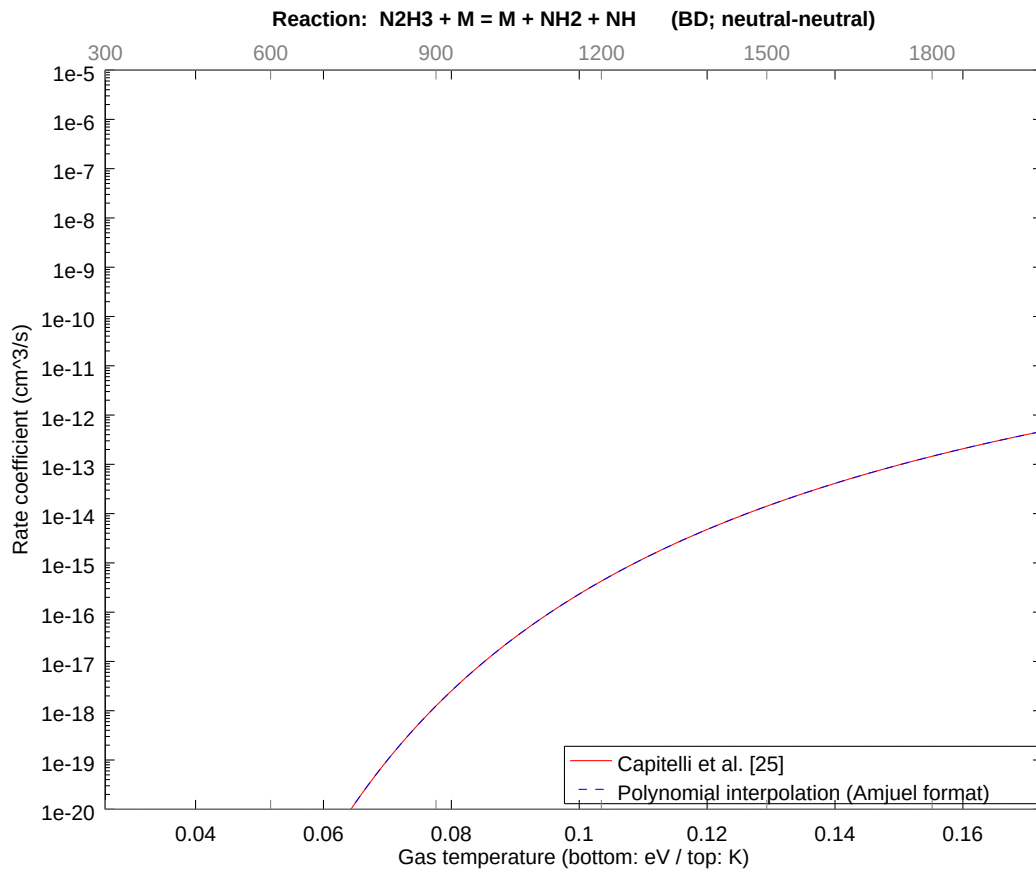
2.115 Reaction BC $\text{N}_2\text{H}_3 + \text{M} = \text{M} + \text{N}_2\text{H}_2 + \text{H}$

b0	-2.042976292036E+01	b1	8.164614287281E-01	b2	-3.122927353608E+00
b3	-1.433944115263E+00	b4	-1.080572472087E+00	b5	-3.373520092273E-01
b6	-8.458475546858E-02	b7	-1.077007036420E-02	b8	-8.011179159485E-04



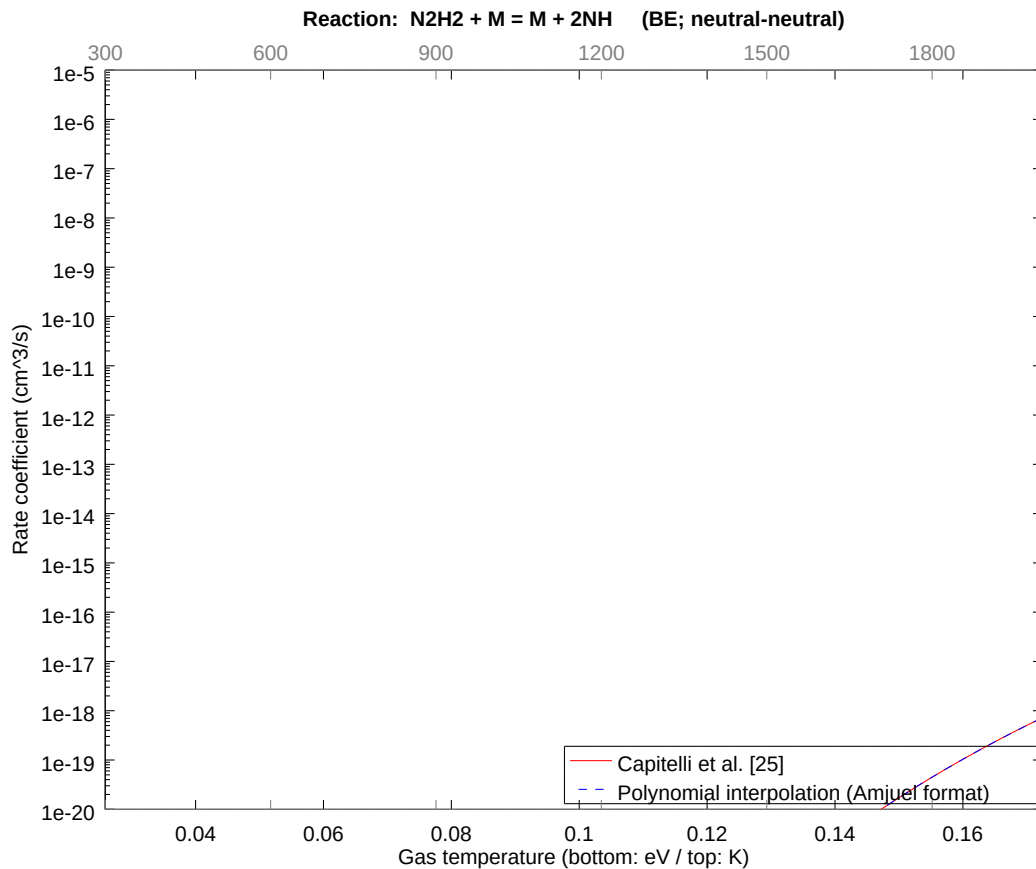
2.116 Reaction BD $\text{N}_2\text{H}_3 + \text{M} = \text{M} + \text{NH}_2 + \text{NH}$

b0	-2.002340925171E+01	b1	6.858276010790E-01	b2	-2.623258975556E+00
b3	-1.204513055552E+00	b4	-9.076808758901E-01	b5	-2.833756875350E-01
b6	-7.105119455069E-02	b7	-9.046859101163E-03	b8	-6.729390491700E-04



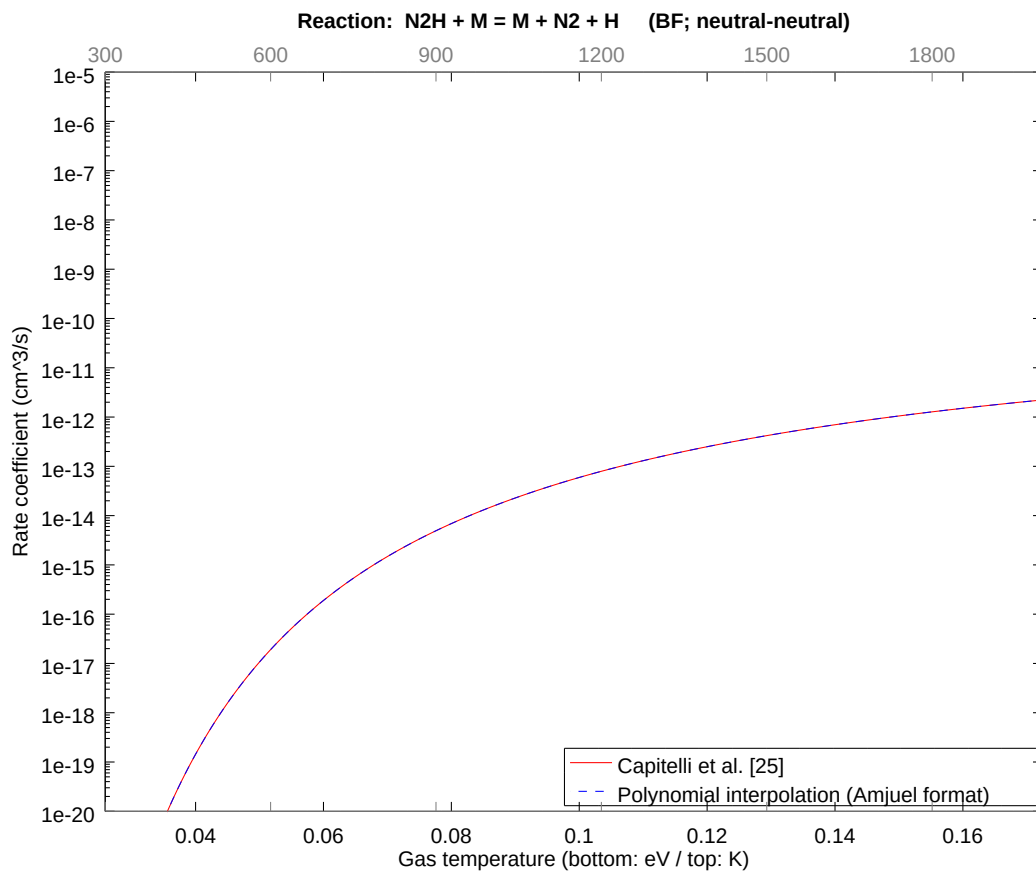
2.117 Reaction BE $\text{N}_2\text{H}_2 + \text{M} = \text{M} + 2\text{NH}$

b0	-2.185144297209E+01	b1	1.632922861522E+00	b2	-6.245854701319E+00
b3	-2.867888225723E+00	b4	-2.161144941771E+00	b5	-6.747040176978E-01
b6	-1.691695107906E-01	b7	-2.154014071243E-02	b8	-1.602235831147E-03



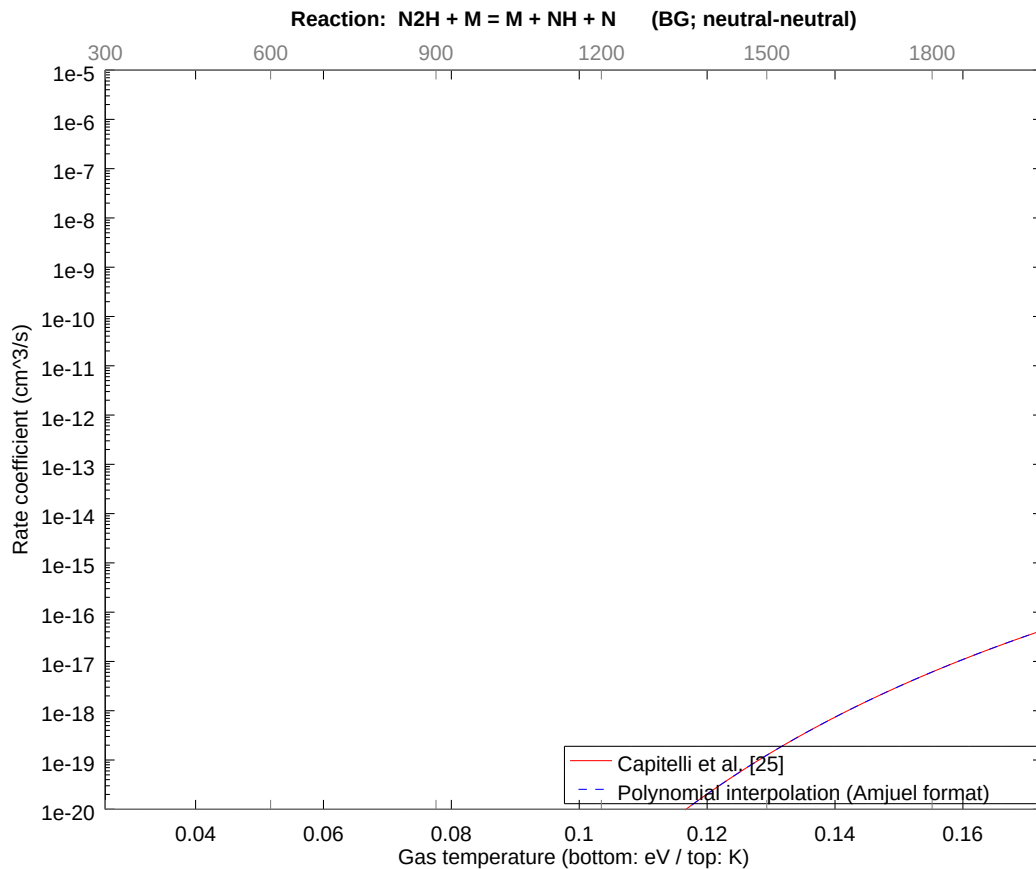
2.118 Reaction BF N2H + M = M + N2 + H

b0	-2.284781263330E+01	b1	3.265845688574E-01	b2	-1.249170944978E+00
b3	-5.735776487824E-01	b4	-4.322289900869E-01	b5	-1.349408040612E-01
b6	-3.383390225510E-02	b7	-4.308028152669E-03	b8	-3.204471666919E-04



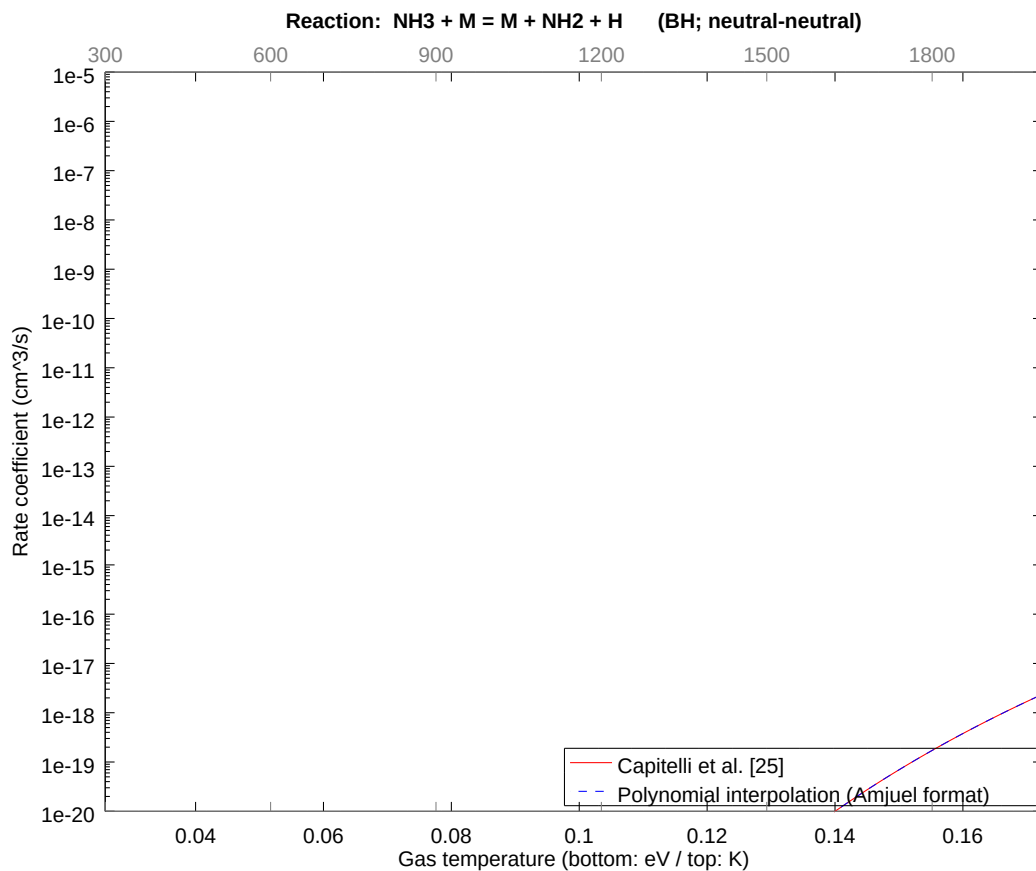
2.119 Reaction BG $\text{N}_2\text{H} + \text{M} = \text{M} + \text{NH} + \text{N}$

b0	-2.374823218580E+01	b1	1.143045995553E+00	b2	-4.372098301507E+00
b3	-2.007521766427E+00	b4	-1.512801463379E+00	b5	-4.722928136757E-01
b6	-1.184186578008E-01	b7	-1.507809852559E-02	b8	-1.121565083068E-03



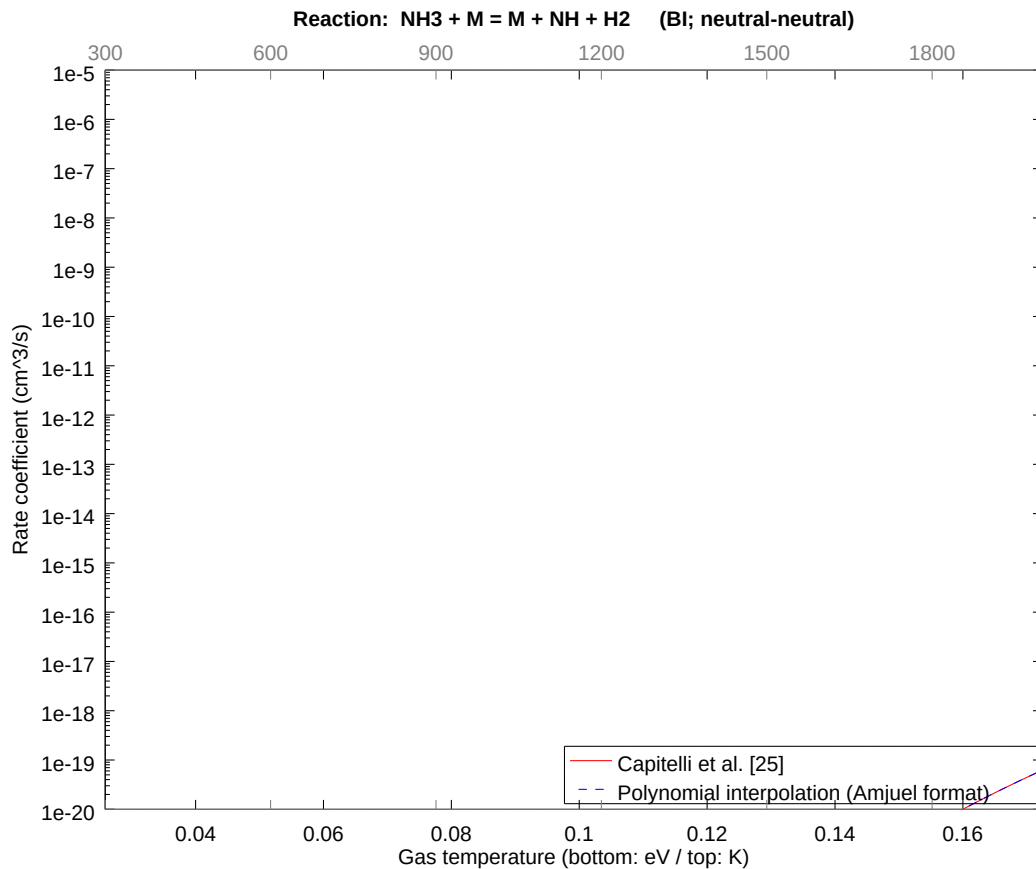
2.120 Reaction BH $\text{NH}_3 + \text{M} = \text{M} + \text{NH}_2 + \text{H}$

b0	-2.178056950826E+01	b1	1.541479169864E+00	b2	-5.896086853409E+00
b3	-2.707286496789E+00	b4	-2.040120830556E+00	b5	-6.369205943611E-01
b6	-1.596960184935E-01	b7	-2.033389286489E-02	b8	-1.512510626084E-03



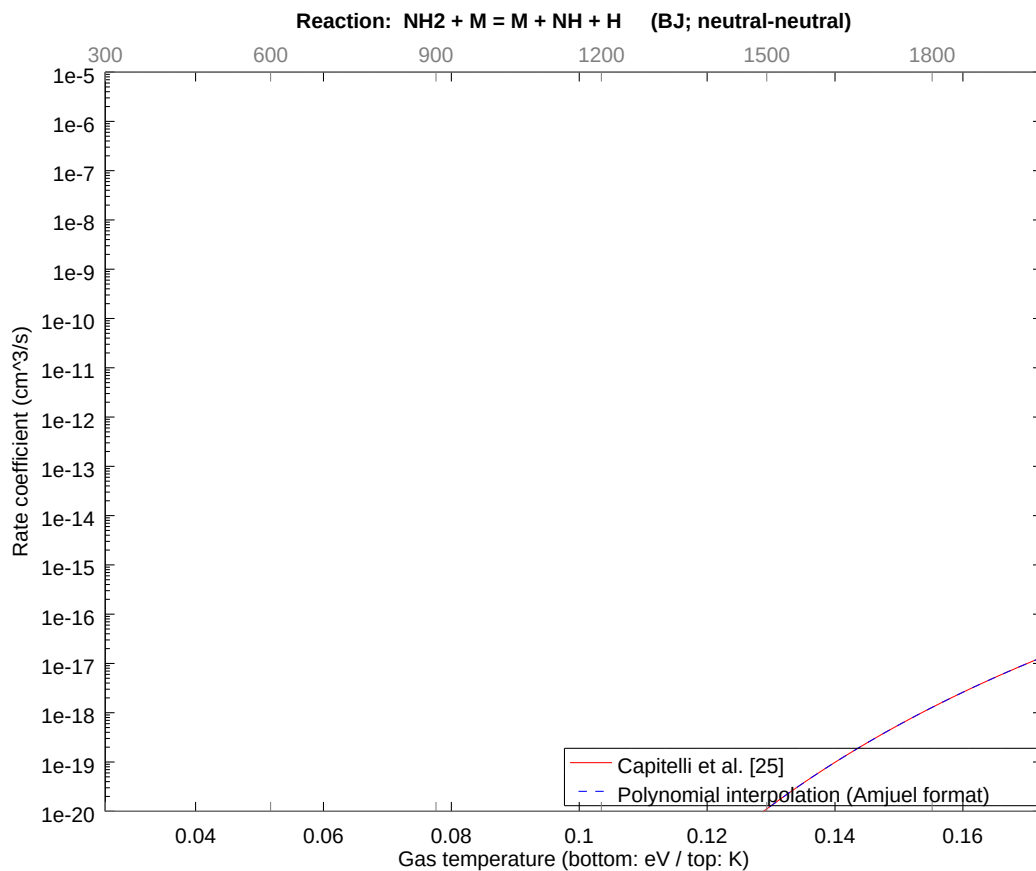
2.121 Reaction BI $\text{NH}_3 + \text{M} = \text{M} + \text{NH} + \text{H}_2$

b0	-2.549792143959E+01	b1	1.534947489548E+00	b2	-5.871103419390E+00
b3	-2.695814932102E+00	b4	-2.031476245129E+00	b5	-6.342217765633E-01
b6	-1.590193401233E-01	b7	-2.024773226688E-02	b8	-1.506101681115E-03



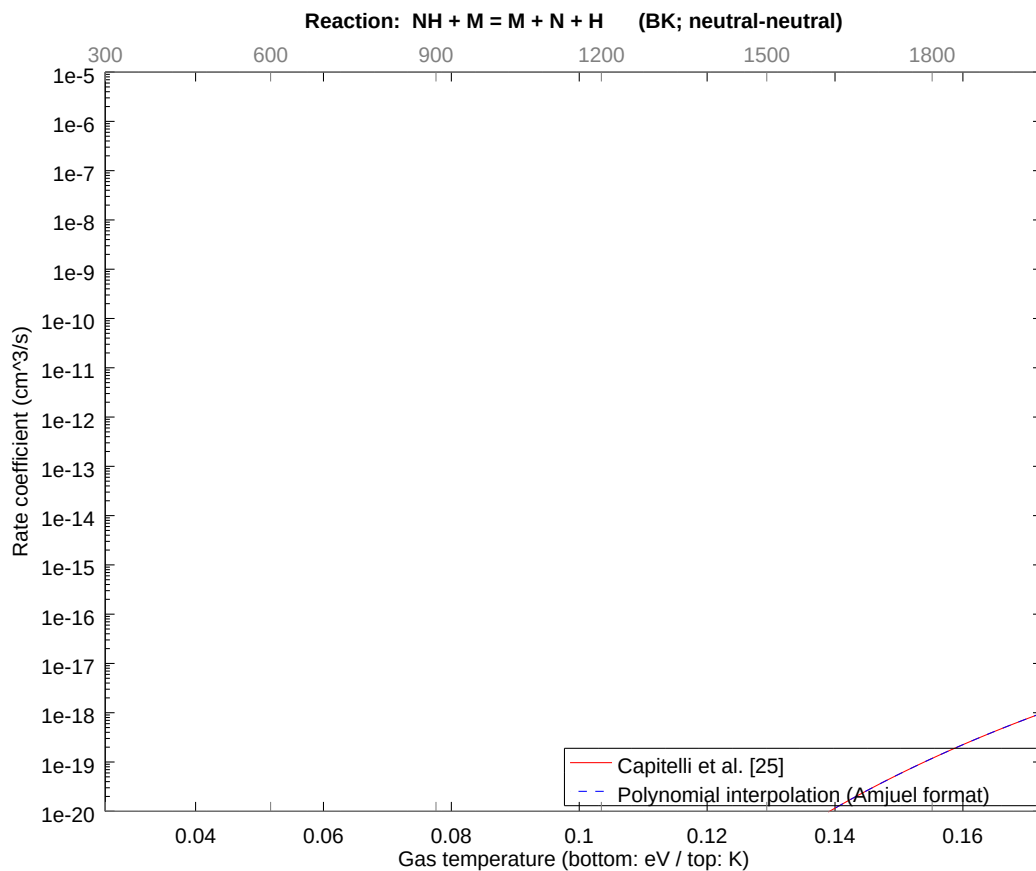
2.122 Reaction BJ $\text{NH}_2 + \text{M} = \text{M} + \text{NH} + \text{H}$

b0	-2.404043242719E+01	b1	-4.977109602235E-01	b2	-5.746186315569E+00
b3	-2.638457160327E+00	b4	-1.988253342970E+00	b5	-6.207276952482E-01
b6	-1.556359497360E-01	b7	-1.981692943540E-02	b8	-1.474056963745E-03



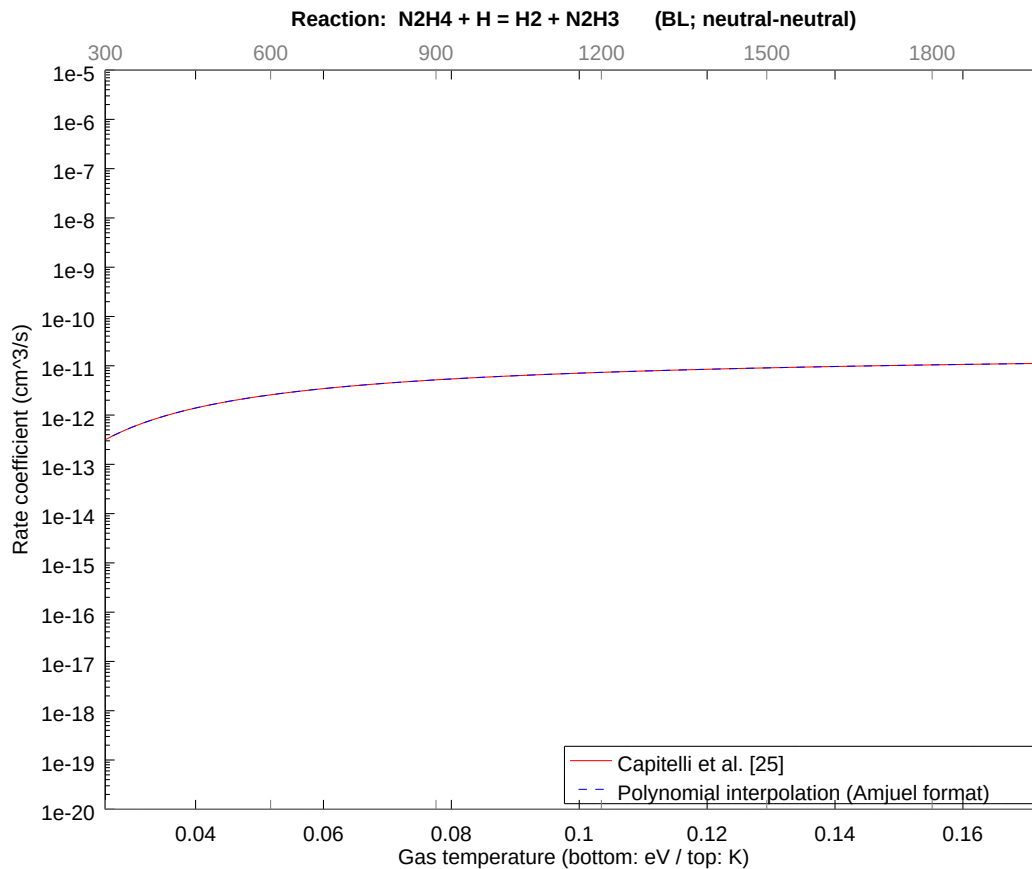
2.123 Reaction BK NH + M = M + N + H

b0	-2.823924894436E+01	b1	-6.283447873423E-01	b2	-5.246517936774E+00
b3	-2.409026100008E+00	b4	-1.815361746457E+00	b5	-5.667513734489E-01
b6	-1.421023887953E-01	b7	-1.809371816955E-02	b8	-1.345878096817E-03



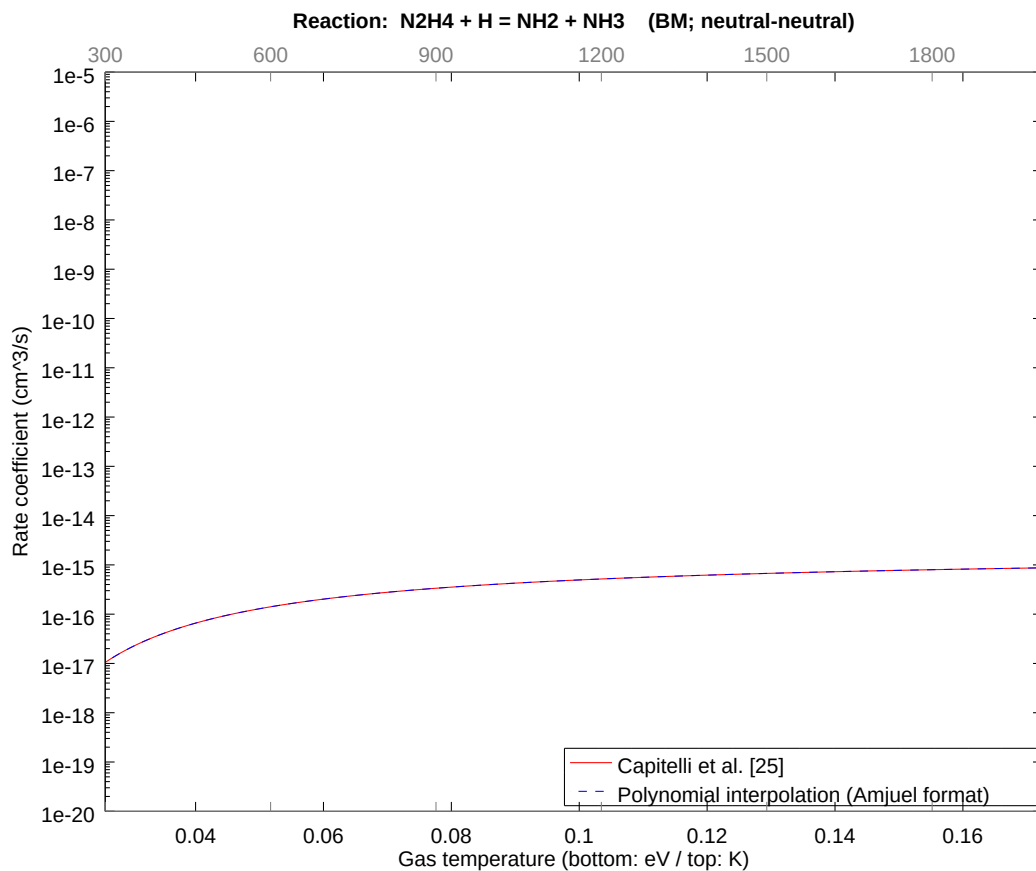
2.124 Reaction BL $\text{N}_2\text{H}_4 + \text{H} = \text{H}_2 + \text{N}_2\text{H}_3$

b0	-2.471450008403E+01	b1	4.114965527105E-02	b2	-1.573955393912E-01
b3	-7.227078380576E-02	b4	-5.446085268013E-02	b5	-1.700254125772E-02
b6	-4.263071667420E-03	b7	-5.428115447081E-04	b8	-4.037634284977E-05



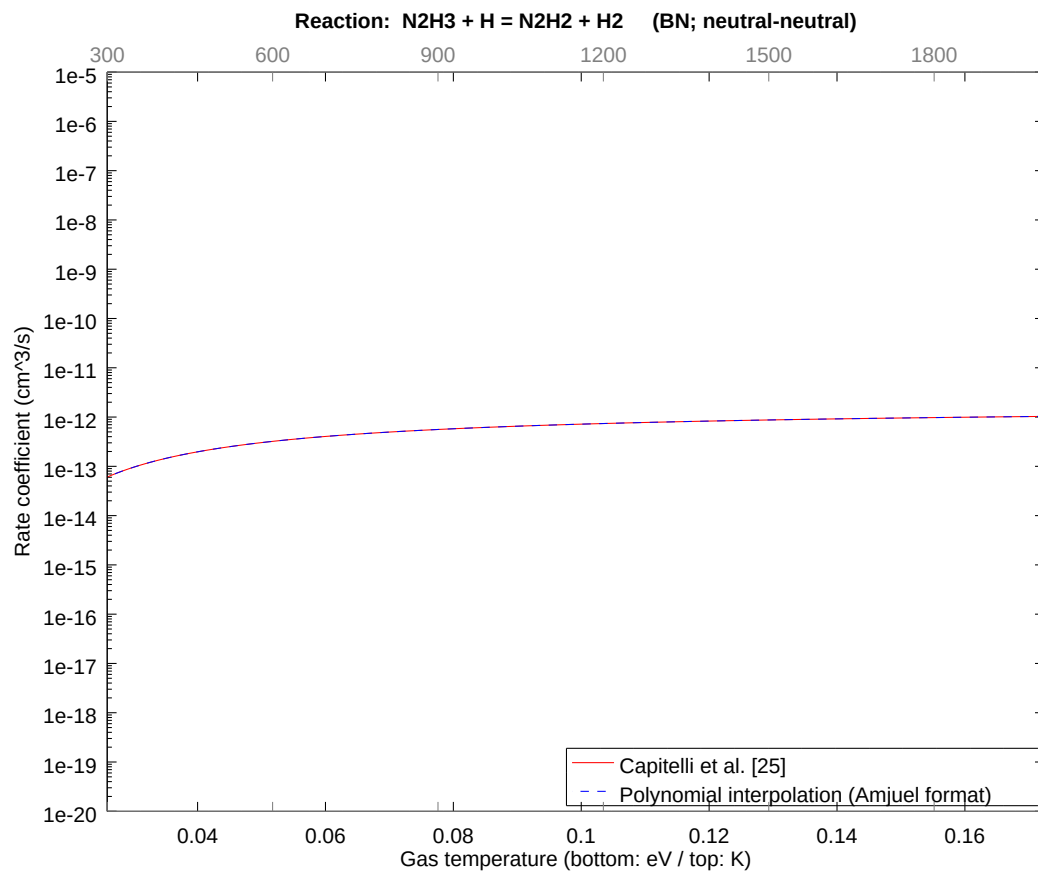
2.125 Reaction BM $\text{N}_2\text{H}_4 + \text{H} = \text{NH}_2 + \text{NH}_3$

b0	-3.405540044124E+01	b1	5.094718741288E-02	b2	-1.948706746692E-01
b3	-8.947811877476E-02	b4	-6.742772508512E-02	b5	-2.105076621879E-02
b6	-5.278088896147E-03	b7	-6.720524067615E-04	b8	-4.998975867108E-05



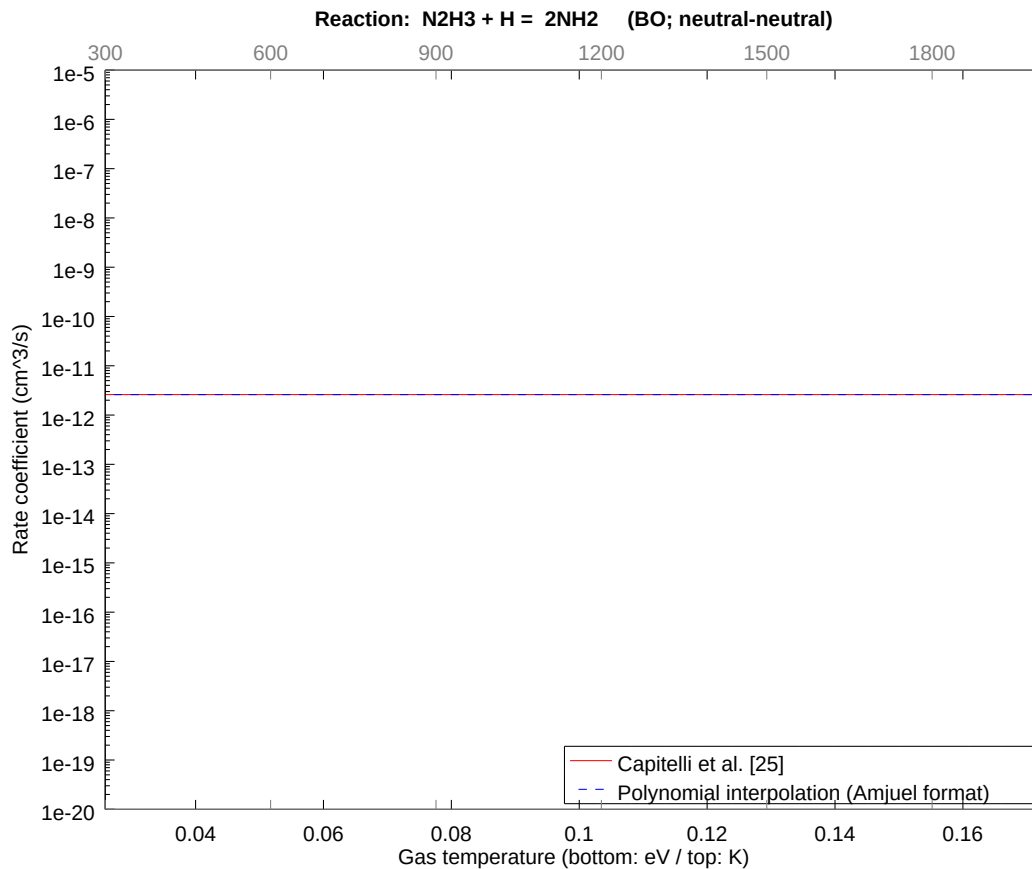
2.126 Reaction BN $\text{N}_2\text{H}_3 + \text{H} = \text{N}_2\text{H}_2 + \text{H}_2$

b0	-2.720198128343E+01	b1	3.265845275819E-02	b2	-1.249170998387E-01
b3	-5.735776876858E-02	b4	-4.322290075117E-02	b5	-1.349408089682E-02
b6	-3.383390310207E-03	b7	-4.308028234413E-04	b8	-3.204471700596E-05



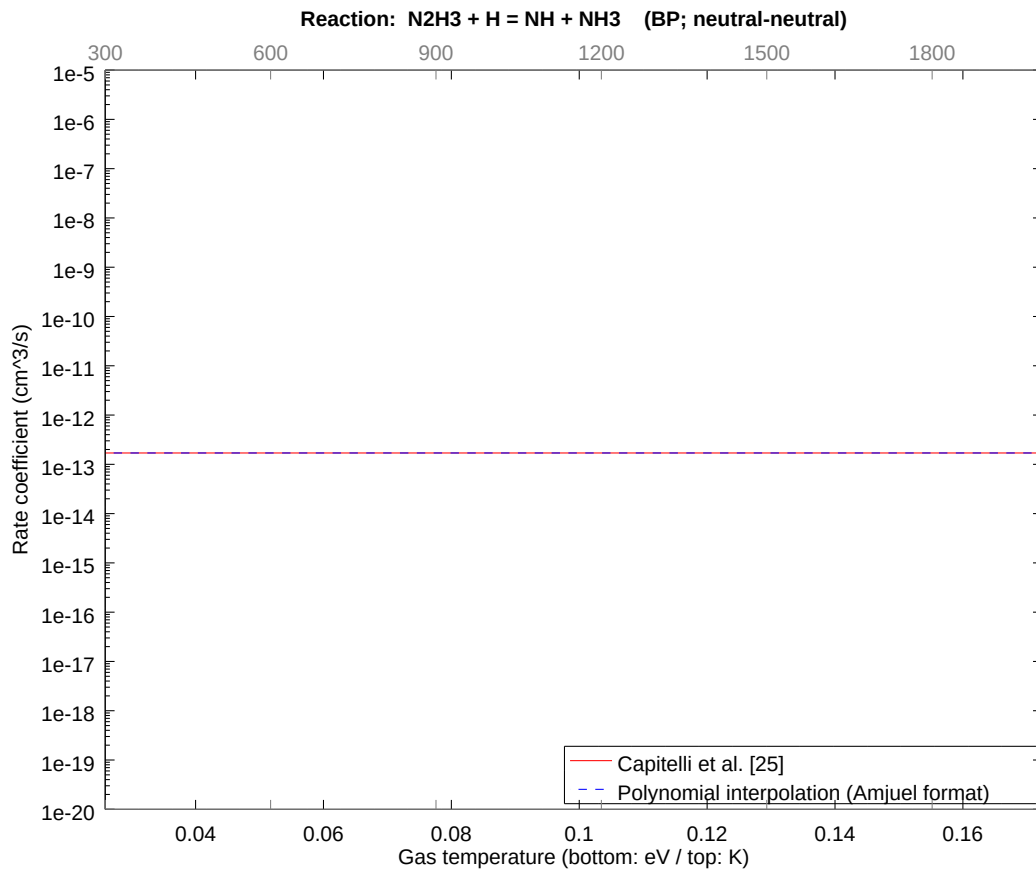
2.127 Reaction BO N2H3 + H = 2NH2

b0	-2.667550967090E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



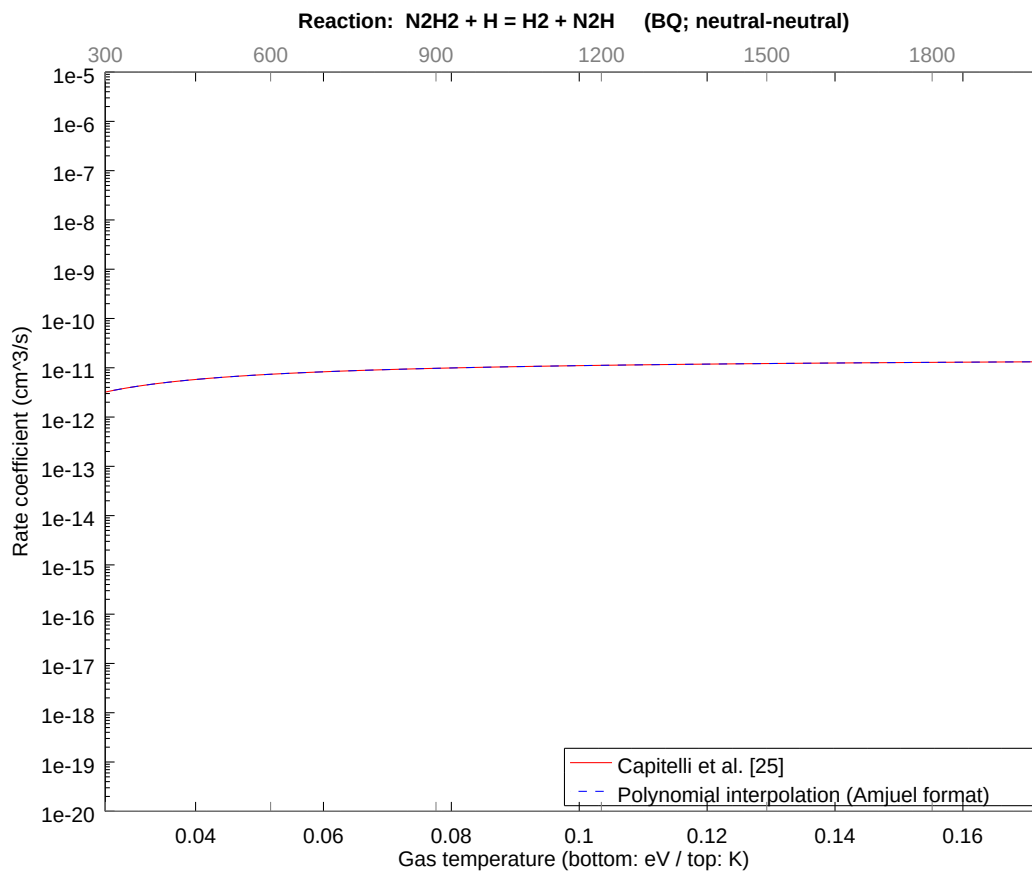
2.128 Reaction BP N2H3 + H = NH + NH3

b0	-2.940297795786E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



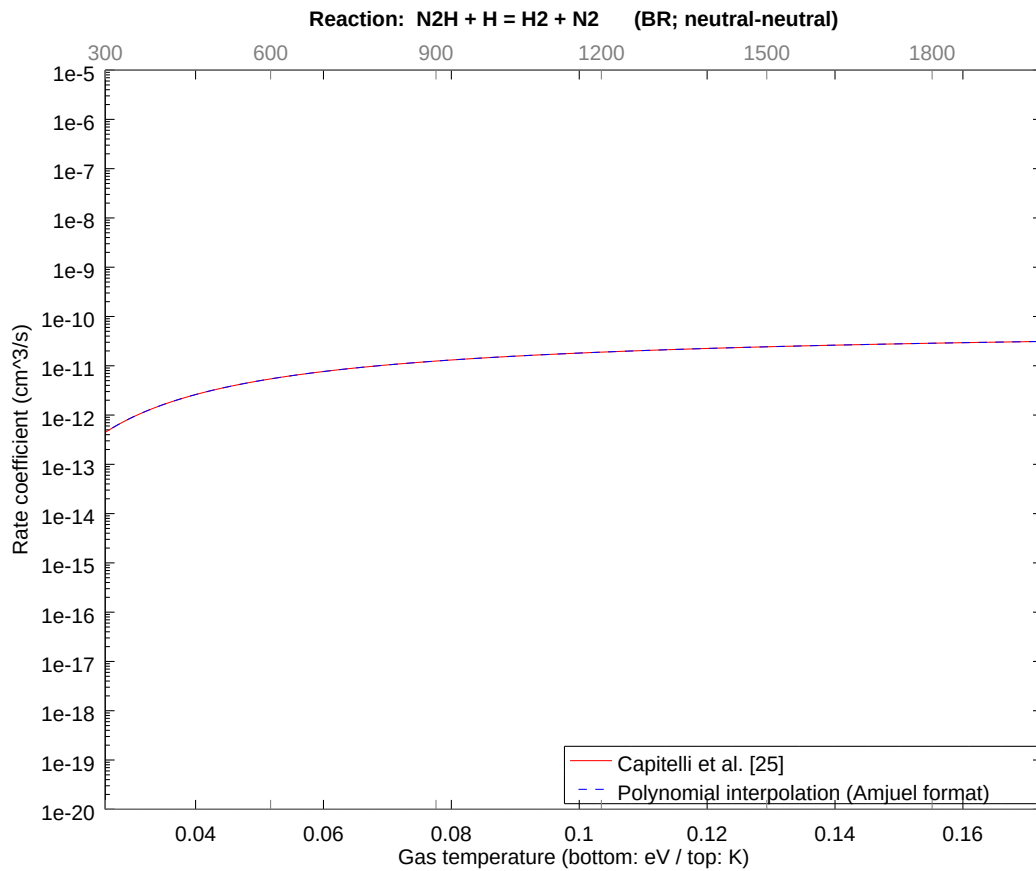
2.129 Reaction BQ N2H2 + H = H2 + N2H

b0	-2.484860198207E+01	b1	1.632922359499E-02	b2	-6.245855355615E-02
b3	-2.867888706383E-02	b4	-2.161145159392E-02	b5	-6.747040798658E-03
b6	-1.691695217315E-03	b7	-2.154014179679E-04	b8	-1.602235877477E-05



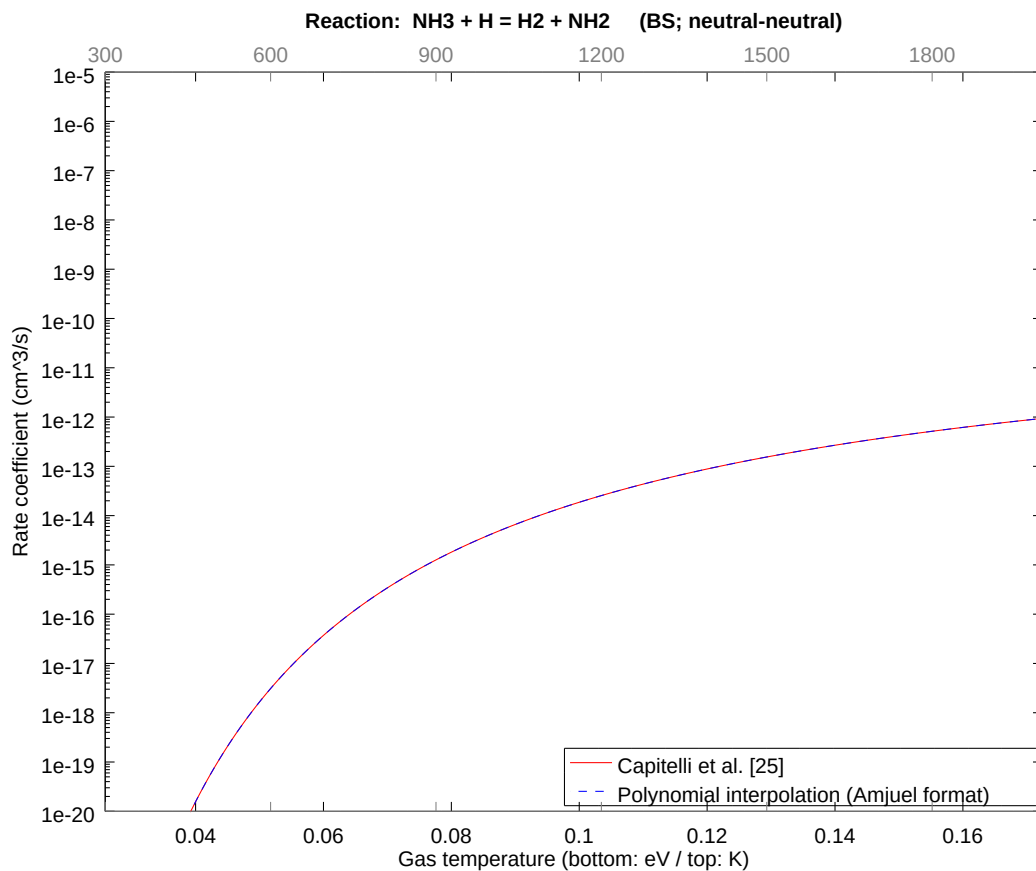
2.130 Reaction BR $\text{N}_2\text{H} + \text{H} = \text{H}_2 + \text{N}_2$

b0	-2.359374900180E+01	b1	4.898767865117E-02	b2	-1.873756508272E-01
b3	-8.603665430488E-02	b4	-6.483435184066E-02	b5	-2.024112161349E-02
b6	-5.075085525879E-03	b7	-6.462042427391E-04	b8	-4.806707591320E-05



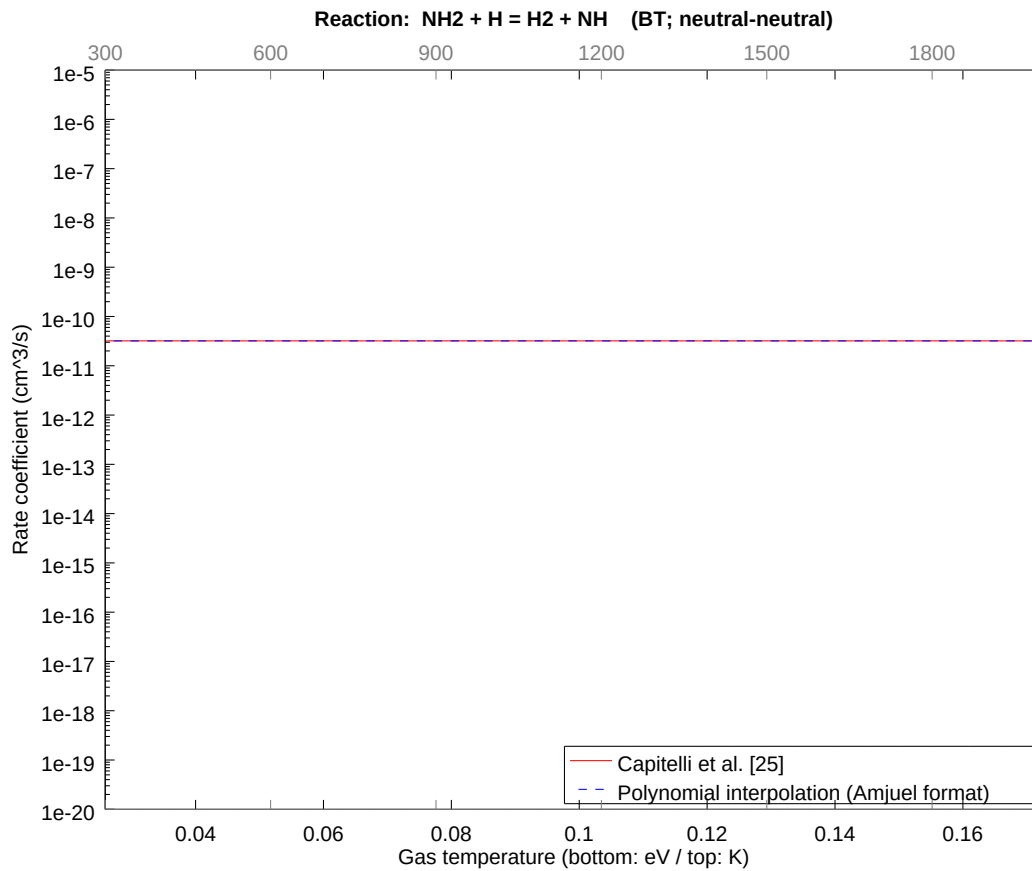
2.131 Reaction BS $\text{NH}_3 + \text{H} = \text{H}_2 + \text{NH}_2$

b0	-2.338310025915E+01	b1	3.533645035060E-01	b2	-1.351602962390E+00
b3	-6.206110158630E-01	b4	-4.676717671848E-01	b5	-1.460059499565E-01
b6	-3.660828223069E-02	b7	-4.661286459931E-03	b8	-3.467238342890E-04



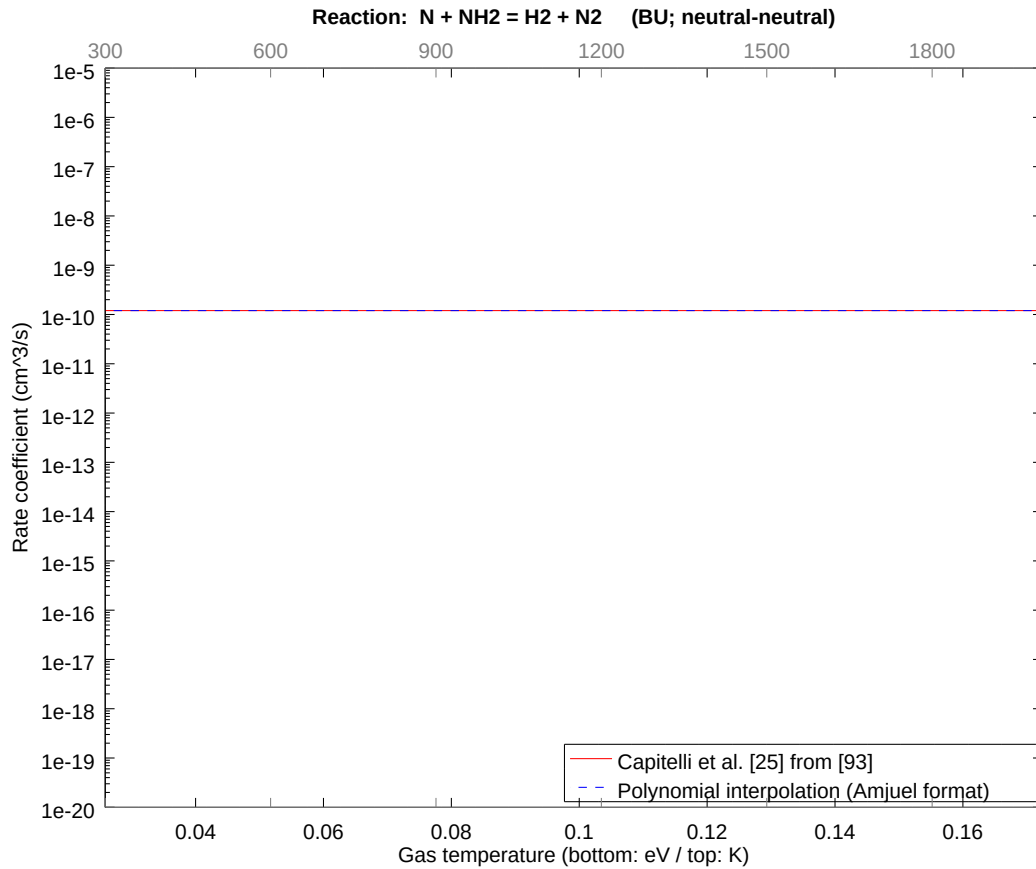
2.132 Reaction BT NH₂ + H = H₂ + NH

b0	-2.416528521313E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



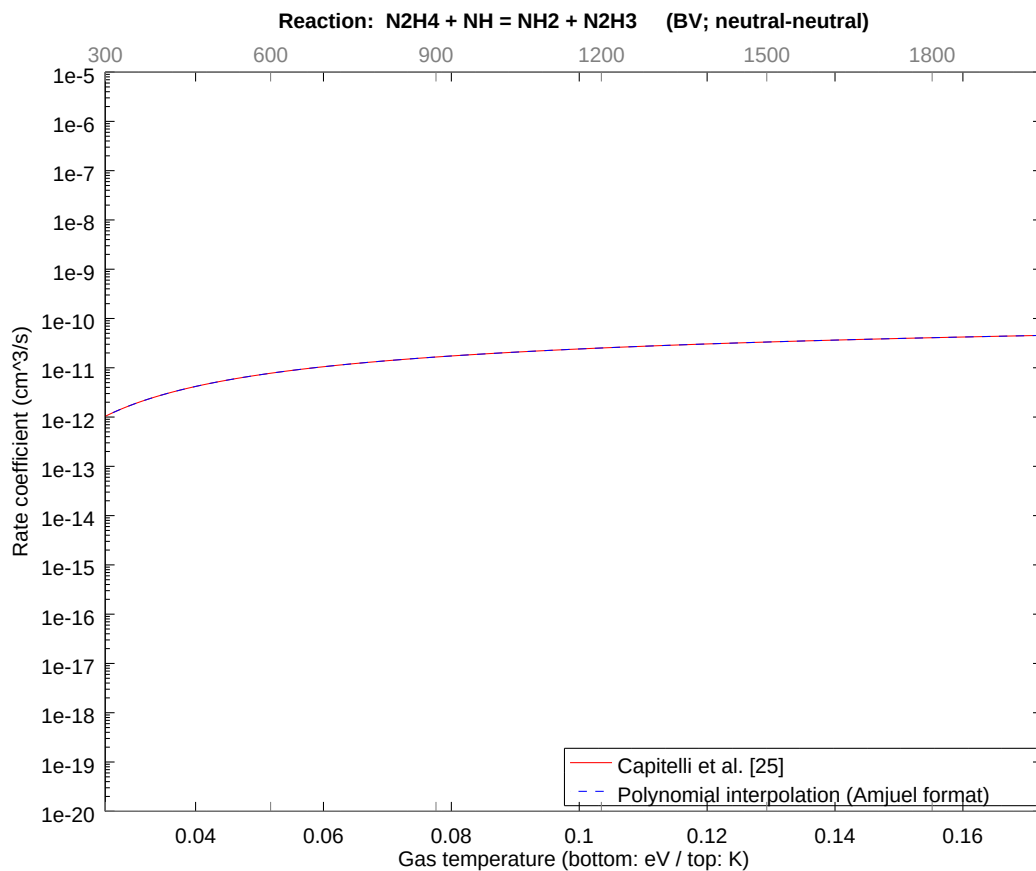
2.133 Reaction BU $\text{N} + \text{NH}_2 = \text{H}_2 + \text{N}_2$

b0	-2.284352937315E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



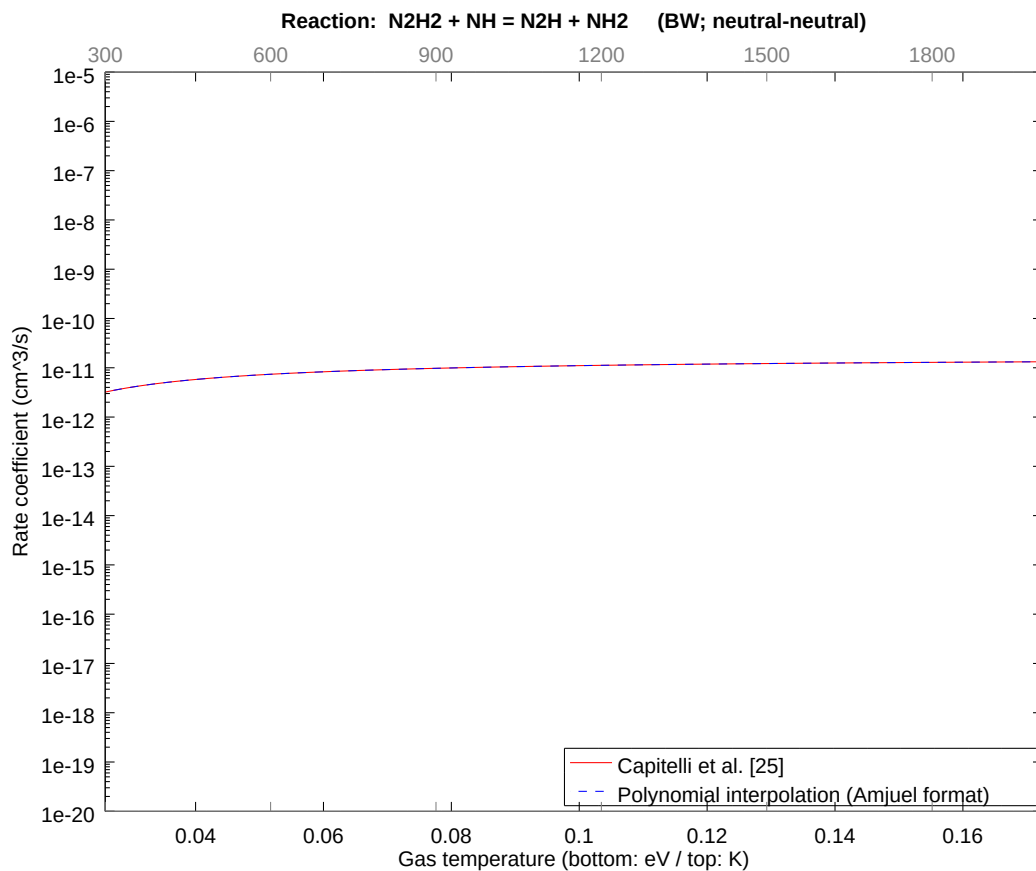
2.134 Reaction BV $\text{N}_2\text{H}_4 + \text{NH} = \text{NH}_2 + \text{N}_2\text{H}_3$

b0	-2.253788555373E+01	b1	5.326584523650E-01	b2	-1.249171003025E-01
b3	-5.735776906945E-02	b4	-4.322290086803E-02	b5	-1.349408092450E-02
b6	-3.383390314085E-03	b7	-4.308028237328E-04	b8	-3.204471701487E-05



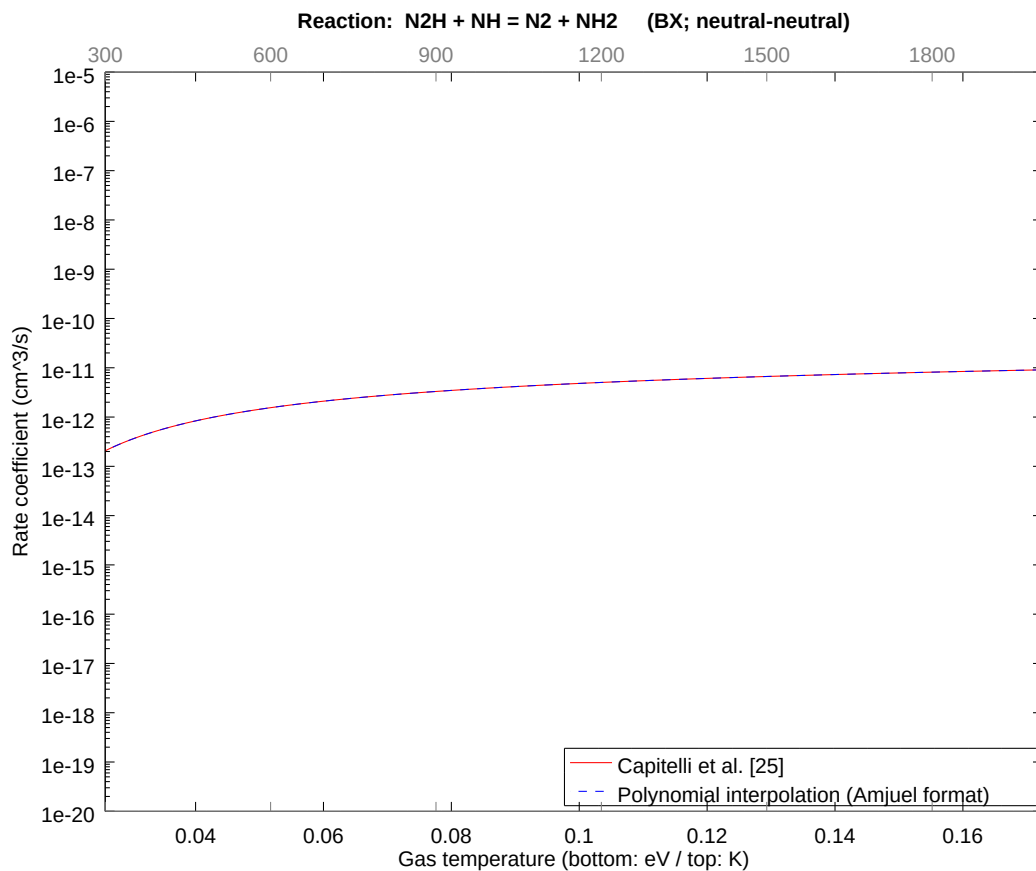
2.135 Reaction BW $\text{N}_2\text{H}_2 + \text{NH} = \text{N}_2\text{H} + \text{NH}_2$

b0	-2.484860198207E+01	b1	1.632922359499E-02	b2	-6.245855355615E-02
b3	-2.867888706383E-02	b4	-2.161145159392E-02	b5	-6.747040798658E-03
b6	-1.691695217315E-03	b7	-2.154014179679E-04	b8	-1.602235877477E-05



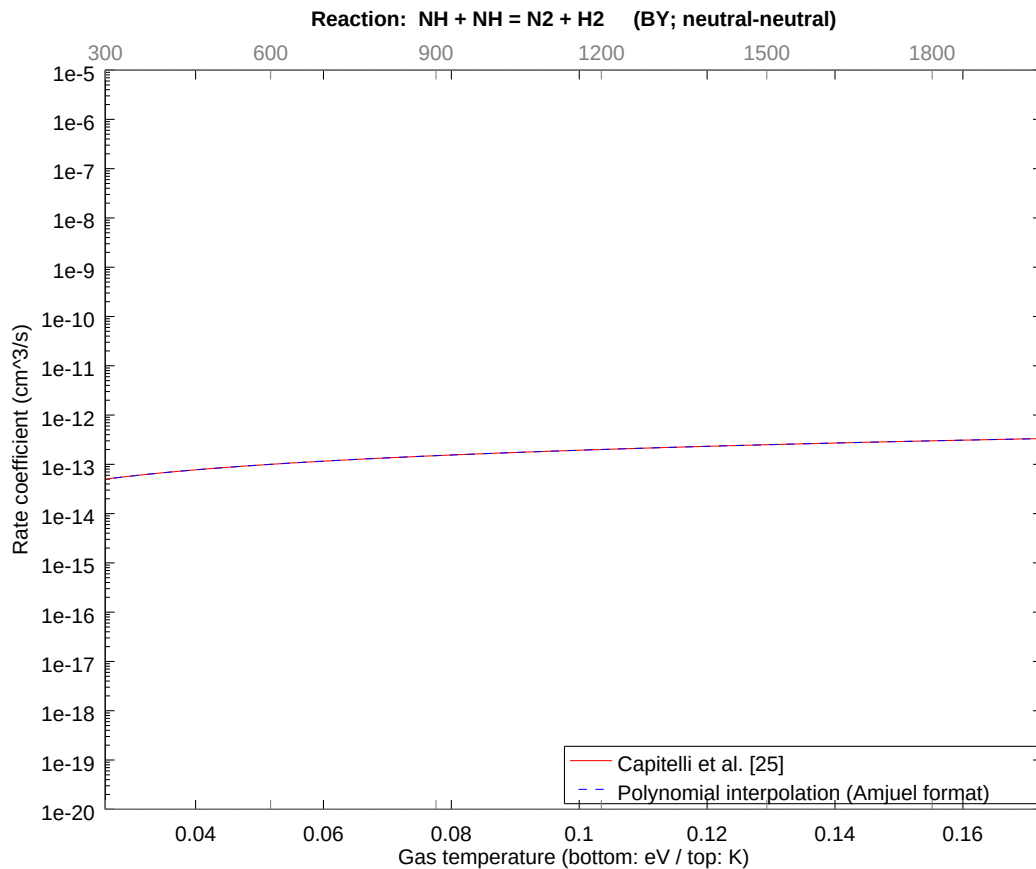
2.136 Reaction BX $\text{N}_2\text{H} + \text{NH} = \text{N}_2 + \text{NH}_2$

b0	-2.414732346361E+01	b1	5.326584602754E-01	b2	-1.249170896738E-01
b3	-5.735776098318E-02	b4	-4.322289705700E-02	b5	-1.349407978481E-02
b6	-3.383390102820E-03	b7	-4.308028015319E-04	b8	-3.204471600200E-05



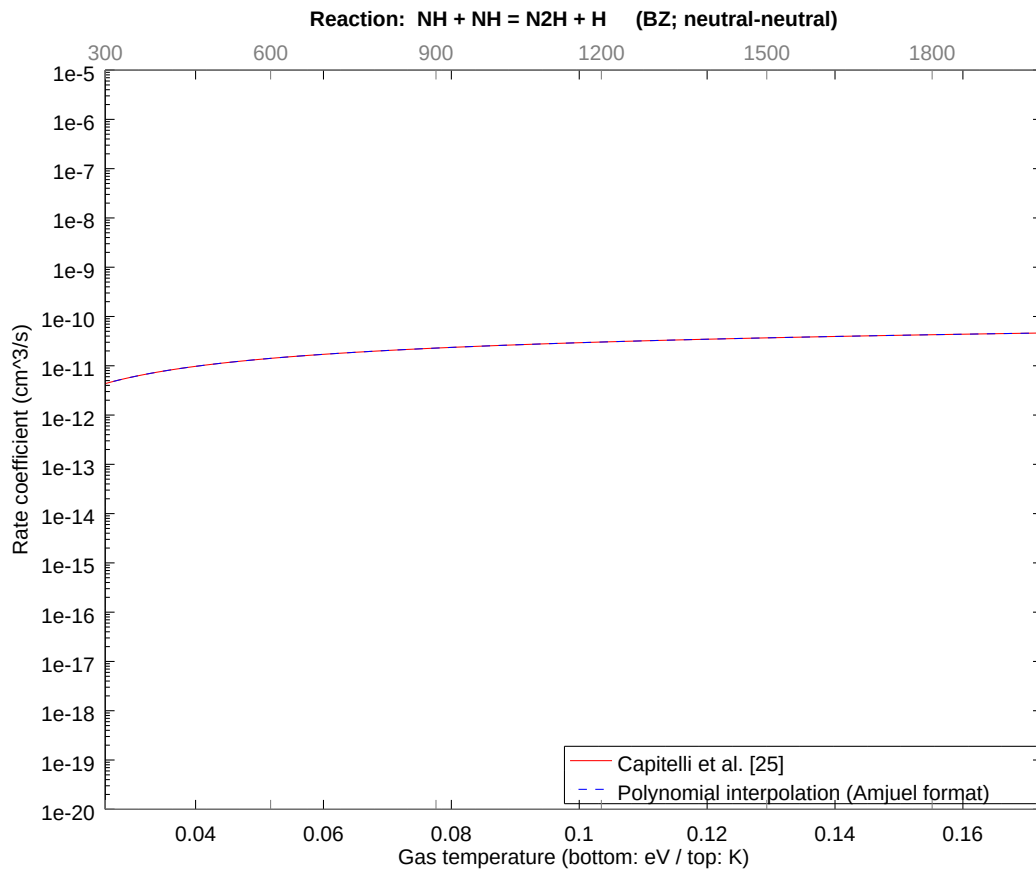
2.137 Reaction BY $\text{NH} + \text{NH} = \text{N}_2 + \text{H}_2$

b0	-2.697189708924E+01	b1	9.999999949711E-01	b2	-6.872542217226E-09
b3	-5.303904524598E-09	b4	-2.527570319855E-09	b5	-7.615316077845E-10
b6	-1.416645555477E-10	b7	-1.487898449876E-11	b8	-6.757021830936E-13



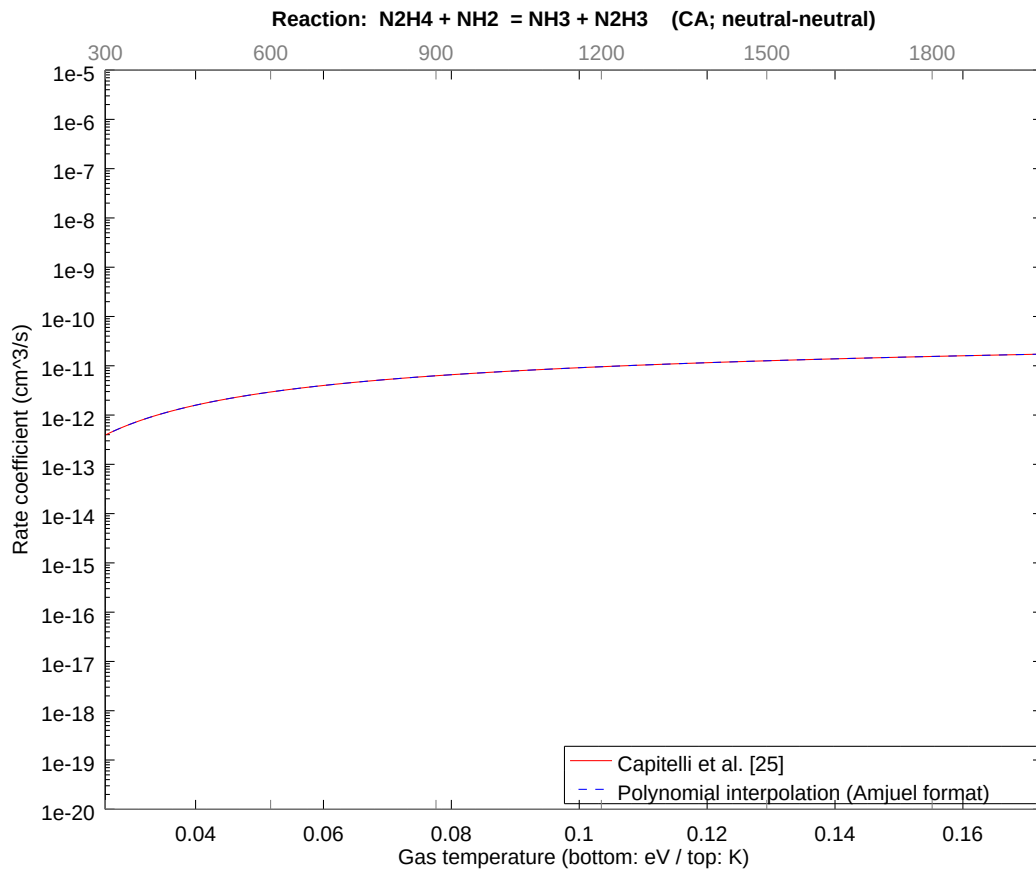
2.138 Reaction BZ $\text{NH} + \text{NH} = \text{N}_2\text{H} + \text{H}$

b0	-2.271889295836E+01	b1	5.163292262952E-01	b2	-6.245855016915E-02
b3	-2.867888467247E-02	b4	-2.161145055453E-02	b5	-6.747040513874E-03
b6	-1.691695169278E-03	b7	-2.154014134061E-04	b8	-1.602235858798E-05



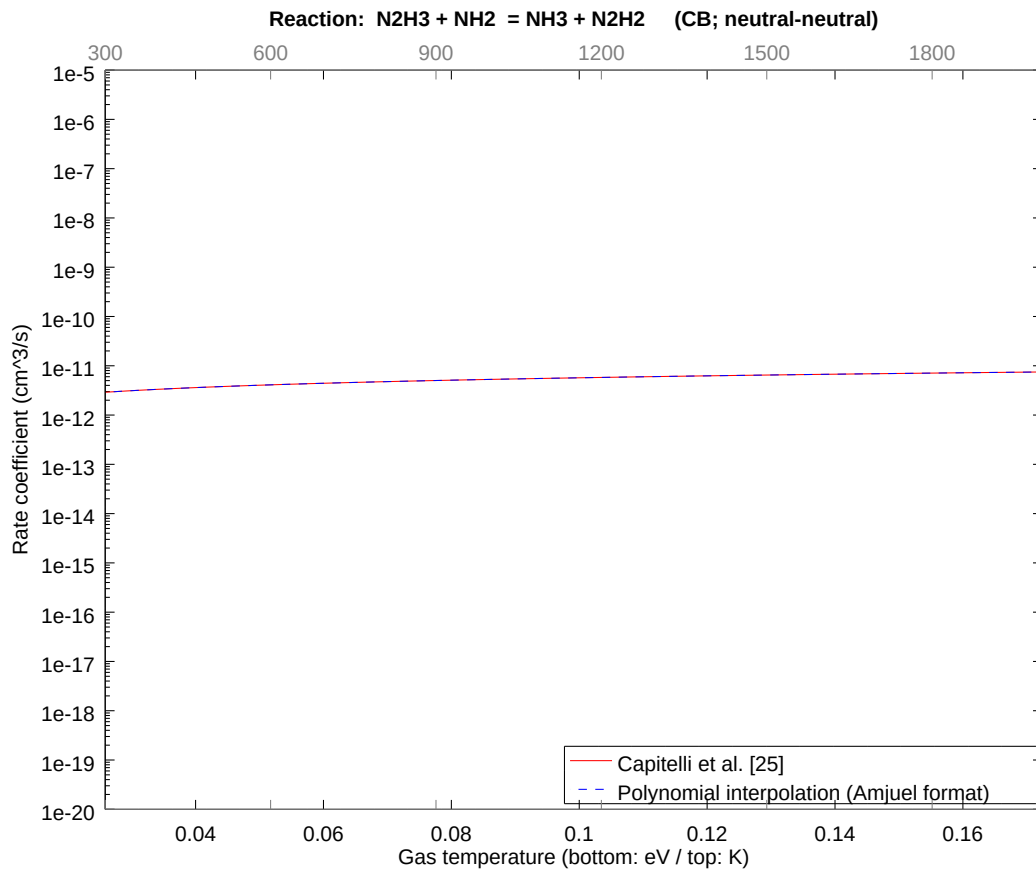
2.139 Reaction CA $\text{N}_2\text{H}_4 + \text{NH}_2 = \text{NH}_3 + \text{N}_2\text{H}_3$

b0	-2.350728610916E+01	b1	5.326584576680E-01	b2	-1.249170933627E-01
b3	-5.735776393498E-02	b4	-4.322289851799E-02	b5	-1.349408024277E-02
b6	-3.383390191598E-03	b7	-4.308028112622E-04	b8	-3.204471646367E-05



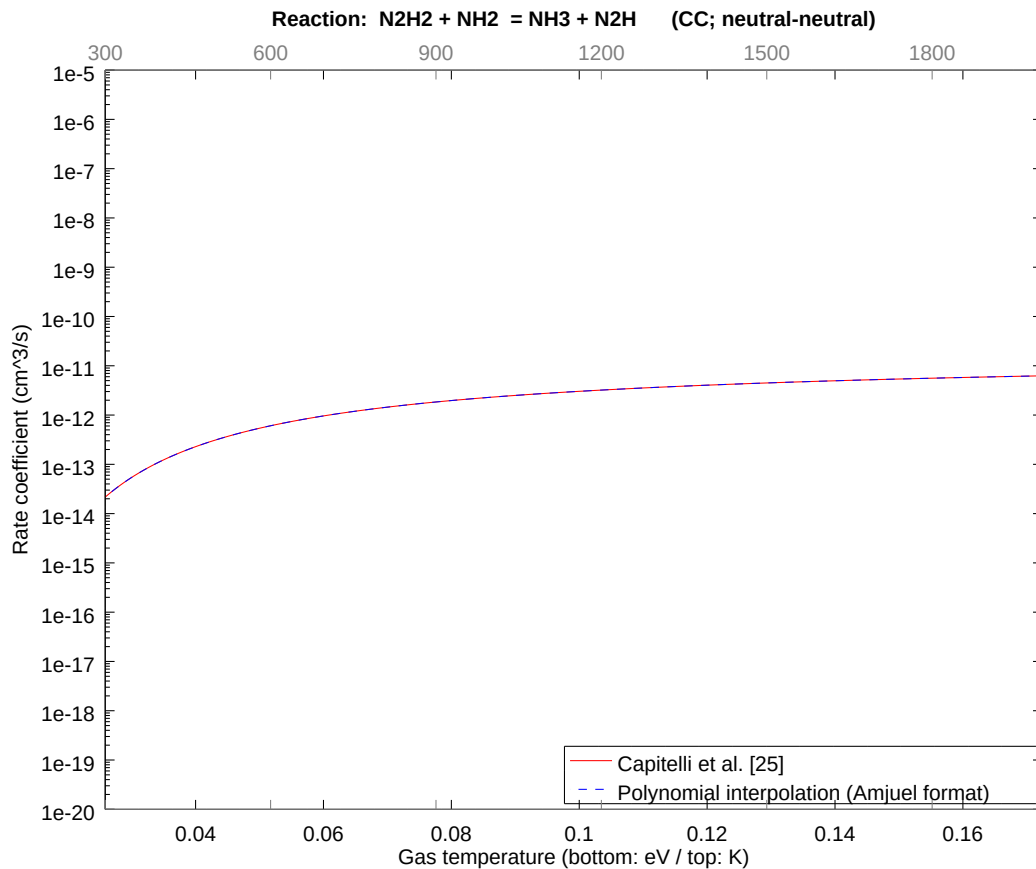
2.140 Reaction CB $\text{N}_2\text{H}_3 + \text{NH}_2 = \text{NH}_3 + \text{N}_2\text{H}_2$

b0	-2.473888223088E+01	b1	4.999999909009E-01	b2	-1.252288180630E-08
b3	-9.740196382601E-09	b4	-4.682862879700E-09	b5	-1.425228340047E-09
b6	-2.682081542161E-10	b7	-2.854101745411E-11	b8	-1.315306849704E-12



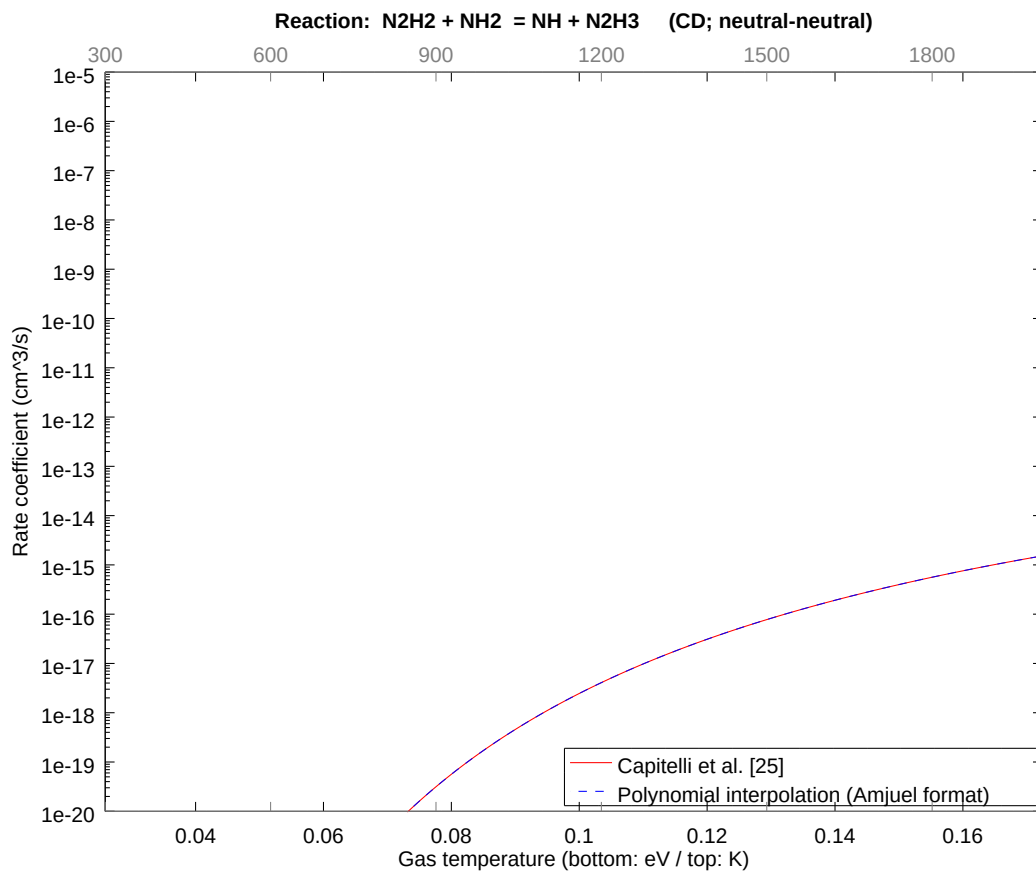
2.141 Reaction CC N2H2 + NH2 = NH3 + N2H

b0	-2.500098460761E+01	b1	6.531690967044E-02	b2	-2.498341942865E-01
b3	-1.147155335862E-01	b4	-8.644579971549E-02	b5	-2.698816128360E-02
b6	-6.766780530776E-03	b7	-8.616056380250E-04	b8	-6.408943363558E-05



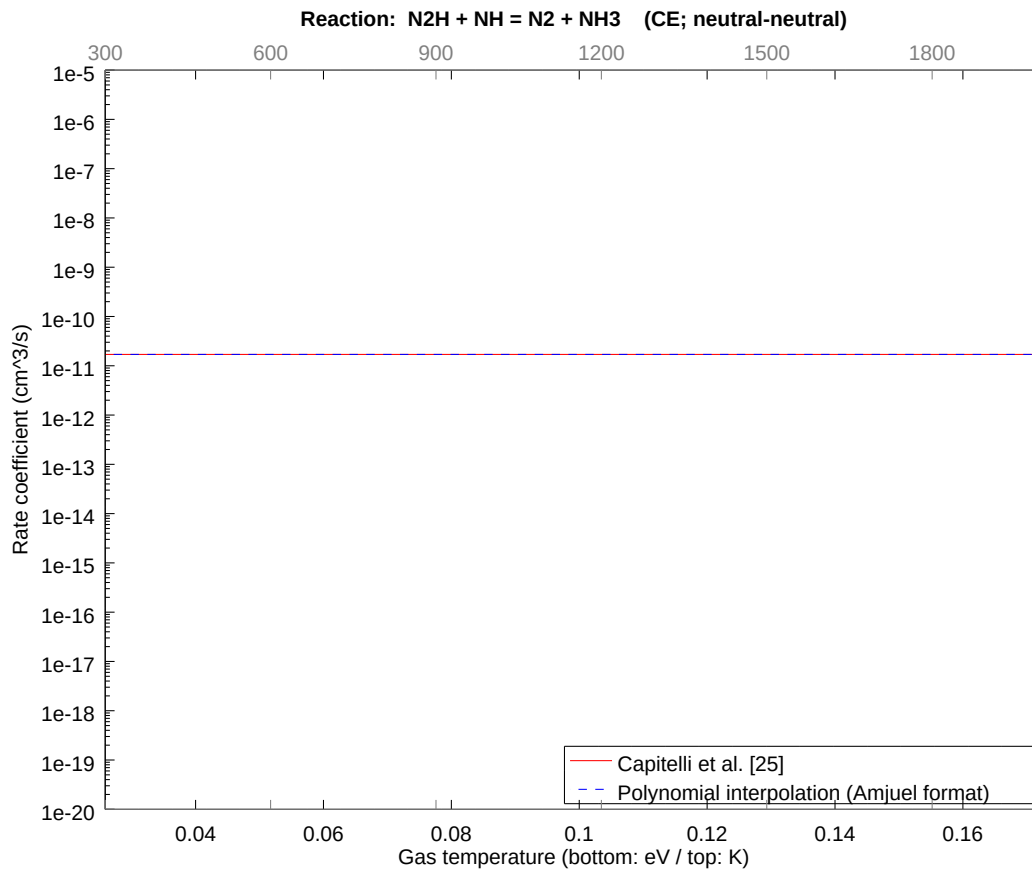
2.142 Reaction CD $\text{N}_2\text{H}_2 + \text{NH}_2 = \text{NH} + \text{N}_2\text{H}_3$

b0	-2.646588531989E+01	b1	1.055193767911E+00	b2	-2.123590605188E+00
b3	-9.750820018544E-01	b4	-7.347892825853E-01	b5	-2.293993667175E-01
b6	-5.751763379524E-02	b7	-7.323647855047E-03	b8	-5.447601831483E-04



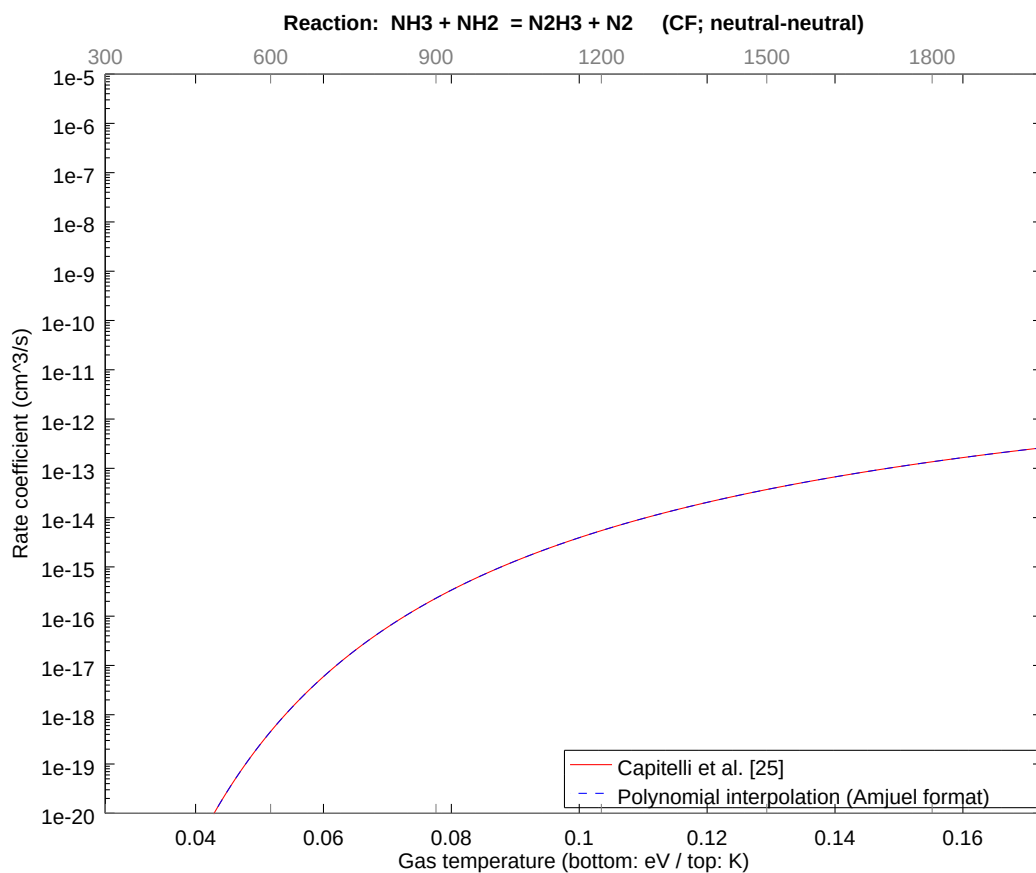
2.143 Reaction CE N2H + NH = N2 + NH3

b0	-2.479780777187E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



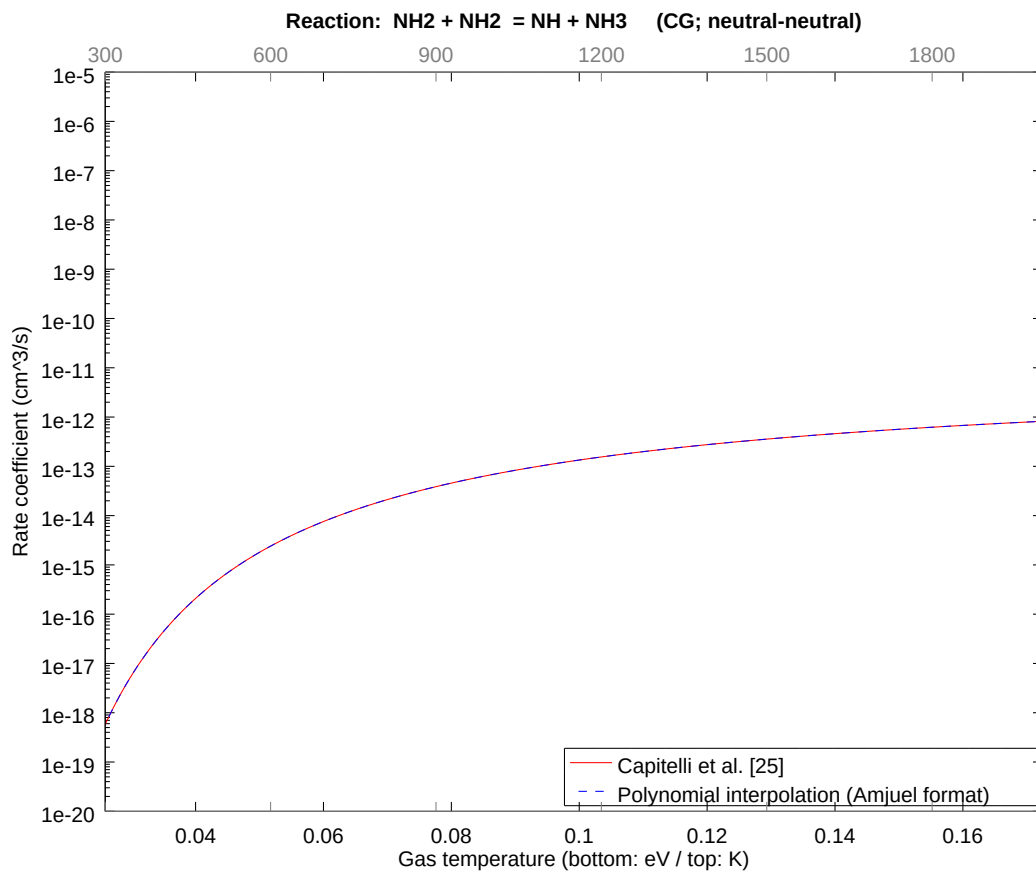
2.144 Reaction CF NH3 + NH2 = N2H3 + N2

b0	-2.377033307596E+01	b1	8.543442558806E-01	b2	-1.355350477056E+00
b3	-6.223317502405E-01	b4	-4.689684548518E-01	b5	-1.464107725851E-01
b6	-3.670978397938E-02	b7	-4.674210549023E-03	b8	-3.476851760129E-04



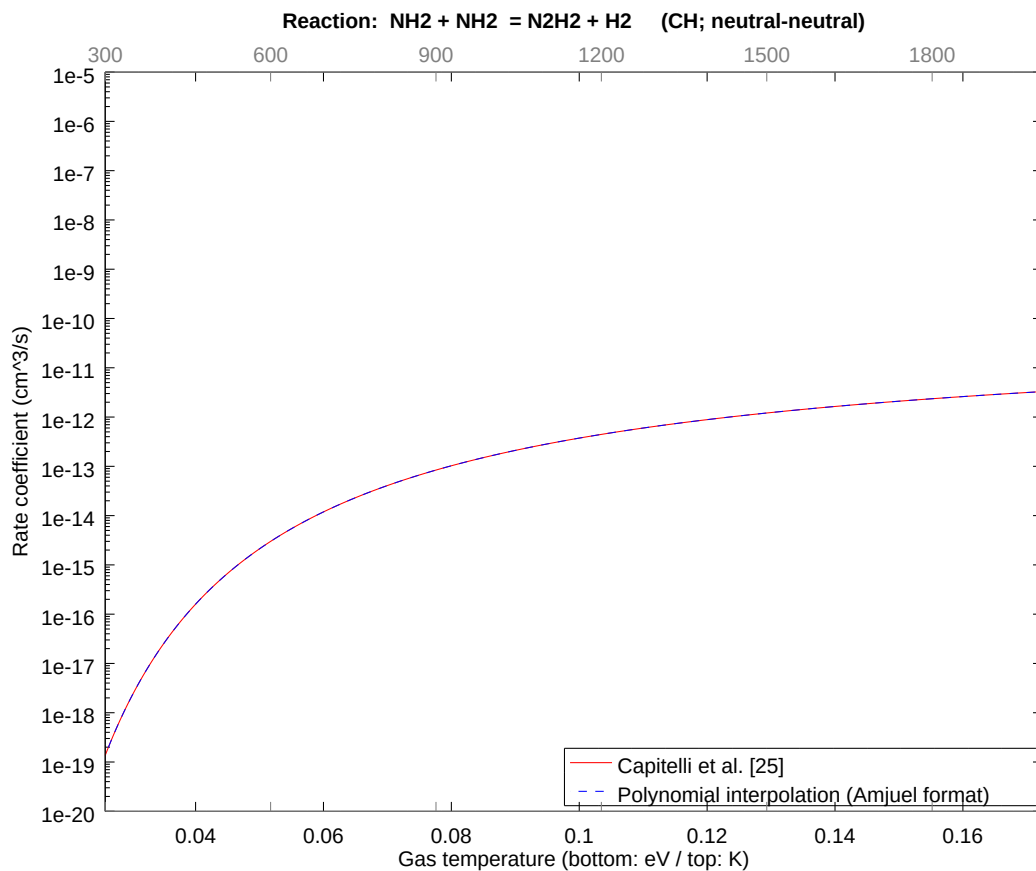
2.145 Reaction CG $\text{NH}_2 + \text{NH}_2 = \text{NH} + \text{NH}_3$

b0	-2.583637811126E+01	b1	1.632922770044E-01	b2	-6.245854823952E-01
b3	-2.867888318610E-01	b4	-2.161144985246E-01	b5	-6.747040305731E-02
b6	-1.691695131471E-02	b7	-2.154014095617E-03	b8	-1.602235842058E-04



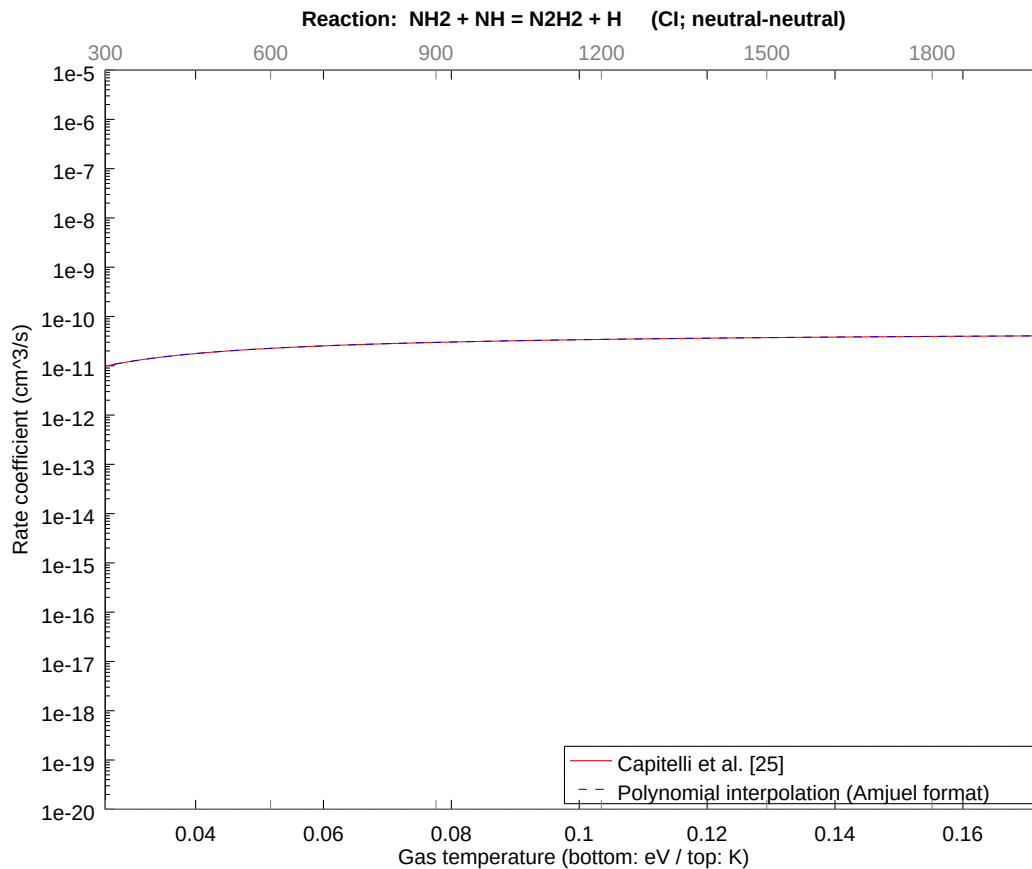
2.146 Reaction $\text{CH NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$

b0	-2.405089687724E+01	b1	1.959507406339E-01	b2	-7.495025678272E-01
b3	-3.441465898399E-01	b4	-2.593373942811E-01	b5	-8.096448249087E-02
b6	-2.030034135994E-02	b7	-2.584816891937E-03	b8	-1.922683000102E-04



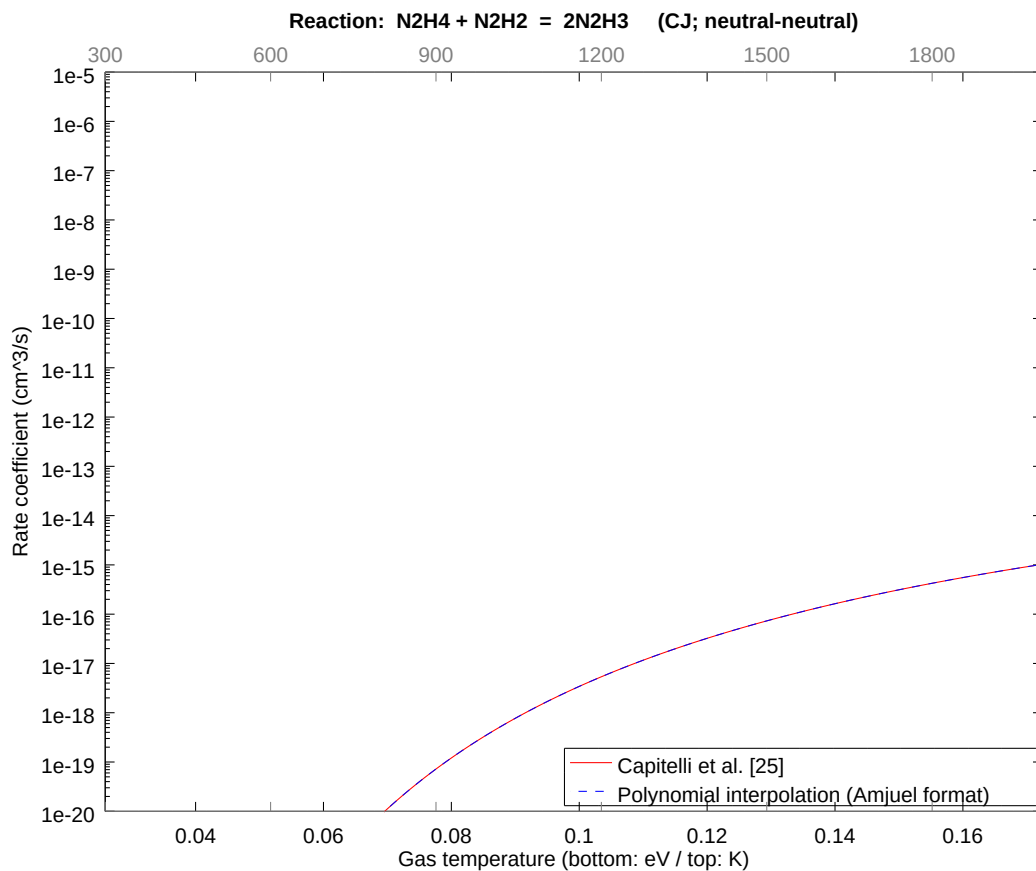
2.147 Reaction CI $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$

b0	-2.373057160757E+01	b1	1.632922319726E-02	b2	-6.245855454798E-02
b3	-2.867888818227E-02	b4	-2.161145229905E-02	b5	-6.747041064367E-03
b6	-1.691695276958E-03	b7	-2.154014253426E-04	b8	-1.602235916207E-05



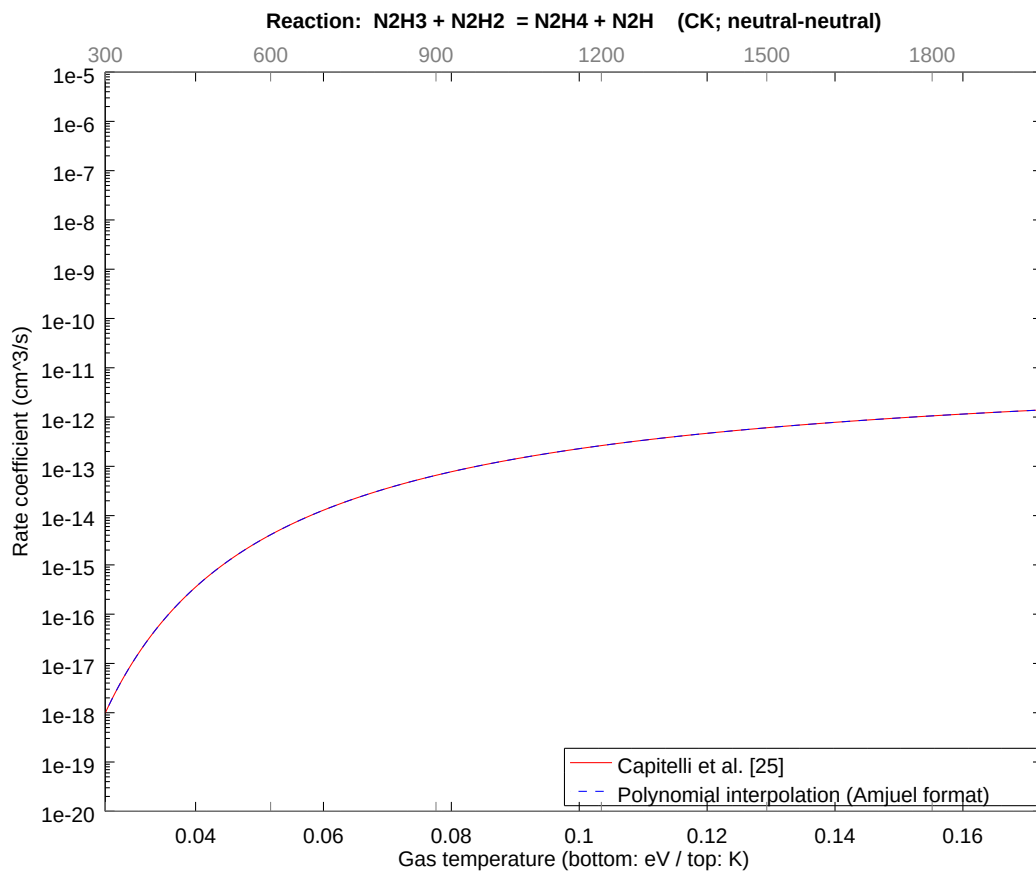
2.148 Reaction CJ $\text{N}_2\text{H}_4 + \text{N}_2\text{H}_2 = 2\text{N}_2\text{H}_3$

b0	-2.765592328912E+01	b1	9.898768553234E-01	b2	-1.873756414556E+00
b3	-8.603664708274E-01	b4	-6.483434839625E-01	b5	-2.024112057243E-01
b6	-5.075085331116E-02	b7	-6.462042221143E-03	b8	-4.806707496634E-04



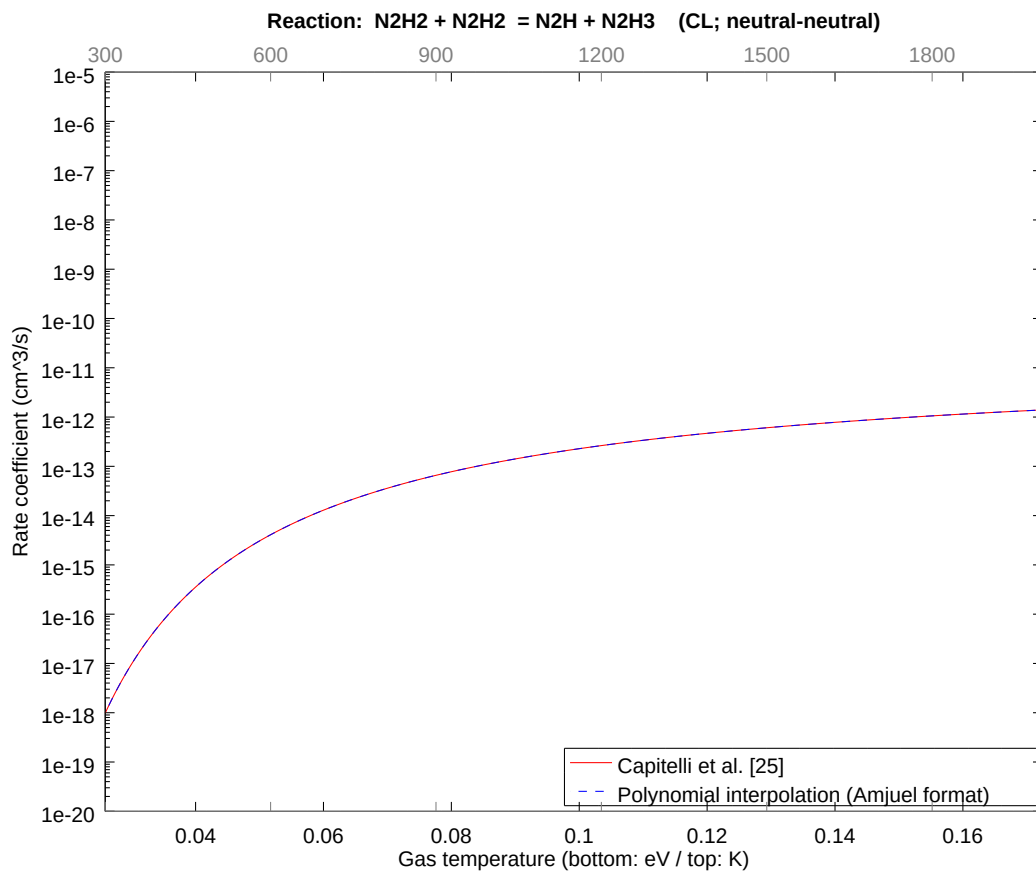
2.149 Reaction CK $\text{N}_2\text{H}_3 + \text{N}_2\text{H}_2 = \text{N}_2\text{H}_4 + \text{N}_2\text{H}$

b0	-2.530574985861E+01	b1	1.632922819981E-01	b2	-6.245854756144E-01
b3	-2.867888266545E-01	b4	-2.161144960521E-01	b5	-6.747040231379E-02
b6	-1.691695117642E-02	b7	-2.154014081072E-03	b8	-1.602235835433E-04



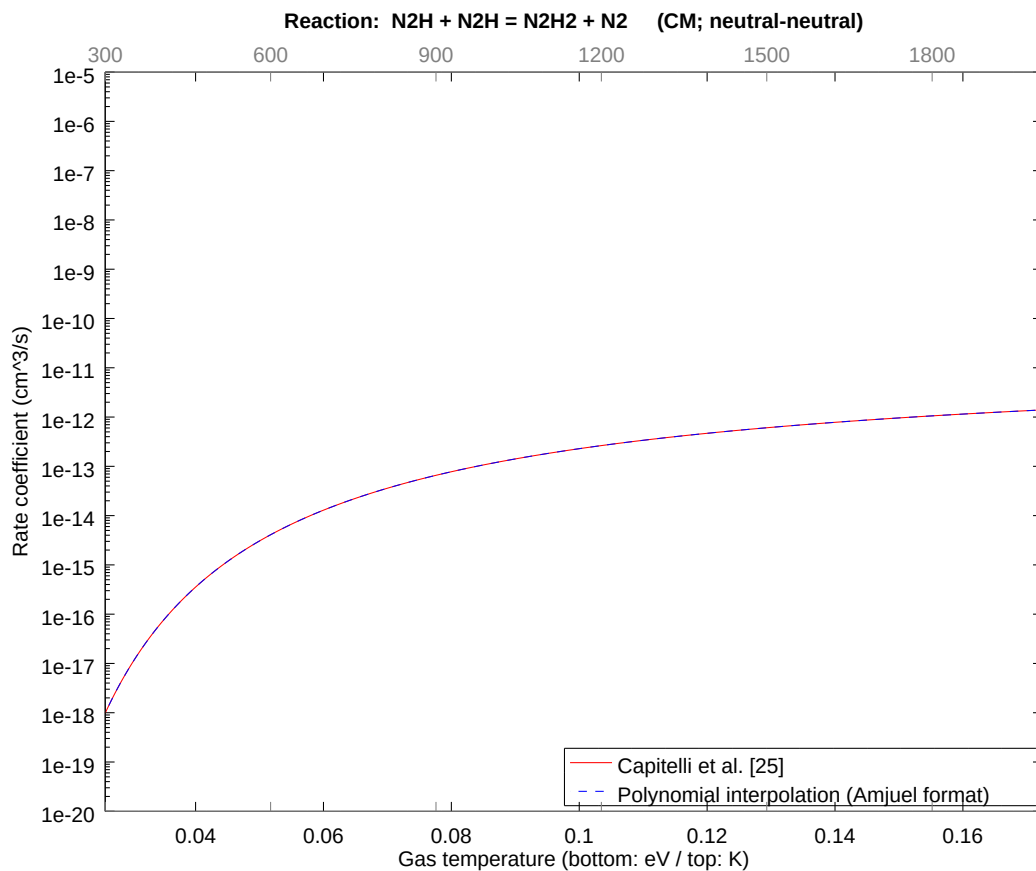
2.150 Reaction CL $\text{N}_2\text{H}_2 + \text{N}_2\text{H}_2 = \text{N}_2\text{H} + \text{N}_2\text{H}_3$

b0	-2.530574985861E+01	b1	1.632922819981E-01	b2	-6.245854756144E-01
b3	-2.867888266545E-01	b4	-2.161144960521E-01	b5	-6.747040231379E-02
b6	-1.691695117642E-02	b7	-2.154014081072E-03	b8	-1.602235835433E-04



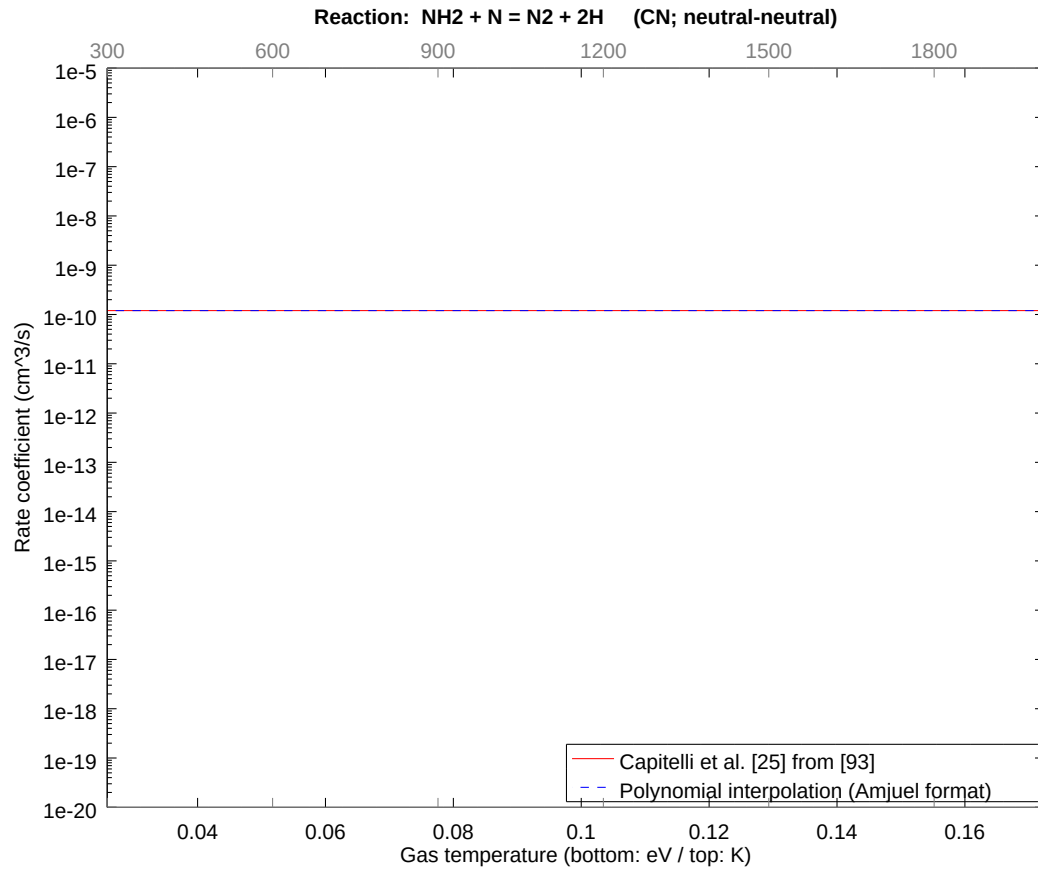
2.151 Reaction CM $\text{N}_2\text{H} + \text{N}_2\text{H} = \text{N}_2\text{H}_2 + \text{N}_2$

b0	-2.530574985861E+01	b1	1.632922819981E-01	b2	-6.245854756144E-01
b3	-2.867888266545E-01	b4	-2.161144960521E-01	b5	-6.747040231379E-02
b6	-1.691695117642E-02	b7	-2.154014081072E-03	b8	-1.602235835433E-04



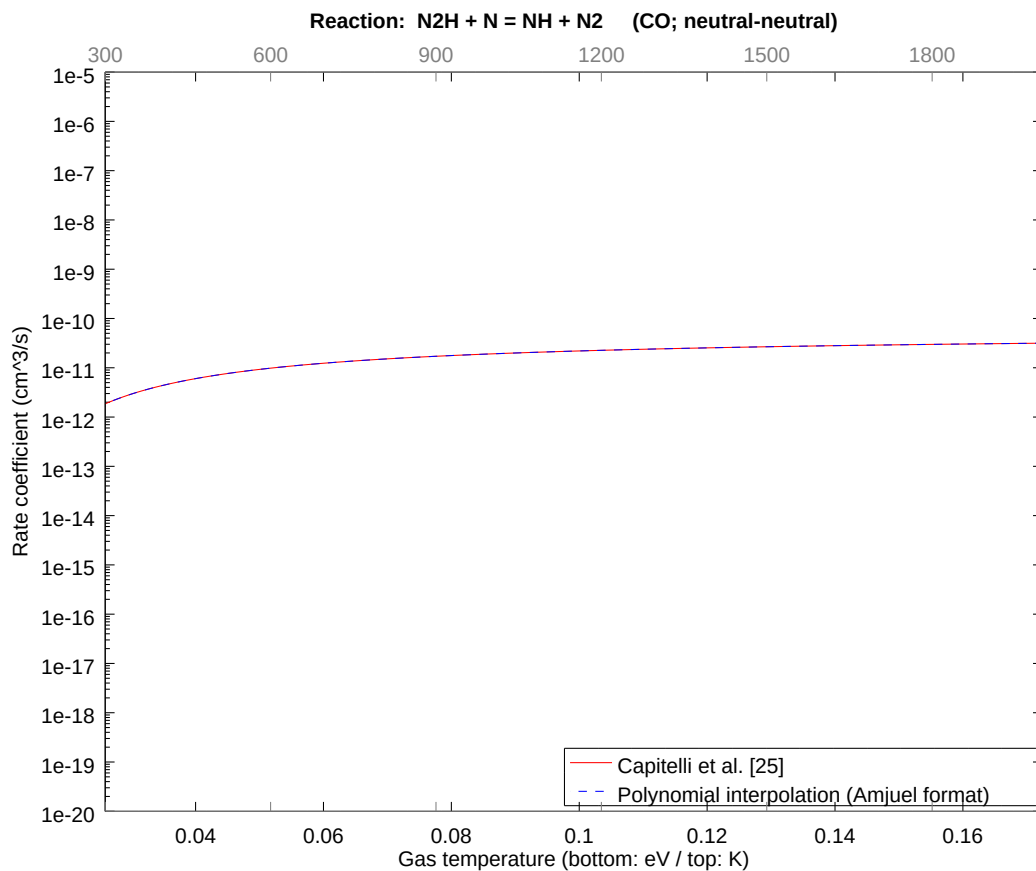
2.152 Reaction CN NH2 + N = N2 + 2H

b0	-2.284352937315E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



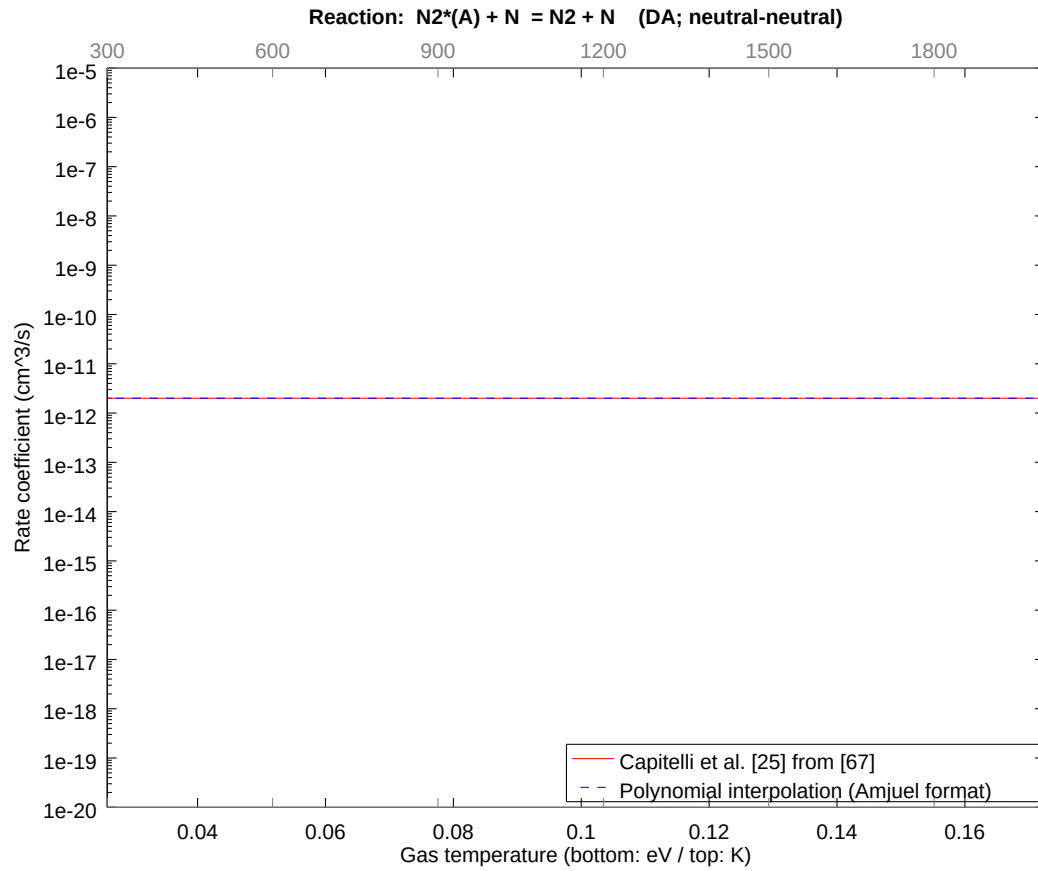
2.153 Reaction CO N2H + N = NH + N2

b0	-2.378136581554E+01	b1	3.265845386299E-02	b2	-1.249170984012E-01
b3	-5.735776771641E-02	b4	-4.322290027883E-02	b5	-1.349408076421E-02
b6	-3.383390287591E-03	b7	-4.308028213161E-04	b8	-3.204471692274E-05



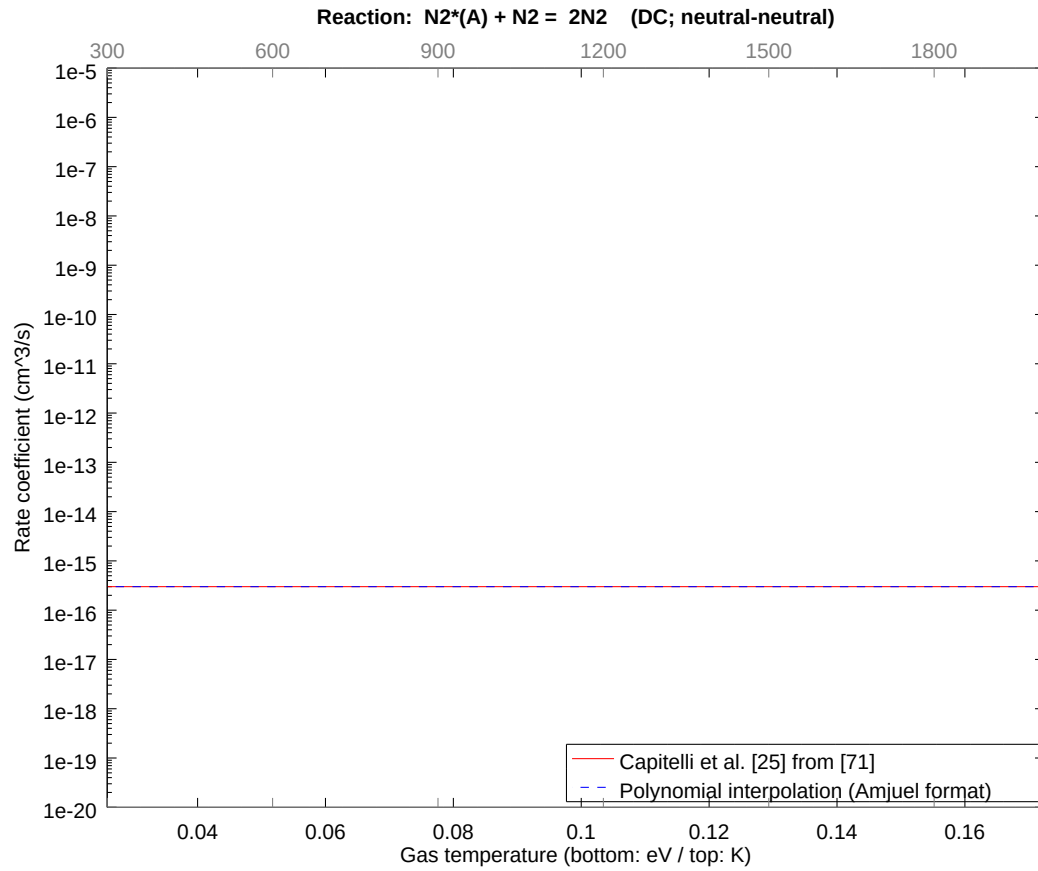
2.154 Reaction DA $\text{N}_2^*(\text{A}) + \text{N} = \text{N}_2 + \text{N}$

b0	-2.693787393537E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



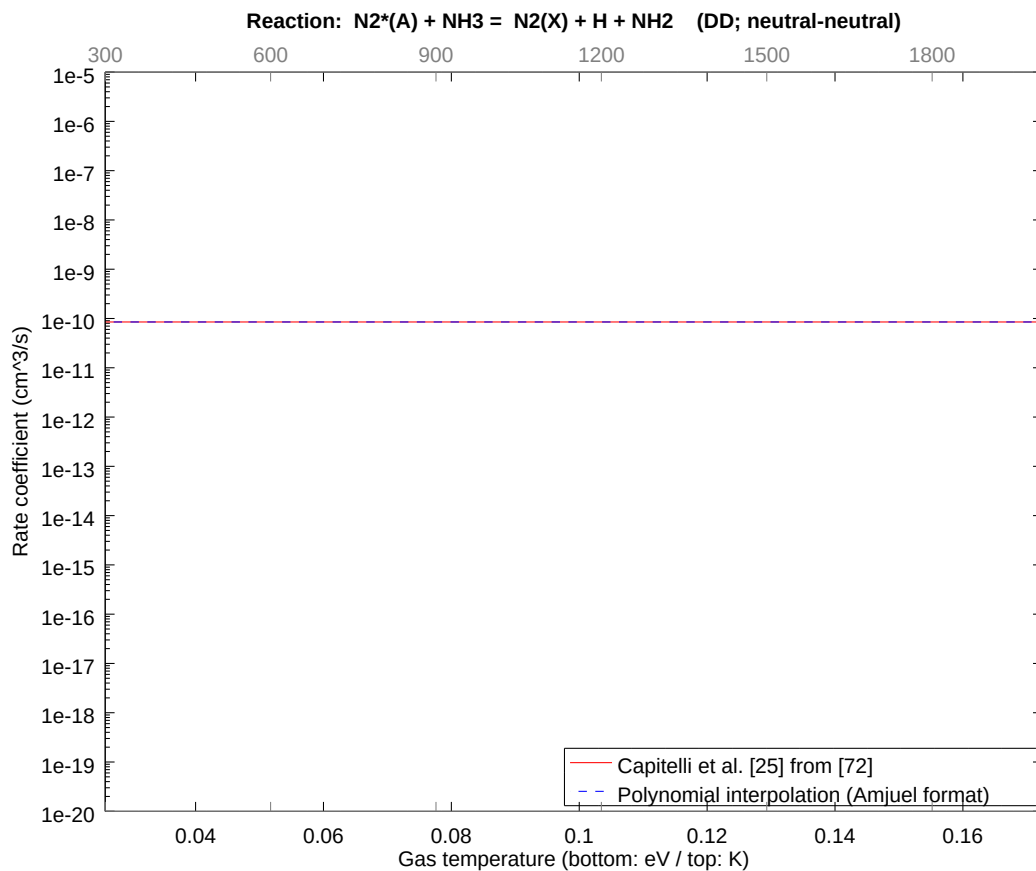
2.155 Reaction DC $\text{N}_2^*(\text{A}) + \text{N}_2 = 2\text{N}_2$

b0	-3.574274919924E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



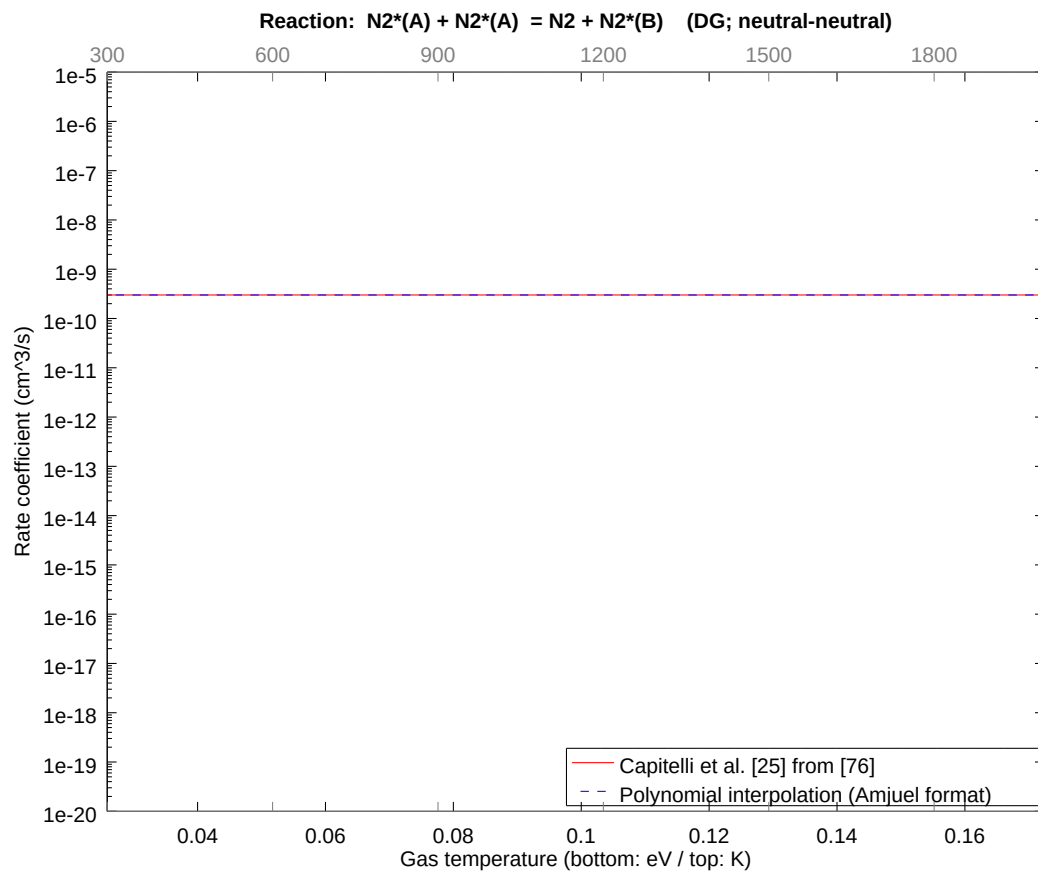
2.156 Reaction DD $\text{N}_2^*(\text{A}) + \text{NH}_3 = \text{N}_2(\text{X}) + \text{H} + \text{NH}_2$

b0	-2.318836985944E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



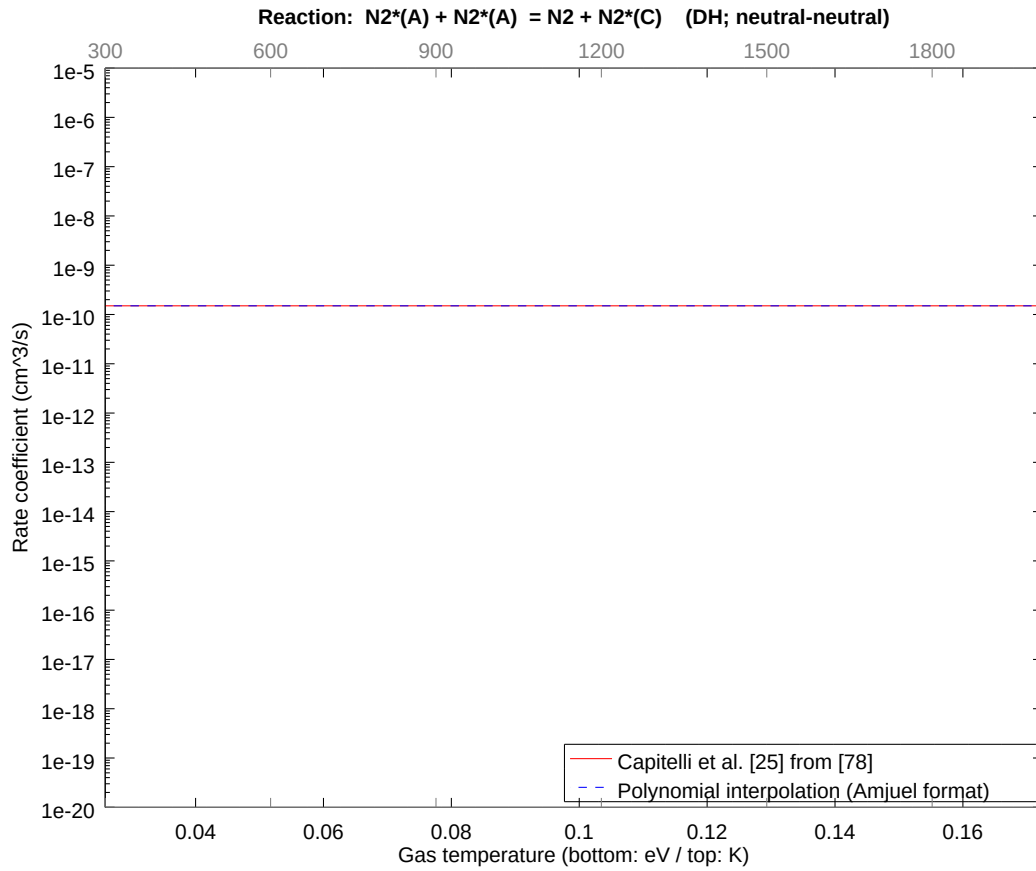
2.157 Reaction DG N2*(A) + N2*(A) = N2 + N2*(B)

b0	-2.192723864127E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



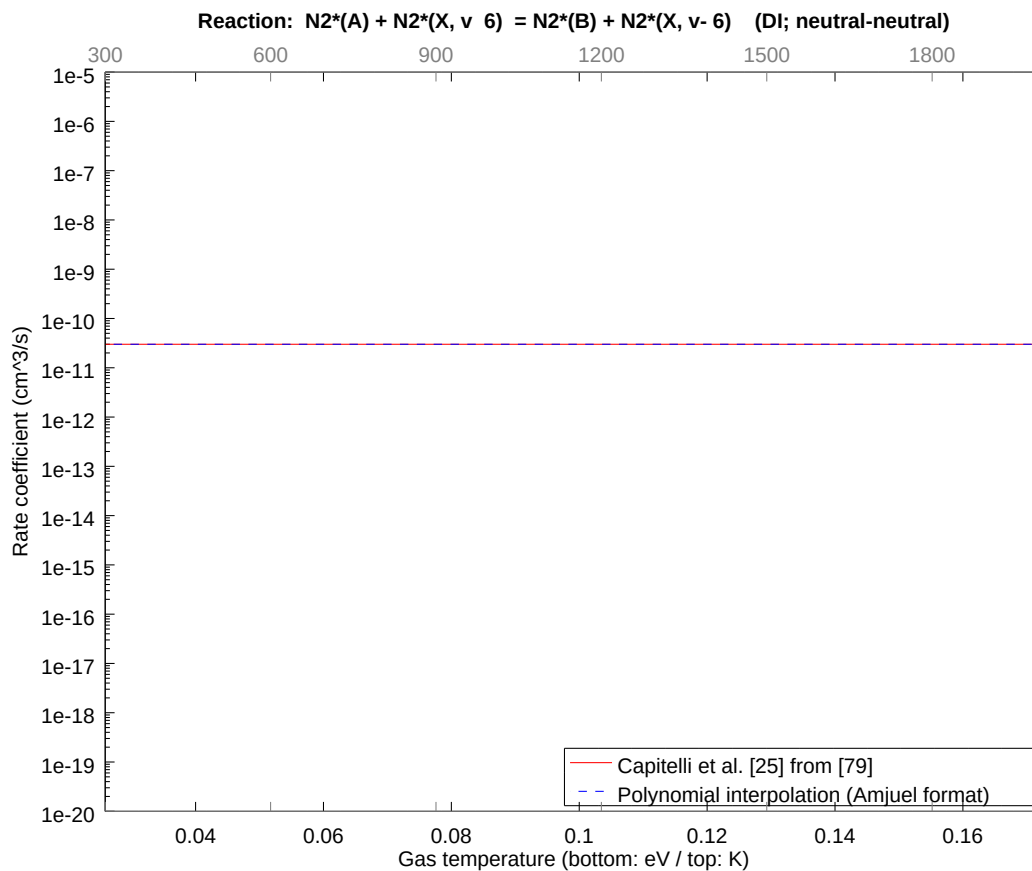
2.158 Reaction DH $\text{N}_2^*(\text{A}) + \text{N}_2^*(\text{A}) = \text{N}_2 + \text{N}_2^*(\text{C})$

b0	-2.262038582183E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



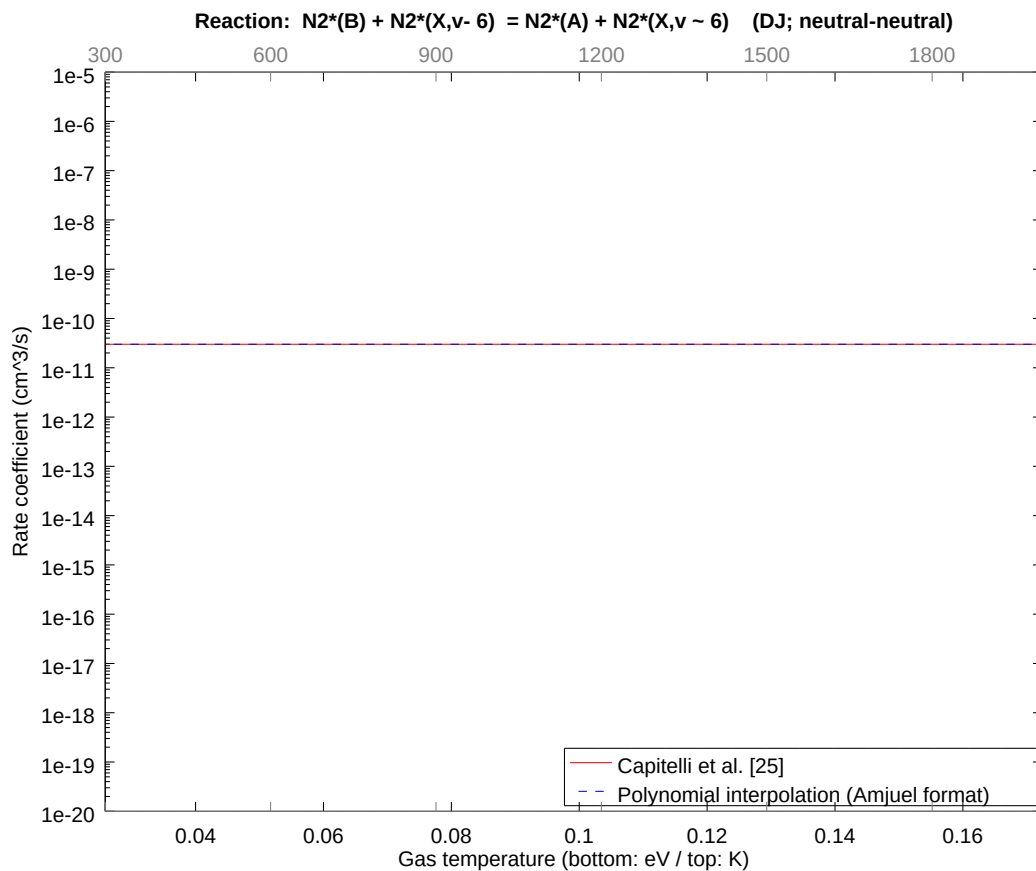
2.159 Reaction DI $\text{N}_2^*(\text{A}) + \text{N}_2^*(\text{X}, v=6) = \text{N}_2^*(\text{B}) + \text{N}_2^*(\text{X}, v=6)$

b0	-2.422982373427E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



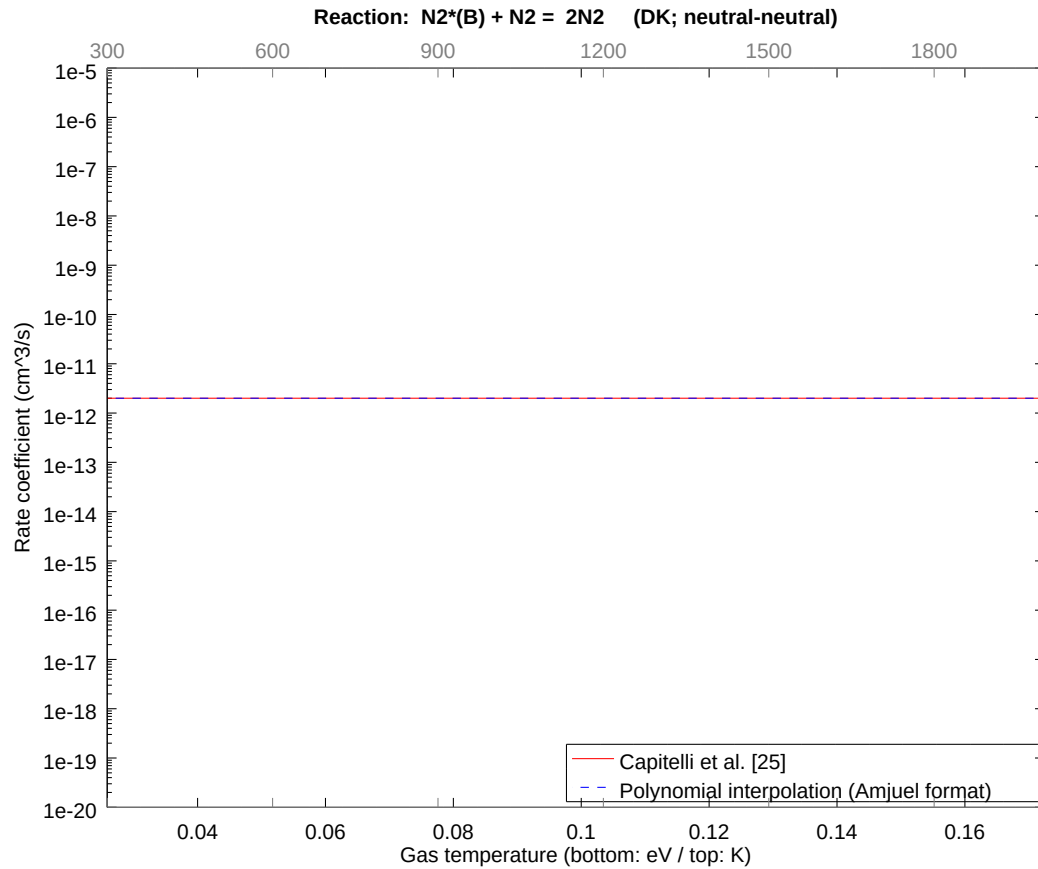
2.160 Reaction DJ $\text{N}_2^*(\text{B}) + \text{N}_2^*(\text{X}, v-6) = \text{N}_2^*(\text{A}) + \text{N}_2^*(\text{X}, v-6)$

b0	-2.422982373427E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



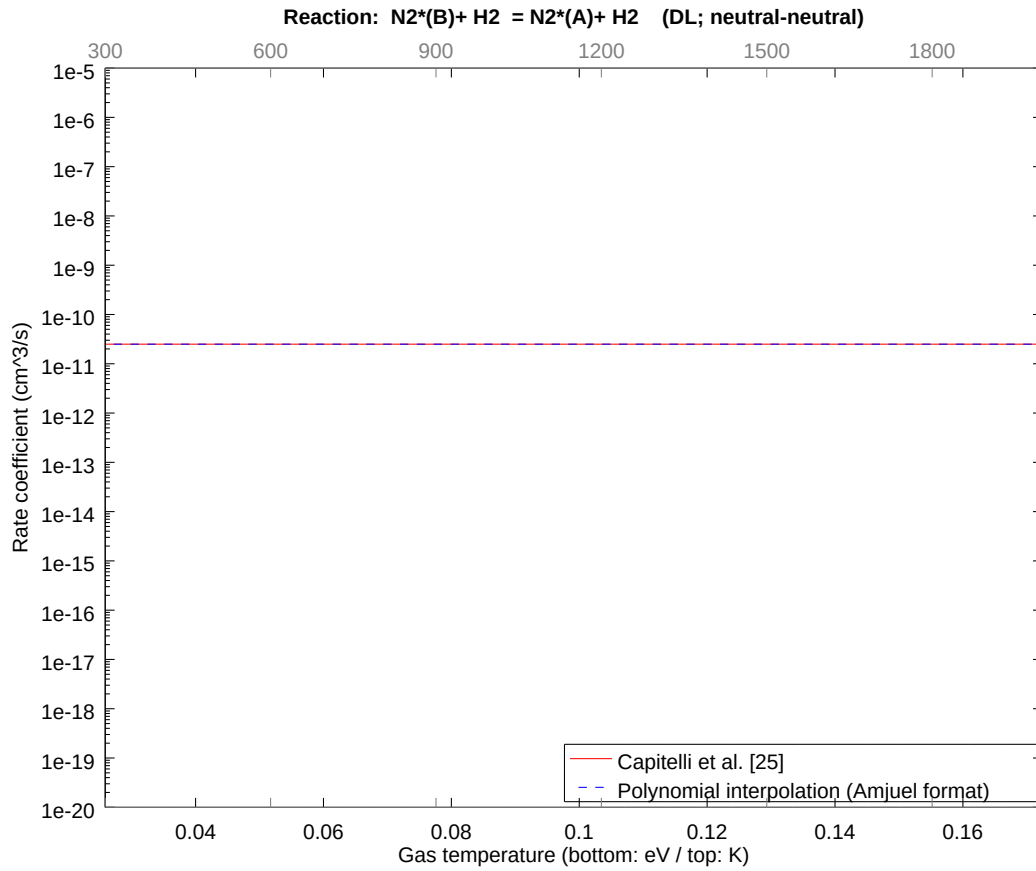
2.161 Reaction DK $\text{N}_2^*(\text{B}) + \text{N}_2 = 2\text{N}_2$

b0	-2.693787393537E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



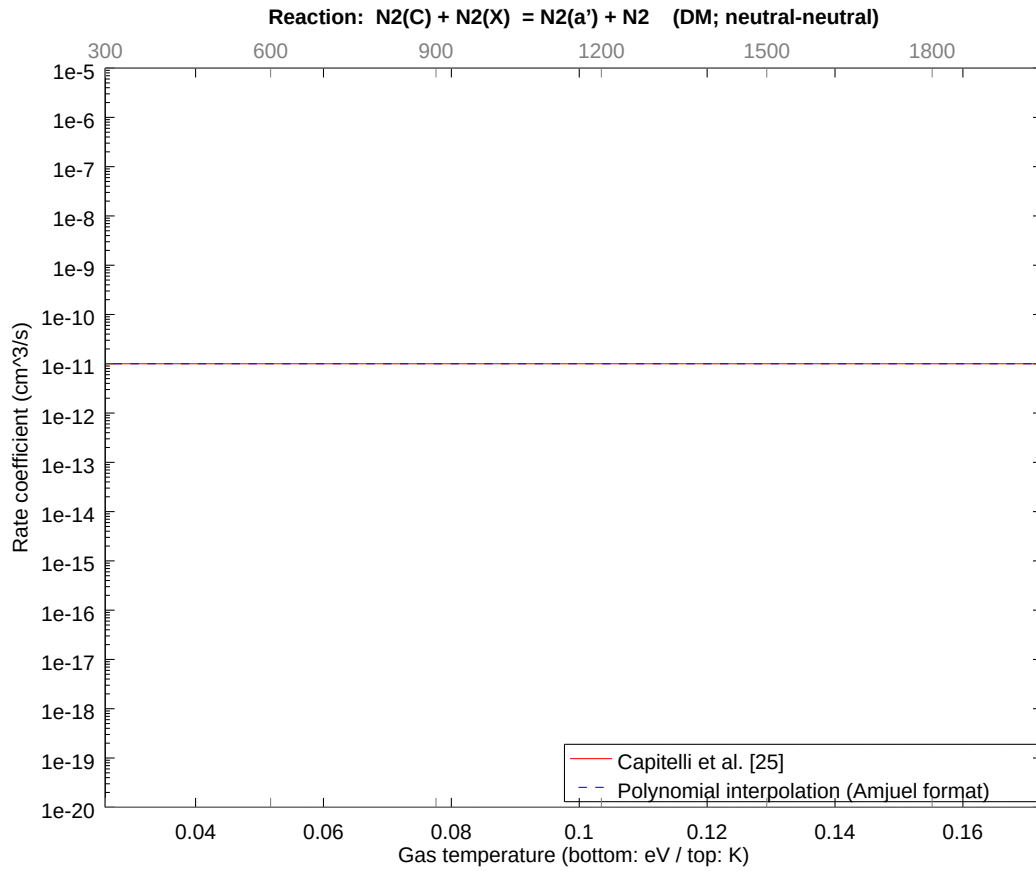
2.162 Reaction DL $\text{N}_2^*(\text{B}) + \text{H}_2 = \text{N}_2^*(\text{A}) + \text{H}_2$

b0	-2.441214529106E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



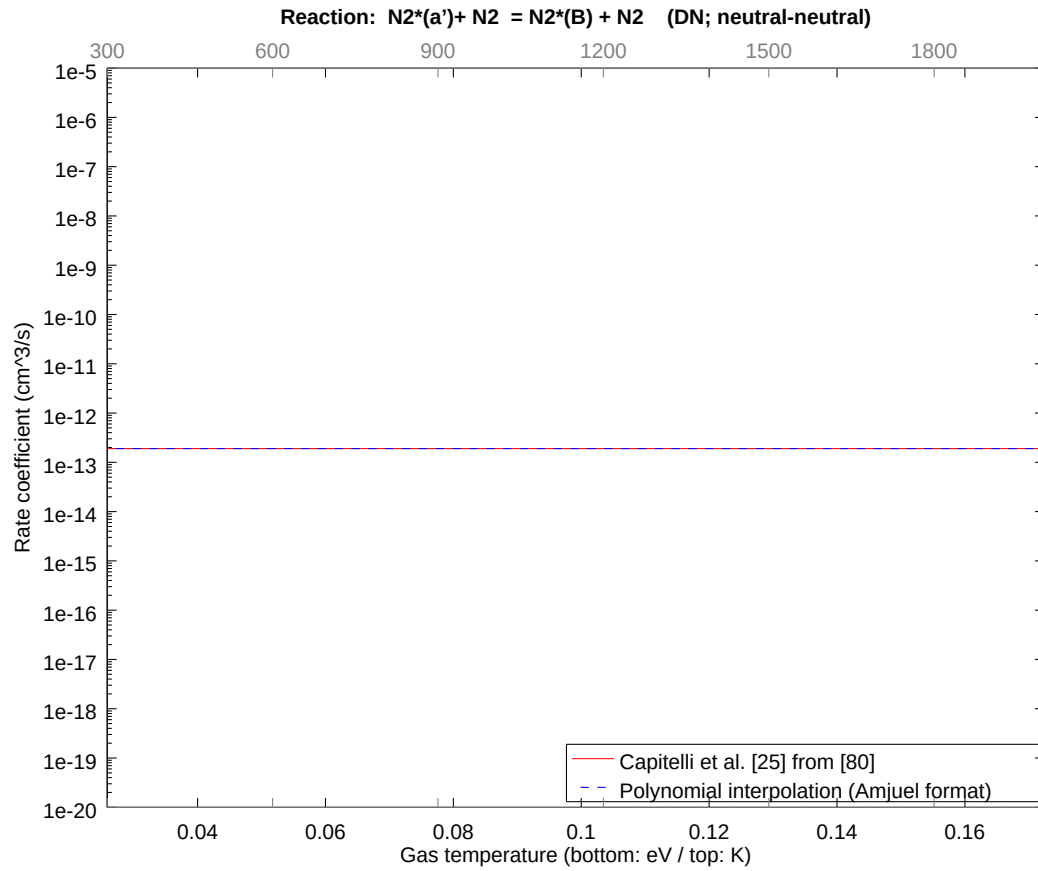
2.163 Reaction DM $\text{N}_2(\text{C}) + \text{N}_2(\text{X}) = \text{N}_2(\text{a}') + \text{N}_2$

b0	-2.532843602293E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



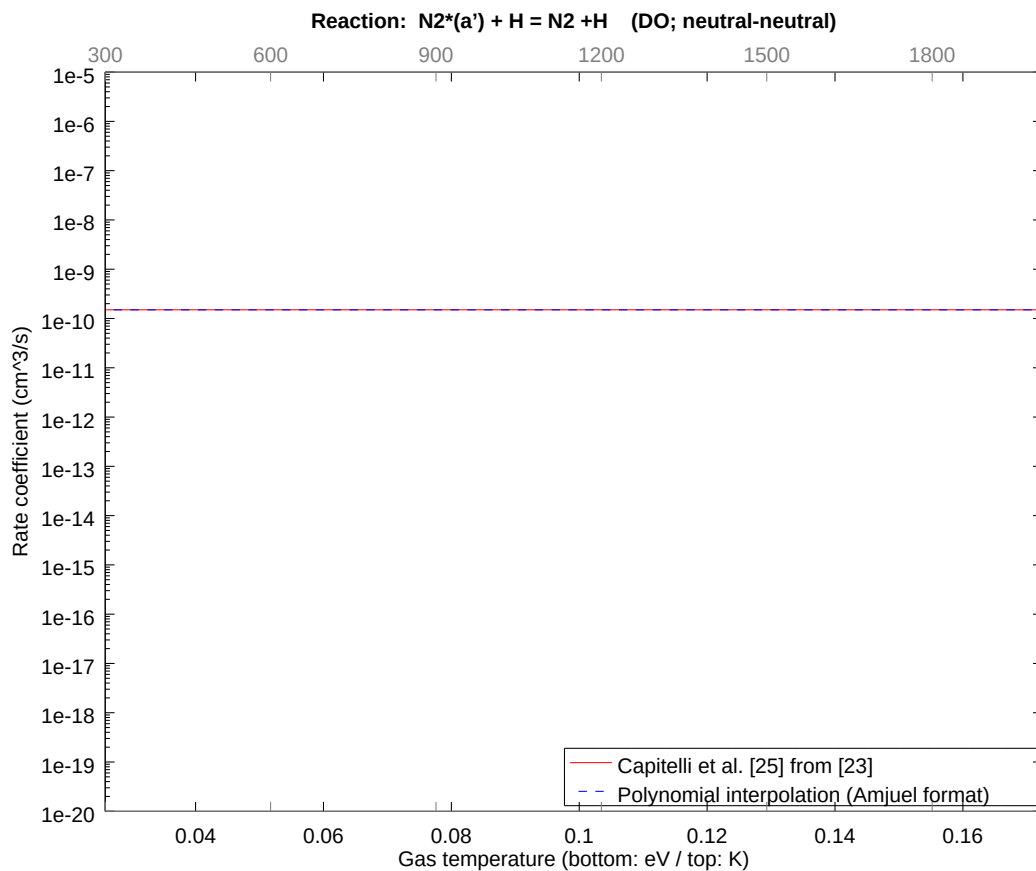
2.164 Reaction DN $\text{N}_2^*(\text{a}') + \text{N}_2 = \text{N}_2^*(\text{B}) + \text{N}_2$

b0	-2.929175232275E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



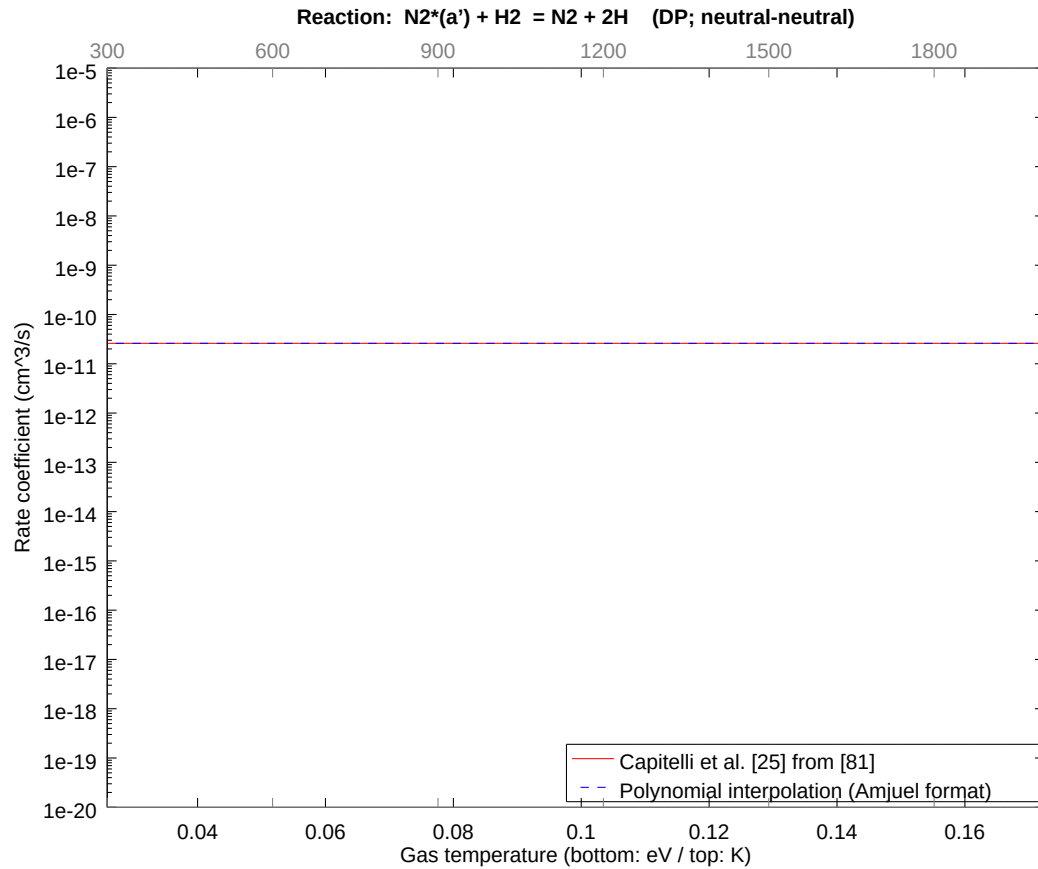
2.165 Reaction DO N2*(a') + H = N2 +H

b0	-2.262038582183E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



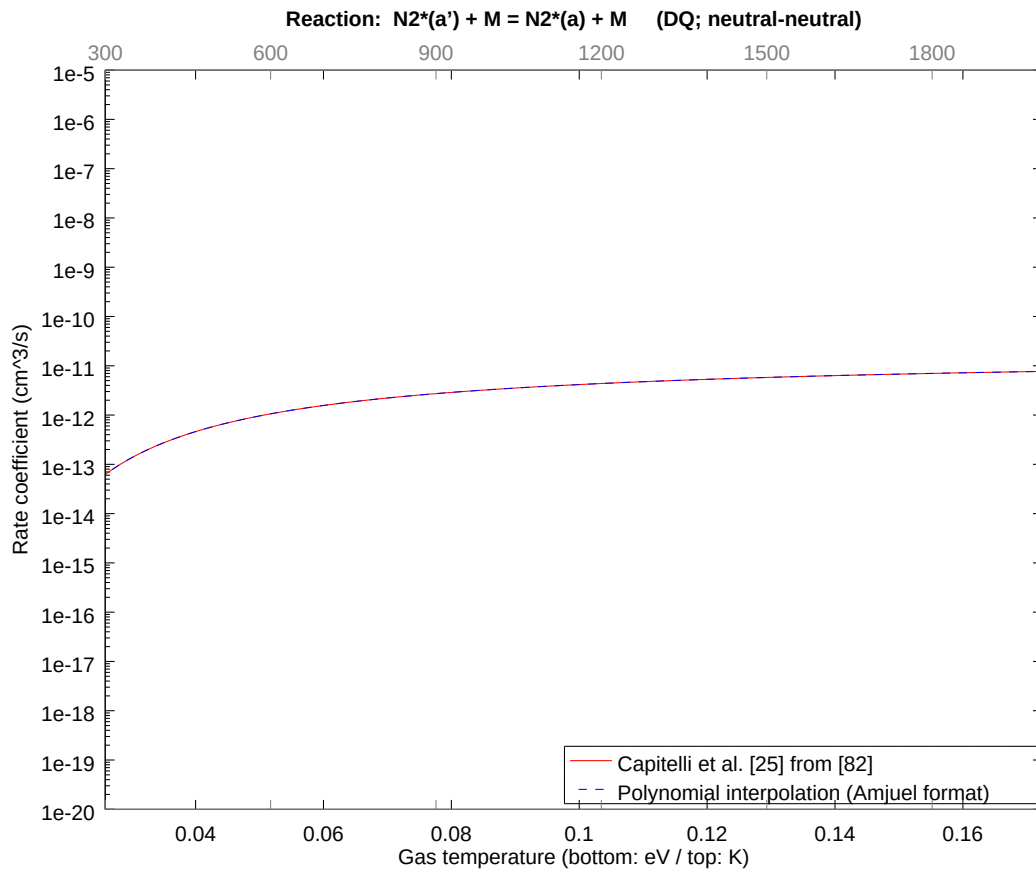
2.166 Reaction DP $\text{N}_2^*(a') + \text{H}_2 = \text{N}_2 + 2\text{H}$

b0	-2.437292457791E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



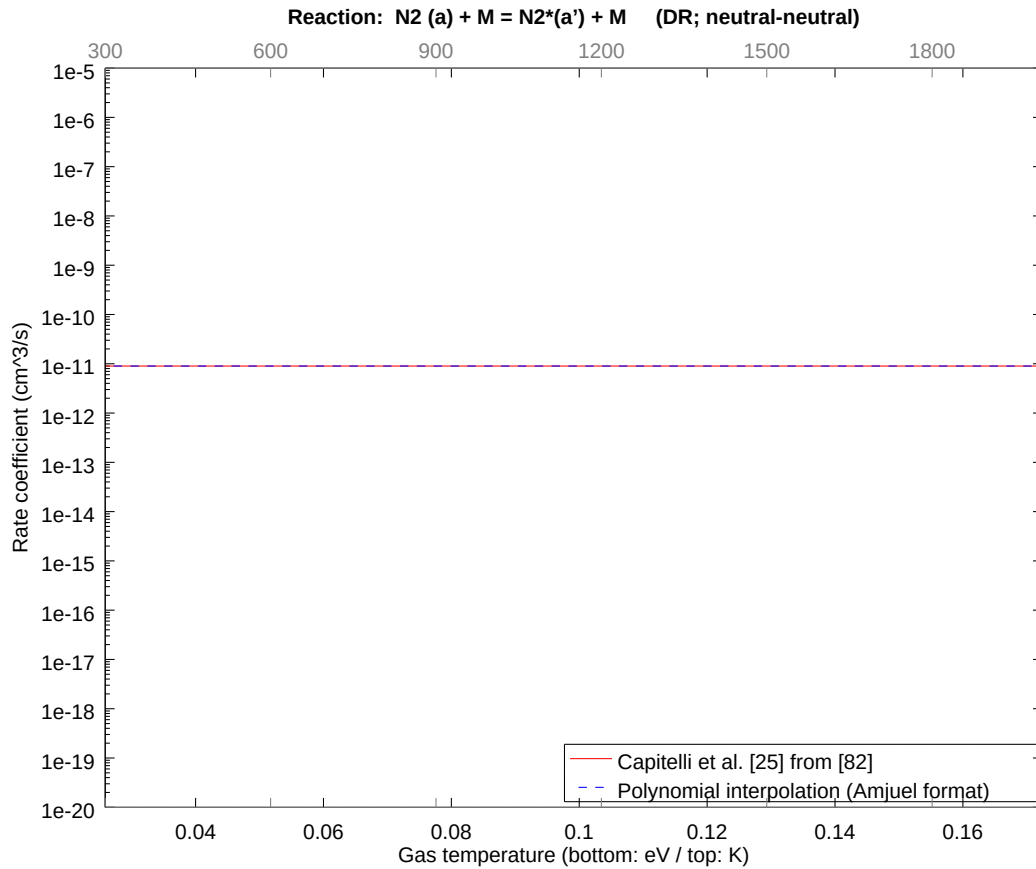
2.167 Reaction DQ $\text{N}_2^*(a') + \text{M} = \text{N}_2^*(a) + \text{M}$

b0	-2.491334966794E+01	b1	5.551937465612E-02	b2	-2.123590632336E-01
b3	-9.750820211534E-02	b4	-7.347892909455E-02	b5	-2.293993689667E-02
b6	-5.751763415969E-03	b7	-7.323647887183E-04	b8	-5.447601843037E-05



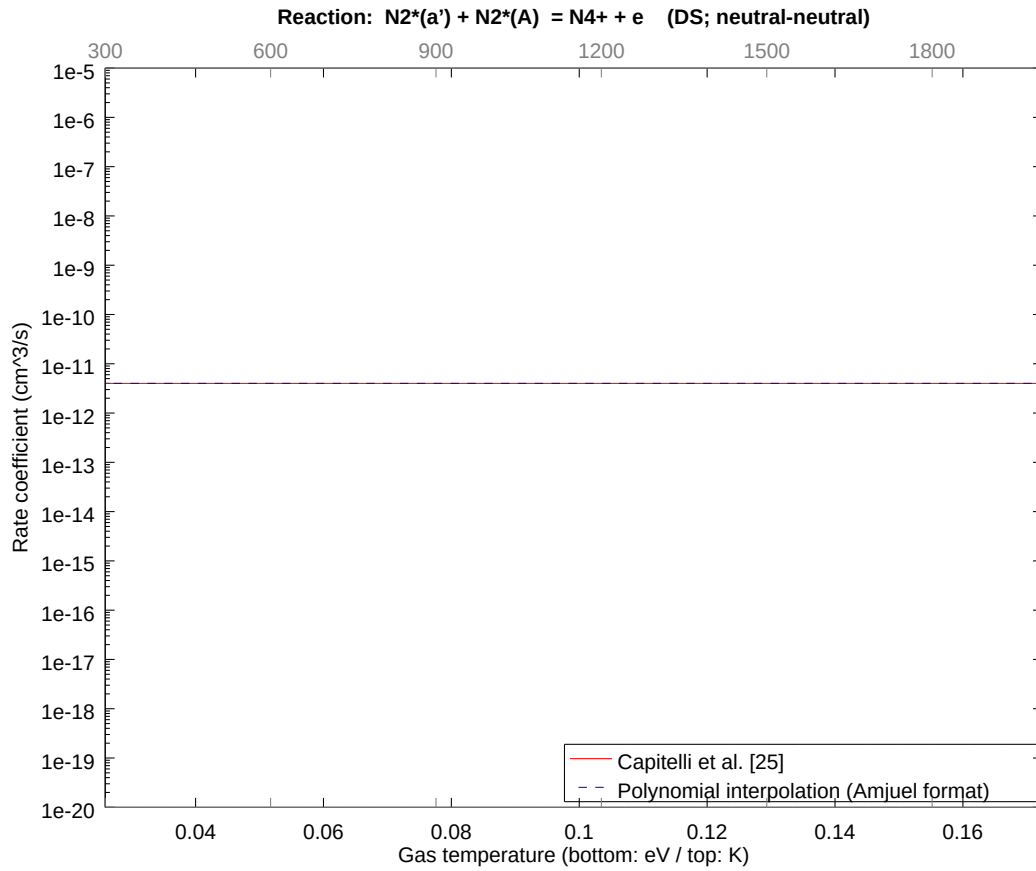
2.168 Reaction DR N2 (a) + M = N2*(a') + M

b0	-2.543379653859E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



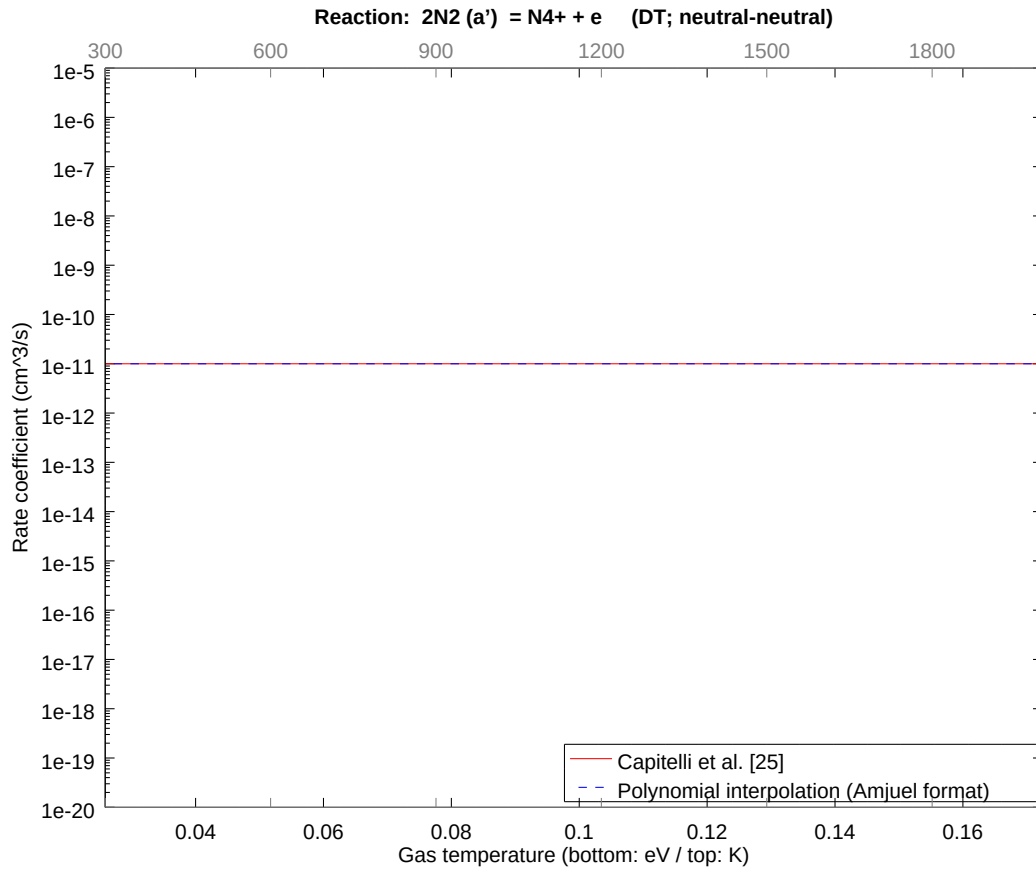
2.169 Reaction DS $\text{N2}^*(\text{a}') + \text{N2}^*(\text{A}) = \text{N4}^+ + \text{e}$

b0	-2.624472675481E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



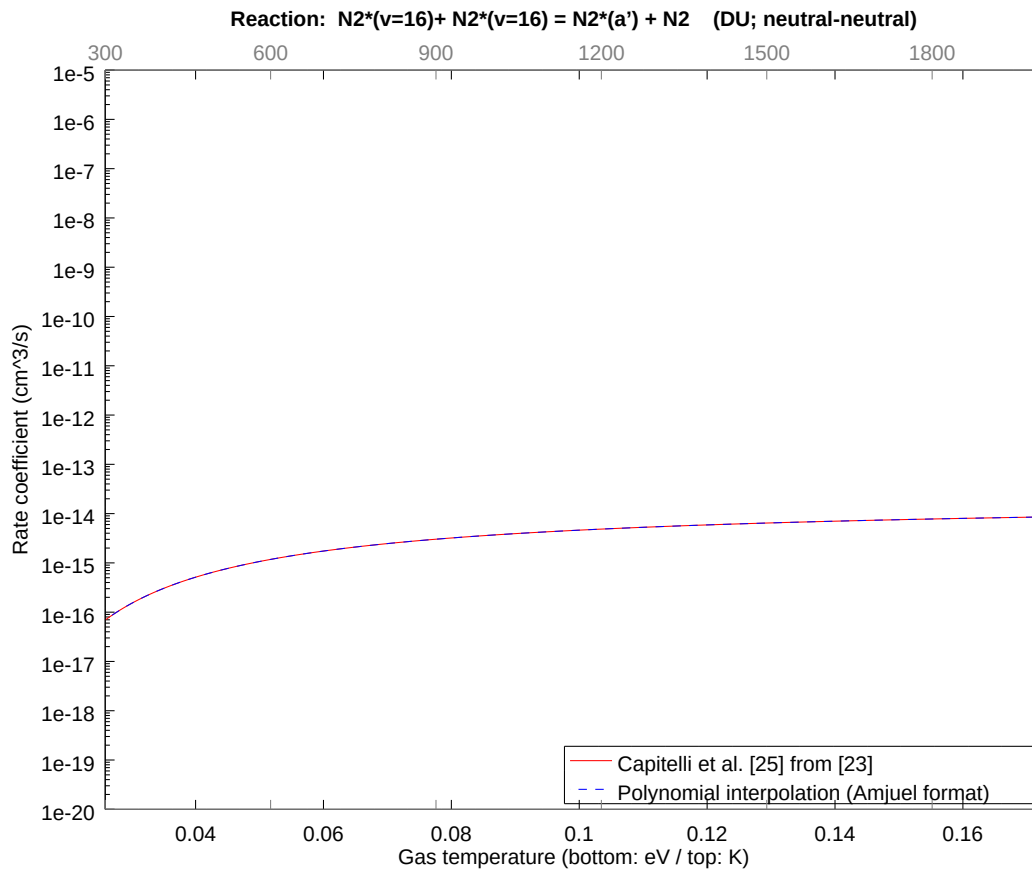
2.170 Reaction DT $2\text{N}_2(\text{a}') = \text{N}_4^+ + \text{e}$

b0	-2.532843602293E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



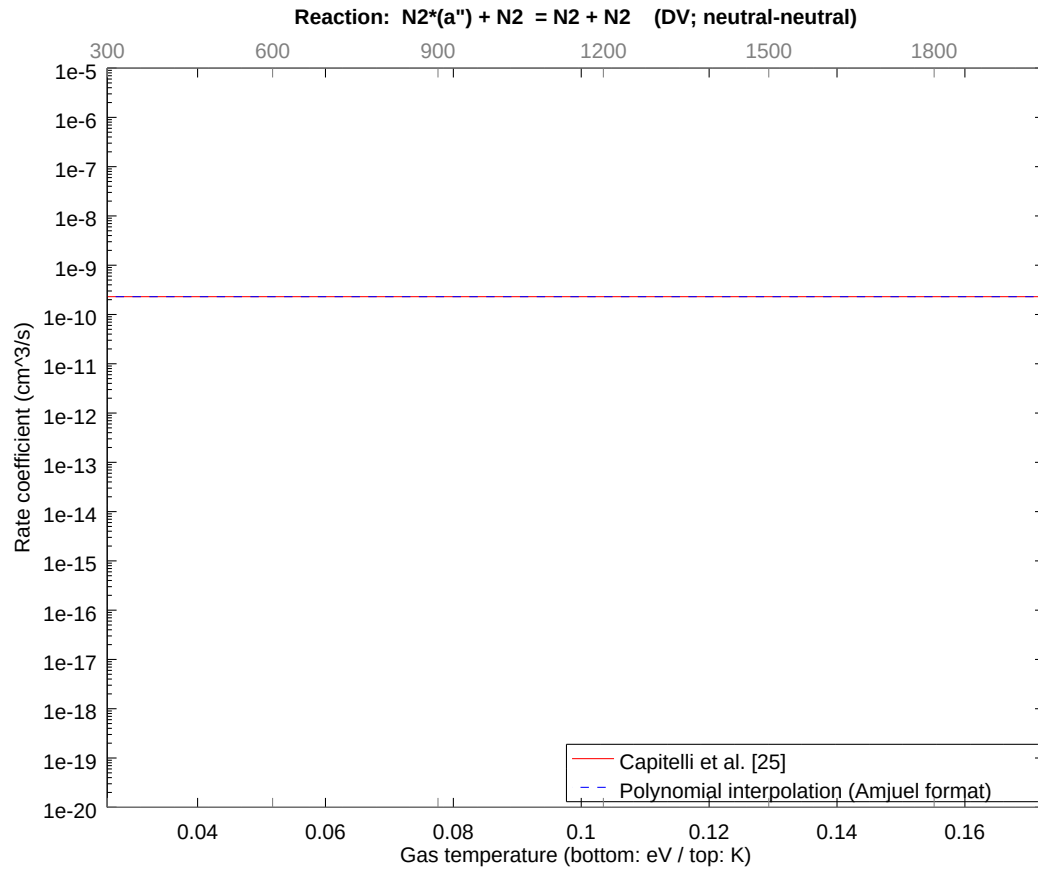
2.171 Reaction DU $\text{N}_2^*(v=16) + \text{N}_2^*(v=16) = \text{N}_2^*(a') + \text{N}_2$

b0	-3.171574443248E+01	b1	5.551937076721E-02	b2	-2.123590686304E-01
b3	-9.750820635699E-02	b4	-7.347893115968E-02	b5	-2.293993753453E-02
b6	-5.751763538053E-03	b7	-7.323648019616E-04	b8	-5.447601905402E-05



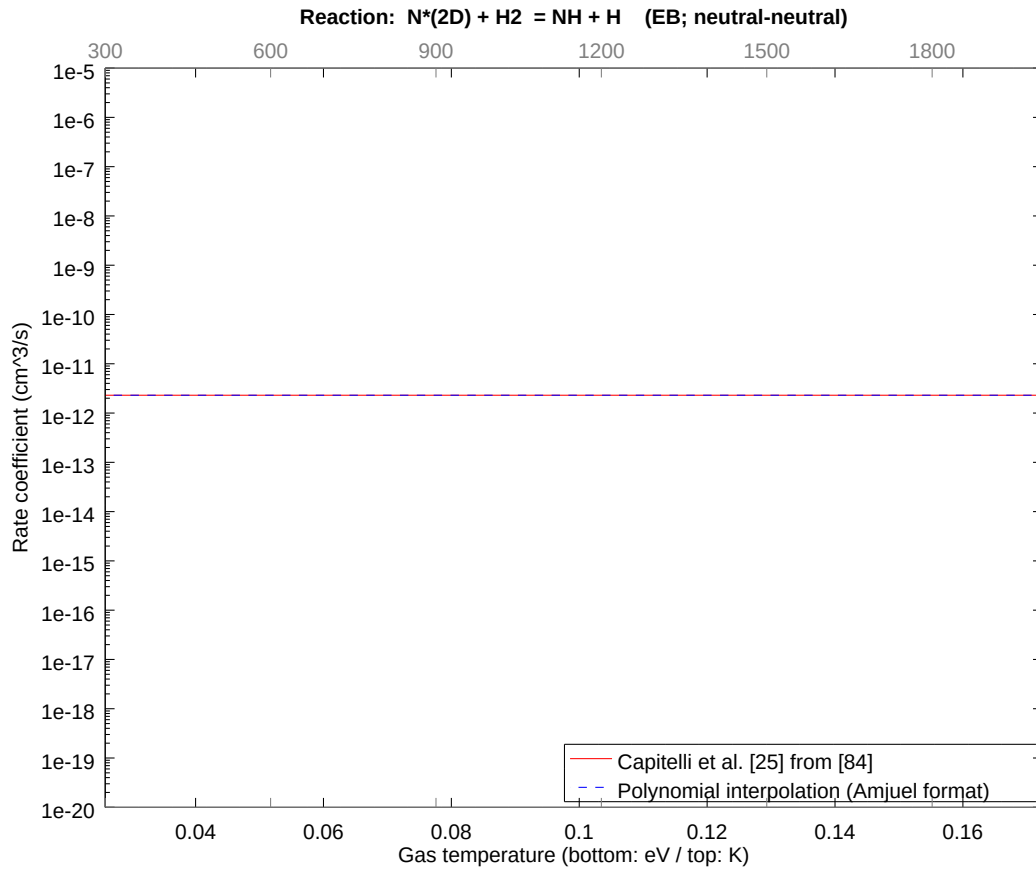
2.172 Reaction DV $\text{N}_2^*(a'') + \text{N}_2 = \text{N}_2 + \text{N}_2$

b0	-2.219294180701E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



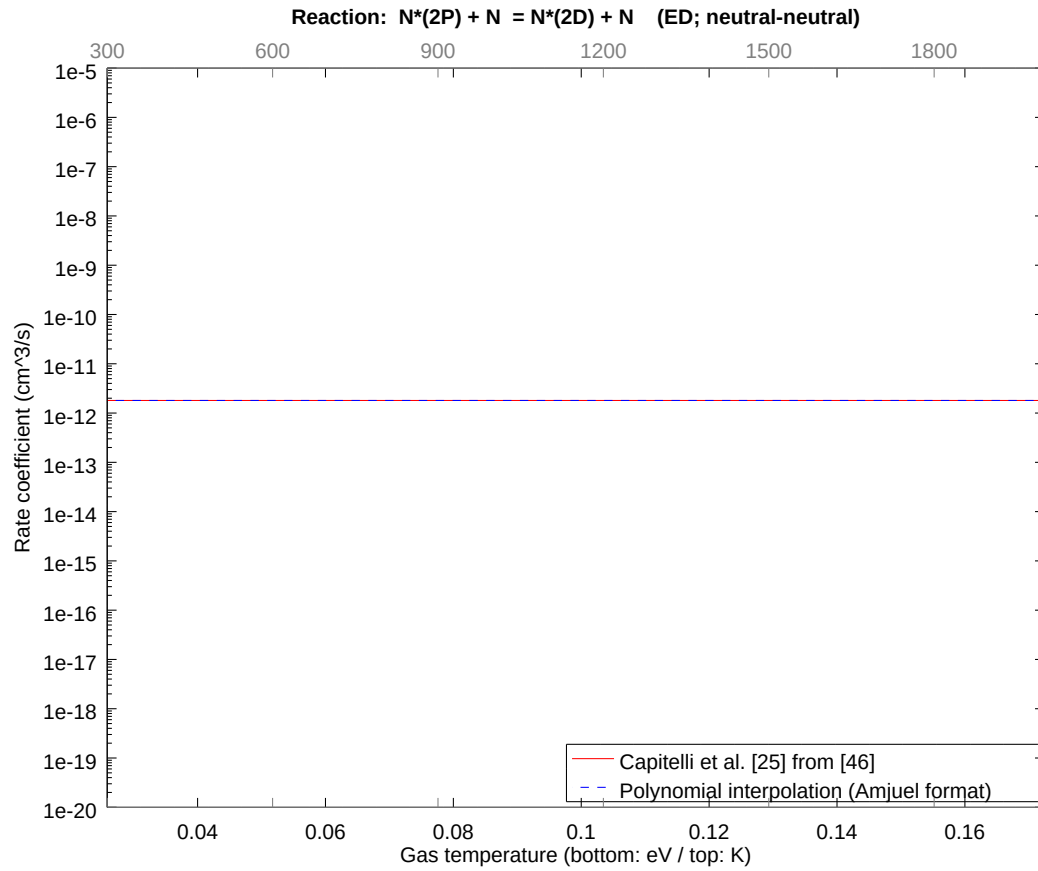
2.173 Reaction EB $\text{N}^*(2\text{D}) + \text{H}_2 = \text{NH} + \text{H}$

b0	-2.679811199299E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



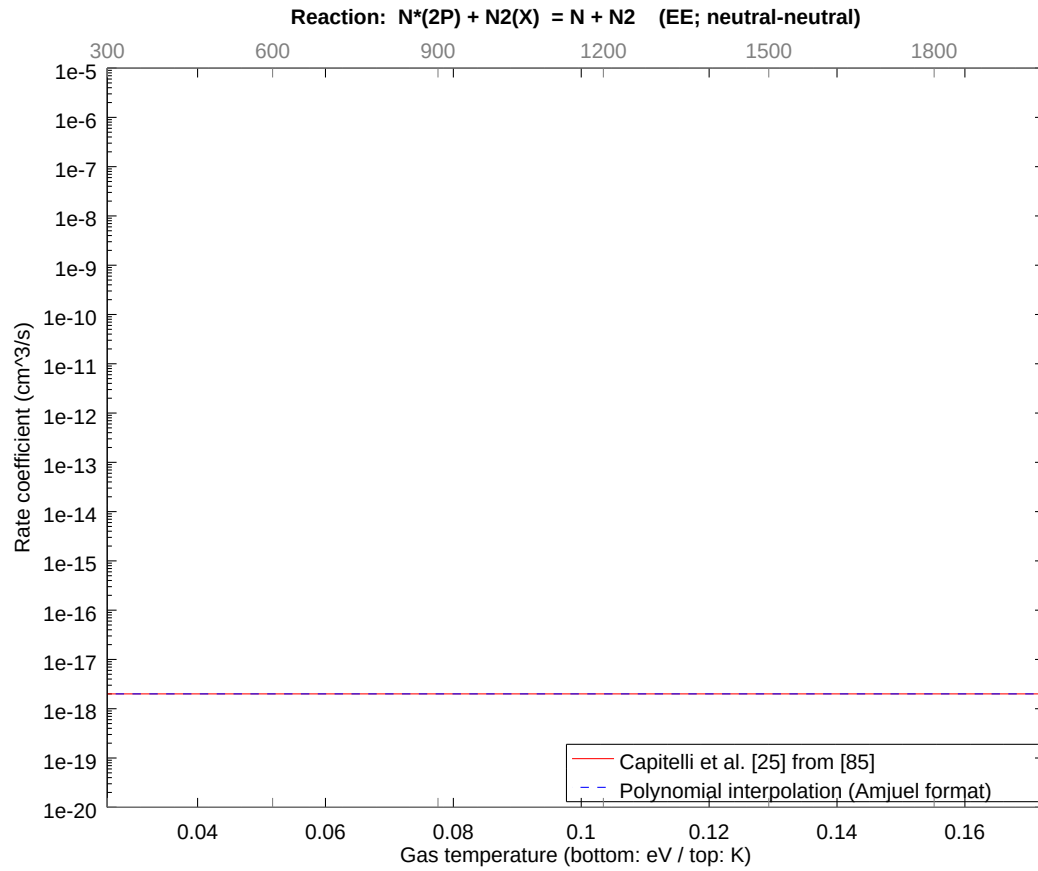
2.174 Reaction ED $\text{N}^*(2\text{P}) + \text{N} = \text{N}^*(2\text{D}) + \text{N}$

b0	-2.704323445103E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



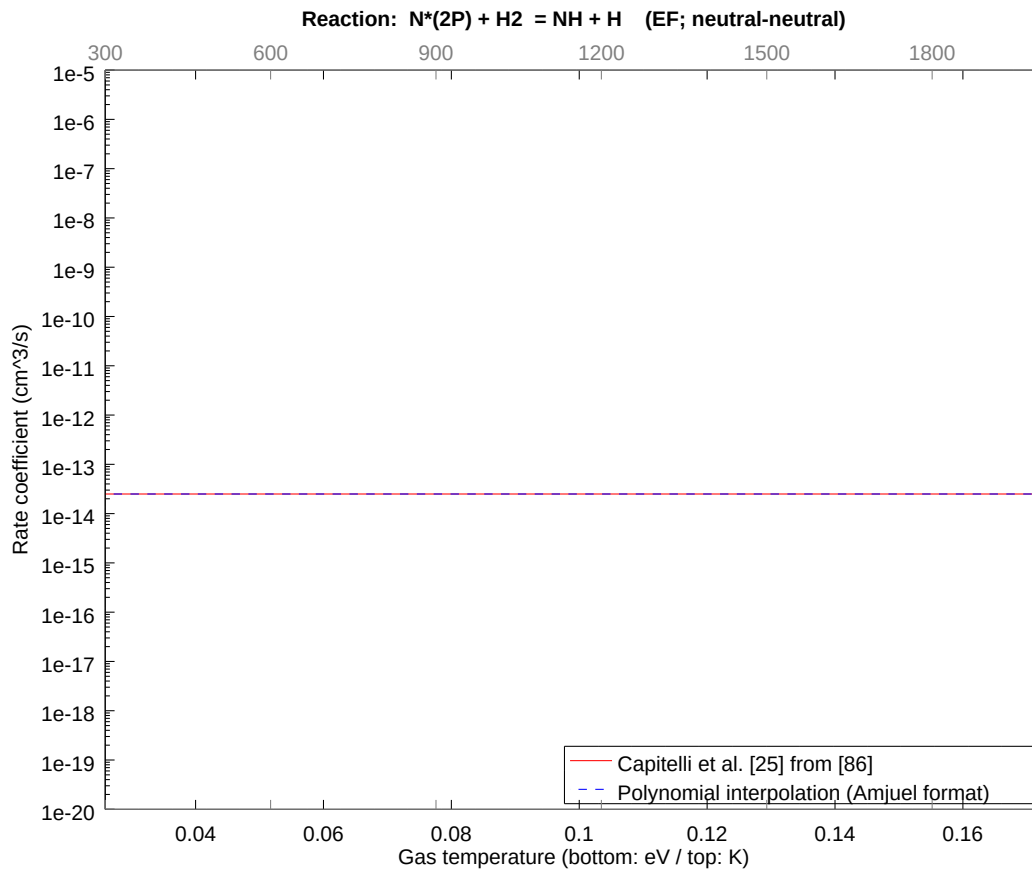
2.175 Reaction EE $\text{N}^*(2\text{P}) + \text{N}_2(\text{X}) = \text{N} + \text{N}_2$

b0	-4.075338449333E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



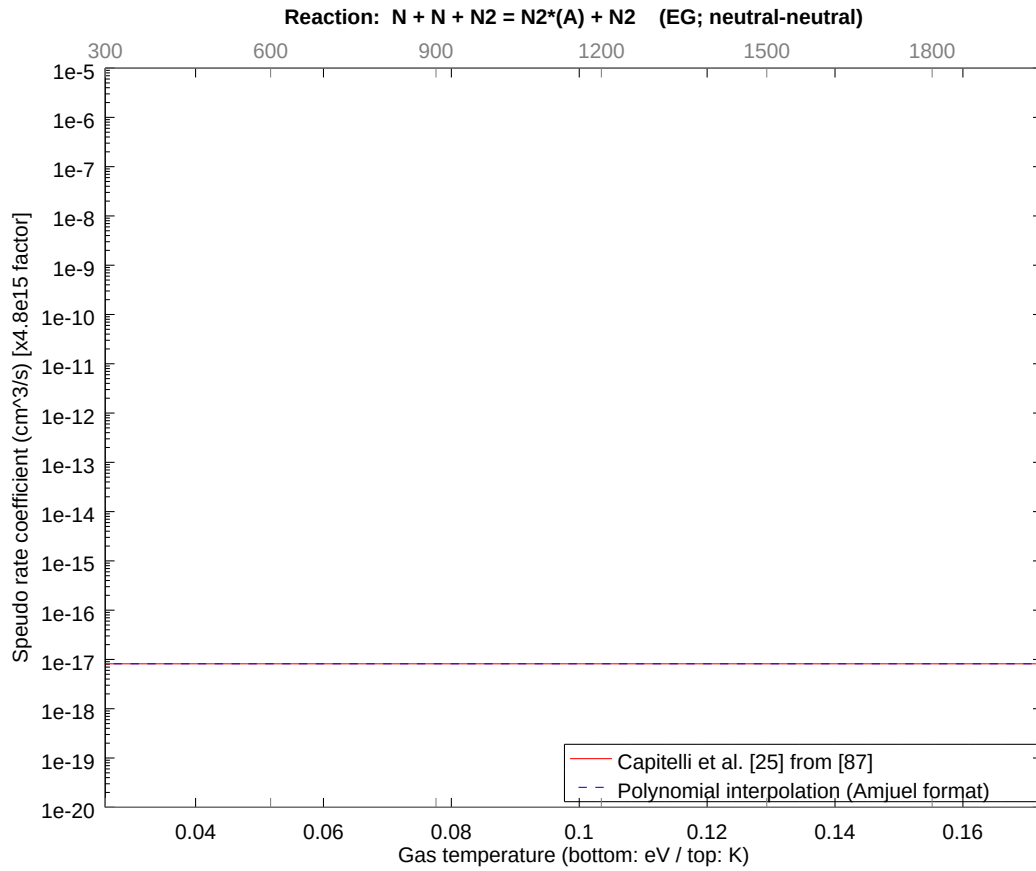
2.176 Reaction EF $\text{N}^*(2\text{P}) + \text{H}_2 = \text{NH} + \text{H}$

b0	-3.131990057004E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



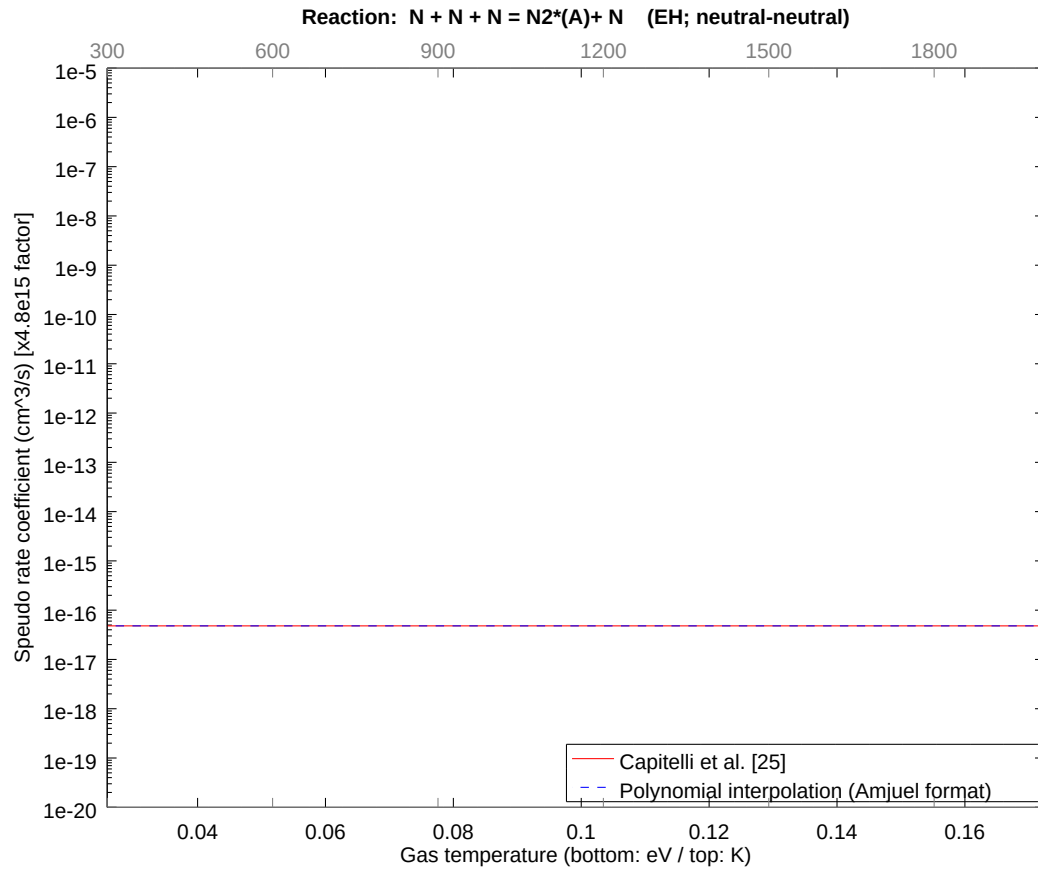
2.177 Reaction EG $\text{N} + \text{N} + \text{N}_2 = \text{N}_2^*(\text{A}) + \text{N}_2$

b0	-3.934728750492E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



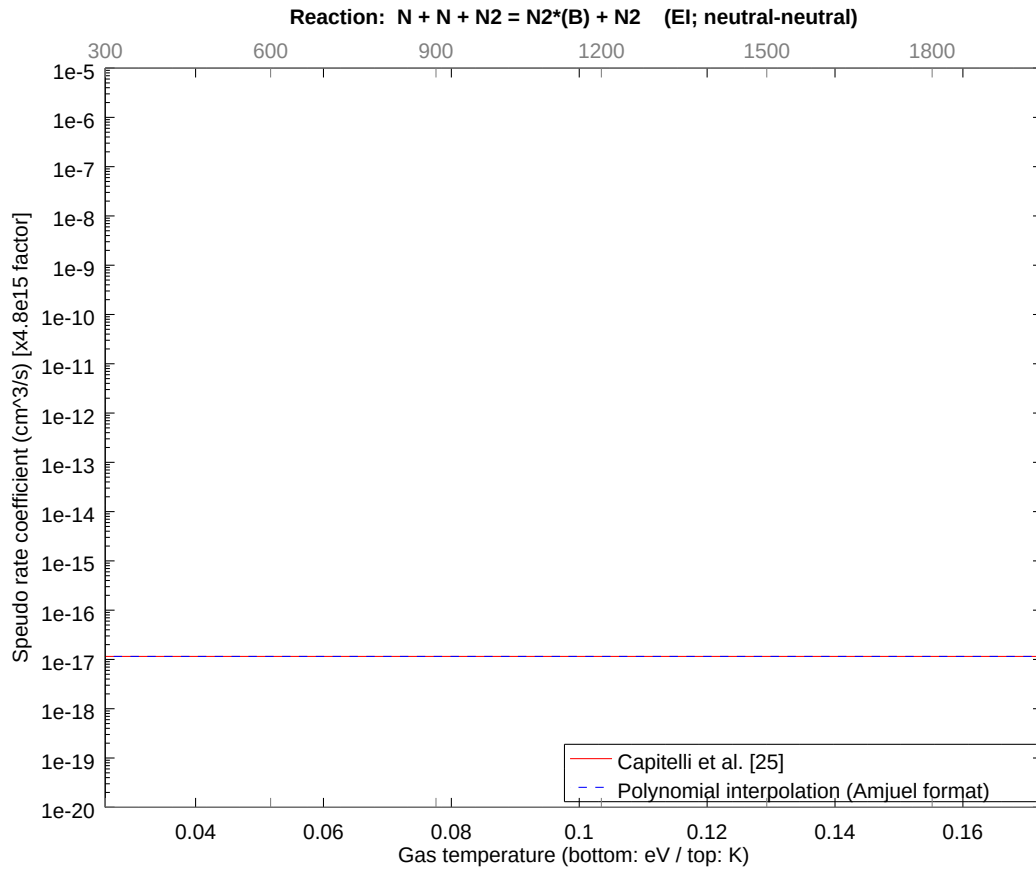
2.178 Reaction EH $\text{N} + \text{N} + \text{N} = \text{N}_2^*(\text{A}) + \text{N}$

b0	-3.757533066298E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



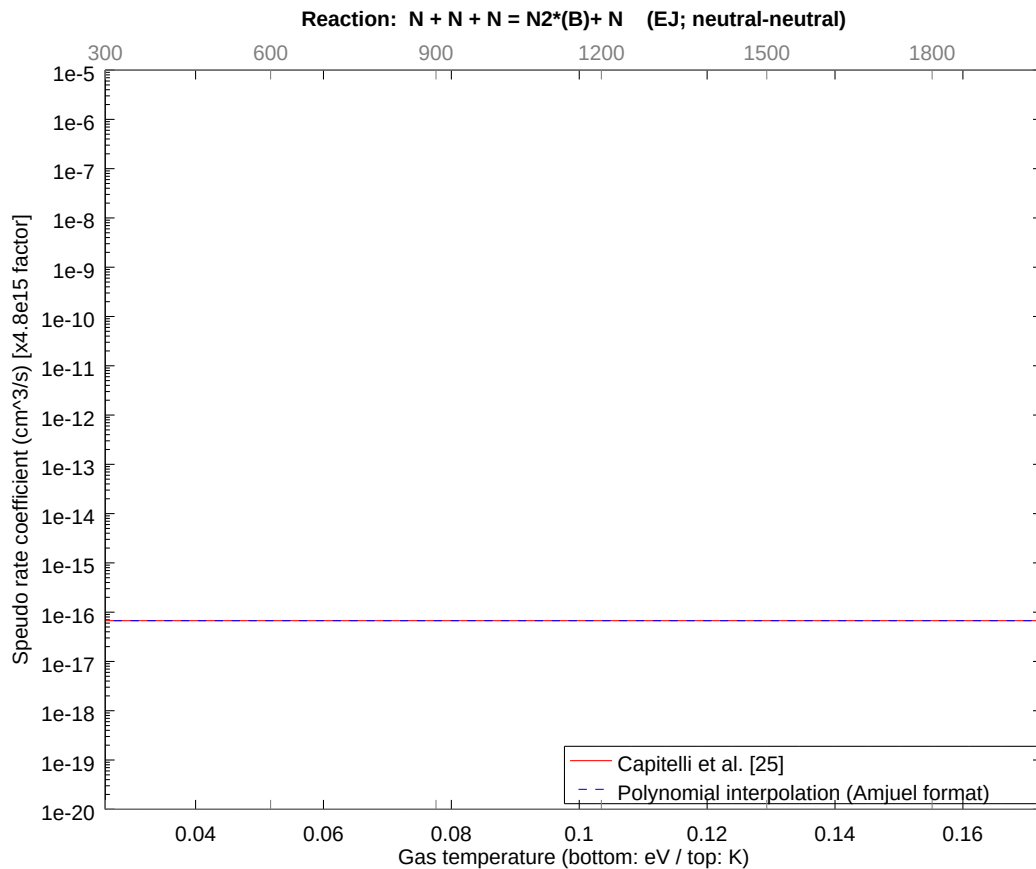
2.179 Reaction EI N + N + N2 = N2*(B) + N2

b0	-3.900244701863E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



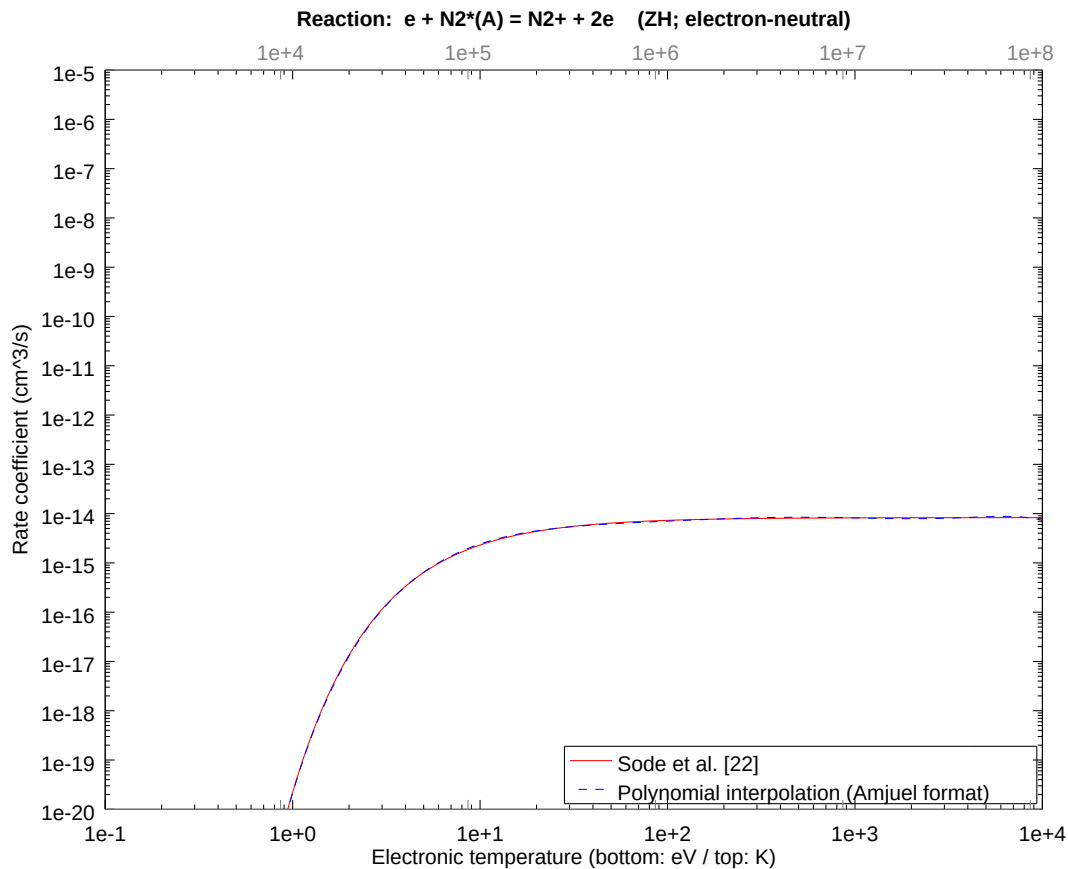
2.180 Reaction EJ $\text{N} + \text{N} + \text{N} = \text{N}_2^*(\text{B}) + \text{N}$

b0	-3.723885842636E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



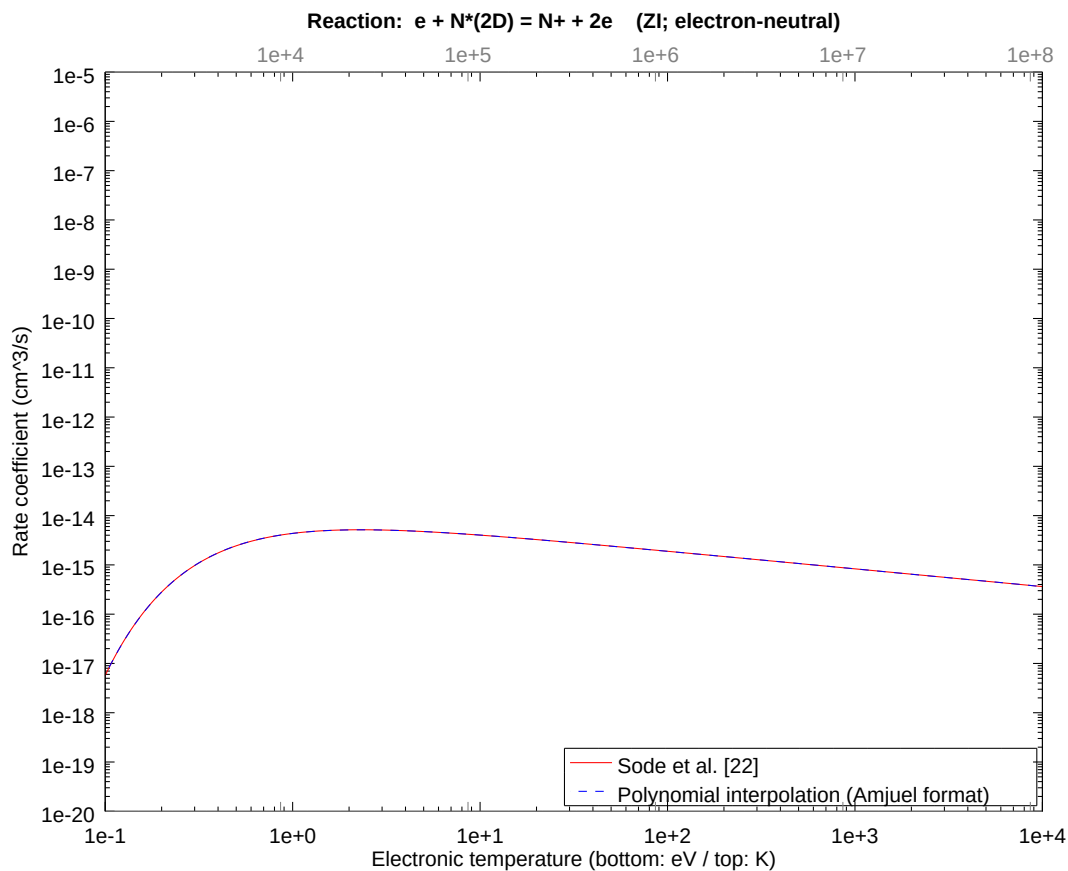
2.181 Reaction ZH $e + N_2^*(A) = N_2 + 2e$

b0	-4.528713119350E+01	b1	1.272883456482E+01	b2	-6.341281325376E+00
b3	2.198832086232E+00	b4	-5.790429011869E-01	b5	1.085334884145E-01
b6	-1.291564834564E-02	b7	8.508444500433E-04	b8	-2.337785141424E-05



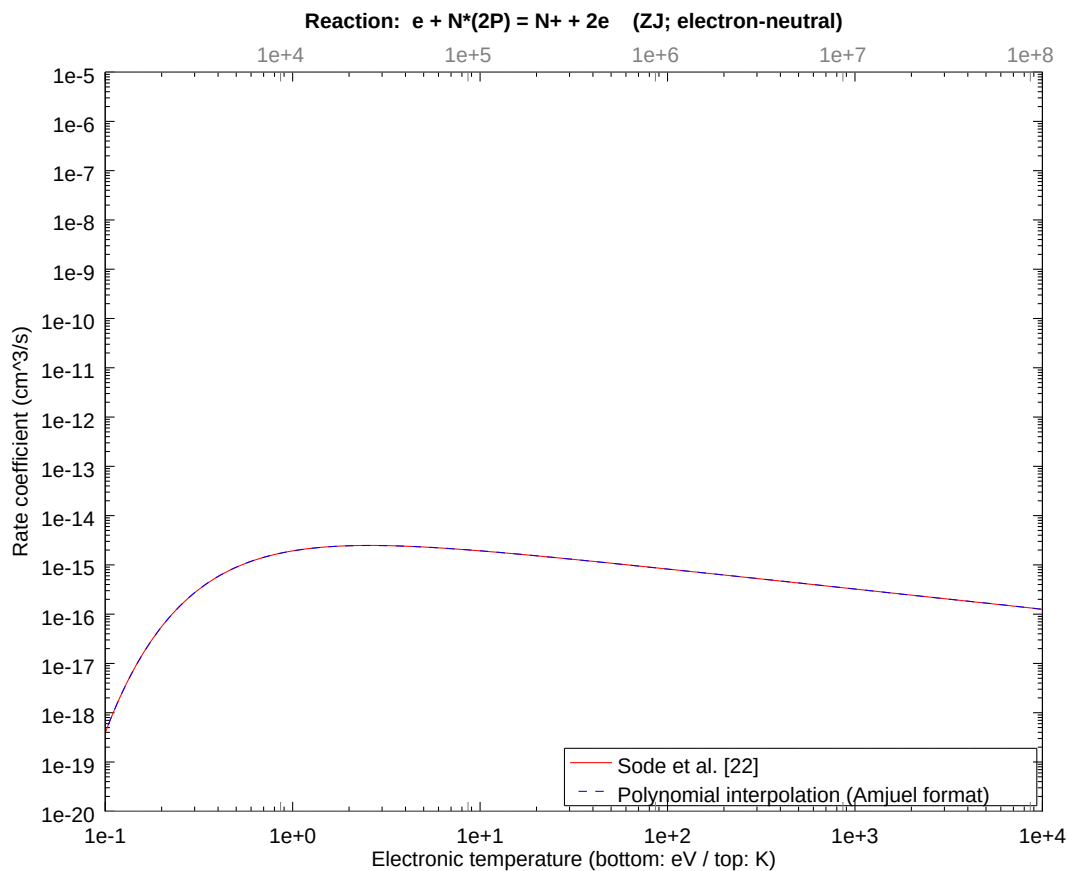
2.182 Reaction ZI $e + N^*(2D) = N + 2e$

b0	-3.306778215551E+01	b1	4.628140723366E-01	b2	-4.099114875437E-01
b3	1.421363420228E-01	b4	-3.743034330102E-02	b5	7.015794033029E-03
b6	-8.348900410347E-04	b7	5.500006959003E-05	b8	-1.511185099207E-06



2.183 Reaction ZJ $e + N^*(2P) = N + 2e$

b0	-3.388604082830E+01	b1	6.309093686186E-01	b2	-5.185627252059E-01
b3	1.798110350890E-01	b4	-4.735163911576E-02	b5	8.875402089977E-03
b6	-1.056186196490E-03	b7	6.957840128859E-05	b8	-1.911740185743E-06



3 Appendix

References

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